

DESIGN REPORT - THE AXOLOTL

(UNIVERSITY AT BUFFALO)



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UB University
at Buffalo

ASME **EFx**

ASME **IAM3D**



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TABLE OF CONTENTS

.....	1
.....	1
.....	1
Table of Contents.....	1
Team organization.....	1
Leadership	2
Group 1 - Chassis.....	2
Group 2 - Movement 1.....	2
Group 3 - payload.....	2
Group 4 - movement 2	2
Electronics team	2
1: Design Requirements and Constraints	13
1.1: Competiiton-Imposed Constraints	13
1.2: Payload Requirements and constraints.....	13
1.2.1: Single-Object Upper Bounds	13
1.2.1.1: Payload Object Constraints	13
1.2.1.2: Payload Object Calculations	13
2: Design Process and rapid prototyping	14
2.1: Overall design methodology	14
2.1.1: System decomposition and Team organization.....	14
2.1.2: Concept Generation and selection	14
2.2: Subsystem level iterative design.....	15
2.3: rapid prototyping strategy	15
3: Assembly.....	16
3.1: exploded view of assembly	16
3.1.1: Movement.....	16
3.1.1.1: Guided sprocket	17
3.1.1.2: Driving sprocket.....	17
3.1.1.3: Driving motor.....	18
3.1.1.4: Tread links.....	18
3.1.1.5: Tread pins	19
3.1.1.6: Sprocket bearing.....	19
3.1.1.7: Inner Driving gear.....	20

3.1.2: Payload.....	21
3.1.2.1: Scoop	21
3.1.2.2: Bearing	22
3.1.2.3: Bearing pin small	22
3.1.2.4: arm	23
3.1.2.5: servo.....	23
3.1.2.6: Long bearing pin	24
3.1.2.7: Long bearing.....	24
3.1.2.8: Base plate	25
3.1.3: Chassis	25
3.1.3.1: Inside Chassis.....	26
3.1.3.2: Outer Chassis.....	26
3.1.3.3: Battery Holder	26
3.1.3.5: Second Floor Battery Holder	26
3.1.3.6: Diagonal Cross bar support	27
3.1.3.7: Crossbar support	28
3.1.3.8: AG Bolt	29
3.1.3.9: AG Nut.....	29
3.1.3.10: Bottom Electrical Plate	30
3.1.3.11: Top Electrical Plate.....	31
3.1.4: Camera.....	31
3.1.4.1: Gimble Brain Stem.....	32
3.1.4.2: Automotive Inspired Snap Fit Connections.....	32
3.1.4.3: Friction Pins	33
3.1.4.4: Lower Gimble Brace.....	33
3.1.4.5: POV Camera	33
3.1.4.6: 40mm Box Axial Fan.....	34
3.1.4.7: Camera Gimble Base	34
3.1.4.8: MG90s Servos	35
4.1.4.9: Upper Gimble Brace.....	35
4.1.4.10: 30mm Box Axial Fan	35
4.1.4.11: 40mm Box blower Fan	36
3.2: Final Design Summary and Overview	36
3.2.1: Final geometry	36

3.2.2: FINAL Statistics	37
3.2.3: Final Justification	37
3.3: Assembly Procedure	37
3.3.1 Modularity	37
3.3.2 Frequently Replaced Parts and Rationale	38
3.3.3 Process to Replace Parts	38
3.3.4 Final Assembly	38
4: Movement design	39
4.1: Track Design	39
4.1.1: Track Overview	39
4.1.2: Track links Design	40
4.1.2.1: LINK TERMINALS	40
4.1.3: Link Pins	42
4.1.3.1: Snap-Fit Connectors	42
4.1.3.2: Threaded Pins	42
4.1.3.3: Friction-Fit Pins	42
4.1.4: Tread Texture	43
4.1.4.1: Initial designs	43
4.1.4.2: Final designs	45
4.1.4.3: texture performance data	45
4.2: Bearing Design	47
4.2.0: Overview	47
4.2.1: Use of Parametrization	48
4.2.1.1: variables affecting bearing geometry	48
4.2.1.2: visual Mathematical Models	48
4.2.1.3: Parametrization in Fusion	48
4.2.2: Bearing variants	48
4.2.3: Bearing test processes and procedures	49
4.2.4: BEARING Challenges	49
4.2.4.1: roller count determination	49
4.2.4.2: Tolerance through constraints	49
4.4: Sprocket Design	50
4.4.1: What are Sprockets	50
4.4.2: challenges	50
4.4.2.1: the challenge of loose sand	50

4.4.2.2: How We Approached Loose Sand	50
4.4.2.3: Designing Around Loose Sand	51
4.4.3: sprocket design and Parametrization	51
4.4.4: sprocket variants	52
4.4.4.1: Driving sprocketS	52
4.4.4.2: Guide sprockets	53
4.4.4.3: Roller sprockets	53
4.4.4.4: IDLER/Tensioning sprocketS	53
4.4.5: Sprocket rationale	53
4.4.5.1: Sprocket setup and placement	53
4.4.5.2: anti sand mechanism and clog prevention	54
5: Payload	54
5.1 Scoop	54
5.1.1: Excavation Method and Reasoning	54
5.1.2: Initial Design	55
5.1.1.1: PROBLEMS with FIRST ITERATION	55
5.1.1.2: SOLUTIONS	55
5.1.3: Updated FOR Aerodynamics	55
5.1.3.1: PROBLEMS	56
5.1.3.2: SOLUTIONS	56
5.1.4: Removal of side walls	56
5.1.4.1: PROBLEMS	56
5.1.4.2: SOLUTIONS	57
5.1.5: Differing angle of attack	57
5.1.5.1: PROBLEMS	57
5.1.5.2: SOLUTIONS	57
5.1.6: GENERATIVE DESIGN	57
5.1.6.1: PROBLEMS WITH CURRENT SCOOP	58
5.1.6.2: BENEFITS OF GENERATIVE DESIGN	58
5.1.6.3: INTEGRATION INTO FINAL DESIGN	58
5.1.7: FINAL DESIGN	58
5.1.7.1: Benefits of current scoop	58
5.1.7.2: JUSTIFICATION OF SCOOP	59
5.2 ARM	59
5.2.1 DESIGN REQUIREMENTS	59

5.2.1.1: DESIGN CONSTRAINTS	60
5.2.1.2: DESIGN OBJECTIVES	60
5.2.2: DESIGN CONSIDERATIONS	60
5.2.2.1: First iteration	61
5.2.2.2: second iteration	61
5.2.2.3: final iteration	62
5.2.3: Generative Design	62
5.2.3.1: PROBLEMS WITH CURRENT ARM	62
5.2.3.2: BENEFITS OF GENERATIVE DESIGN.....	62
5.2.3.3: integration into final design	62
5.2.4: CALCULATIONS	63
5.2.4.1: DIAGRAM	63
5.2.4.2: EQUATIONS	64
5.3 Servo Motor Integration	64
5.3.1: MOUNTING	64
5.3.1.1: Chassis Mount	64
5.3.1.2: Scoop mount.....	65
5.3.2: Torque and mechanical interface	65
5.3.2.1: Torque.....	65
5.3.2.2: Mechanical interface.....	66
5.3.3: SERVO CHOICE AND CALCULATED FORCE	66
5.3.3.1: SERVO CHOICE.....	66
5.3.3.2: calculated force	66
6: Chassis	67
6.1: chassis reasoning.....	67
6.2: Chassis strength	67
6.3: Chassis crossmember properties	67
6.4: Design changes - crossmembers, plates, shelving system	68
6.5: chassis connections, screws, pins, etc	68
6.6: bracket	69
6.6.1: early bracket design	69
6.6.2: early issues & changes	69
6.7: Track-Enclosed Battery Container	69
6.7.1: Battery Container Reasoning.....	69
6.7.2: Battery Container Parametrization	69

6.7.3: Hooks	69
6.7.3.1: Why We Separated Battery Container Hooks from Chassis	69
6.7.3.2: Initial Designs	70
6.7.3.3: WHY WE used hooks	70
6.7.4: Battery Container Design	71
6.7.4.1: Lightweight Design	71
6.7.4.2: Overhangs and geometry of Container Holes	71
7: Camera	71
7.1: Camera Gimbal	71
7.1.0: Reasoning for implementation	71
7.1.1: Use case and Overview	72
7.1.1.1: Camera Gimbal Inspiration	72
7.1.1.2: Final Camera Gimbal Design	72
7.1.2: Use of Generative Design to Mitigate Vibrations	72
7.1.2.1: Vibrations and their Effects on the Camera System	72
7.1.2.2: Approach and Mitigation Strategies to Vibration	73
7.1.2.3: Application of Generative Design Towards Mitigating Vibration	73
7.1.2.4: Testing and FEA Results	73
7.1.3: Control	73
8: Electronics	75
8.1: Component Selection	75
8.1.1: Selection Criteria	75
8.1.2: Component Selection Table	76
8.1.3: Component Calibration and Testing	77
8.2: Diagrams	78
8.2.1: Final Electronic Design	78
8.2.1.1: Final Electrical Schematic	78
8.2.1.2: Final PCB (2d)	79
8.2.1.3: Final PCB (3D)	80
8.2.1.4: Electrical Diagram (2D, SIMPLIFIED)	81
8.2.1.5: Electrical Diagram Including PCB (3D)	83
8.3: Electronic Challenges	84
8.3.1: Electronics Budget Issues	84
8.3.2: Transmitter Issues	84
8.3.3: Autorouting Limitations	84

9: DFAM	85
9.1: DFAM INTRODUCTION	85
9.1.1: WHY DFAM MATTER FOR OUR ROVER	85
9.1.2: Lab Equipment	85
9.1.2.1: Available 3D printers.....	85
9.1.2.2: machining Tool Availability	86
9.1.2.3: HANDHELD Tools	86
9.1.2.4: Post Processing capabilities.....	86
9.1.2.5: Applications	87
9.2: Working Around Layer Lines	87
9.2.1: print orientation.....	87
9.3: working with Supports and Material Optimization.....	88
9.3.1: SUPPORT	88
9.3.1.1: Support Structure.....	88
9.3.1.2: Overhang Reduction.....	88
9.3.2: MATERIALS	88
9.3.2.1: Material strengths	88
9.3.2.2: Print settings	88
9.1.3.1: Tolerances	89
10: DFMA	89
10.1: Introduction to DFMA	89
10.1.1: DFA	89
10.1.2: DFM	89
10.2: DFMA and our Rover	89
10.2.1	90
11: Testing.....	91
11.1: Testing incentives/reasons/why we tested	91
11.2: How We Tested	91
11.2.1: Sandbox.....	91
11.2.2: Rigs.....	92
11.2.2.1: SCOOP TESTING RIG.....	92
11.2.2.2: Component Temperature Testing Rig	92
11.2.2.3: General Electronics Testing Rig	92
11.3: Materials Testing Results and Final Material Selection.....	93
11.3.0: Material Selection Criteria	93

11.3.1: Tread Link Material Testing	93
11.3.2: Tread Link Connector Pin Material Testing	94
11.3.3: Battery Holder Material Testing	95
11.3.4: Drive Sprocket Material Testing	96
11.3.5: Drive Sprocket Insert Material Testing	97
11.3.6: Final Snap-Fit Bearing Material Testing	99
11.3.7: Outer Sandwich Tread Chassis Bracket Material Testing.....	100
11.3.8: Inner Sandwich Tread Chassis Bracket Material Testing	102
11.3.9: Guide Sprocket Material Testing	104
11.3.10: Top Electronics Shelf Material Testing.....	104
11.3.11: Middle Electronics Shelf Material Testing	106
11.3.12: Bottom Electronics Shelf Material Testing.....	108
11.3.13: CHASSIS HOLDING PINS Material Testing	109
11.3.14: camera Gimbal Material Testing	110
11.3.15: RIGID HORIZONTAL CHASSIS CROSS-BAR Material Testing	111
11.3.16: RIGID DIAGONAL CHASSIS CROSS-BAR Material Testing	112
11.3.17: Scoop Material Testing	114
11.3.18: ARM TO CHASSIS MOUNT Material Testing	115
11.3.19: Arm Material Testing	117
11.3.20: Inventoried Structural Filament Properties Data Table	118
11.3.21: Final Material Selection	119
11.4.1: Cross-Member Testing	120
11.4.1.1: Horizontal Cross-Member Testing.....	120
11.4.1.2: Diagonal Cross-Member Testing	121
11.4.1.3: Bolt connection testing.....	122
11.4.2: Bracket Testing.....	123
11.4.2.1: Inner Bracket Testing.....	123
11.4.2.2: Outer Bracket Testing	124
11.4.3: Chassis Rack Testing	125
11.4.3.1: Top Rack/Payload Mount Testing	125
11.4.3.2: Mid Rack Testing	126
11.4.3.3: Bottom Rack Testing	127
11.5: MOvement Testing Results	129
11.5.1: Bearing Testing	129

11.5.1.1: Bearing Roller Count Testing	129
11.5.1.2: Bearing Load Testing (Snapping/X Axis)	130
11.5.2: Tread Testing Results.....	131
11.5.3: Motor Testing Results.....	131
11.5.3.1: Current versus load torque	131
11.5.3.3: Power Consumption vs load.....	132
11.6: Payload Testing Results	134
11.6.1: Scoop Testing	134
11.6.1.1: Scoop Testing Methods	134
11.6.1.2: Scoop Testing Data.....	134
11.6.1.3: Scoop Interpretation.....	135
11.6.2: Arm Testing	135
11.7: Electronics Testing Results	136
11.7.1: Electronics Temperature Testing	136
11.7.1.1: Electronic Temperature Testing Reasoning	136
11.7.1.2: Electronic Temperature Testing Data	136
11.7.2.3: Electronic Temperature Testing Interpretation	138
11.7.2: Camera Transmitter Range Testing.....	138
11.7.2.1: Camera Transmitter Range Testing Approach.....	138
11.7.2.2: Camera Transmitter Range Testing Data	139
11.7.3: Camera Gimbal Thermal Testing.....	140
11.7.3.1: Camera Gimbal Thermal Testing Methods	140
11.7.3.2: Camera Gimbal Thermal Testing Data	140
11.7.3.3: Camera Gimbal Thermal Testing Intepretation.....	141
11.7.4: Camera Gimbal Cycle Testing	141
11.7.4.1: Camera Gimbal work Cycle Testing Methods and Definition.....	141
11.7.4.2: Camera Gimbal work Cycle Testing Data	142
11.7.4.3: Camera Gimbal work Cycle Interpretation	142
11.8: Testing Challenges.....	143
12: Risk Simulation, assessment, and mitigation.....	144
12.1: Movement Component Simulation.....	144
12.1.1: Movement Simulation Purpose & Relevance.....	144
12.1.2: Movement Simulation by Component	144
12.1.2.1: Bearing Static Stress Simulation	144
12.1.2.2: Tread Pin Static Stress Simulation	146

12.1.2.3: Full Tread Static Stress Simulation	146
12.1.2.4: Sprocket Static Stress Simulation	146
12.2: Payload Component Simulation	147
12.3: Chassis Component Simulation.....	148
12.3.1: Outer bracket static simulation	148
12.3.1.1: outer bracket static simulation method and reasoning.....	148
12.4.1: Horizontal crossmember static simulation	150
12.5.1: Diagonal crossmember static simulation.....	151
12.4: Camera Gimbal Component Simulation.....	153
12.4.1: Camera Gimbal Static Stress Simulations	153
12.4.1.1: Camera Gimbal Brace Static Simulation Method and Reasoning	153
12.4.1.2: Camera Gimbal Brace Static Simulation (PLA+)	154
12.4.1.3: Camera Gimbal Base Verdict	154
12.4.2: Camera Gimbal Modal Simulations.....	154
12.4.2.1: Camera Gimbal Base Modal Simulation Method and Reasoning.....	154
12.4.2.2: Camera Gimbal Base Modal Simulation (95A, PLA+)	155
12.4.2.3: Camera Gimbal Base Modal Simulation (PLA+ Only)	156
12.4.2.4: Camera Gimbal Base verdict	156
13: Report Conclusion	157
13.1: Main Challenges.....	157
13.2: WHAT we would do differently	157

1: DESIGN REQUIREMENTS AND CONSTRAINTS

1.1: COMPETITION-IMPOSED CONSTRAINTS

The ASME IAM3D R.O.V.E.R competition defined a set of rules that emphasized compact system integration, additive manufacturing, and remote operation. They incentivized a push for entirely additively manufactured components, guiding the team towards multifunctional printed structures. These constraints shaped the rover architecture and informed major tradeoffs through the design process.

1.2: PAYLOAD REQUIREMENTS AND CONSTRAINTS

1.2.1: SINGLE-OBJECT UPPER BOUNDS

1.2.1.1: PAYLOAD OBJECT CONSTRAINTS

Maximum size of objects given by rules = 1cm x 1cm x 1cm to 10.16cm x 10.16cm x 10.16cm; of varying materials. There are no restrictions on object material, but the rules did mention that they would be composed of “various materials, rocks, and metals”, meaning that we can assume most are common/household and so we can exclude rare earth materials and aerospace alloys.

1.2.1.2: PAYLOAD OBJECT CALCULATIONS

With maximum resource size being $(10.16\text{cm})^3 = \text{Volume} = 1048\text{cm}^3$, it's expected that we can find the potential maximum size of objects simply by assigning a material to a cube of that volume. However, this is not the case; after going down this path initially, we tried lead at 11.34g/cm^3 * the volume of the cube, and we reached 11.88kg, which is simply not practical in an ASME student competition where most accessible materials are generic polymers and the maximum vehicle size is $(30\text{cm})^3$. Even with less heavy, more common materials like steel (block reaches 8.3kg) and aluminum (2.8kg). Realistically, none of these objects are excavatable, much less liftable by the extremely few hobby-grade servos in our budget. Furthermore, most objects are going to be less than the maximum bounding box and not fully solid; more likely to be a fraction of the density of the metal block from earlier in an irregular shape.

In short, the rulebook does not imply solid metal blocks; therefore, we can assume an average density of 1.0g/cm^3 (~water) and apply that to the maximum payload bounding box to receive an average of ~0.31kg.

2: DESIGN PROCESS AND RAPID PROTOTYPING

The rover was designed using iterative engineering processes. This includes repeated cycles of modeling, prototyping, and testing of ideas and concepts. Given the complexity of the entire rover, it was divided into four major subsystems: movement, payload, chassis, and electronics.

2.1: OVERALL DESIGN METHODOLOGY

The methodology for our entire design process was to break it up into many smaller sections so people could compartmentalize tasks more efficiently. Smaller teams were created to work within themselves and help speed up the overall process.

2.1.1: SYSTEM DECOMPOSITION AND TEAM ORGANIZATION

The rover can be separated into four primary subsystems: Chassis, Movement, Payload, and Electronics subsystems. While the mechanical subsystems were the first to be developed, the electronics subsystem was developed retrospectively to integrate with the assembled mechanical system to complete the final design.

- **Movement:**
The movement of the subsystem is defined by the components organized to allow the rover to traverse the terrain it may encounter. Parts of this subsystem include treads, sprockets, bearings, etc.
- **Payload:**
The payload subsystem is designated as all components used or developed to interact with objects. This includes excavating, picking up, transporting, and depositing objects into the bucket. Parts of this subsystem include arm, scoop, servo mounts, etc.
- **Chassis:**
The chassis subsystem serves as a structural backbone connecting both movement and payload subsystems. Parts of this subsystem include cross bars, chassis side bodies, and various mounts connecting electronics to the rover, such as battery housing, etc.
- **Electronics:**
The electronics subsystem contains every piece of electronics going throughout the rover body and supplies power and controls the rover. Parts of this subsystem include wiring, servos, motors, batteries, camera, fans, etc.

2.1.2: CONCEPT GENERATION AND SELECTION

The concept generation and selection was a critical factor in the design process as it would establish the framework to guide future development. Every aspect of the design process was developed early in the setup as it was necessary to continue the rest of the research and design.

At the very start of the design process, large meetings were held to brainstorm ideas to create ideas relating to rover design and what each member thought was the best approach. To determine the ideal design for the rover, a vote was conducted to select the current designs for the rover. This ended up being a single arm and scoop instrument, tank treads with linked tracks, and chassis framework to support other subsystems.

2.2: SUBSYSTEM LEVEL ITERATIVE DESIGN

Each sublevel iteration is broken down into smaller components to be worked within the sublevel itself. This is to create maximum efficiency while one person works on one part; another person will work on another part within the subsystem.

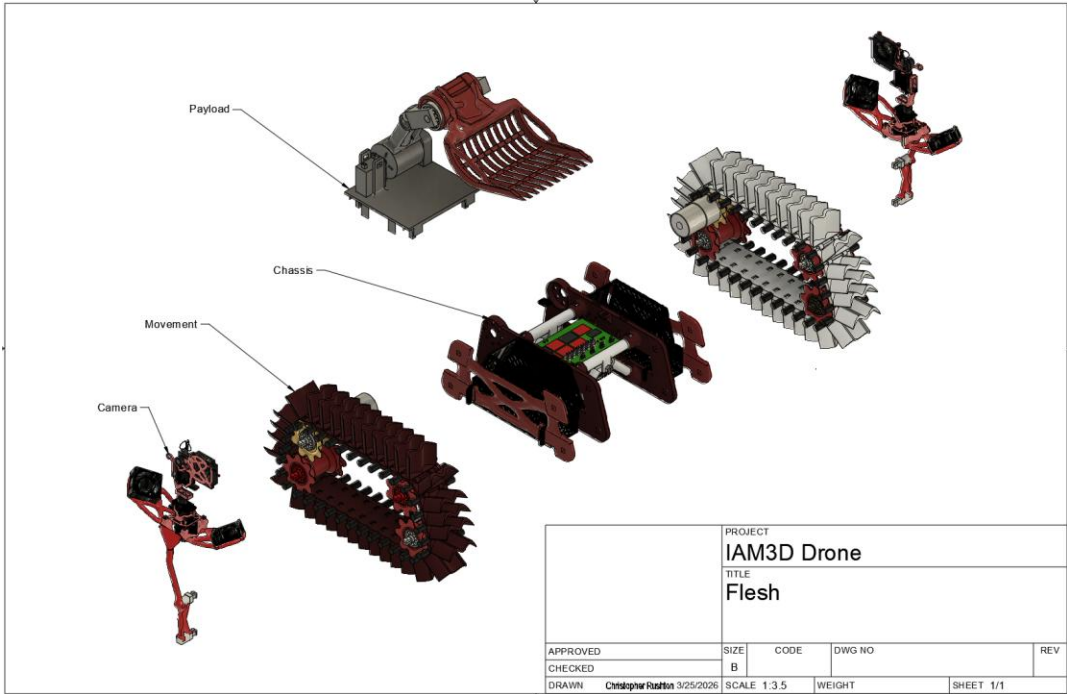
- The movement iteration was made by first designing the tread links. After tread links the sprocket and bearings could be worked independently of each other as these were not related systems but worked together.
- Payload iteration was very easy to break down into smaller parts because of universal connections. Early on we created a universal joint that was to be used between all parts on the arm and allowed us to be able to interchange pieces at will.
- Chassis was designed to be the connection between payload and movement. Because of this, all the designs were based on the finished payload and movement design.
- Electronics were made to wire every electrical component on the rover to achieve its goal. The electronics worked last once every other part was completed.

2.3: RAPID PROTOTYPING STRATEGY

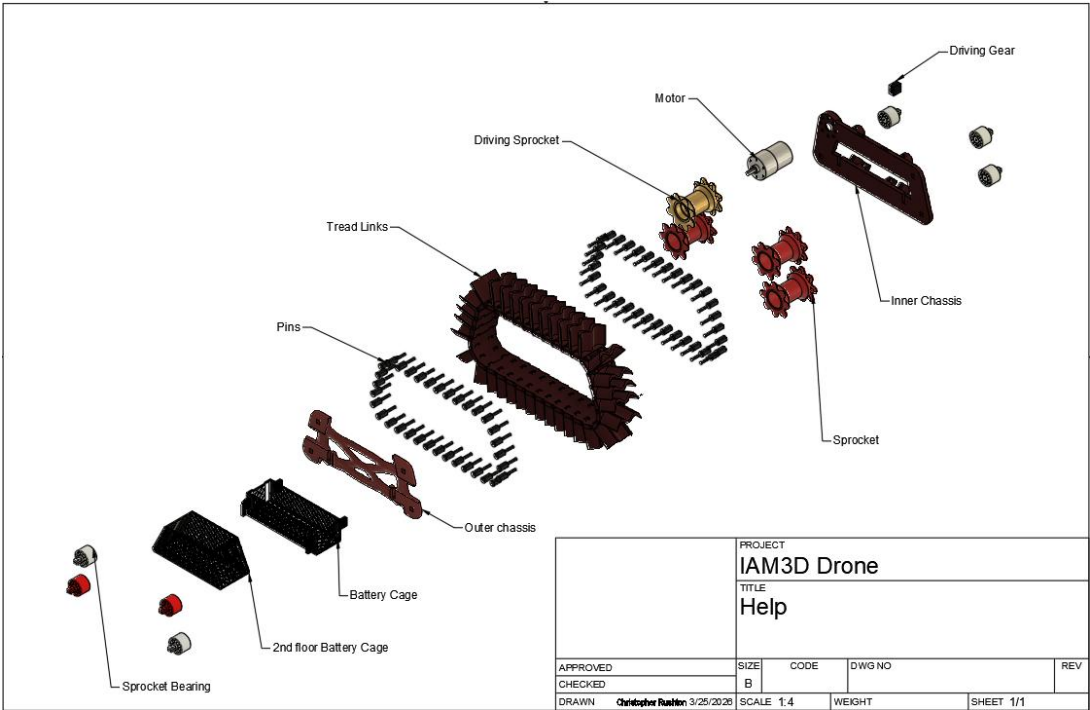
The strategies used involved rapid prototyping, specifically high-fidelity, where 3D models would be created and then printed out to test and investigate if each model worked as per expectations. The models that performed the best were kept and worked on to finalize and make better. This technique helped speed up the design process to get it ready for the competition. Prototypes of the treads and payloads were the first to be developed before moving on to developing the chassis and electrical subsystems.

3: ASSEMBLY

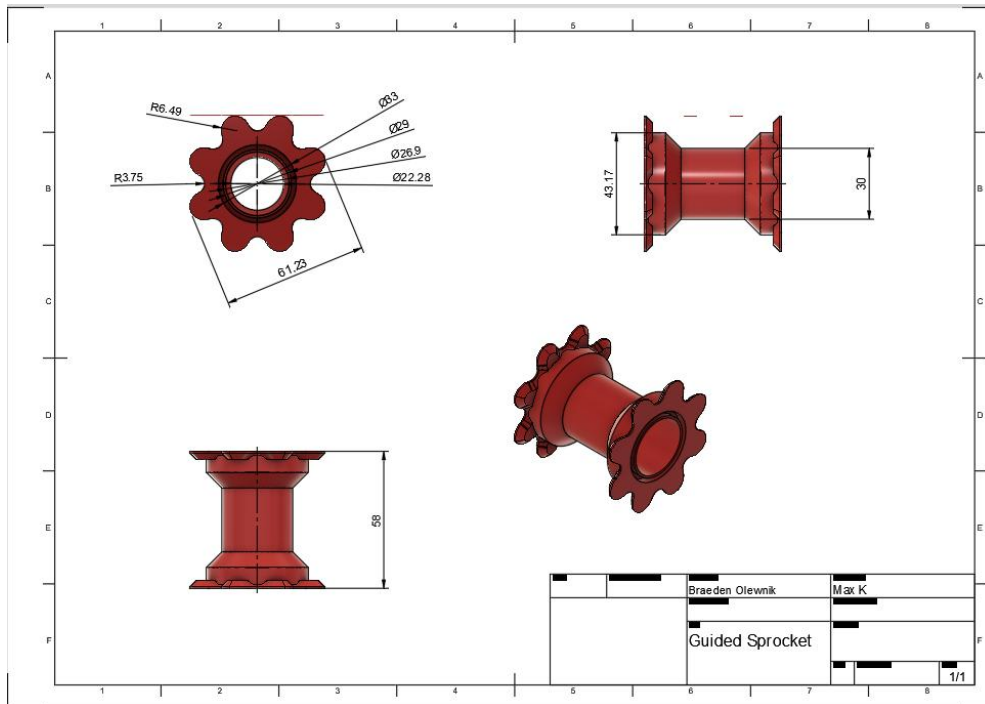
3.1: EXPLODED VIEW OF ASSEMBLY



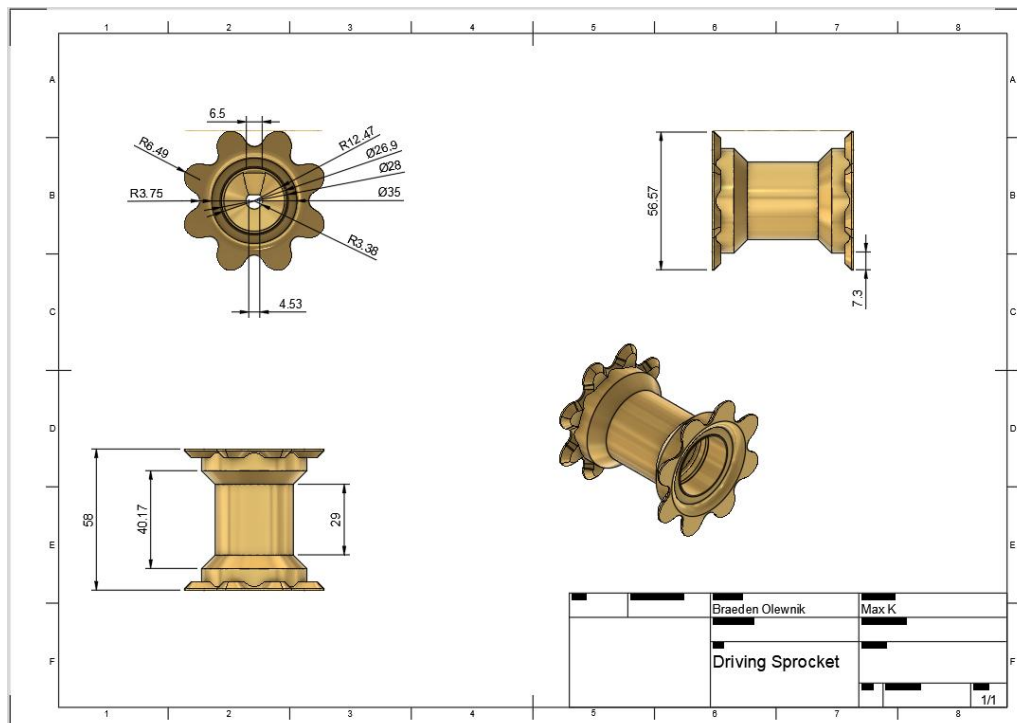
3.1.1: MOVEMENT



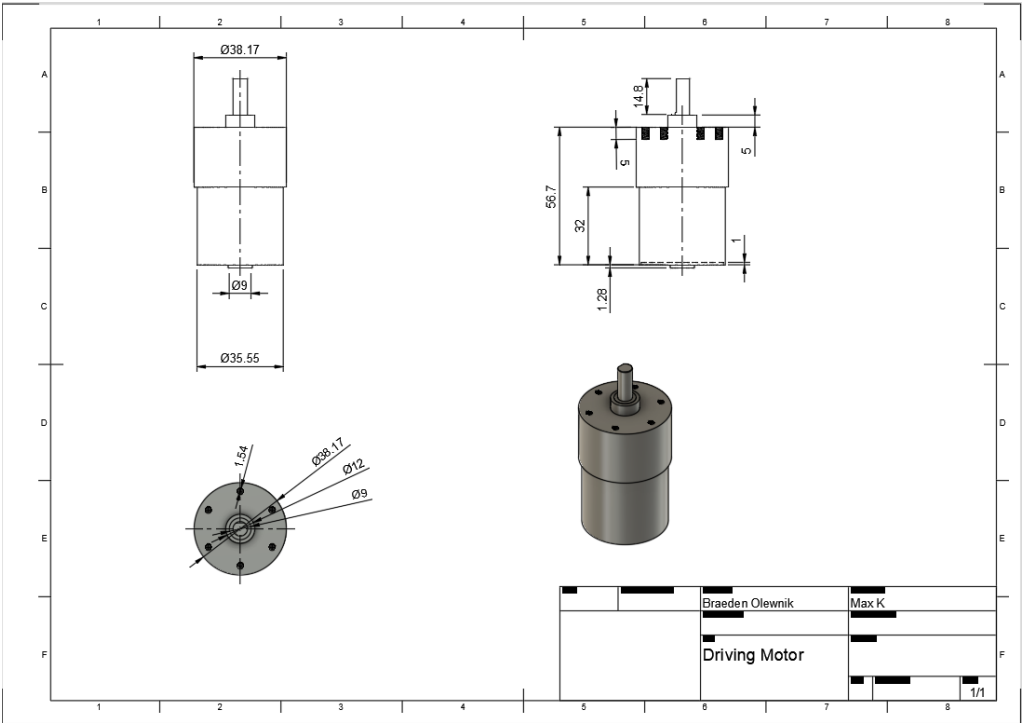
3.1.1.1: GUIDED SPROCKET



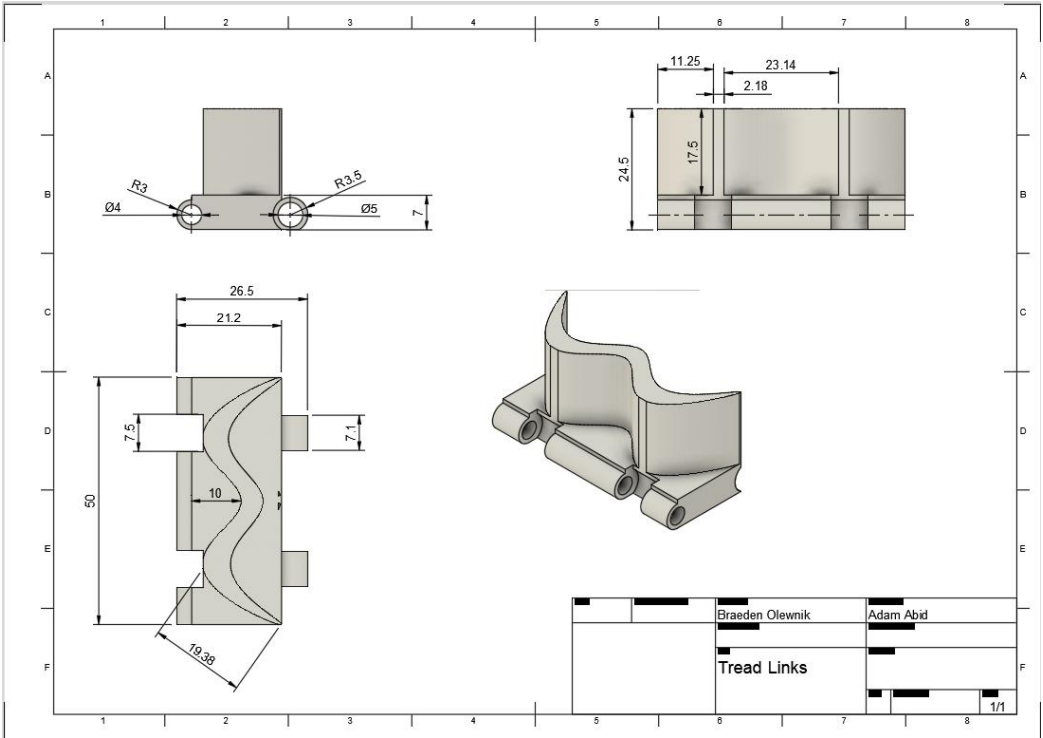
3.1.1.2: DRIVING SPROCKET



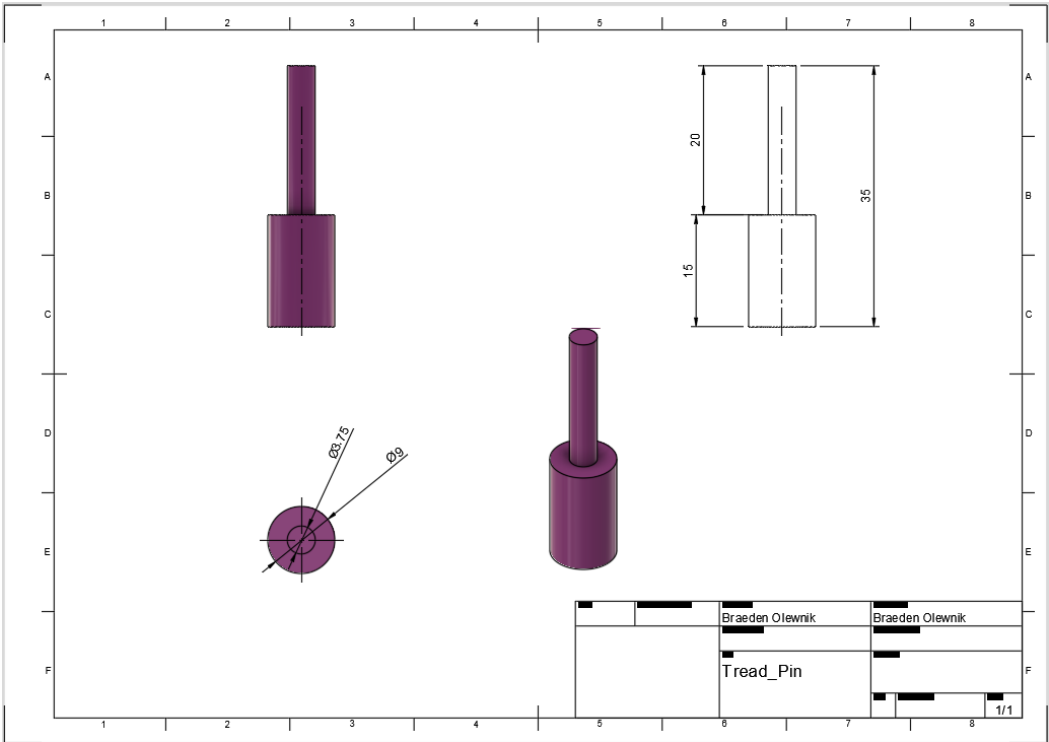
3.1.1.3: DRIVING MOTOR



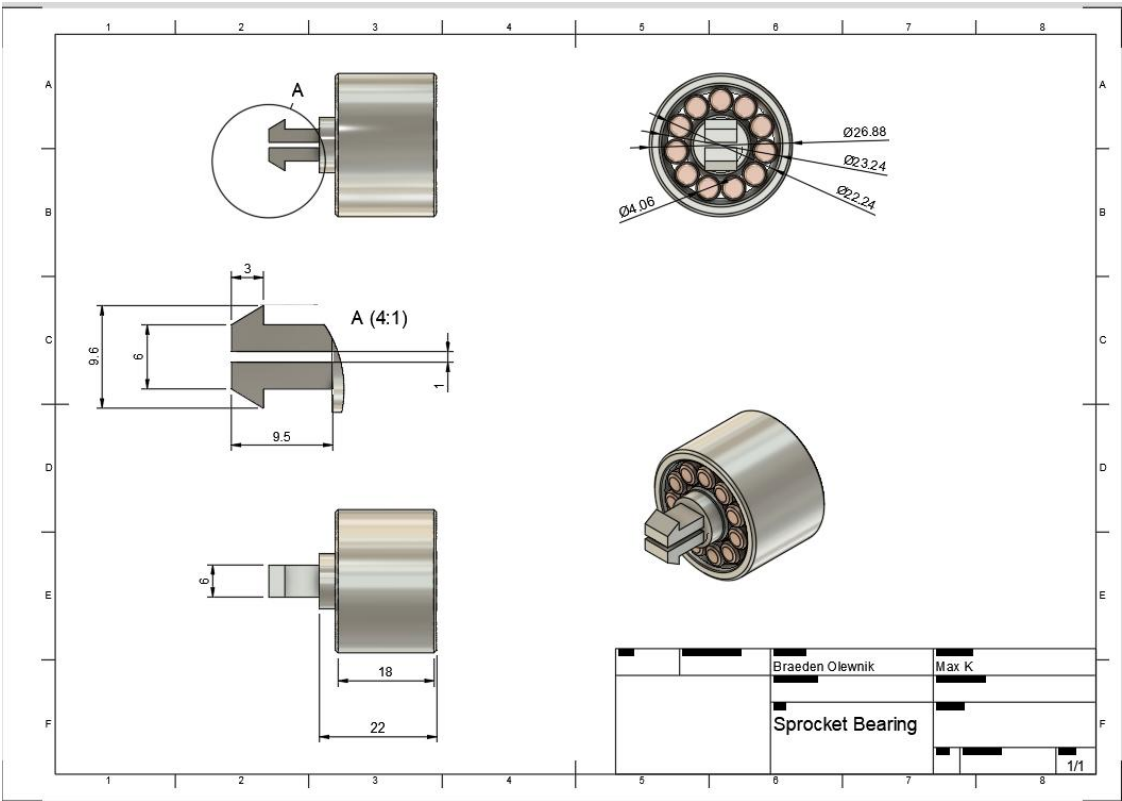
3.1.1.4: TREAD LINKS



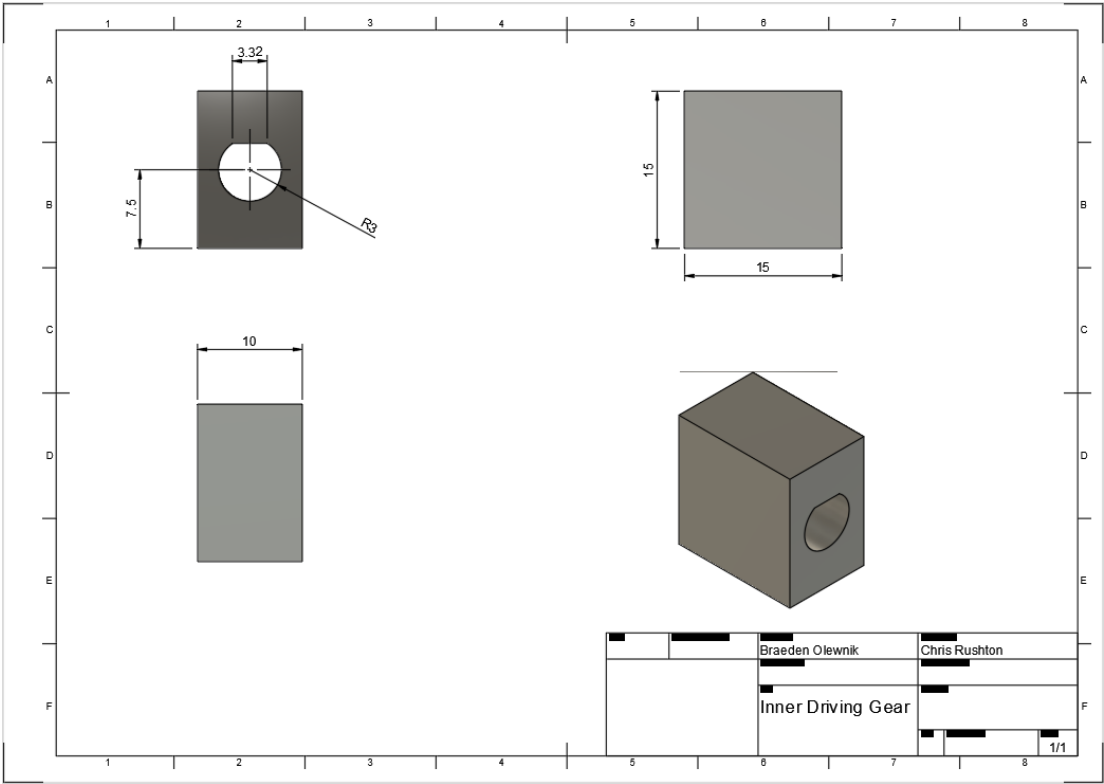
3.1.1.5: TREAD PINS



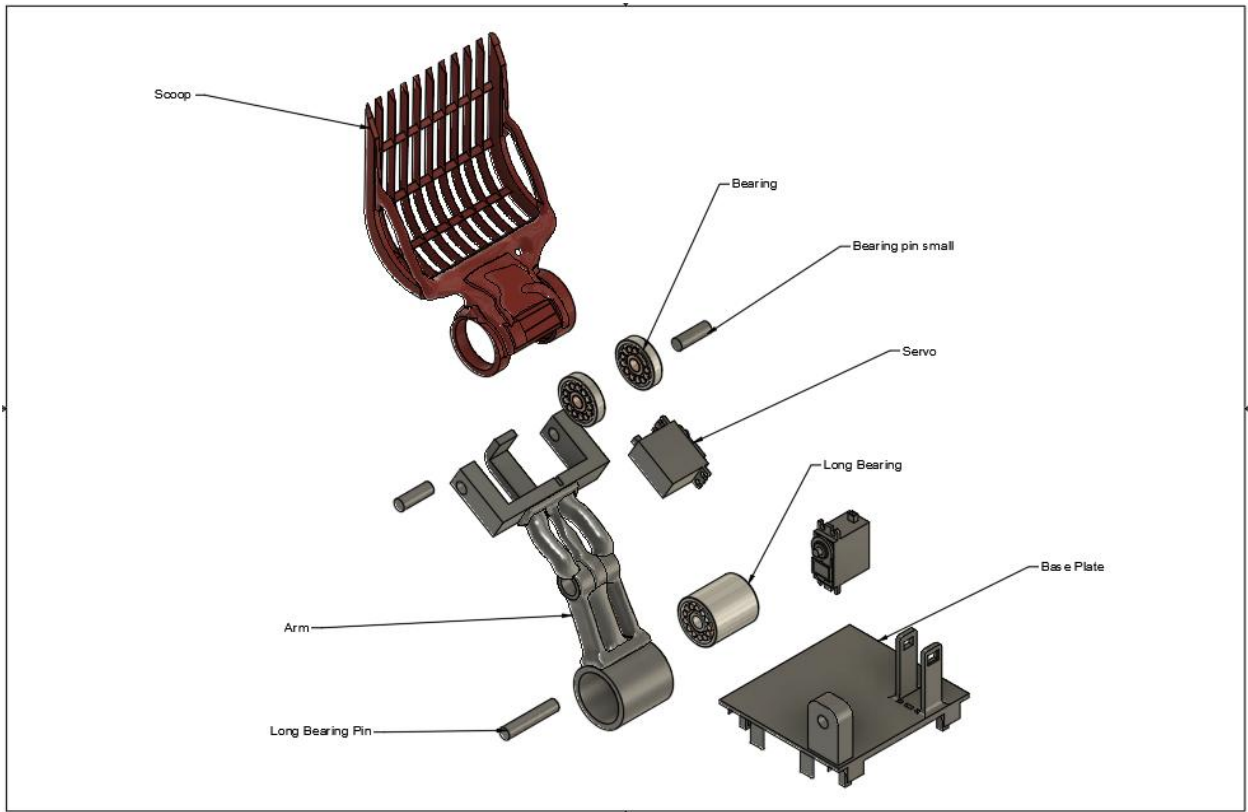
3.1.1.6: SPROCKET BEARING



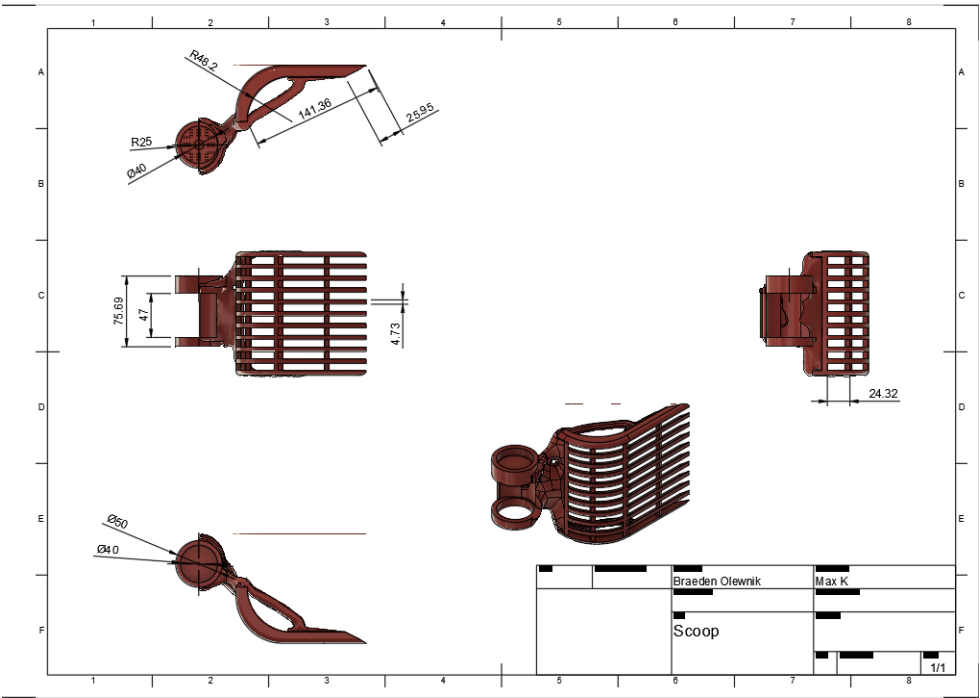
3.1.1.7: INNER DRIVING GEAR



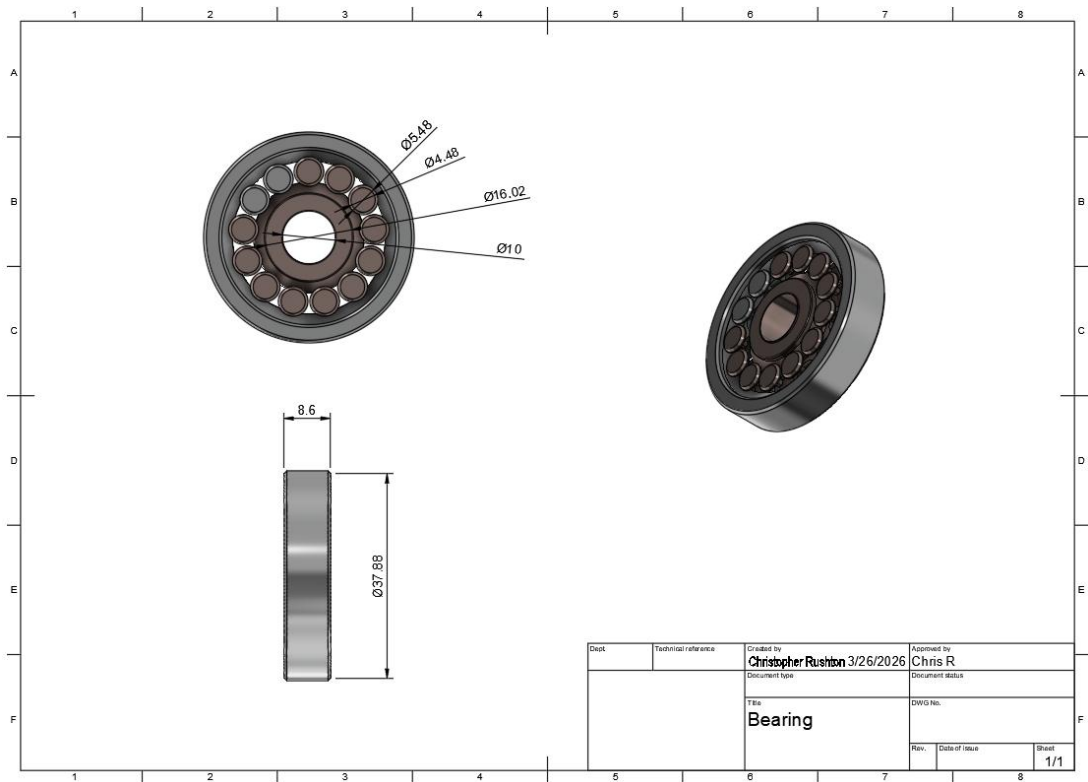
3.1.2: PAYLOAD



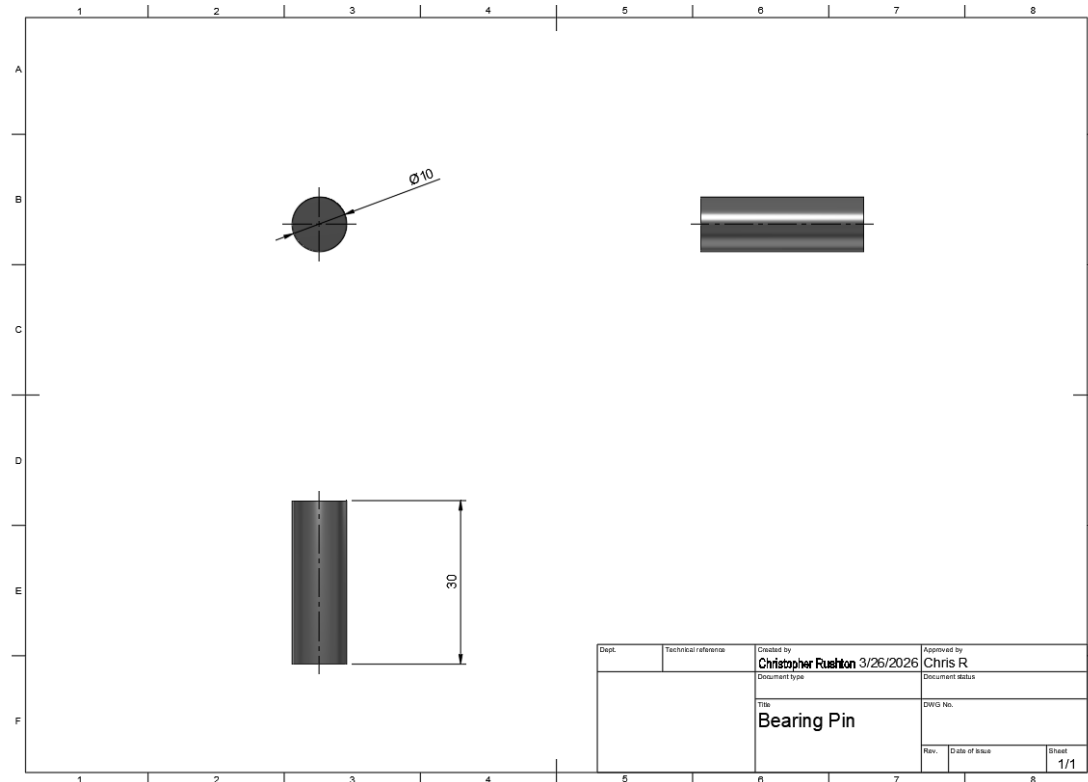
3.1.2.1: SCOOP



3.1.2.2: BEARING



3.1.2.3: BEARING PIN SMALL



[illegible]

Technical drawing of a servo motor, showing multiple views and dimensions. The drawing is oriented on a grid with horizontal coordinates 1 through 8 and vertical coordinates A through F.

Top View (A-B): Shows a rectangular servo motor with a width of 20.08 and a height of 88. A small feature on the top has a height of 4.1.

Front View (C-E): Shows the front of the servo motor. Key dimensions include:

- Top mounting holes: $\varnothing 8$ and $\varnothing 11$.
- Central shaft hole: $\varnothing 13$.
- Radius: R_{10} .
- Internal width: 16.02.
- Distance from top to internal width: 11.
- Bottom mounting holes: $R_{2.2}$.
- Distance from bottom to internal width: 2.5.
- Distance from bottom to bottom mounting holes: 7.5.

Right Side View (A-C): Shows the right side of the servo motor. Key dimensions include:

- Overall height: 55.04.
- Distance from bottom to bottom mounting holes: 2.5.
- Distance from bottom to bottom mounting holes: 7.5.
- Distance from bottom to bottom mounting holes: 2.5.
- Distance from bottom to bottom mounting holes: 7.5.

Bottom View (E-F): Shows the bottom of the servo motor. Key dimensions include:

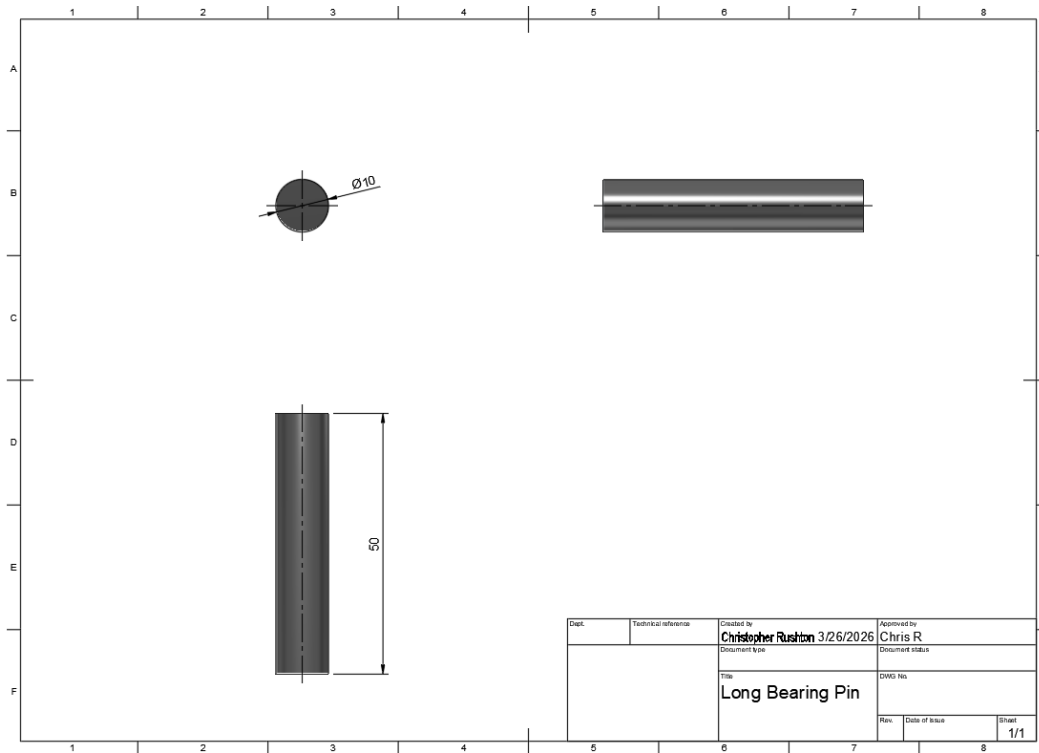
- Distance from bottom to bottom mounting holes: 2.5.
- Distance from bottom to bottom mounting holes: 7.5.
- Distance from bottom to bottom mounting holes: 2.5.
- Distance from bottom to bottom mounting holes: 7.5.

Isometric View (A-C): Shows the servo motor in an isometric perspective.

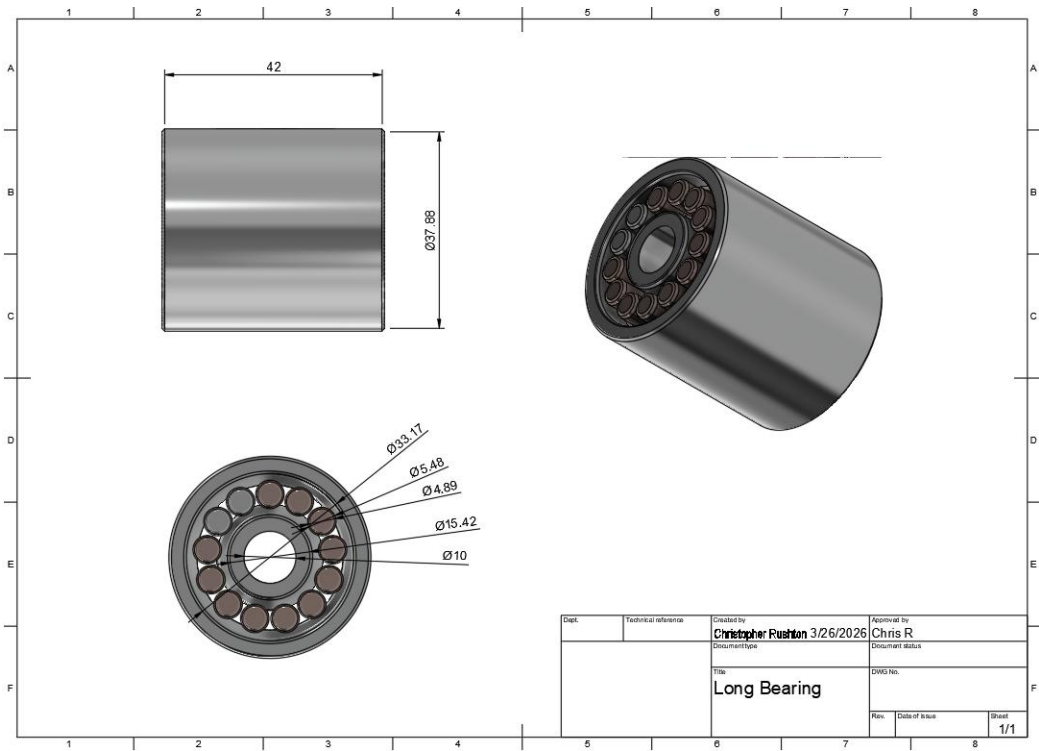
Bottom Table:

	Christopher Ruckton 3/26/2026	Chris R
	Servo	

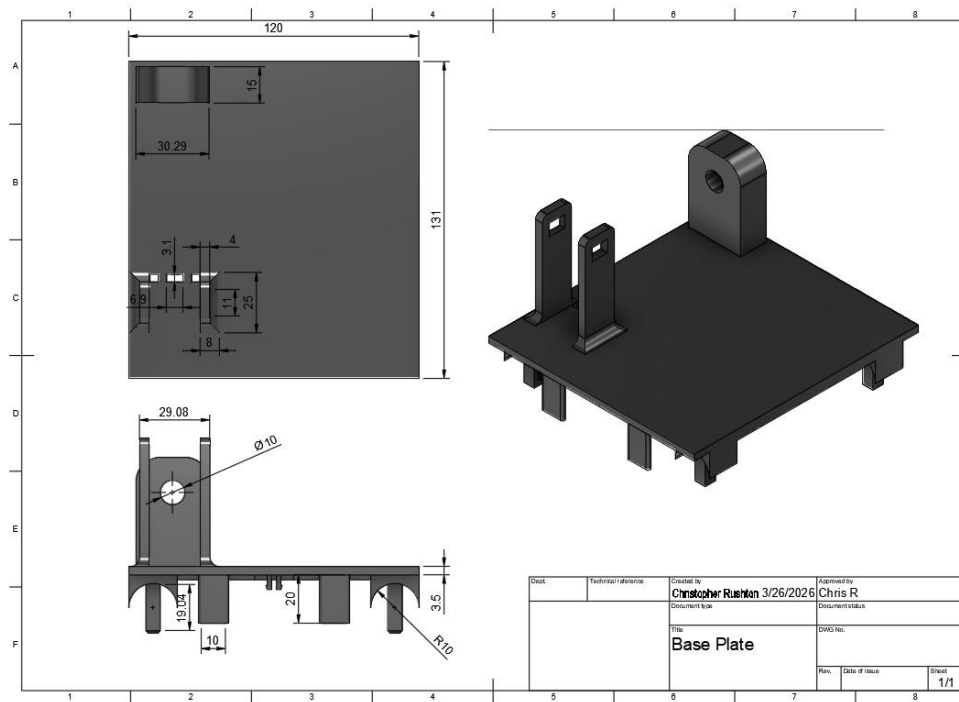
3.1.2.6: LONG BEARING PIN



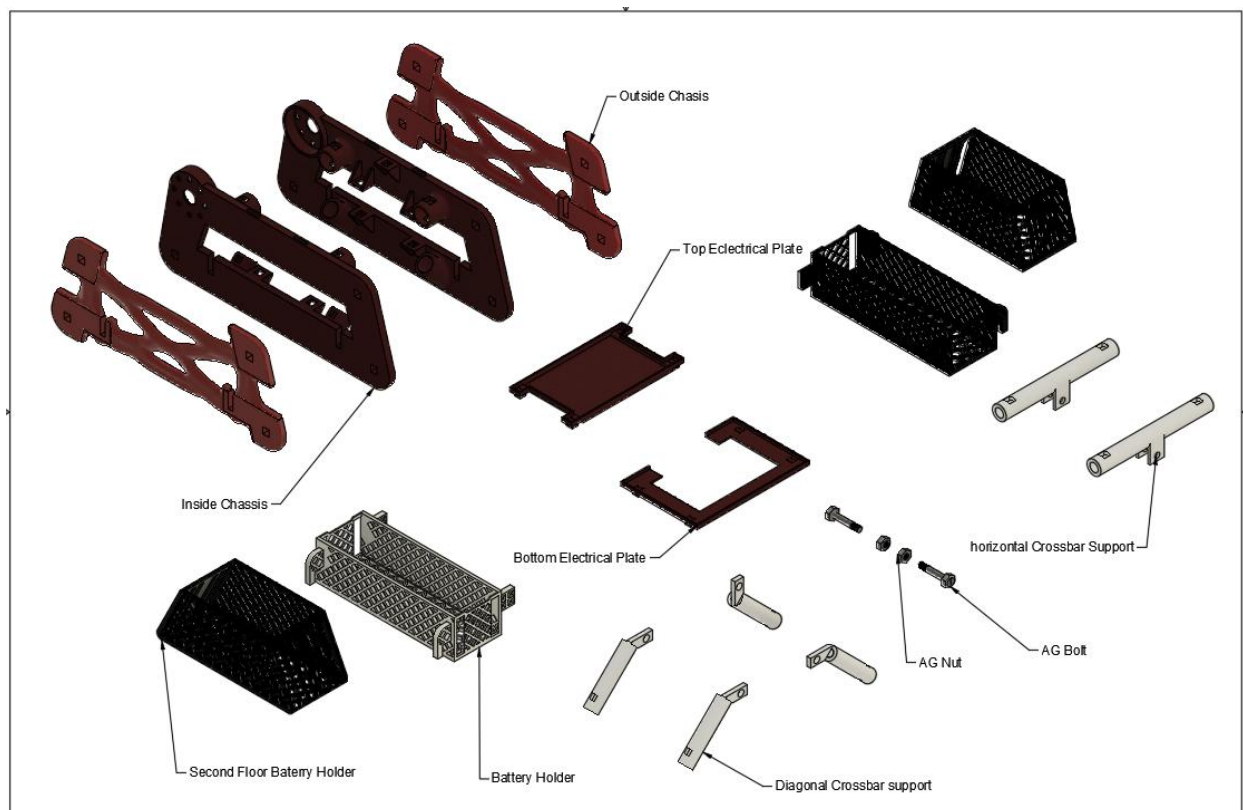
3.1.2.7: LONG BEARING



3.1.2.8: BASE PLATE

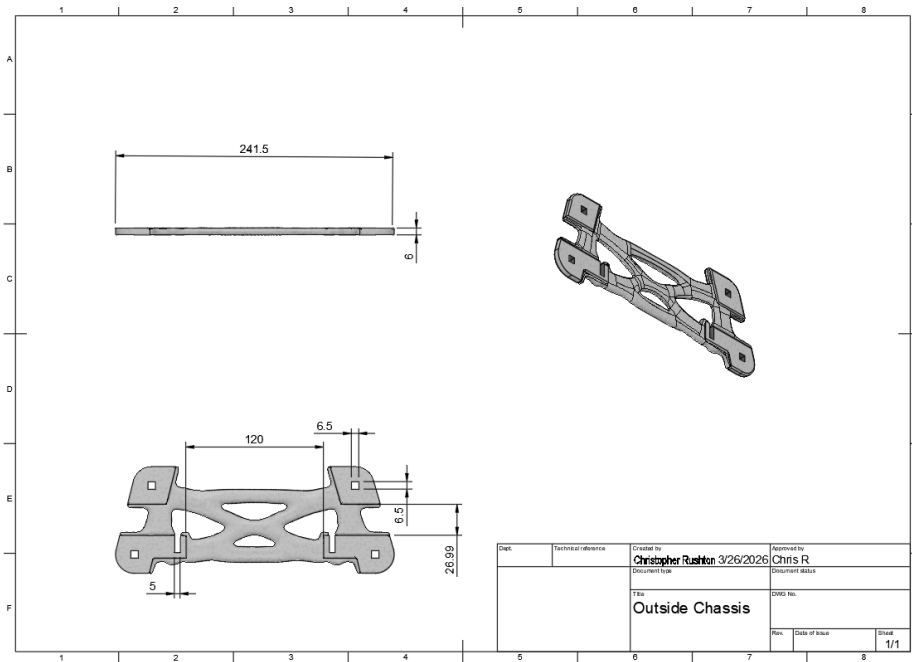


3.1.3: CHASSIS



3.1.3.1: INSIDE CHASSIS

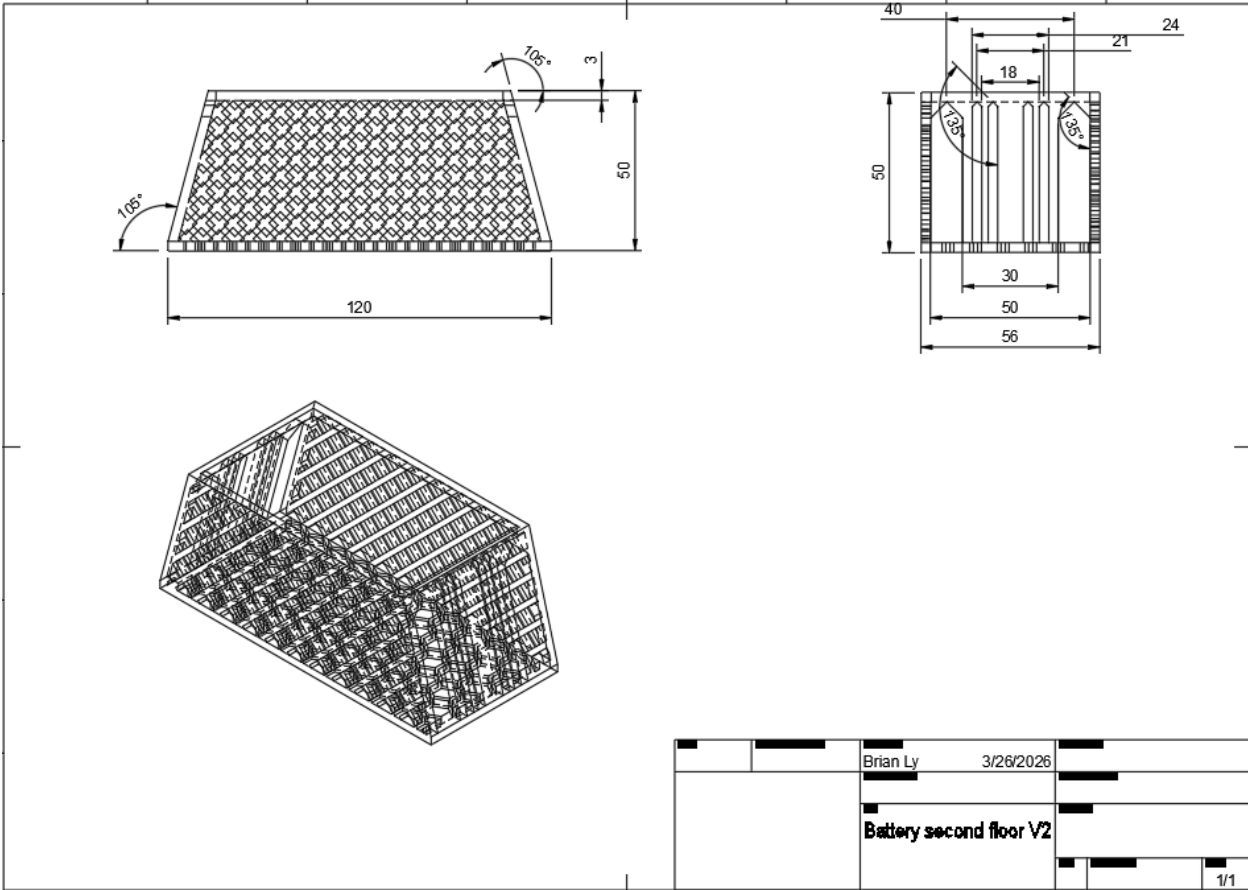
3.1.3.2: OUTER CHASSIS



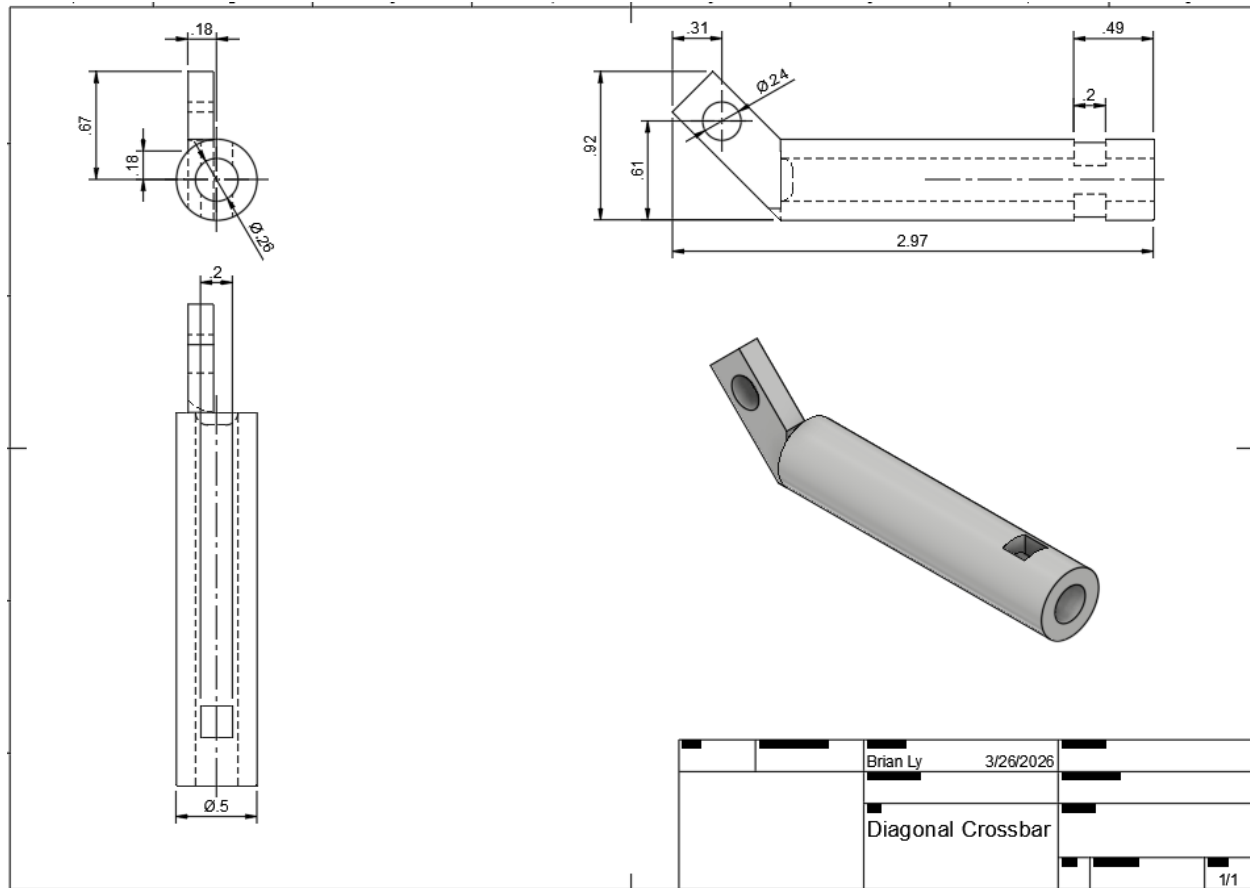
3.1.3.3: BATTERY HOLDER



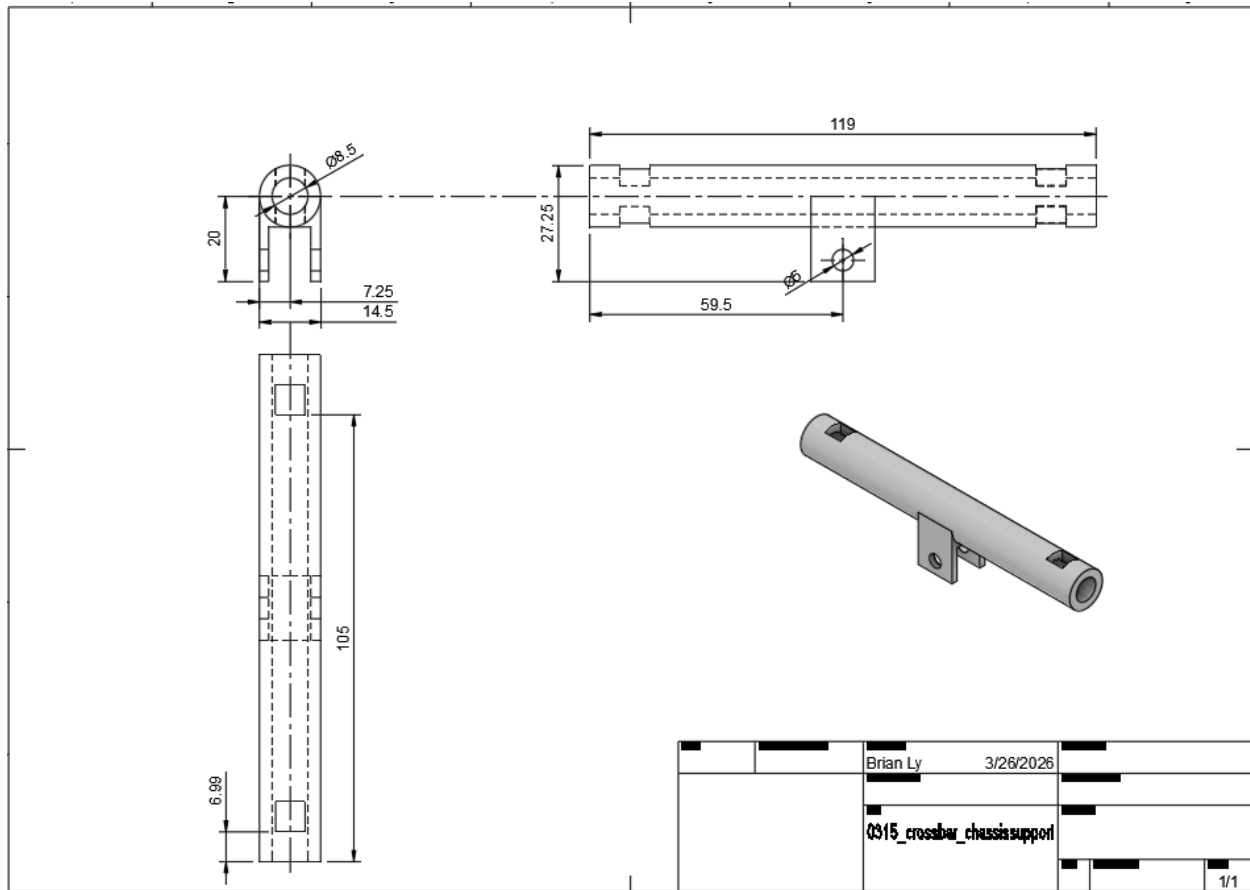
3.1.3.5: SECOND FLOOR BATTERY HOLDER



3.1.3.6: DIAGONAL CROSS BAR SUPPORT

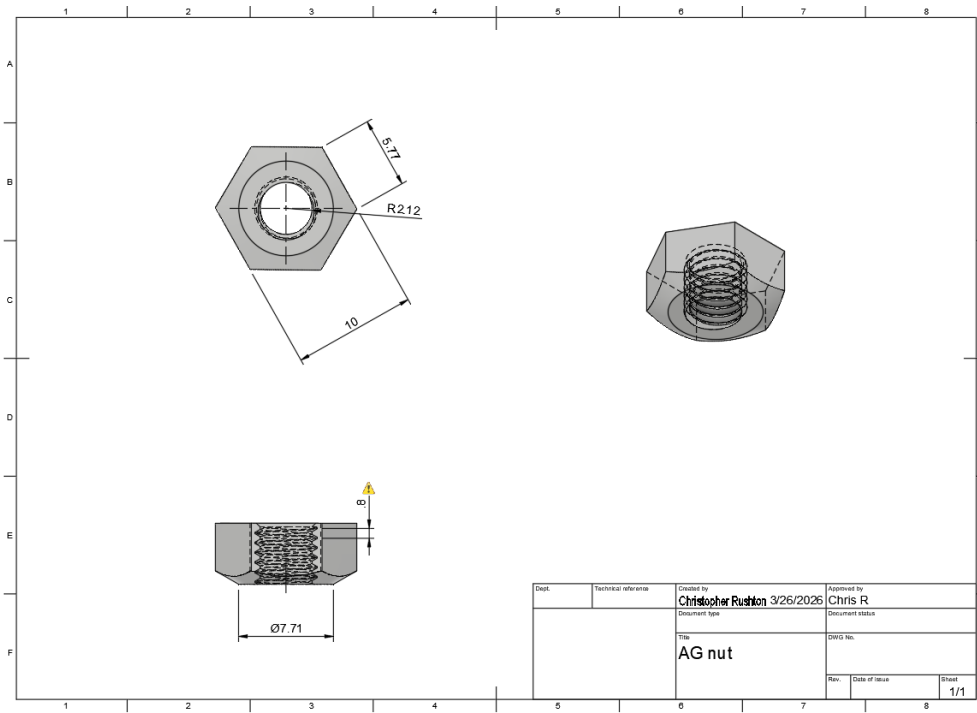


3.1.3.7: CROSSBAR SUPPORT

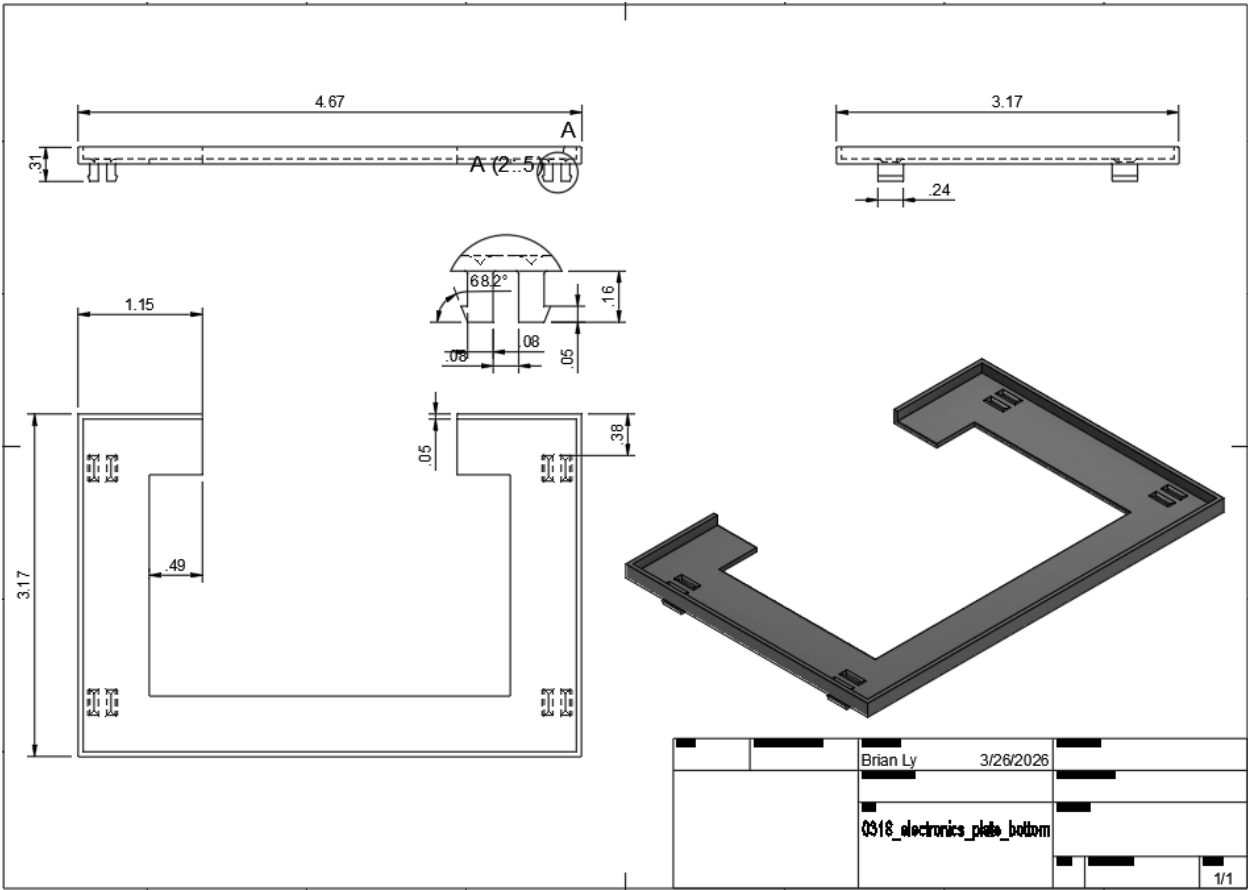


3.1.3.8: AG BOLT

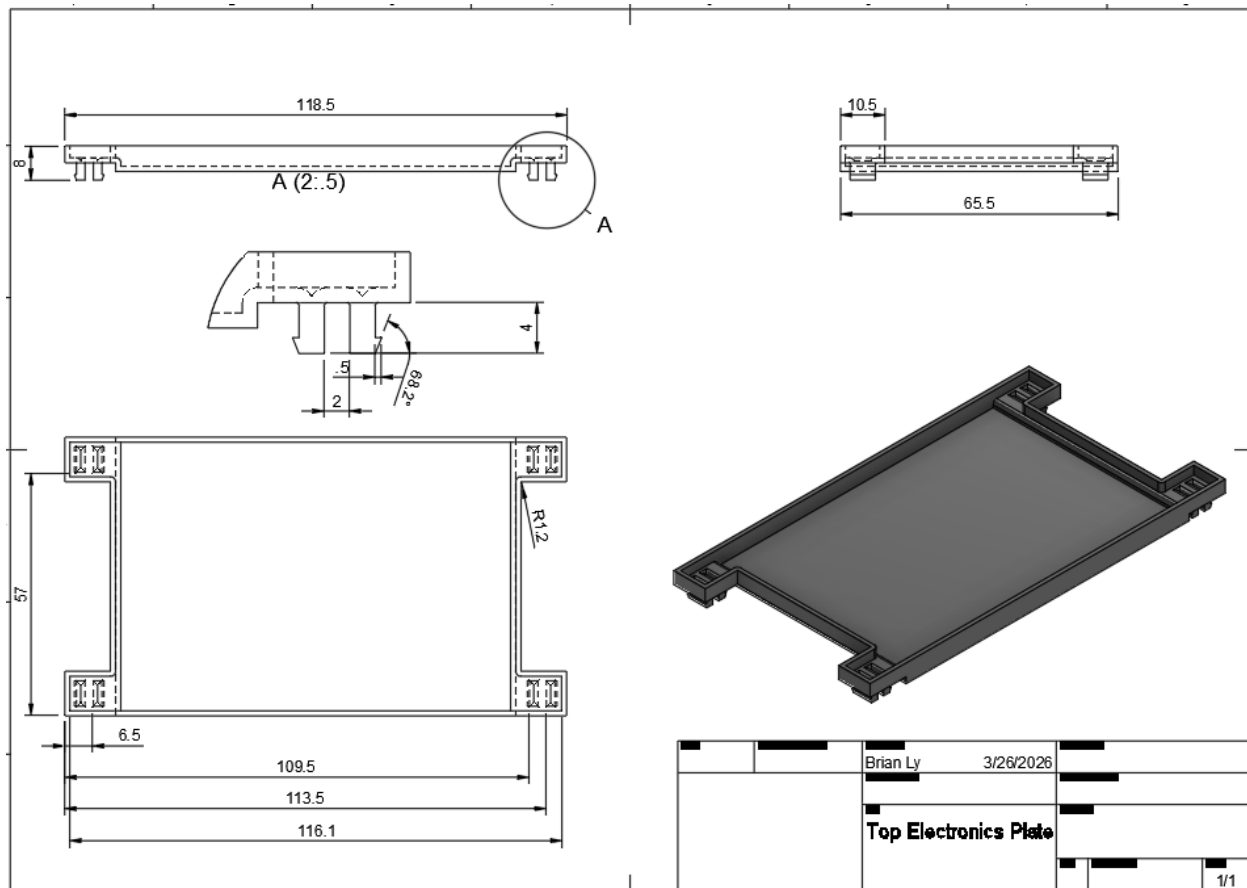
3.1.3.9: AG NUT



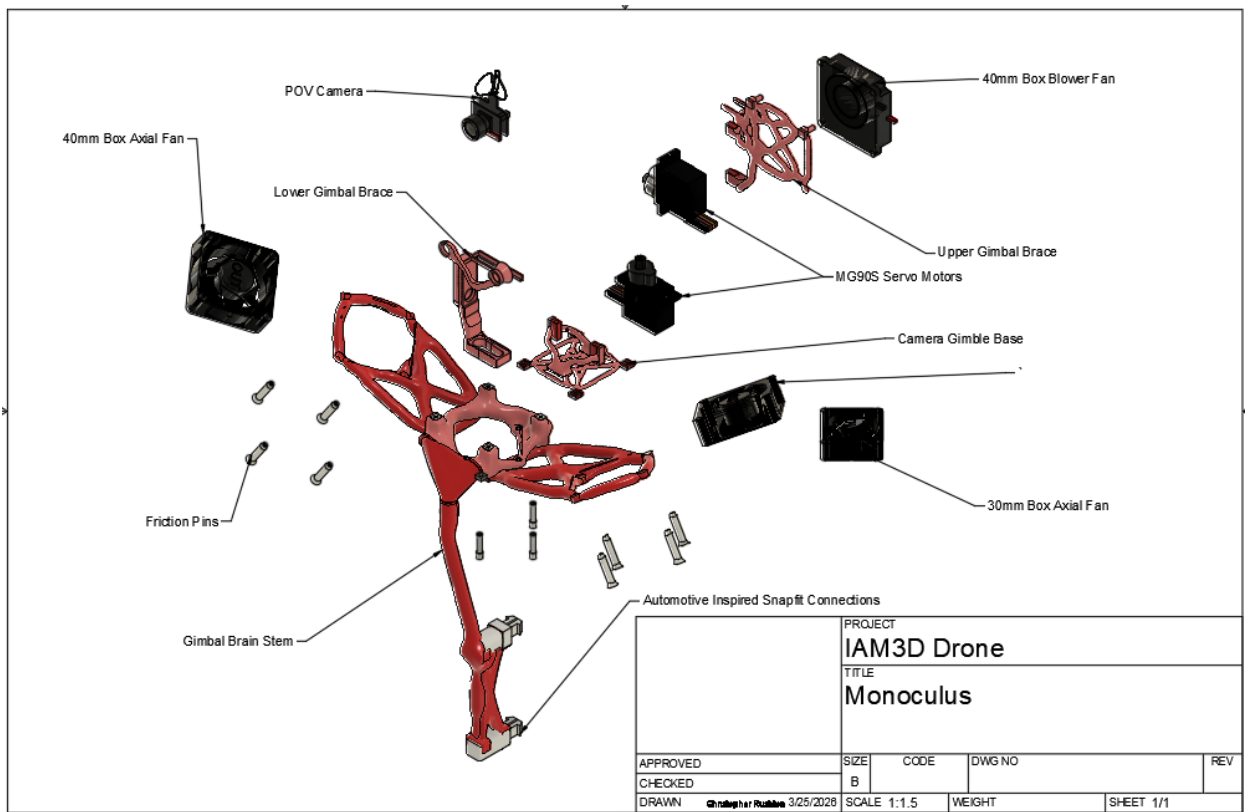
3.1.3.10: BOTTOM ELECTRICAL PLATE



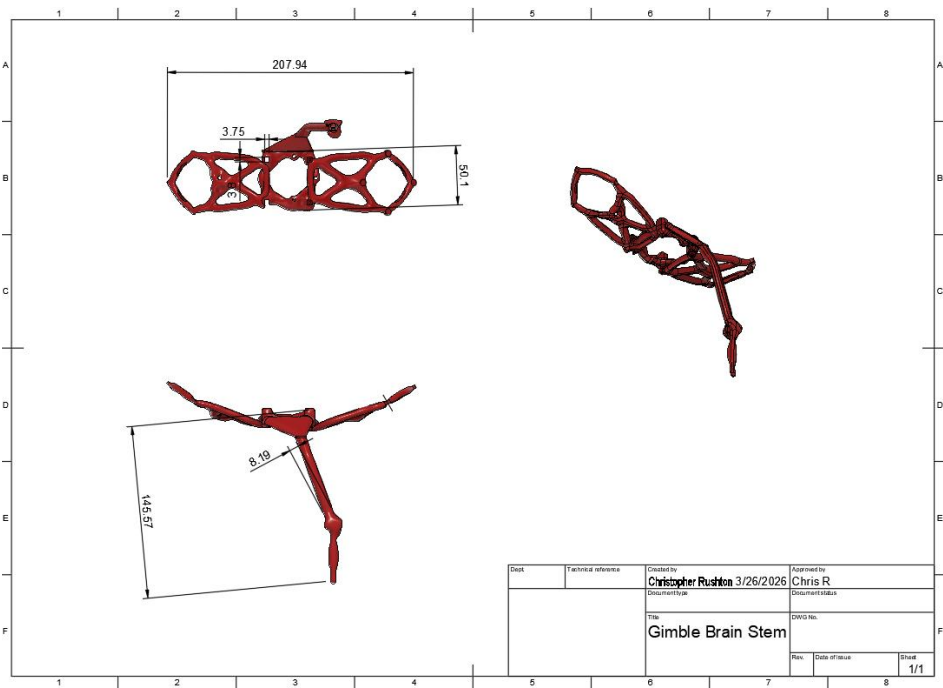
3.1.3.11: TOP ELECTRICAL PLATE



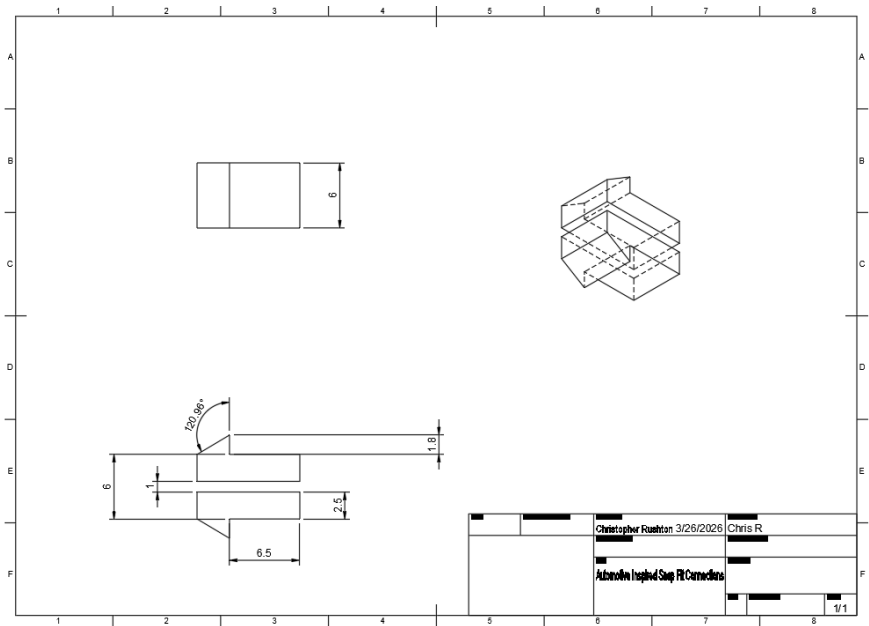
3.1.4: CAMERA



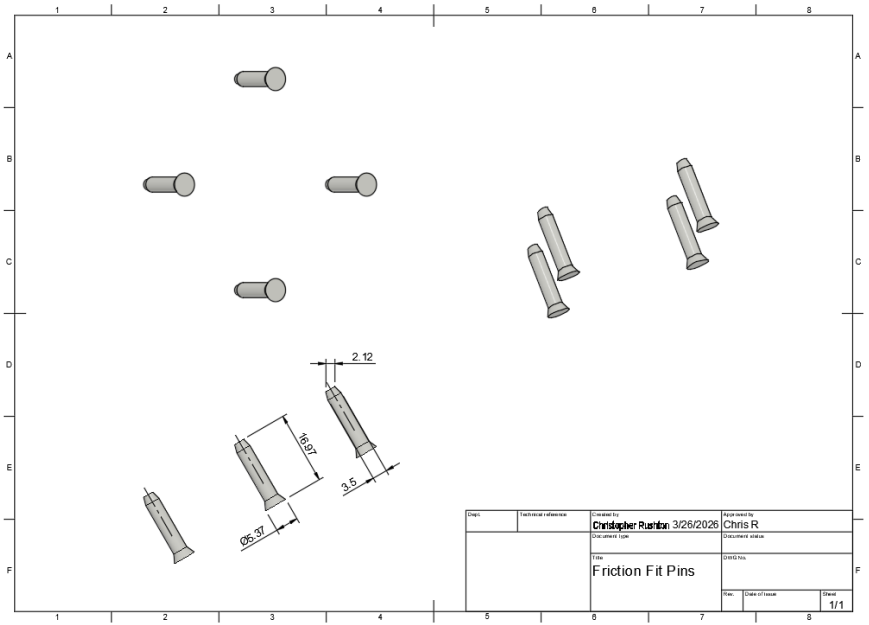
3.1.4.1: GIMBLE BRAIN STEM



3.1.4.2: AUTOMOTIVE INSPIRED SNAP FIT CONNECTIONS

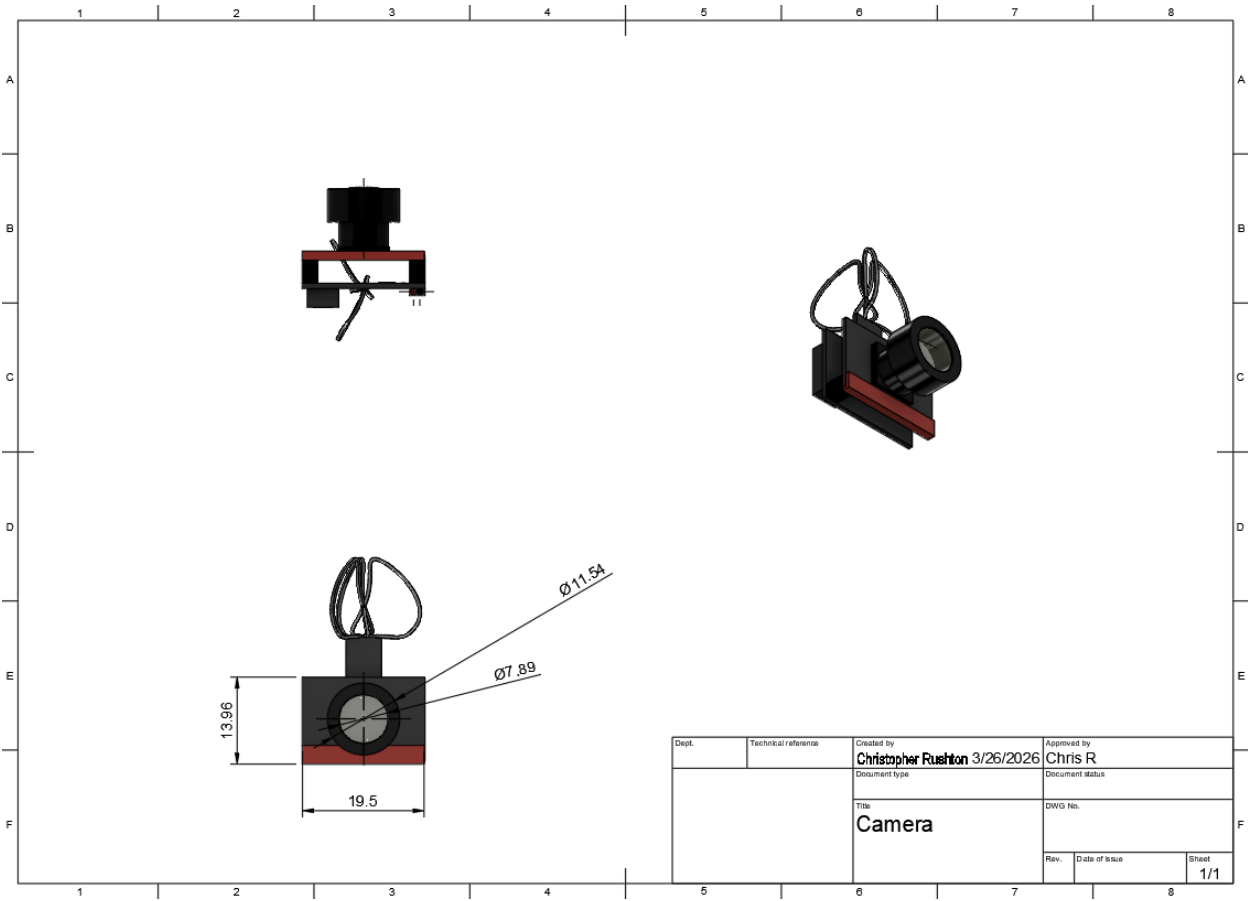


3.1.4.3: FRICTION PINS

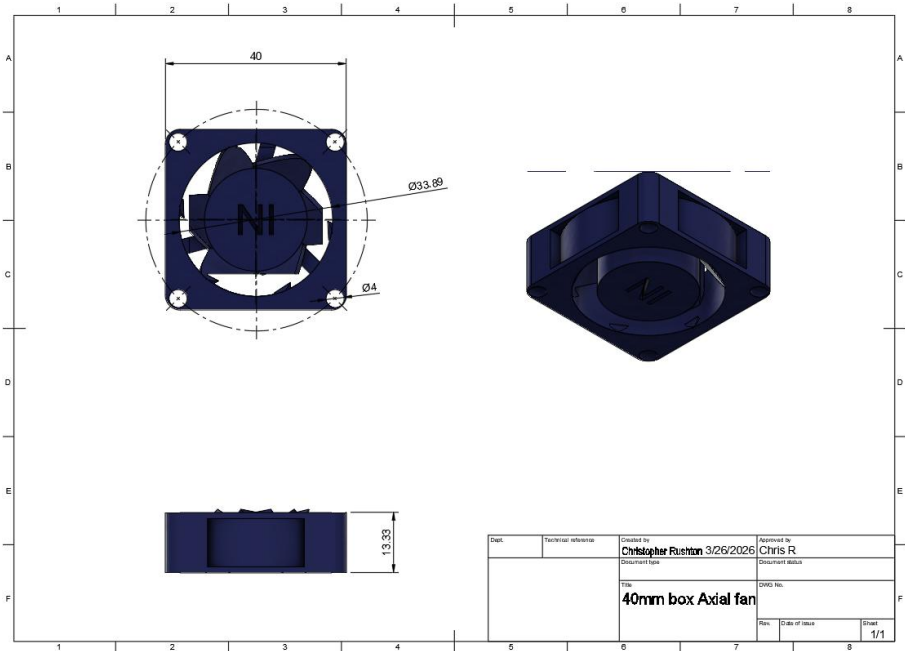


3.1.4.4: LOWER GIMBLE BRACE

3.1.4.5: POV CAMERA

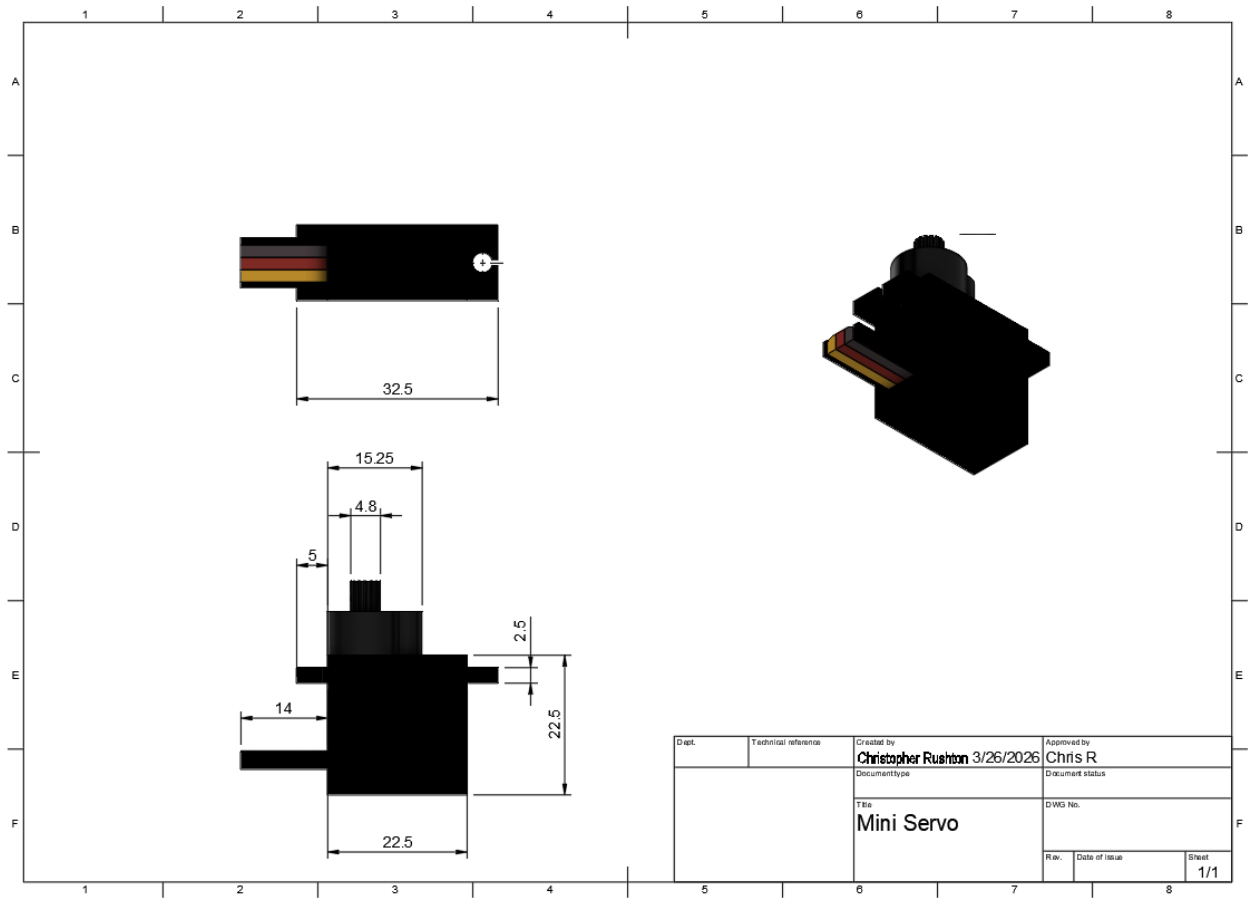


3.1.4.6: 40MM BOX AXIAL FAN



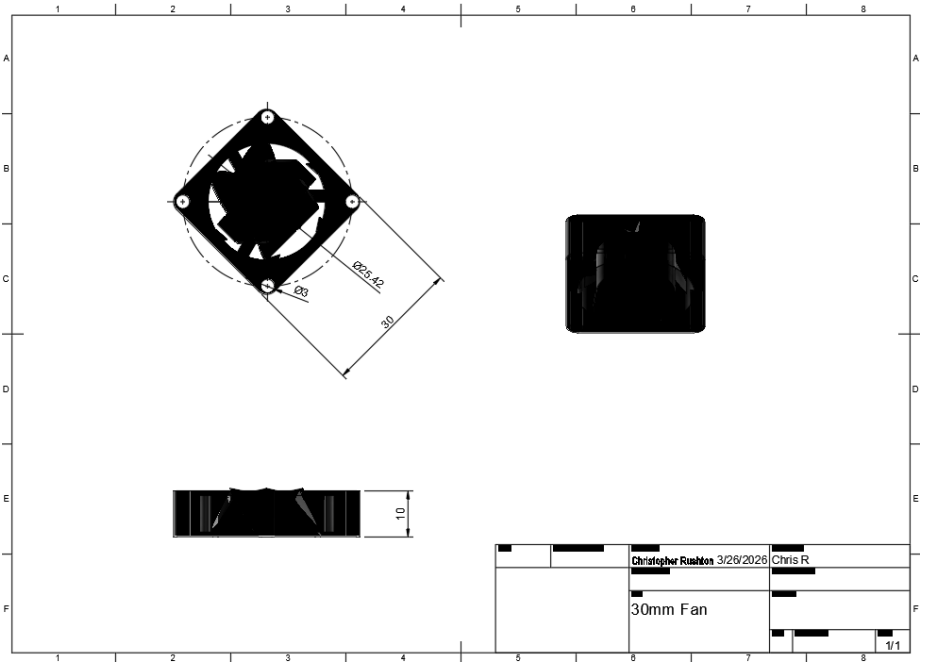
3.1.4.7: CAMERA GIMBLE BASE

3.1.4.8: MG90S SERVOS

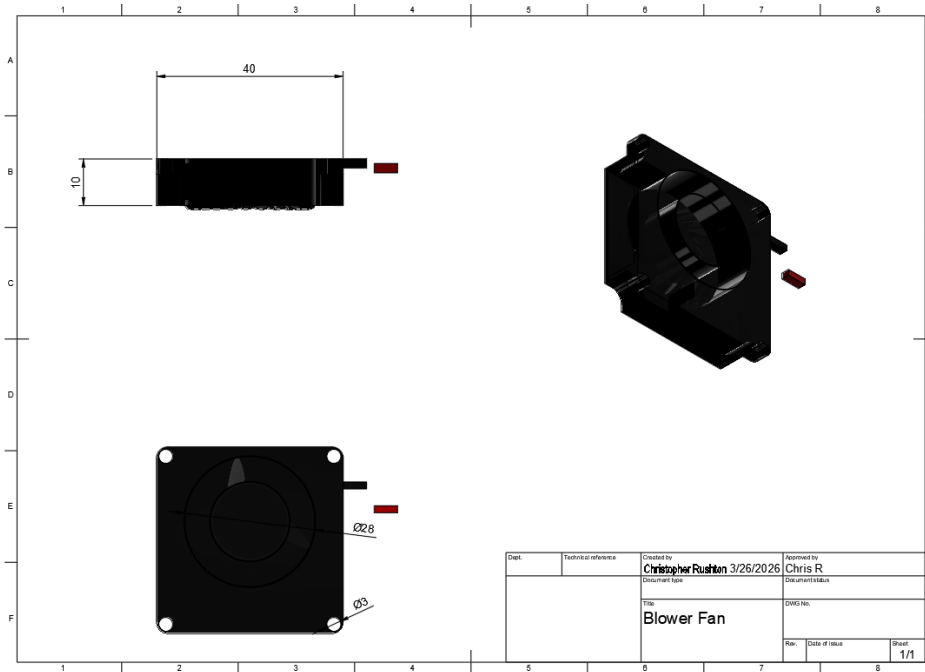


4.1.4.9: UPPER GIMBLE BRACE

4.1.4.10: 30MM BOX AXIAL FAN



4.1.4.11: 40MM BOX BLOWER FAN



3.2: FINAL DESIGN SUMMARY AND OVERVIEW

3.2.1: FINAL GEOMETRY

(low center gravity chassis, dual layer internal housing with a top and bottom electrical plate, symmetric tread based drive system, track links designed to maximize surface area contact for improved traction in sand terrain. The Gimbal brain stem allows for an unobstructed field of view)

3.2.2: FINAL STATISTICS

The final configuration of the rover is a highly optimized mass of 1.85 kg including all battery system and the specialized payload. This lightweight rover is built in a compact of 280mm x 280mm x 290mm,

(total mass-all battery system, dimensions-L x W x H, Drive systems- symmetric dual tread drive, Max Motor RPM, Operating voltage-12V?, Total component count-unique 3D printed parts assembled, Material composition-PLA etc

Category	Metric
Total Mass	1.85 kg (including all battery systems and payload)
Dimensions	280mm (L) x 280mm (W) x 190mm (H)
Ground Clearance	55mm at the lowest chassis point
Drive System	Symmetric dual-tread drive with 15mm wide track links
Max Motor Speed	6,000 RPM (Effective top speed: 1.2 m/s)
Operating Voltage	11.1V (Primary); Regulated 5.6V (Gimbal/Logic)
Component Count	62 unique 3D-printed parts
Primary Materials	Carbon Fiber PLA+, Standard PLA+, and TPU (Treads)
Incline Capacity	40° Slope (tested on granular/sand surfaces)

3.2.3: FINAL JUSTIFICATION

3.3: ASSEMBLY PROCEDURE

3.3.1 MODULARITY

One of the core strategies we employed in our rover design process was the allowance of constant and rapid iterative design changes. We achieved this by opting for a highly modular prototype, which permitted individual components, such as treads, to be swapped, upgraded, and redesigned independently without requiring a total overhaul.

3.3.2 FREQUENTLY REPLACED PARTS AND RATIONALE

Although overall durability of the rover is of the utmost importance, certain components can be omitted from this criterion. For example, we understand the volatility of the treads and tread pins. These parts are under constant heavy torque loads, shear stress, bearing stress, dynamic vibrations, axial stress, and wear over time. With four motors running at a maximum of 6000 rpm, these treads and pins will experience significant wear over their life cycle. In addition to these concerns, sand might enter the internal components of the rover, causing localized grinding within the contacting parts. With this in mind, we must constantly replace the treads and pins over time. By replenishing these components frequently, the rover should experience no issues.

3.3.3 PROCESS TO REPLACE PARTS

3.3.4 FINAL ASSEMBLY

4: MOVEMENT DESIGN

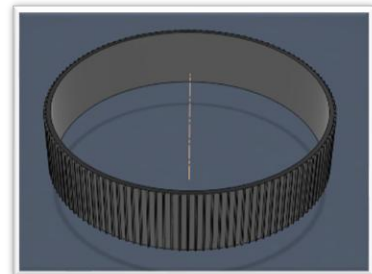
The movement design was researched upon and selected during the initial stages of the rover design project. Our researched options included wheels, Archimedes screws on either side, hexapod legs, and treads. Out of all these options, we found that either wheels or treads would perform better in sand compared to all other options. The idea to manufacture the wheels, however, was quickly eliminated by our team due to technical difficulties as well as potential possibility of poor performance in the sand, therefore we shifted our focus onto designing tracks. Additionally, we chose treads in response to our payload being an arm and scoop mechanism, considering that treads would allow for more surface area contact with the sand and therefore more torque would be translated into the sand, allowing us to maneuver through it easier with our payload.

This section is split into multiple parts: Tracks that were designed and tested by our subsystem layout plan as well as sprockets and bearings.

4.1: TRACK DESIGN

4.1.1: TRACK OVERVIEW

The original design of the track included printing a single continuous “belt” out of the TPU which would be rotated using a sprocket. However, during the development of the sprocket our team realized two issues with our plan, first that slippage of the “belt” was a high possibility and second issue was the fact that sand can get between the sprocket and the belt, which would lead to entire dislocation of the belt. Our team also ran into fitting issues with certain bends in the belt. Therefore, our team initiated the development of the tracks which consisted of interconnected links.



When designing the tracks, we analyzed the existing solutions of military and heavy equipment. We investigated the undercarriage of an excavator which includes links that are connected via a pin. And the sprocket would interact with it on the inside of the link. The friction-fit link was inspired by such design, instead of adding a pin between the links, we explored the idea that 3D prints can be flexible, which allowed us to connect the links in a more reliable way.

Another solution our team researched was an all-terrain tank or tank tread. The tank tread links have a pin between the links, and they are free to rotate with the connection being the pivot point which is at the end of each link. The advantages of such a design is that if implemented correctly it allows for a smoother engagement with the sprocket and it also allows us to shift the interaction between the sprocket and the links on the outside parts of the link. This research allowed us to initiate the development of multiple variations of the links which helped us with optimization design.





Finally, we have explored the variations of tanks that are specifically designed with a more specific type of operation environment in mind. Meaning they have special adaptations that might make it worse for overall use, but much better in certain environments.

Our specific interest landed on the tank that can perform good is on sand and water which is called an amphibian design. We have considered this fact when iterating the

4.1.2: TRACK LINKS DESIGN

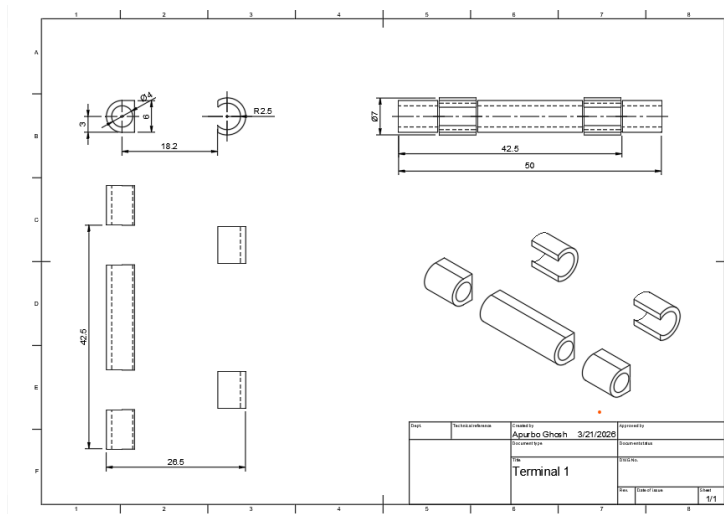
We started the design of the tracks by studying the basics of a simple chain design. First, we discovered that the distance between each link provides valuable information for the design of the sprocket. Therefore, we made a commitment to normalizing it to 20mm. Therefore, the designs of our links have been focused around that number. After determining the kind of tread we wanted, we started the development of two separate designs with two different modes of connection. Those competing ideas allowed us to develop a piece that can be compared with another and allowed us to find an optimized design.

4.1.2.1: LINK TERMINALS

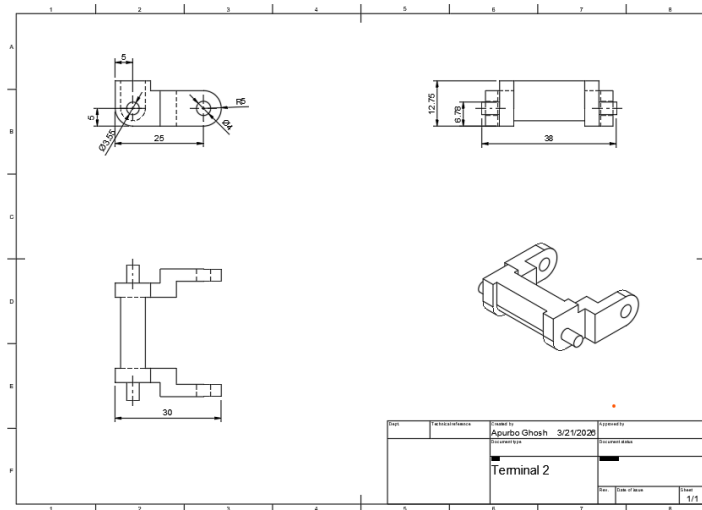
The terminals are the points at which our treads link with each other. It is integral that these points are well designed, since if they jam then the tanks won't be able to move at all, or will have extreme friction during movement. The main factors we considered when designing these terminals were:

- Low friction between other terminals during motion
- How they would prevent loose sand accumulation inside of terminals
- How the terminals would connect to each other
- Simple, but strong connections

After considering these factors, we came up with two designs that we thought met our criteria:



This was the first linkage design we came up with. It's relatively simple, using a friction-fit pin (Section 4.1.3.3) to create a connection between the treads. The basic idea is that the front of one terminal will line up with the back of another so that two pins, on opposite sides, can reach through the outer chamber into the inner chamber and hold the treads together. A key factor in this design is the difference in diameter between the front and rear of the terminals. If these two sections were the same size, then the treads would not be able to move since friction would be applied on all parts of the connection, and rotation about the pins would be restricted. (Section 4.1.3.3)



This is our second design for the linkage system. This design is slightly more complicated in terms of design but is conceptually the same idea as the first. The overall distance between the terminals is about the same in both drawings (20 mm). In this case, the friction does not come from an external force, but an internal one. The snap-fit between terminals is what holds the links together. (Section 4.1.3.1) The motion is slightly smoother in this design since the treads are held together with much stronger frictional forces. This system is also more compact due to the lack of pins on the side, which allows us to increase the size of the pads on top of the track.

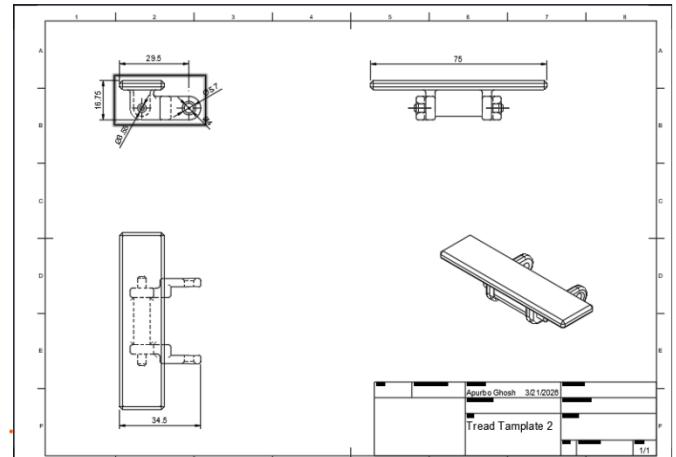
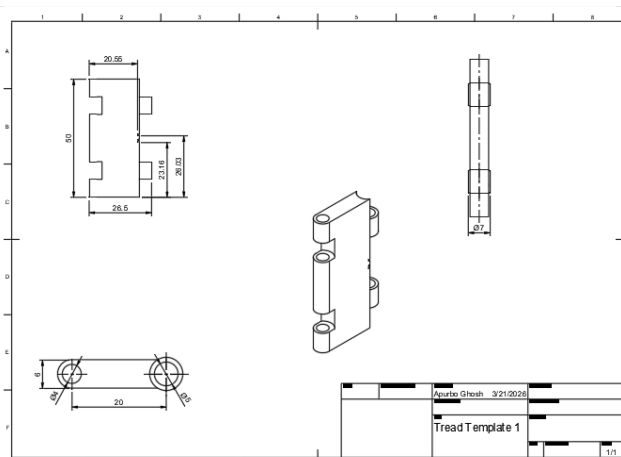
This increased surface area is very crucial, and something we tried to maximize in our design (4.1.4) The main drawback to this system is the sprocket, which the treads rotate about, this system is much more susceptible to sand being stuck within the load points with the sprocket. This excess friction is important to our future design decisions (Section 4.4.2.1)

4.1.2.2: LINKS FINAL DESIGNS

The link pads are the sections of the treads on top of the terminals. They are the sections on top of which the patterns are applied. There isn't as much design that goes into this as the terminals and sprockets. The pads are more of a byproduct of those designs. They must fit into the constraints, and they serve as the canvas on top of which we apply our texture (Section 4.1.4) The main factors that went into the design of these pads were:

- *High surface area*
- Streamlined shapes.
- Low print time/material cost

These are the finalized link templates we ended up using for all our tread designs:

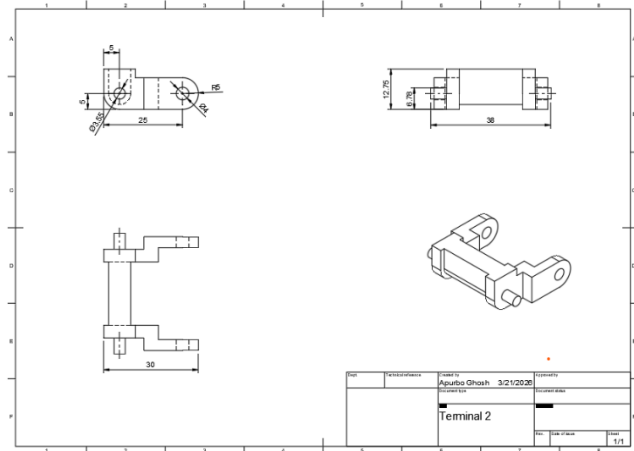
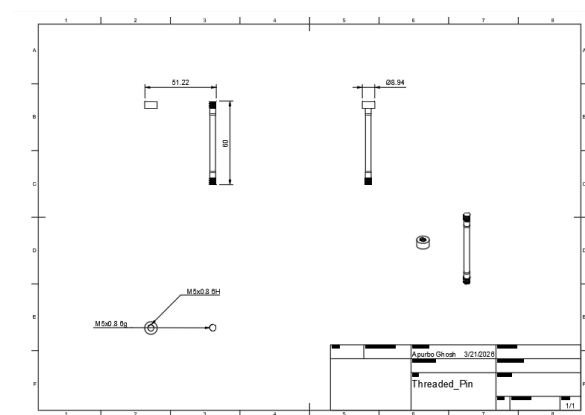


4.1.3: LINK PINS

The design number one includes a separate component, which is the pin that connects the two links together. Our team started with printing a long thin cylinder which would provide just enough tolerance to hold it together and provide just enough clearance to allow for free rotation.

4.1.3.1: SNAP-FIT CONNECTORS

Early designs used snapfit connectors. Although functional and durable, we found that lubricant is necessary to make them move properly, and in an environment of sand, this was clearly unusable.



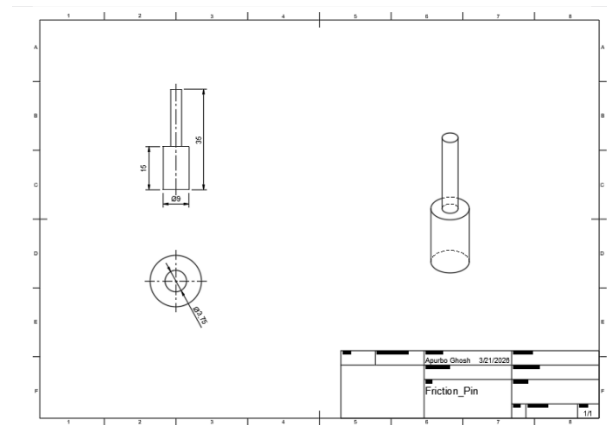
4.1.3.2: THREADED PINS

Our second design included threaded pins with a nut on each end. Functional and well adapted to sand, but we found that our printers are only capable of

printing one at a time, vertically, with the smallest nozzle we have. The biggest issue here was that by printing vertically the layer lines made it easy to snap, and when considering that they were printed at a smaller nozzle size (smaller nozzle = more layer lines, weaker), we found that they were unusable for our design.

4.1.3.3: FRICTION-FIT PINS

Friction fit pins are our most current design. These are printed horizontally, mitigating layer lines, while also interchangeable and easy to replace. Additionally, their contact with the second link is relatively isolated and they are unaffected by sand after testing.

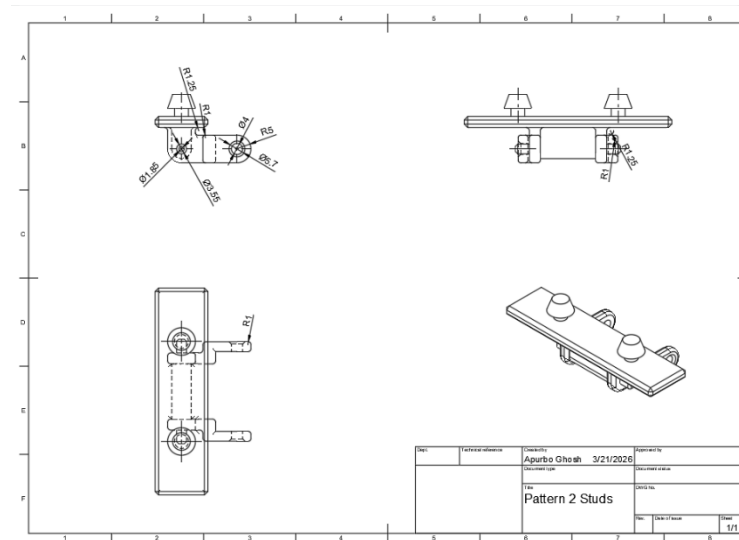


4.1.4: TREAD TEXTURE

Tread texture refers to the extrusion tangent to the link path.

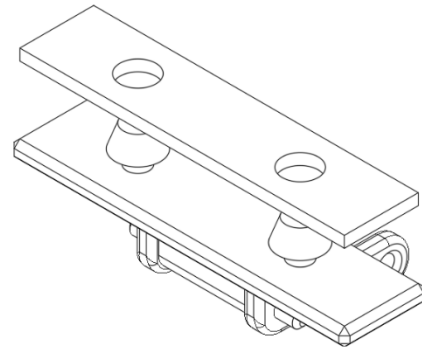
4.1.4.1: INITIAL DESIGNS

The texture on top of the treads is one of the most crucial parts of the design. Even if our links are extremely smooth, and our motors generate massive amounts of torque, this won't mean much if we can't transmit all that torque efficiently to the ground, or if the tank sinks into the sand. Going into the initial design process, we weren't completely sure how to maximize traction while properly distributing pressure. This led us to take inspiration from different fields where traction and weight distribution are integral to designs. Here are all our initial designs:



Studs-

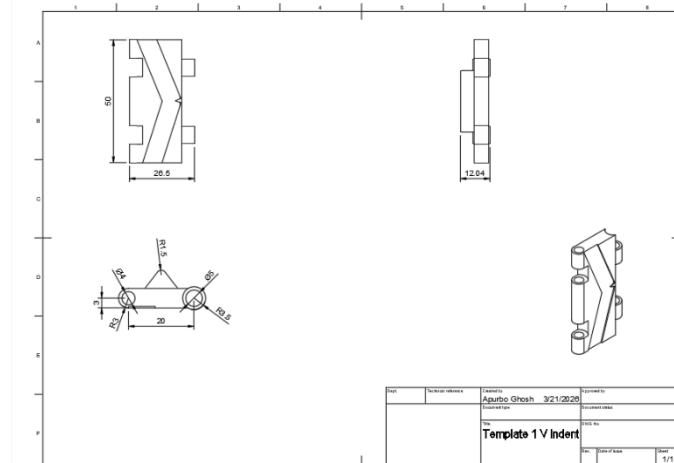
The illustrated diagram was the first design we came up with. Here, we found inspiration in football cleats, which have studs protruding out of the sole which grip onto the turf and prevent slippage during rapid acceleration. Unique to this design, PLA Flex Pro at high infill percentages was necessary to mitigate stud sharing.



Additionally, we implemented TPU flexible pads that were able to snap onto the top of this design and cushion some of the downforce (illustrated on the right); although, after limited testing, we found that this option was best suited for a mix of grassy and rocky environments rather than the loose sand we were working with.

V-indent:

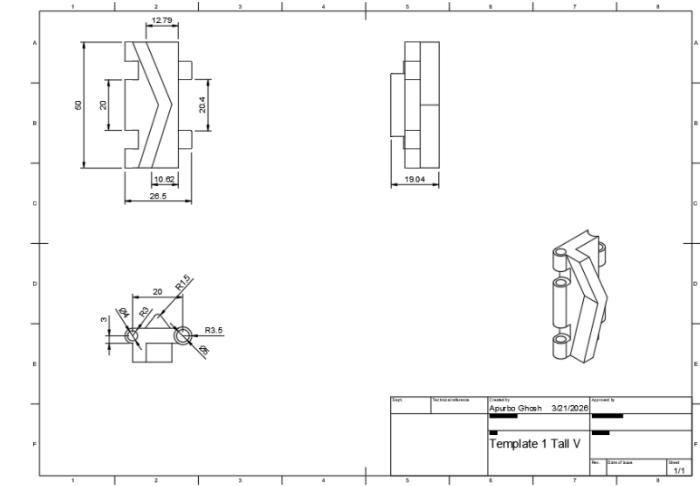
This is another early design we made, but for the first template. We chose this pattern since we saw it while searching for unique tank tread designs. Many of the future patterns we made took heavy inspiration from tanks, just like this one. After testing these treads, we quickly learned that they were very poor. The reason this didn't work for our rover, but it did for the tank we based it off was because of the difference in the depth of the indent. The V shape was much deeper on the pattern we saw than on our own tread. The reason for this matter is because this difference affects the amount of sand we can displace. Something we also originally ignored in our design process was sand displacement. After doing research, we



realized that loose sand behaves like fluids in many ways. It flows around surfaces; it exerts a force like surface tension, and objects usually sink in it. This means the best way to deal with it is how we deal with other fluids. Generating traction through displacement. The more sand we can displace, the more force will propel the tread and therefore rover forward. This can simply be related to Newton's 3rd Law (equal/opposite force pairs)

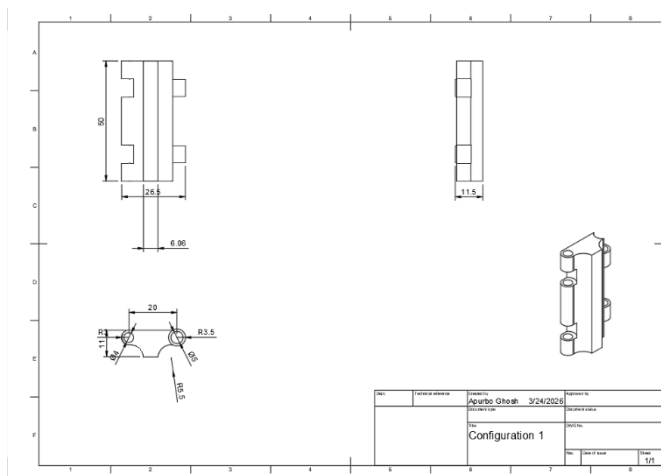
Tall V:

After our previous blunders we tried a similar pattern to the previous tread, but with an extrusion even greater than that of our first prototype.

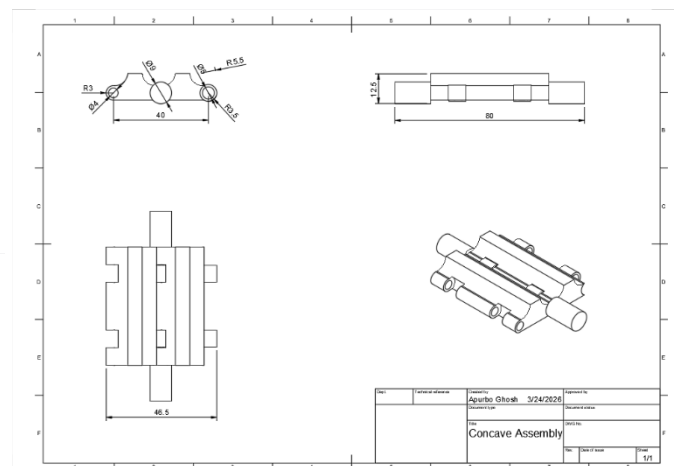


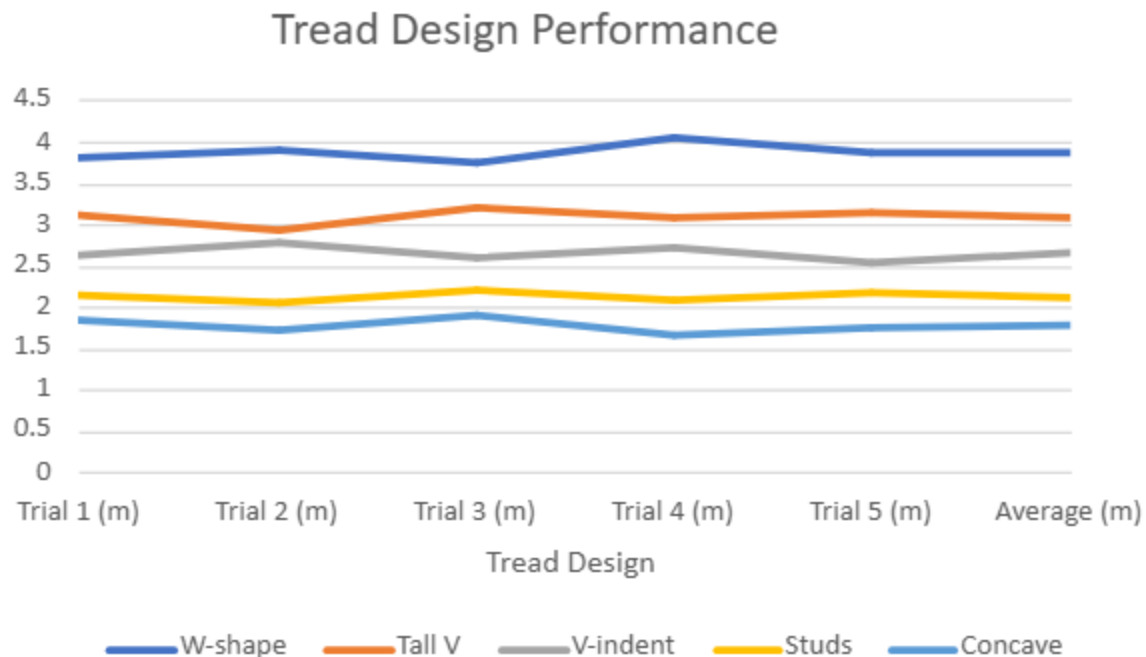
Although this model was certainly an improvement. This was our first tread design that maintained stable movement over the sand for a prolonged duration. Although this was our first sign of a successful design, we know there were many things that could be improved. The tank would dig itself into the sand and since sand acts as a fluid, as we previously discussed, the deeper into the sand we go, the more pressure is applied on our treads from all directions. This means our motor will have to expend more of its power output on overcoming this force.

Another tank tread design we saw while researching was one where extrusions were joined together to make concave assemblies. The idea behind this design was that the larger, flat surface of these treads would distribute the weight better and stop the tank from falling into the sand. Additionally, the area between the treads was large enough that the volume of sand displaced would generate a lot of traction.



This is how the joined assembly would end up looking. After testing, we noticed that this design didn't work as well as we expected. The amount of sand displaced wasn't as high as we thought it would be. We would have to increase the height of the extrusions for this to be more effective.





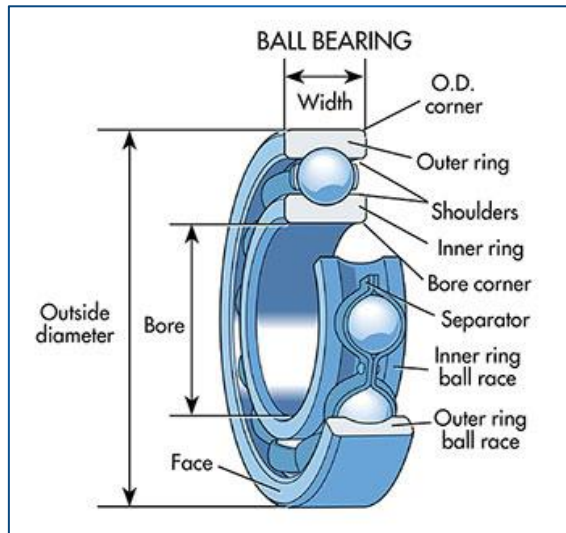
After rigorous testing, the W-shape tread design performed the best out of all iterations. By travelling the farthest given a constant torque in a set time, the W-shape tread design gives us the greatest power efficiency per distance travelled. This is important to preserve the battery for the other important components of the rover.

The W-shape tread design excelled because of its large surface area, translating the most rotational torque into linear motion. Because the competition field is composed of fine sand, the terrain acts as almost as a fluid under higher velocities. Slippage is minimized compared to previous designs because of the increased contact area with the sand at all times.

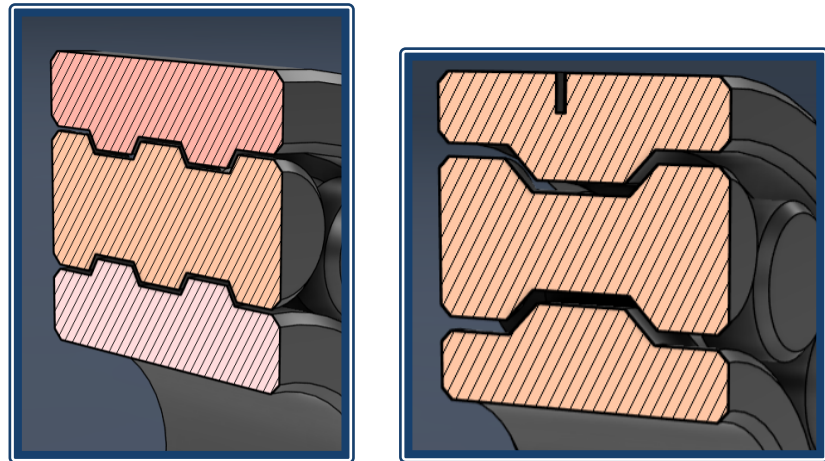
4.2: BEARING DESIGN

4.2.0: OVERVIEW

The nature of our design includes a lot of rotating parts, and typically two 3D printed elements can handle only limited stresses. So, our team was bound to find a solution to prevent excessive wear as well as increase the efficiency of the design. One solution that most real-life mechanisms utilize is bearings. They include some rolling elements as well as inner and outer casings. Typical bearings most people are familiar with include small balls as rolling elements and two ring raceways. The way bearings work they change the sliding friction to rolling friction which has much less resistance, therefore provides more smooth movement.



Instead of purchasing a fixed diameter metal ball bearing, we decided to explore the idea of 3D printed bearings, as it is easier to modify and embed into the design. Additionally, rapid manufacturing provides a quick way to iterate the most suitable designs, allowing us to test bearings with different fixed variations.



We tried to print regular balls however it didn't work as it is extremely hard to achieve the level of precision needed to have a smooth small ball. Therefore, we have switched our focus on rolling bearings which include rolling cylinders instead of regular balls as rolling elements which would be much easier to manufacture - Design for manufacturability. Off the shelf roller bearings are manufactured in different parts and require assembly. However, given the abilities of additive manufacturing, we were able to skip assembly altogether by printing our bearings in place.

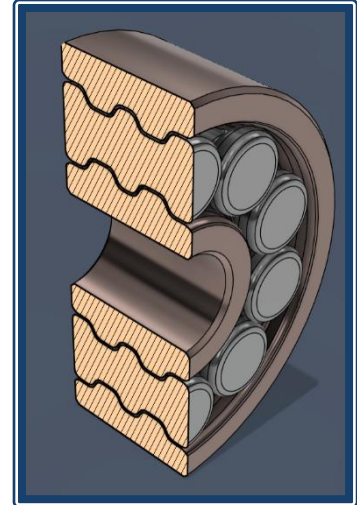
The initial design and testing revealed that to achieve a suitable design we would need a way to secure the cylinders and prevent them from falling out and to prevent them from having too much freedom of movement. So, we increased the width of the cylinder closer to the edge, which provided enough

resistance to prevent each cylinder from falling out. Then we realized that by having only one indentation we are allowing a lot of movement from side to side therefore we have added additional indentation in the middle which allowed for more smooth movement, while adding only small amounts of additional friction.

We then implemented additional fillets and chamfers to provide better print quality and smoothness of motion, which allowed us to come very close to the final model.

4.2.1: USE OF PARAMETRIZATION

Parametrization is a tool that allows us to assign constraints and geometries by applying equations on existing ones.



4.2.1.1: VARIABLES AFFECTING BEARING GEOMETRY

During design stages of the bearing, we have identified a very important feature that would need to be incorporated and considered in our model is a way to quickly change one or multiple parameters, without any difficulties. Therefore, we have utilized built-in parameters feature which allowed us to design the entire model around multiple parameters. We have identified multiple key features that we were interested in modifying, including **Inner** and **Outer Diameter**. Initially, our entire design was focused on those parameters. However, later we have developed many other parameters such as **Tolerance** which is essential when dealing with different printers and materials, as well as **Width** which allowed us to experiment with the optimal size and shape of the rollers, and finally the **Number of Rollers** which proved to be essential when testing and identifying the optimal number.

4.2.1.2: VISUAL MATHEMATICAL MODELS

The design of the bearing included multiple mathematical models which allowed us to precisely control how the final model behaves when changing certain parameters.

Mathematical Models include equations as well as utilizing geometrical properties to achieve a certain property. The following links a Desmos simulation modeling SOME (not all) of the variables in the bearing design including tolerance, inner/outer diameters of the inner/outer rings, and tolerance: [Desmos Simulation of Bearings](https://www.desmos.com/calculator/gafvgajqyb) (https://www.desmos.com/calculator/gafvgajqyb)

4.2.1.3: PARAMETRIZATION IN FUSION

Our main tool for variability and parametrization was achieved by utilizing more powerful resources such as fusion modeling software. It allowed us to change the width of the bearing which proved to be crucial when developing the variations of bearings, sometimes we needed a small bearing and sometimes we needed a thicker to provide smoother even with higher friction. Depending on the situation we needed different variety of bearing therefore parametrized design was playing a very crucial role during various iterations and different stages of our design.

4.2.2: BEARING VARIANTS

Parametrized design allowed us to explore a variety of Bearings which opened the possibility of testing and optimizing the design. We have explored different Wall Thicknesses as well as variety of numbers of rollers with constant Wall Thickness and tested the efficiency and durability of each as depicted in the next section. Additionally, we have explored a variety of fittings that would incorporate the bearing into the actual assembly. We developed various options including snap fit, friction fit and threaded fit. With thorough testing and consideration, we concluded that the snap fit was the most optimal design, considering manufacturing and other constraints.

4.2.3: BEARING TEST PROCESSES AND PROCEDURES

We have tested various materials and different types of bearings to optimize the design and engineer the bearing that would suit our needs the best,

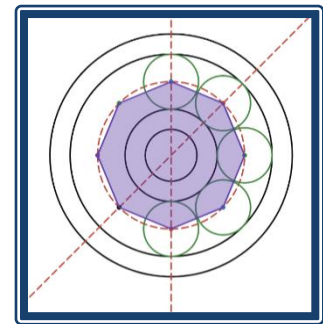
We tested the bearings using the compressed an air compressor and tested how long it would spin and seeing whether it breaks and whether there's any other visible damage to the bearing.

4.2.4: BEARING CHALLENGES

The challenges our team encountered mostly included figuring out the math behind the design. Mathematical modeling plays a huge role in the development of parametrized models thus we will explore a couple instances when it helps us solve a challenge during the development.

4.2.4.1: ROLLER COUNT DETERMINATION

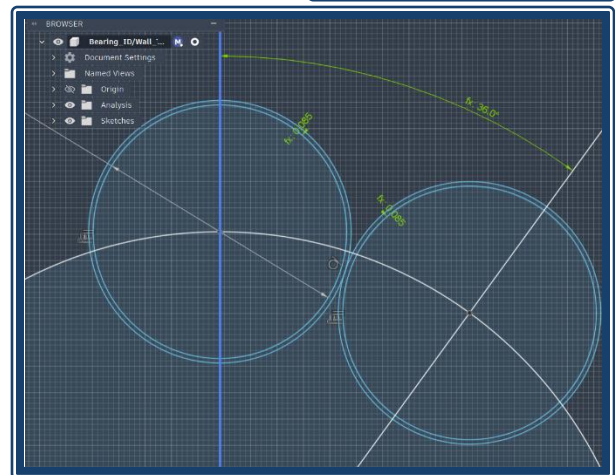
When dealing with variable number of rollers (N) that would need to be calculated utilizing software, it is easy to overapproximate and have non-integer values for number of rollers. Therefore, after some research we have utilized a floor function which when utilized properly gives the closest approximation of the number of rollers for given diameter. The given polygon showcases the approximation that has occurred, and utilizing additional steps we were able to constraint the roller diameter to the precise values needed for proper model.



4.2.4.2: TOLERANCE THROUGH CONSTRAINTS

Additionally, we used CAD to help add tolerance in a precise and controllable way. Utilizing the angle that is dependent on the maximum number of rollers to constraint the two circles together to achieve maximum allowable diameter and then offsetting the two circles allows us to precisely control the distances between all rollers.

This in addition, the ability to control the distance between the roller and the walls of the bearing allowed us to have a single parameter that controls the entire model which helped us to produce a bearing with optimal tolerance.



4.4: SPROCKET DESIGN

This section discusses the types of sprockets, how we used mathematics to model them, how we designed them, challenges we faced during testing, and how we iterated our sprockets around those challenges.

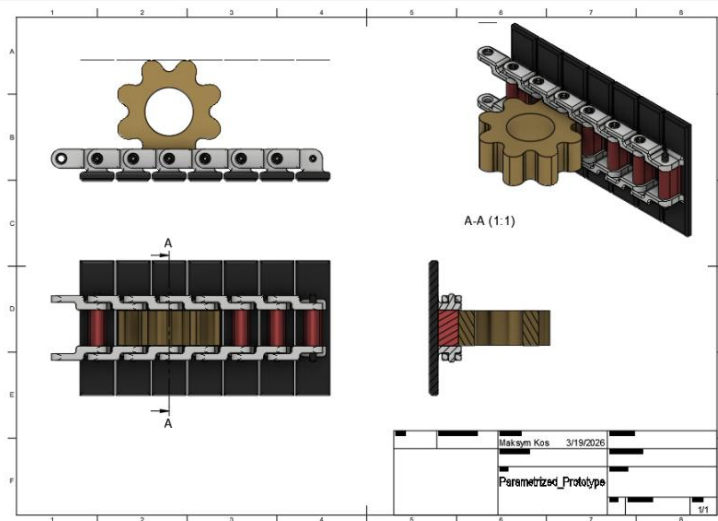
4.4.1: WHAT ARE SPROCKETS

Sprockets are essentially grooved wheels shaped like teeth, specifically designed to work with chains. They make machine motion easier, more efficient and functional. While sprockets are similar to gears, unlike gears they can only interlink with chains. Sprockets are commonly used in many applications such as bicycles, motorbikes, the chain drive of trains and various industrial equipment.

4.4.2: CHALLENGES

Common challenges that we encountered during the testing of sprockets are listed below

4.4.2.1: THE CHALLENGE OF LOOSE SAND



Loose sand was the main constraint we designed sprockets around. Most 3D printed/RC vehicles have extreme trouble with loose sand and immediately break down; with even full-scale military and industrial vehicles requiring specialized geometries to keep traction.

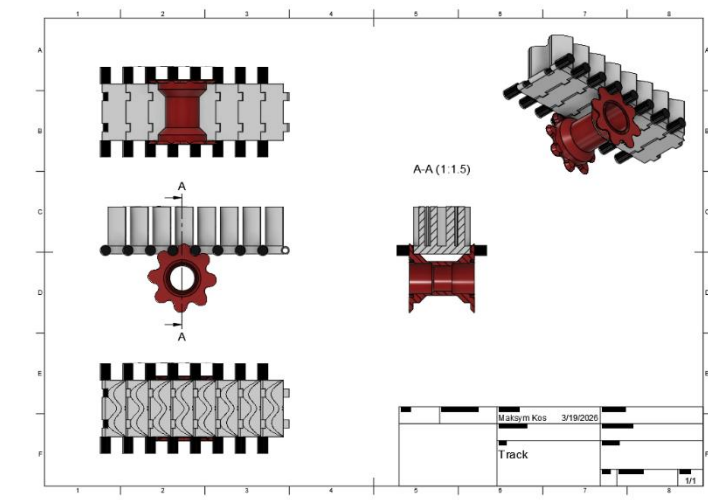
Our original design used inner gears as illustrated, but as we kept iterating, we noticed that these designs struggled with loose sand and were prone to clogging very quickly during testing (we built a sandbox) due to the nature of loose sand.

4.4.2.2: HOW WE APPROACHED LOOSE SAND

Designing sprockets around said substrate required us not to think of sand as a variant of “normal,” clumped sand but rather a fundamentally different medium in line with a liquid than solid ground. In simplest terms, we can say that “normal” sand supports load, and loose sand collapses under it.

[Original Sprocket \(~V34\)](#)

4.4.2.3: DESIGNING AROUND LOOSE SAND



Designing around this we moved the link-gear interaction from the center to the outer perimeter and added a chamfer towards the center. As can be seen in the illustration, this allows sand to flow through the middle or the side of the gear, allowing the tread to run without clogging.

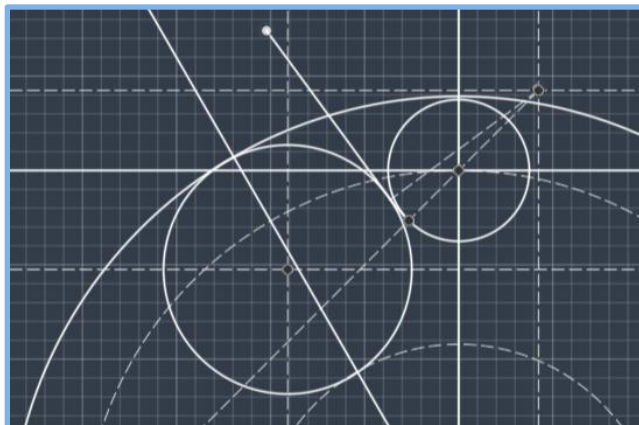
As a side effect of said redesign, sprocket tooth geometry had to be strong and/or thick enough to support the weight of the rover without snapping. While we partially fixed this with our material choice for the sprockets,

Final Sprocket (~V60?)

we went printing the sprockets vertically (tooth face-down) to maximize the strength of the teeth. This was a compromise to maximize tooth strength without making them too bulky; however, this led to sprockets being snapped in half easily along the axle axis. To compensate, we increased the inner gear thickness from 1.2mm to 2mm, fixing the issue.

4.4.3: SPROCKET DESIGN AND PARAMETRIZATION

We utilized an ANSI standard for designing a sprocket to ensure we have the most optimal transmission of movement, which would increase the efficiency of our system.



The method of designing included the utilization of existing constants and geometry to ensure that the motion is getting transmitted smoothly, which with additional changes to suit our design, allowed us to have an optimal model for the sprocket.

By using equations and parametrizing them within the CAD design, we were able to make a model that can be modified very quickly by just adjusting single variables which allowed us to

test various solutions rapidly.

The equations we have used include:

$$Ds_Roller_Diameter_w_Tolerance = 1.0005 * Dr_Roller_Diameter + 0.003$$

- $\text{AngleBetweenRollers} = (180 / N_Number_Of_Teeth)$
- $\text{PD_Pitch_Circle_Diameter} = (\text{Pitch} / (\sin(\text{AngleBetweenRollers})))$
- $\text{Ds_Roller_Diameter_w_Tolerance} / 2$

A_Angle	deg	$35 \text{ deg} + (60 / N_Number_Of_Teeth) * 1 \text{ deg}$	45.0
B_Angle	deg	$18 \text{ deg} - (56 / N_Number_Of_Teeth) * 1 \text{ deg}$	8.7
ac_Line	mm	$0.8 * \text{Dr_Roller_Diameter}$	6.00
M_Offset	mm	$(0.8 * \text{Dr_Roller_Diameter}) * \cos(\text{A_Angle})$	4.243
T_Offset	mm	$(0.8 * \text{Dr_Roller_Diameter}) * \sin(\text{A_Angle})$	4.243
× E_Length	mm	$1.3025 * \text{Dr_Roller_Diameter} + 0.0015 \text{ mm}$	9.77
xy_Arc_Length	mm	$(2.605 * \text{Dr_Roller_Diameter} + 0.003 \text{ mm}) * \sin(\dots)$	1.476
C_Angle	deg	$17 \text{ deg} - (64 / N_Number_Of_Teeth) * 1 \text{ deg}$	6.3
yz_Line	mm	$\text{Dr_Roller_Diameter} * (1.4 * \sin(\text{C_Angle}) - 0.8 * \dots)$	0.254
ab_Line	mm	$1.4 * \text{Dr_Roller_Diameter}$	10.50
W_Offset	mm	$\text{ab_Line} * \cos(\text{AngleBetweenRollers})$	9.093
V_Offset	mm	$\text{ab_Line} * \sin(\text{AngleBetweenRollers})$	5.25
F_Constant	mm	$\text{Dr_Roller_Diameter} * (1.4 * \cos(\text{C_Angle}) + 0.8 * \dots)$	6.597
H_Constant	mm	$\text{sqrt}(((\text{F_Constant}) * (\text{F_Constant}))) - (1.4 * \text{Dr_Roller_Diameter})$	6.578
S_Constant	mm	$(\text{Pitch} / 2) * \cos(\text{AngleBetweenRollers}) + \text{H_Constant}$	11.949

Those equations were utilized to create the geometry of the tooth.

That produced geometry would provide the smoothest transmission of motion as it utilizes the standardized constants and equations.

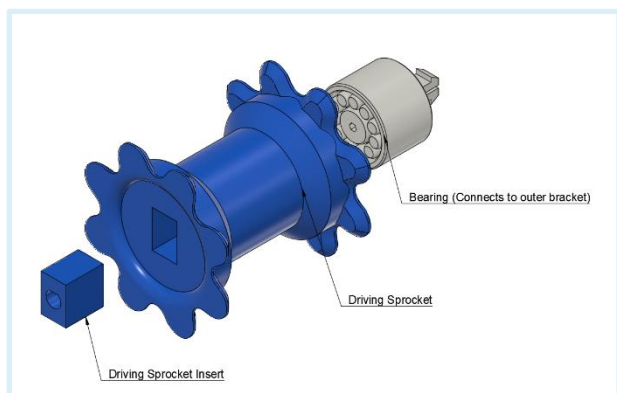
4.4.4: SPROCKET VARIANTS

The nature of treads leads to four main sprocket variants on most relevant vehicles (construction, military, industrial). Each variant is suited towards different applications and serves a unique purpose in the context of IAM3D guidelines (Loose sand over a 2.4m square area from the report, few hills or bumps).

4.4.4.1: DRIVING SPROCKETS

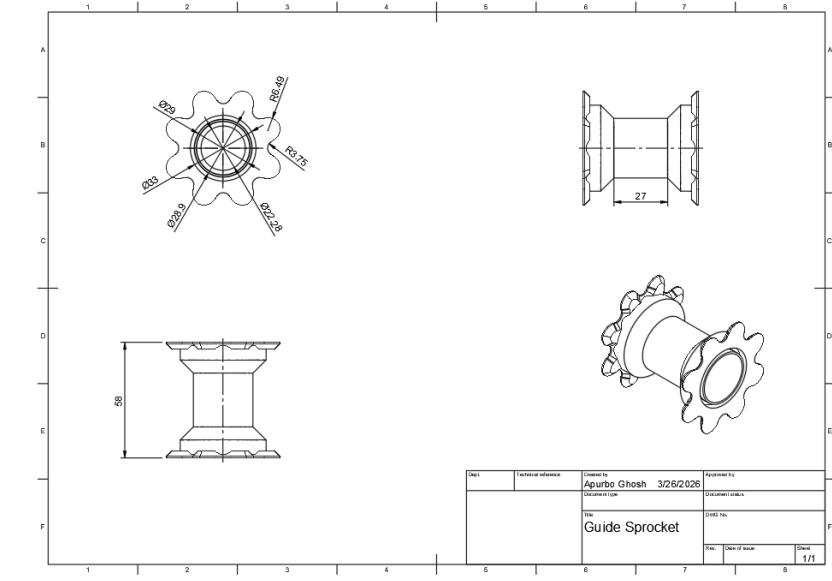
Driving sprockets are powered sprockets coupled with the motor/gearbox and directly engage the track lugs to transmit torque and make the tread move. They also define the effective pitch of the tread system and require adequate tensioning to operate. As is essential for any practical vehicle, the addition of a driving sprocket for each motor is trivial.

Our design for the driving sprockets includes three parts: The sprocket itself, a snap-fit bearing to attach to the outer bracket, and an insert for the driving sprocket. Initially, our driving sprockets were printed at 15% and directly connected to the motor; however, they had a tendency to strip rapidly during testing, only lasting around a minute. Due to restrictions on filament use (budget is incredibly low), we decided to split the sprocket into the gear itself (still at 15% infill), and an insert printed at 100% infill. This change increased the keying so that the insert itself doesn't get stripped because of the high infill, and the insert can't physically strip the gear due to the dramatically increased torque transmitting geometry.



4.4.4.2: GUIDE SPROCKETS

Guide sprockets have teeth but are NOT motor driven, unlike the driving sprockets. There are 3 guiding sprockets on each side of our movement assembly and only one for driving. This is because having 4 separate motors on each side would not only be inefficient, but it would also make it hard for us to time the motors, so the torques aren't disruptive. The main purpose of the guides is to provide our tracks with tension. They also serve the purpose of aligning the tracks and controlling sideloading (a direct result of misalignment).

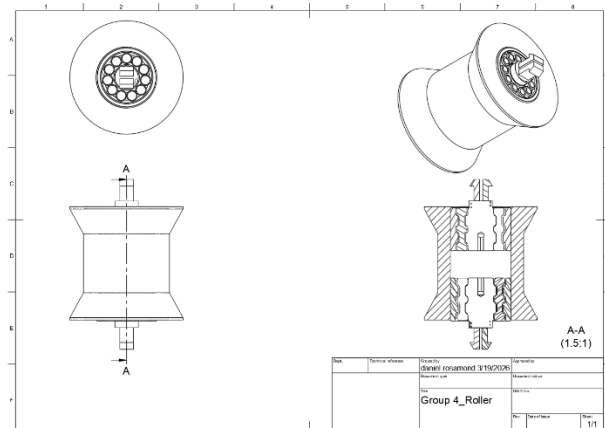


4.4.4.3: ROLLER SPROCKETS

Roller sprockets are essentially guided by sprockets without teeth. If guide sprockets are the teeth of an ordinary working man (where the track represents food), think of roller sprockets as an old guy with dentures. While stronger and typically better for high-speed applications, they don't grip onto the food/track and therefore transmit torque as well as guide sprockets.

4.4.4.4: IDLER/TENSIONING SPROCKETS

Idler sprockets are typically used to maintain tension throughout a track system. Whether they are toothed or smooth like a roller sprocket depends on the style and design of the track. In that sense their main function is purely for tension, and while you can transmit torque through them, but that torque is not usually usable drive torque and more just torque to keep them from spinning freely. For our final design we used rigid tracks which didn't require tension, eliminating the need for idler gears.

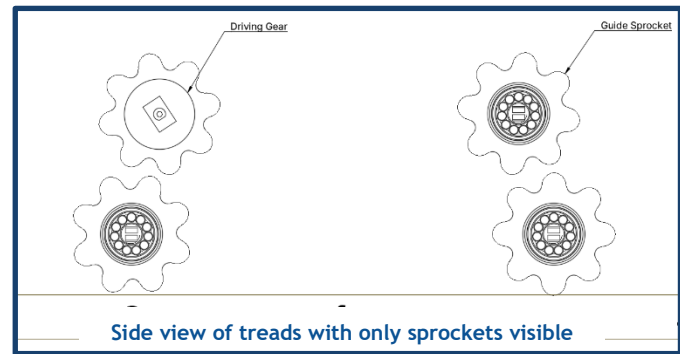


4.4.5: SPROCKET RATIONALE

IAM3D guidelines suggested sandy terrain with an area of approximately 2.4m x 2.4m. Further questions consulted through the official ASME student section discord gave us additional information of the type of sand and terrain, leading us to redesigning around loose sand and minimal terrain elevation changes.

4.4.5.1: SPROCKET SETUP AND PLACEMENT

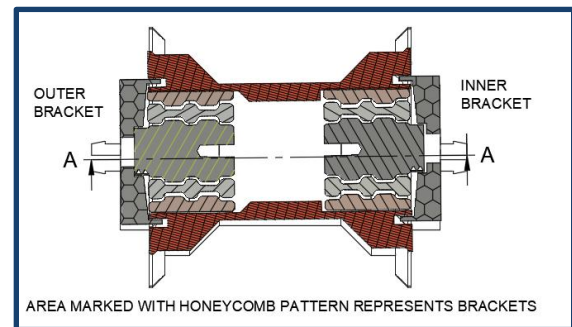
Considering that information, we decided to pursue one sprocket drive and three guide sprockets. Due to our treads being fully rigid, we did not require a tensioning sprocket, as our gear teeth were long enough to accommodate for any small displacement, and assembly uses pins for the links, meaning we did not have to stretch the treads at all to put it together.



Roller sprockets were unnecessary due to the nature of sand and the limited distance we needed to cross. Our motors are high torque, low speed, and the competition area is very small in comparison to the vehicle. This meant that we wanted to prioritize transmitting all that torque onto the surface rather than load bearing, so normal guide sprockets were the optimal choice. Originally, choosing only guide sprockets for a vehicle designed for sand presents the challenge of keeping sand from clogging the system, but as our sprockets were designed with sand in mind and only engage with the outer area and pins on the links, this challenge was mitigated easily.

4.4.5.2: ANTI SAND MECHANISM AND CLOG PREVENTION

Another big issue as a result of sand was the effect of it on our bearings. Due to the open gaps in the bearings and the type of sand being loose, we found that our sprockets often halted midway through testing due to the bearings being full of sand. This was a much greater problem than originally thought, especially due to the type of lubricant being gel WD40 and therefore more prone to clinging to sand. Our solution for this was a labyrinth mechanism, where in theory it's possible for sand to get in, but the distance to the actual bearing from the outside is significantly increased, with grains of sand requiring multiple directional changes and passing through narrow gaps.



Anti-Sand Mechanism

Although not fully effective (Sand gets in the bearings after about an hour of runtime on average on the rover), we don't really need it to be fully effective as the competition guidelines explicitly state that the competition itself only lasts five minutes, well within the abilities of our anti sand mechanism as long as replaced with new bearings right before.

5: PAYLOAD

The Payload consists of the scoop, arm, and includes every piece made to help those complete the goal of moving and depositing.

5.1 SCOOP

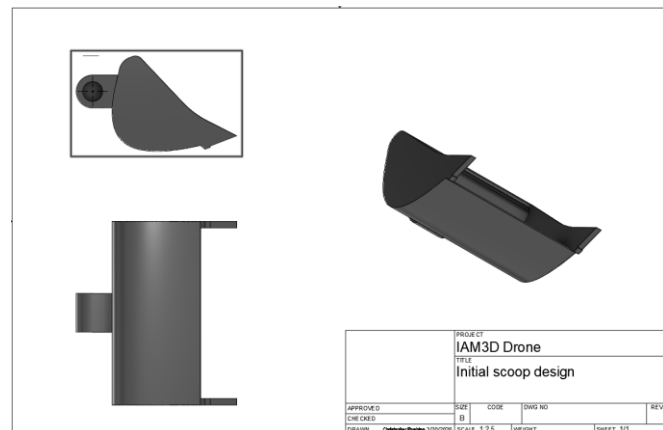
This section goes in depth on the design of the scoop, issues relating to the displacement of sand, how we solved those issues, the testing done for each scoop design, the use of generative design, and the final scoop we decided on.

5.1.1: EXCAVATION METHOD AND REASONING

The environment given to us by ASME was known to be that of sand with objects buried in various depths into said sand. This left us with the decision of how we could efficiently retrieve the objects and lift them into the bucket 20cm off the ground. We decided on a scoop to be the main mechanism early on for delivery of objects picked up by the rover. We came to this plan due to it being the most straight forward method for digging in the sand.

5.1.2: INITIAL DESIGN

Our initial design for the scoop was to make it as large as possible while still fitting within the design constraints of our rover. We decided on this because it allowed us to pick up the most objects and allowed us to have a larger margin of error when attempting to pick up said objects. The scoop was approximately 15 cm tall and 28 cm wide, featuring a parabolic side profile allowing it to glide through sand and pick up objects effectively.

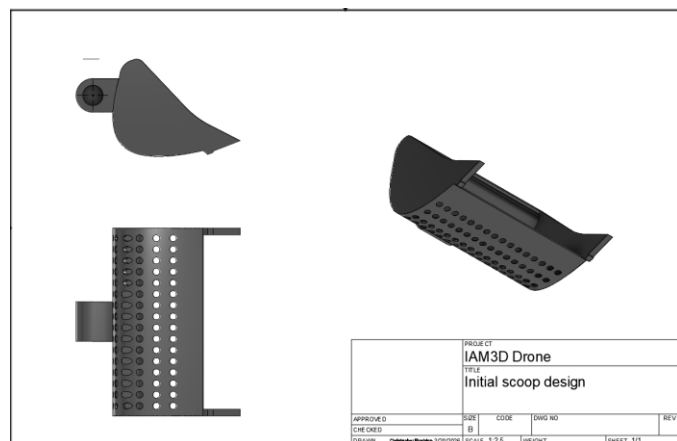


5.1.1.1: PROBLEMS WITH FIRST ITERATION

The biggest issue with the first design build was the buildup of sand within the scoop which resulted in a higher difficulty in lifting objects, gliding through sand, and moving when deployed. These issues made it nearly impossible to accomplish the goal set forth by this project effectively. Additionally, when printed with lower infill, the scoop would no longer be able to support the weight of the objects and the shear force applied by pushing through sand would cause it to break.

5.1.1.2: SOLUTIONS

The way we addressed these problems was by adding several holes to the scoop hoping that it would work like a sieve and allow for the sand to flow through the scoop rather than get stuck. Additionally, we only printed 100% infill when using this design for the scoop as it allowed us to bypass all strength issues.



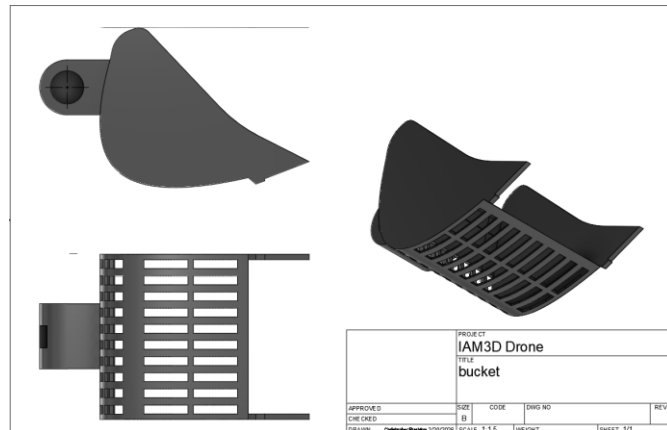
5.1.3: UPDATED FOR AERODYNAMICS

We quickly noticed that with the buildup of sand and overall friction caused by the sand we would not be able to effectively lift, move, or gather items without a way to remove the sand. We found that the biggest factor affecting all of these was the surface area the scoop was covering. Because of this, we decided to shrink the surface area as much as possible by reducing the width to 15.5 cm and adding a sieve. We worked under the

assumption that while the rover was moving sand could be seen as a liquid gliding around the rover, so we made it as aerodynamic as possible.

5.1.3.1: PROBLEMS

The biggest problem that we saw with this scoop design was once again the inability to push through sand and lift objects. Additionally, we were running into the same problem of strength as when the scoop would push through the sand some of the cross supports on the sieve would break.

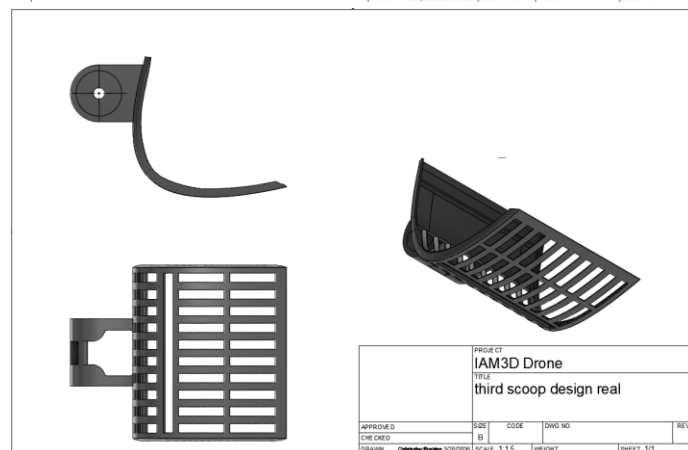
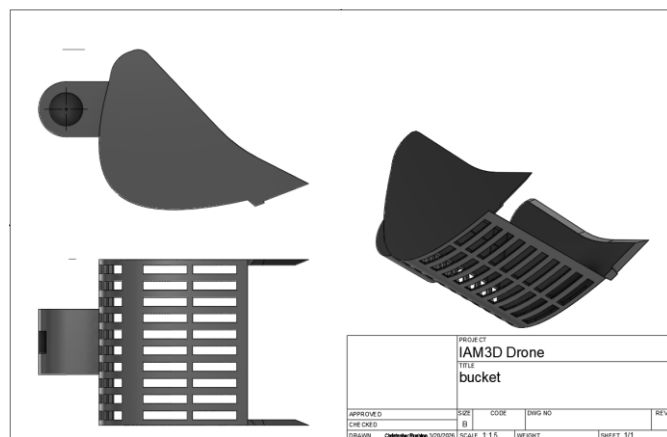


5.1.3.2: SOLUTIONS

To solve these problems, we decided that the scoop was not aerodynamic enough, so we added numerous chaffers to multiple edges on the scoop which pushed sand to the side as the scoop moved through it. Additionally, we held to the same standard of only printing with 100% infill to improve the strength of the scoop and stop the cross bar supports in the sieve from breaking.

5.1.4: REMOVAL OF SIDE WALLS

As previously mentioned, we wanted to reduce the surface area of the scoop, so we decided to remove the side walls that covered each side from the top of scoop to tip. The belief was that this removal would allow it to glide better through sand since it now only had the bottom and back to catch the sand. Furthermore, we filled the edges of the design to allow for more aerodynamics as the scoop flows through the sand. Additionally, we realized that a large portion of the attachment would not be necessary, so we hollowed out the connecting piece to remove weight from the design and allow for a higher carrying capacity.

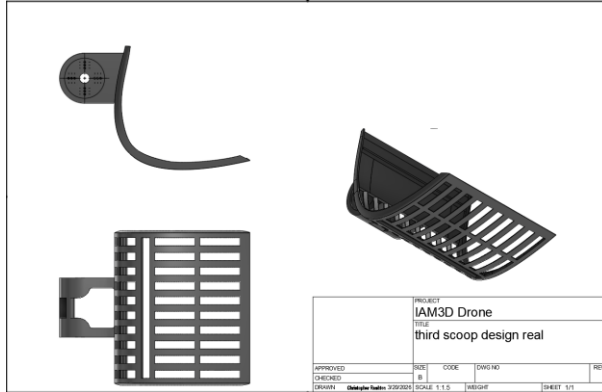


5.1.4.1: PROBLEMS

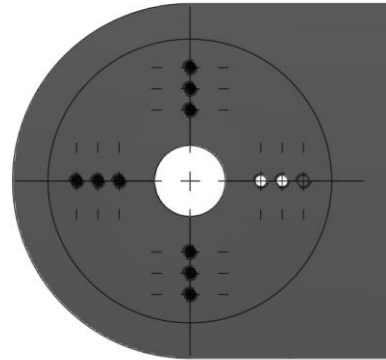
We realized with this design we would need a way to physically rotate the scoop once picked up because of the parabolic nature of the back end otherwise there would be no way to deposit the items. Additionally, we continued to have problems with the sieve bending and cross supports breaking when carrying an object with too much weight. A new problem we experienced was the scoop bending in on itself because of the pressure when driving forward under the sand.

5.1.4.2: SOLUTIONS

We added holes to the attachment which are designed to fit perfectly with the servo mounts we have. We would connect the servo to this, and it would be able to rotate the scoop at a full 180 degrees while attached to our arm.

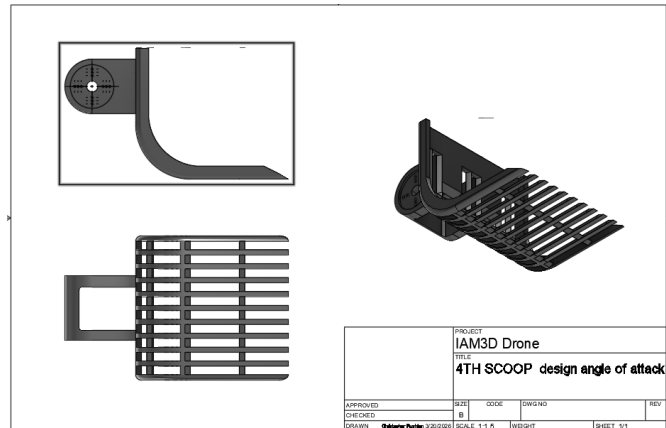


To solve the problem of the scoop bending in on itself we printed out a scoop with a 0.6mm nozzle and changed layer lines to 0.1mm which greatly improved strength and rigidity.



5.1.5: DIFFERING ANGLE OF ATTACK

We decided on making alternative designs with angles of attack closer to 0 degrees so when fully extended, the scoop would be flat and level while pushing through sand to solve the bending issue we had from previous designs where the scoop would bend in on itself. We also expanded the sieve to the upper sections on the back of the scoop to improve the flow of sand through the scoop.



5.1.5.1: PROBLEMS

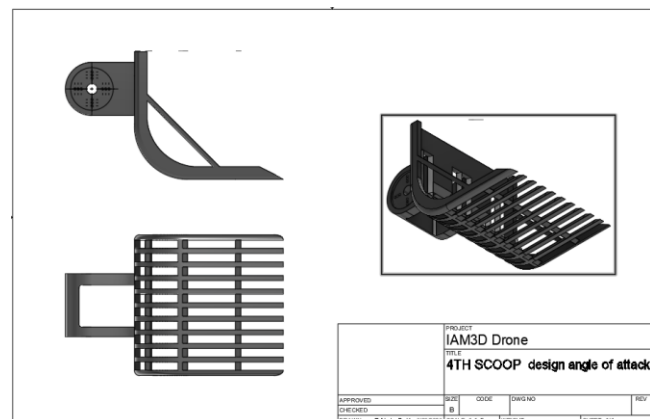
We continued to have the same problem of the scoop bending in on itself albeit less severe in this case due of the difference in design, however, it was still a hinderance. Since it would slightly bend in on itself when the scoop would run into objects below the sand, it would sometimes push the objects further down where we could not retrieve them and may break the sieve cross supports in the process.

5.1.5.2: SOLUTIONS

The solution to this problem was added a diagonal support from the back of the scoop down to the bottom which allowed for the bending to be entirely removed and fixed all previous problems relating to objects being pushed further down and the sieve breaking.

5.1.6: GENERATIVE DESIGN

This section discusses the use of generative design to address loading and weight challenges in the scoop, along with its integration into the final design.



5.1.6.1: PROBLEMS WITH CURRENT SCOOP

The current scoop we have included a lowered angle of attack, diagonal support, a sieve, and servo mount. However, it remains heavy, specifically in the back and still faces loading issues when faced with complex loads.

5.1.6.2: BENEFITS OF GENERATIVE DESIGN

Some of the many benefits related to generative design include weight reduction, displacement of complex loads, and smoothed designs. Weight reduction allows us to reduce the total material used, which reduces the weight of the design while allowing us to increase our carrying capacity. Displacement of complex loads allows us to increase the factor of safety with respect to what our scoop is expected to lift while moving since we will have more evenly distributed loads. Smoothed out designs allows us to not worry about objects getting stuck in our design due to the surface will being smoothed out and pieces flowing along them.

5.1.6.3: INTEGRATION INTO FINAL DESIGN

The way we integrate generative design into our final design is by generating the connection between servo and scoop. By doing this we can eliminate the biggest problem we have, which is the weight in those areas. Additionally, it solves our other problems of complex loads as that would be the most likely breaking points during use, which will increase the amount we can carry.

Generative Design



Old Design



5.1.7: FINAL DESIGN

The final scoop design incorporates features from all of the previous designs. This includes a reduced angle of attack, so it is parallel to the sandy plane. Improved structural support from the diagonal cross support, allowed the sieve to dissipate the sand more efficiently. Integrated servo attachment allowed for rotation of the scoop, and generative design was used to establish a more evenly distributed load when moving. Importantly, the servo mount was made to be placed inside of our final design for better rotation.

5.1.7.1: BENEFITS OF CURRENT SCOOP

- SIEVE

- The sieve we added in allowed us to fix the problem of sand building and pushing objects out of the way of the scoop. Additionally, with the integration of the sieve, we no longer had to worry about the stresses on the arm and scoop that were caused by the additional weight of the sand. This allowed us to collect objects in the sand effectively for the first time.
- **ANGLE OF ATTACK**
 - The reduced angle of attack incorporated into our design allows us to reduce bending moment pressure from the forward push into sand. Additionally, it will no longer push objects down if they are just below the parabolic curve of other designs.
- **CROSS SUPPORT**
 - Bending stress from the forward push into the sand would often cause the front sieve to curve inwards. This would create numerous problems in terms of structural support and bring about caution in regard to potentially breaking the scoop and limiting our ability to pick up objects. The addition of cross support from the back to the sieve serves to mitigate these problems and makes it possible to treat them as negligible in our final design.
- **SERVO ATTACHMENT INTEGRATION**
 - Depositing objects into the bucket at a height of 20cm above the ground height with the scoop would be an issue because initially there was no way to rotate the scoop itself. Adding a servo attachment mitigates this issue as the scoop could now rotate as much as needed to deposit the items.
- **GENERATIVE DESIGN**
 - Weight was a massive problem with our scoop design; this was mostly caused by the necessity of printing at 100% infill for structural issues. Additionally, we had issues with complex loads when the scoop was moving its common break points were at the attachment of servo mount to scoop. Using generative design solved both problems by reducing total filament used and helping us distribute the complex load more evenly.

5.1.7.2: JUSTIFICATION OF SCOOP

The final design of the scoop was selected because it meets all functional requirements laid out by the ASME competition as well as achieving club goals. This led us to design one of the most critical pieces: our sieve. With it, we were able to minimize surface area by simply removing the sides and adding cross support for structural rigidity, allowing us to reduce the overall area and therefore reduce the total kinetic friction. Our realization to add the servo mount was necessary because it allows us to rotate the scoop to dump objects into the bucket. An angle of attack parallel to the sand was selected as this allowed the objects to be better collected. All these innovations were important, but the overarching problem of weight and distribution of complex loading still existed. Generative design solved all these issues by drastically reducing the total weight of the scoop by decreasing filament used and distributing the load more efficiently using organic designs.

5.2 ARM

The function of the arm is to serve as the primary segway between the servo and the scoop. Its only function is to act as a lever.

5.2.1 DESIGN REQUIREMENTS

The initial constraints of the arm began with the plotting of our axis of rotation mounted on the chassis along with the axis of rotation where the arm meets the scoop. The second step consists of constraining the height difference, as the arm-scoop axis must sit lower while the chassis-arm axis is

mounted high (>89.5mm, measured the bottom of the chassis frame). Then we extruded bodies that resembled the existing chassis body (payload plate, cross members, inside chassis plates). Now, with our necessary faces we want to connect and our constrained bodies to avoid we can run generative design to maximize strength and minimize material while doubling as an effective weight-reduced and sand-resistant payload arm.

With the position we mounted the arm onto the chassis (positioned towards the back end of the chassis on top of the payload plate), our arm constraints paired with generative design created an arm 158mm long, not including the scoop. The positioning of our arm is fully intentional, as we planned to deposit all items into the bucket by reversing into the face of the bucket and rotating our arm backwards 110-140 degrees from rest while rotating our scoop. Our collected items will slide off the scoop due to the incline and drop into the bucket.

5.2.1.1: DESIGN CONSTRAINTS

Early on we decided on design constraints for the arm based on the rules provided to us and personal club outlines. These constraints are important because they narrow the overall scope of our design into something more optimal than we can focus on.

- Area of scoop and arm assembly
 - Fit an area of 15cm tall x 30 cm wide x 30 cm long. This will allow enough space for the tank treads we designed for movement. Additionally, this gives us an end goal and space to work within.
- Support Maximum Shear Possible
 - This is important because we need to make sure the arm does not break upon the point of force applied at the joint between the arm and the scoop. We assume the joint connecting the scoop and arm will have a point of force acting on it since the shear force will be transferred along the scoop.

5.2.1.2: DESIGN OBJECTIVES

The objectives of the arm as laid out by ASME rules and club objectives set fourth numerous requirements. Based on these requirements, we will design our rover to fit all criteria and allow for maximum performance in the following categories:

- Minimize weight
 - We need to minimize weight as this allows us to neglect more shear force since the weight of the arm will be less effective towards the summation equations performed later. Additionally, this will help with the moment applied by our servo at the base of the arm later.
- Maximize Strength
 - Maximization of strength is hugely important because we need to make sure the arm can support the shear stress it will experience from the point of force of the scoop joint. However, we will need to balance this with weight to ensure the best outcome.
- Minimize Area
 - We need to minimize the area taken up by the arm and scoop system. This is because we still need to add a camera to the back of our rover to see the first person's point of view. The camera will be added after the assembly is complete and will fit around it so it's important to use the least area possible.

5.2.2: DESIGN CONSIDERATIONS

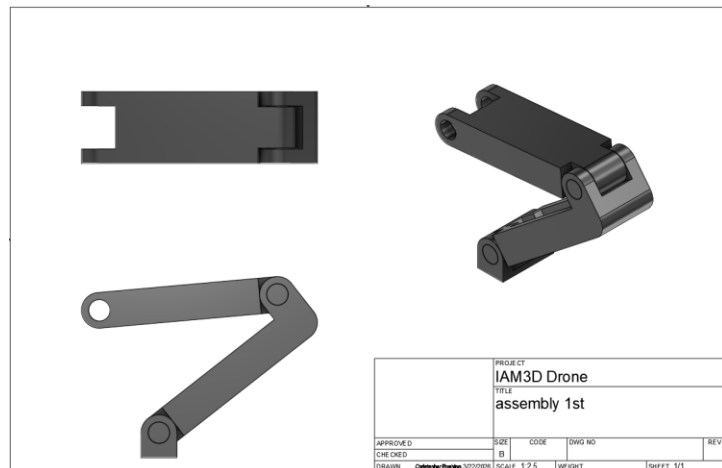
Many iterations were taken into consideration, and we used the strategy of rapid prototyping to select the best and most promising.

5.2.2.1: FIRST ITERATION

Our first iteration of the arm was very quickly put together and done to get a quick idea of what we may want future designs to look like. Because of this it had a lot of issues that will be discussed and dealt with in the future.

5.2.2.1.1: DESIGN

The first design assembly was one with 2 separate arm pieces. We initially modeled it based off the fact that our arms as humans have two sections the forearm and arm. It fits within the area requirements and allows us to maintain the reach we would need for the 20cm bucket height requirement. Furthermore, it featured a universal connection joint designed to be interchangeable if they had the joint.



5.2.2.1.2: ARM ASSEMBLY PROBLEMS

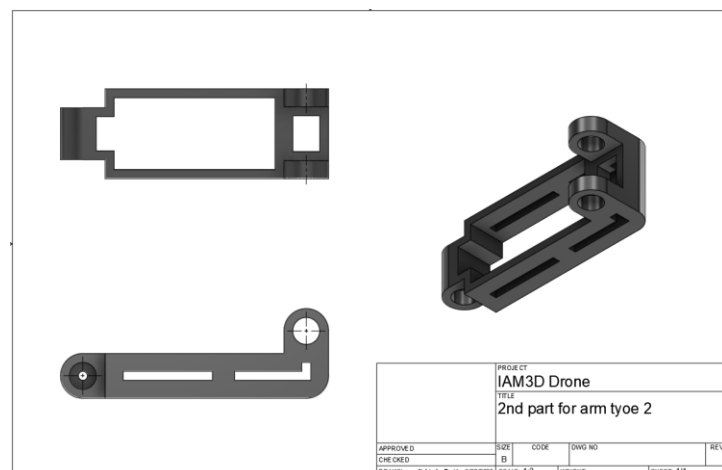
The biggest problem with this design was the size and complexity of it. When you add 3 servos, it makes the electrical work substantially more complicated and heavier. Each servo weighs about 55 grams which, when fully extended along with the weight of each arm segment, would overload the bottom servo and not allow for it to lift the assembly. Additionally, the arm segments were 30 cm long, which would allow very little room for error when designing the scoop and additional segments.

5.2.2.2: SECOND ITERATION

The second iteration of the arm was more reasonable in terms of simplicity and some fundamental designs like the single lever system. However, it still had a major problem with weight and size that would need to be addressed.

5.2.2.2.1: DESIGN

We were able to shrink the overall size of the assembly allowing for the scoop to fit with ample room for error. Additionally, we shrunk the total filament used and weight of the arm substantially, which allowed us to effectively test the servo's ability to lift the arm. Furthermore, the strength within the arm was still very strong and would easily be able to hold and shear stress applied by the point for at the end connection of the arm. We also kept the concept of the universal joint attachment system.



5.2.2.2.2: PROBLEMS

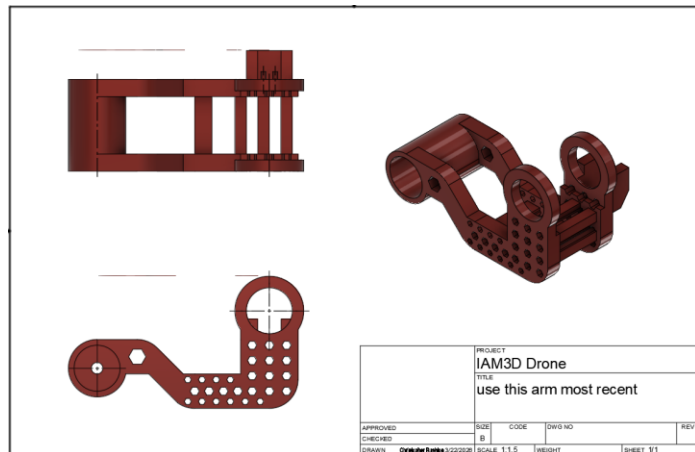
The main problem with this design is the size of the arm; at this point in the design process, we assumed the arm should be a full 30 cm in length to allow for maximum error in depositing the items. However, we soon realized it would cause too much stress to have the arm be that long as even an extra few centimeters drastically reduced the servo's ability to lift the arm. Furthermore, when the arm is longer it inherently will weigh more, so reducing the length would fix both problems.

5.2.2.3: ARM FINAL ITERATION

Our final iteration of the arm solved all the main concerns and allowed us to have an effective test using the arm and scoop assembly. However, weight was still a concern, and the minimization of it was a top priority.

5.2.2.3.1: ARM DESIGN

The final design we made solved all the problems we previously had. We keep the universal joint that allows us to reduce printing and continue interchangeability. The main addition added to this new design is the servo mount that allows us a design spot to keep the servo in for the scoop attachment. Additionally, the total length of this arm is only 20 cm, which gives us substantially more flexibility for the servo at the bottom attachment.



5.2.2.3.2: ARM PROBLEMS

The weight of this arm is still greater than desired, although the total design is very specialized. Additionally, there was a concern with the complex loading of the arm as it moves and lifts the scoop. This is because the arm has numerous sharp angles that would not distribute the load very well internally.

5.2.3: GENERATIVE ARM DESIGN

Applying loads and running generative design allows us to condense our design.

5.2.3.1: PROBLEMS WITH CURRENT ARM

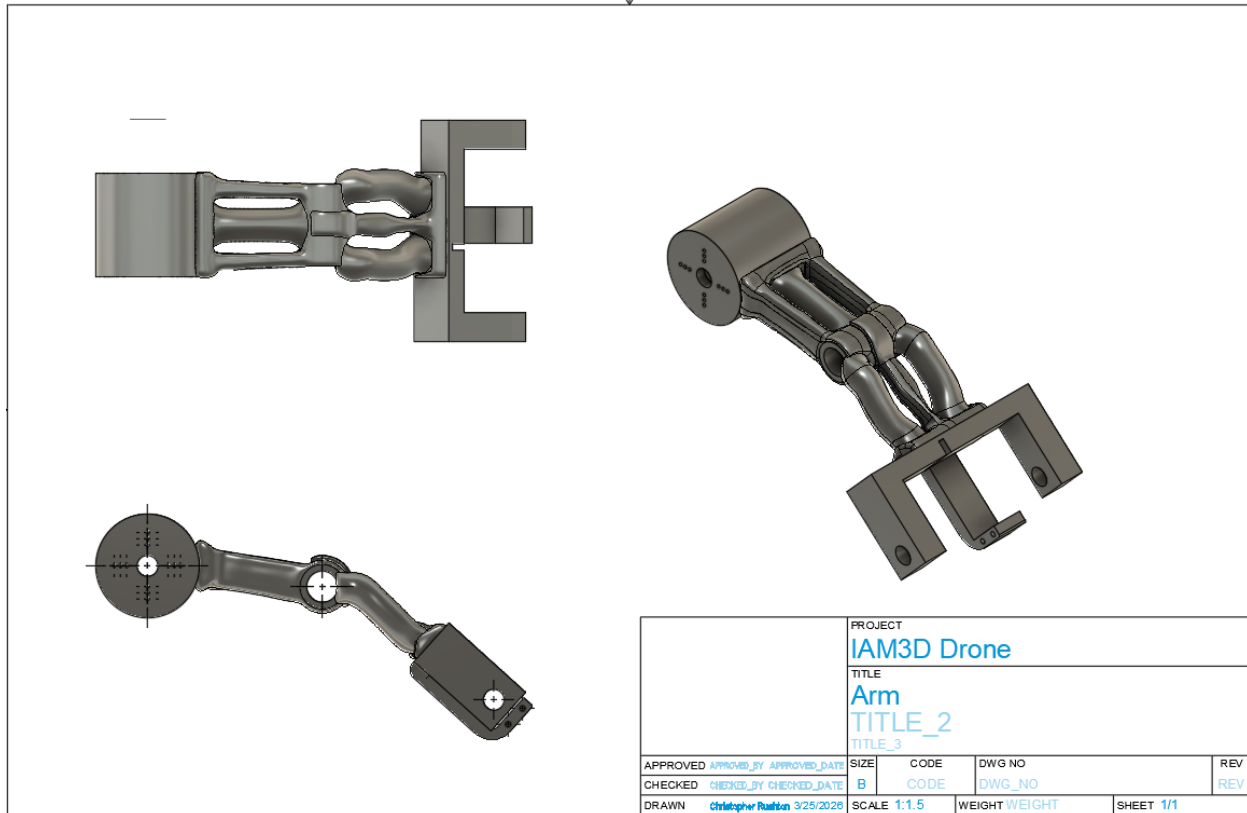
The two main problems with the current arm design are the weight and the internal loading stemming from complex loading. Weight has been the biggest factor when designing the arm as we want to minimize it as much as possible to allow for the focus of the bottom servo to lifting the scoop. Additionally, we are facing a new problem of the distribution of load internally within the scoop. We want it to be as evenly distributed as possible so we can minimize the weight and be certain of its integrity.

5.2.3.2: BENEFITS OF GENERATIVE DESIGN

When generative design is performed, all unnecessary filaments are removed and only the parts sectioned off are kept. Because of this we can be assured that we are not keeping any overly bulky design and all only essential pieces. Furthermore, we can be assured that the distributed loading will be as equally distributed as possible in the final design.

5.2.3.3: INTEGRATION INTO FINAL DESIGN

The way we integrated this into the final design is by generating the design between the bottom servo connection and the joint scoop connection. By doing this, it eliminates all excess materials and allows us to get the minimum weight while getting the strongest result under load. It solves our biggest problem while preserving our design so we can ensure it fits into our assembly. Furthermore, we changed the initial base to have better rotation and added a servo mount on the end the scoop will fit over.



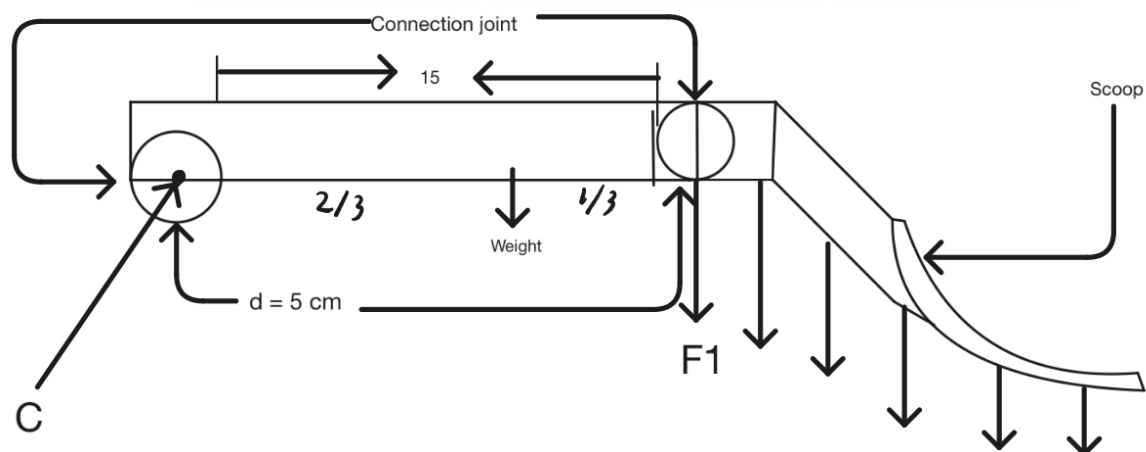
5.2.4: ARM CALCULATIONS

Based on the data we have gathered, we performed calculations on the various forces acting on the arm and scoop assembly. The point of these calculations was to determine the maximum moment necessary to apply to the arm from the servos and the total shear it would experience.

Important note: the final weight of the scoop was 173.0 grams; servos were 55.00 grams and arm was 162.2 grams.

5.2.4.1: DIAGRAM

The diagram listed below is made to show the forces acting on the arm and scoop assembly. These include the moment at C, the shear forces the total scoop experiences, and the reaction forces at C. Additionally the Weight and point force F1 are included. With F1 acting as a point of force given that all the force acting on the scoop would be transferred to that joint connection. Additionally, we did these calculations based on only the horizontal distance at the maximum stress point of full expansion parallel to sand.



5.2.4.2: EQUATIONS

The calculation for the vertical support reaction at point C is shown below and the applied force in kg as a response to the loading on the right side of the diagram. Total would be 3.83 Newtons. We assumed a 0.5 kg load for the point force F1 which will be combined with weight of scoop

$$\sum F_y = C_y - F_1 - W = 0$$

$$C_y = 0.173 + 0.055 + 0.5 + 0.1622 = 0.3902 \text{ kg}$$

The below equation shows the moment that would need to be applied at point C to support an excess load of 0.5 kg in the scoop. Our servo must be able to output 16.7 kg*cm in terms of torque.

$$\sum (M)_c = F_1 \cdot 20 + W \cdot \left(\frac{2}{3}\right) \cdot 20 =$$

$$M_c = (0.173 + 0.5 + 0.055) \cdot 20 + 0.1622 \cdot \frac{2}{3} \cdot 20$$

$$M_c = 16.7 \text{ kg*cm}$$

5.3 SERVO MOTOR INTEGRATION

This section will cover all servo selection, mounting process, integration into the design, range of motion, and the interface used to connect the servos.

5.3.1: MOUNTING

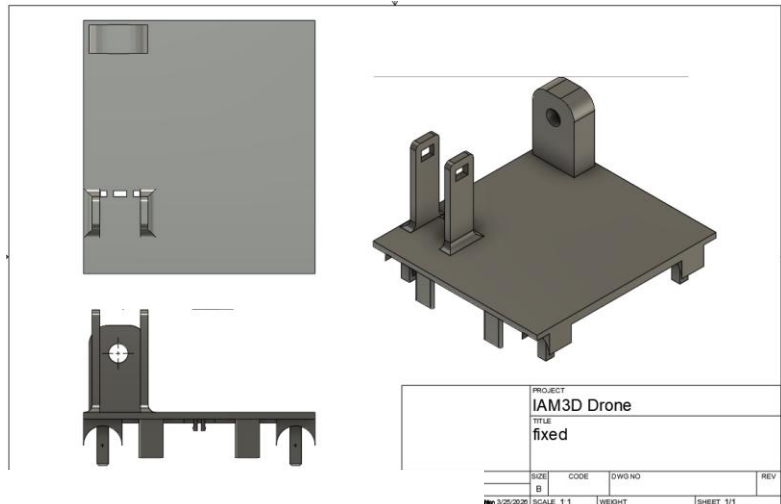
The mounting of servos was done through physical structures.

5.3.1.1: CHASSIS MOUNT

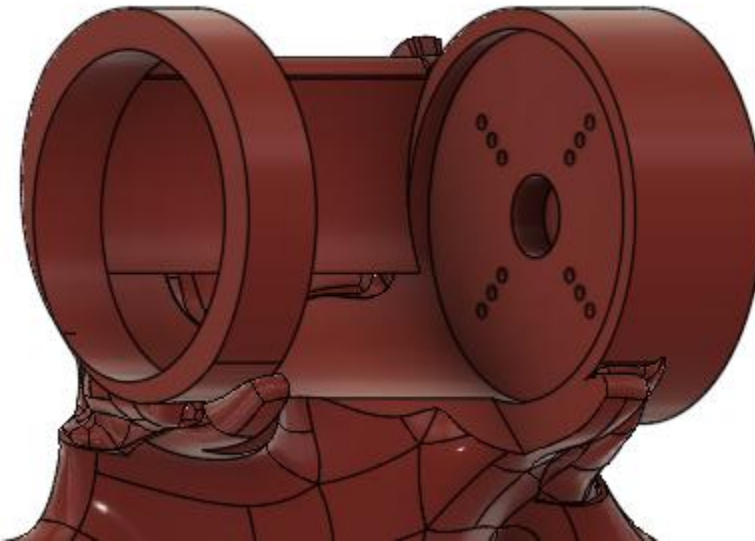
The chassis servo mount was done by adding two poles to hold the servo in place. The arm will be right next to it and rotated based on this servo. See payload assembly section: 3.2.1

5.3.1.2: SCOOP MOUNT

The servo mount for the scoop was put inside the connection joint. The servo would be screwed on and held in place by a mount based on the



arm. See assembly section 3.1.2



5.3.2: TORQUE AND MECHANICAL INTERFACE

Torque calculations are based on the power of each servo and the mechanical interface is the connection for the servos to their rotational pieces.

5.3.2.1: TORQUE

Torque is an essential calculation for our use of the arm to lift the scoop, arm, and payload. Servos available to us include

- MG996R
 $T = 11 \times 0.0981 = 1.079\text{N}$
 $F = 1.079 / 0.2 = 5.40\text{ N}$
 $\text{mass} = 5.40 / 9.81 = 0.55\text{ kg}$
- BILDA 5 turn (0504)
 $T = 5.4 \times 0.0981 = 0.53\text{N}$
 $F = 0.53 / 0.2 = 2.65\text{N}$
 $\text{mass} = 2.65 / 9.81 = 0.27\text{kg}$
- BILDA 300 degree (0002)

$$T = 21.6 * 0.0981 = 2.12\text{N}$$

$$F = 2.12 / 0.2 = 10.6\text{N}$$

$$\text{mass} = 10.6 / 9.81 = 1.08\text{kg}$$

5.3.2.2: MECHANICAL INTERFACE

The mechanical interface to connect the servos to the objects they are rotating consists of 12 screwed holes. This design allows us to securely fit the servos into place and receive the maximum mechanical output.

5.3.3: SERVO CHOICE AND CALCULATED FORCE

5.3.3.1: SERVO CHOICE

The choice of servos was driven by the output they could produce. Our most powerful servo, the BILDA 300 degree (0002), was decided for the arm because of its ability to lift a maximum of 1.08 kg. The scoop servo was picked to be the MG996R as it had the second most torque of 0.55kg.

5.3.3.2: CALCULATED FORCE

$$\sum (M)_c = F_1 * 20 + W * (2/3) * 20 =$$

$$21.6\text{kg} * \text{cm} = F_1 * 20 + 0.1622 * 2/3 * 20$$

$$F_1 = 0.97186 \text{ kg}$$

$$\text{Weight of objects} = 0.97186 - 0.173 - 0.055 = 0.74386\text{kg}$$

This is the final calculated max load we can lift with our payload assembly.

6: CHASSIS

This section will go over the progression of our chassis system, including the initial designs, calculations, and our final design choices.

6.1: CHASSIS REASONING

Our chassis was tested in multiple parts. Initially, the chassis was one solid printed part, but our design kept increasing in size, beyond the capabilities of our current working printers. We conducted static force analysis on our horizontal and diagonal cross members first as they endured the most load due to the magnitude of mass on the outside of the chassis. This created a strenuous effect traveling through the chassis into the members causing a static tensile load on each member.

6.2: CHASSIS STRENGTH

We figured out our chassis needs to withstand the weight of every existing part of our drivetrain and payload systems, along with the electronics. For peace of mind, we designed a multipart crossmember system that can withstand more-than-enough force cycles safely. Below are the derivations of the section properties, relating axial stress to max weight, shear stress of the pin, and bearing stress due to the pin.

We have two sets of crossmember-systems. Thus, our total weight can be split in half, denoted as W_{each} , diverted onto each system assuming exact symmetry and ideal situations. With that assumption, we can relate the diagonal member's axial force to weight by $P_d = \frac{W_{each}}{2 \sin(45)} = 0.354 * W_{total}$. Substituting this into the axial stress formula $\sigma_d = \frac{P_d}{A_d} = \frac{0.354W_{total}}{52.78}$. For most basic PLA+, the maximum allowable axial stress is generally 45 MPa. Plugging in 45 MPa for σ_d , we solve for weight and yield $6709 \text{ N} = 684 \text{ kg}$. As I mentioned, this frame design is absolutely overkill for this robot. Now we'll look at the linkage bolt that sandwiches all 3 members in each set. Once again assuming 45 MPa PLA+ and $V_d = P_d = 0.354W$, we can also calculate the area of our bolt ($d=5\text{mm}$) to be $A_{bolt} = \frac{\pi d^2}{4} = 19.635 \text{ mm}^2$. Knowing our bolt is double shear, we say $\tau_{bolt} = \frac{V}{2A_b} = \frac{0.354W}{2(19.635)} = 0.00901W$. Assuming allowable bolt shear to be 100 MPa, $W = 11100 \text{ N} \rightarrow m_{max} = 1132 \text{ kg}$. Again, overkill. For bearing stress, once again assuming $P_d = P = 0.354W$, our plate thickness $t = 5 \text{ mm}$, and our bolt diameter $d = 5 \text{ mm}$, we setup $\sigma_{bearing} = \frac{P}{td} = \frac{0.354W}{5*5} = 0.01416W \text{ MPa}$. Once again, using PLA+ $\sigma_{max} = 45 \text{ MPa}$, $W = 3178 \text{ N} \rightarrow m_{max} = 324 \text{ kg}$. For our governing result, we can assume the center bolt joint is governed by the bearing in the 5mm plate with a max chassis mass of 324 kg assuming ideal symmetrical conditions.

6.3: CHASSIS CROSSMEMBER PROPERTIES

Section Properties and Assumptions:

$$\text{Area: } A = \frac{\pi}{4} (D_o^2 - D_i^2)$$

$$\text{Bending inertia: } I = \frac{\pi}{64} (D_o^4 - D_i^4)$$

$$\text{Polar moment: } J = \frac{\pi}{32} (D_o^4 - D_i^4)$$

Section Modulus: $S = \frac{2I}{D_o}$

Horizontal Crossmember: ($D_o = 14.5 \text{ mm}$, $D_i = 11.5 \text{ mm}$, $L = 119 \text{ mm}$)

Area: $A_h = 61.26 \text{ mm}^2$

Bending inertia: $I_h = 1311.37 \text{ mm}^4$

Polar moment: $J_h = 2622.74 \text{ mm}^4$

Section modulus: $S_h = 180.88 \text{ mm}^3$

Diagonal Crossmember: ($D_o = 12.7 \text{ mm}$, $D_i = 9.7 \text{ mm}$, $L = 58.5 \text{ mm}$)

Area: $A_d = 52.78 \text{ mm}^2$

Bending inertia: $I_d = 842.41 \text{ mm}^4$

Polar moment: $J_d = 1684.83 \text{ mm}^4$

Section modulus: $S_d = 132.66 \text{ mm}^3$

6.4: DESIGN CHANGES - CROSSMEMBERS, PLATES, SHELVING SYSTEM

The crossmember idea spawned in as soon as the overall design changed to a multipart chassis with plates. We decided to make them overkill (aka each hollow cylinder having a wall thickness of 1.5mm), as we planned to mount our entire payload and other subsystems onto them, before we made the multi-level shelf system. Our plates progressed from one full flat chassis plate to one full triangular chassis plate, to our multipart system. With the new system, we integrated a shelving system to hold all our electronics and our payload subsystems on our chassis while retaining structural integrity. We bottom-mounted all our electronics on our 2 lower-level shelves while the payload is mounted to our well-structured top plate, more about that plate below (see 6.5).



6.5: CHASSIS CONNECTIONS, SCREWS, PINS, ETC

Our horizontal crossmembers are pinned into the chassis plate sleeves with a friction-fit 7mm x 7mm pin. These pins are a part of the top plate that mounts the payload arm. The pins are one of the 4 separate kinds of chassis connections the plate was designed to have.

1. Horizontal crossmember pins
2. Horizontal crossmember sleeve supports
3. Click-in tabs (into the chassis plate)
4. Inserting slots (into the chassis plate)

Our diagonal crossmembers are pinned into the 45-degree angled plate sleeves with a friction-fit 5mm x 5mm pin. The pair of diagonal crossmembers meet in the bottom middle tab of the horizontal member where they are all bolted together.

6.6: BRACKET

Our bracket design has progressively improved in terms of material optimization, structural integrity, modularity, and aesthetically.

6.6.1: EARLY BRACKET DESIGN

Initially our inner brackets were constructed as one whole part. As time progressed, each subsystem grew inevitably, outgrowing the bracket. This spawned the multipart chassis design with two inner plates, two outer plates, and our beloved crossmember and shelf system. Our new bracket design had initial problems including threads stripping and rendering useless and sprockets rubbing on the frame.

6.6.2: EARLY ISSUES & CHANGES

The biggest initial issue was printing, as our current printers' build plates did not permit the upscaled chassis to be printed in a reasonable orientation. We experimented with a TPU chassis, but that idea quickly ended after the first iteration. We landed with the current shaped frame eventually, but the inner frame was 9 mm thick.

6.7: TRACK-ENCLOSED BATTERY CONTAINER

6.7.1: BATTERY CONTAINER REASONING

The battery container securely mounts the power source by enclosing the battery entirely within the unused space of the track system. This placement of the battery frees up crucial room needed for other components like the payload and electronics, ultimately maximizing the available space and contributing to an organized and efficient overall design.

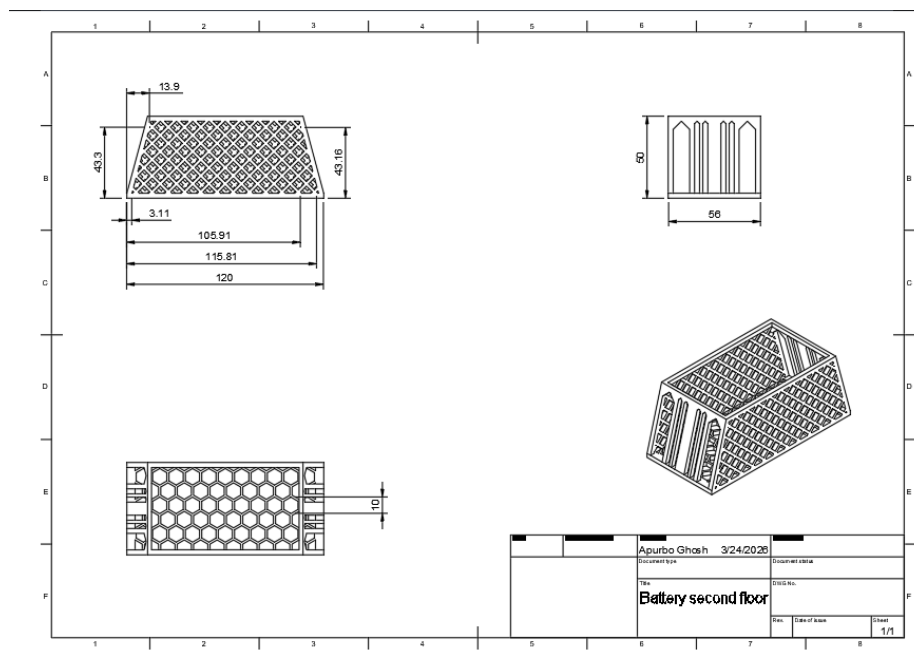
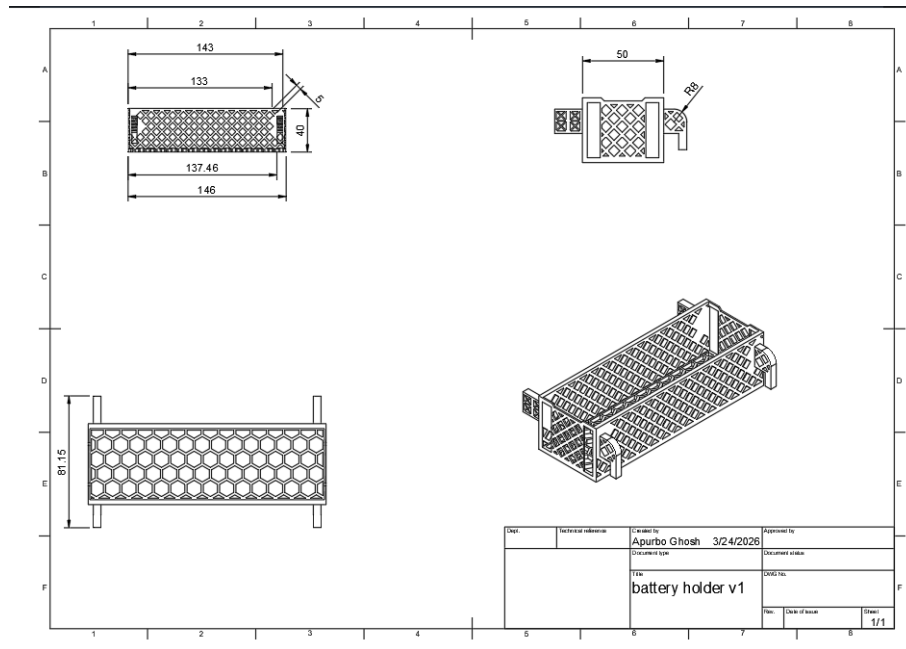
6.7.2: BATTERY CONTAINER PARAMETRIZATION

We used parametrization like how we did for our bearings (section 4.2.1.3) so that we could easily change the container as the bracket changed shape and size. Our variables were relatively simple, including bracket length and height, with the equations simply converting said variables into scalars and modifying the container by those scalars.

Interestingly, the holes are portrayed on top of the container AFTER the parametrization variables are assigned. This means that we can change the hole shape at

6.7.3: HOOKS

6.7.3.1: WHY WE SEPARATED BATTERY CONTAINER HOOKS FROM CHASSIS



6.7.3.2: INITIAL DESIGNS

A pin mounting system was the initial design idea due to its ability to create a mechanical lock and strictly hold the battery holder with the bracket in place. However, this method has proved to have drawbacks due to its need to align small pins that could be lost inside the treads.

6.7.3.3: WHY WE USED HOOKS

To avoid the issues presented with the pin mounting system, a new design was developed where the battery holder has downward faced hooks on the side. This eliminates the need for loose and strict

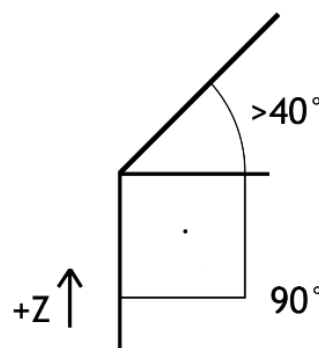
hole alignment needed for pins, it speeds up assembly time while relying on battery's weight and bracket's design tolerances to make sure it has a secure fit during competition.

Initially, we started with straight, rectangular "hooks" that practically sat our basket in place horizontally, but we noticed the bracket had excessive vibration and popped upwards and out of track on occasion. We revised the inside link to a snug-fit tolerance hook that sets into the inside chassis plate. The outside hooks chassis remained rectangular tabs as the snug-fit was sufficient for vibration control.

6.7.4: BATTERY CONTAINER DESIGN

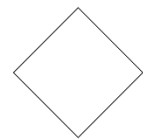
6.7.4.1: LIGHTWEIGHT DESIGN

To optimize enclosed containers, the battery holder features diamond shaped cutouts for the purpose of lightweighting and thermal management. These holes significantly reduce the battery holder's mass and conserve 3D printing filaments required to print them. Crucially, these holes also create passive ventilation, allowing air to flow continuously around the battery, this can dissipate the heat generated while the rover is running and keep our batteries from overheating and making sure the system stays safe and efficient throughout the competition.



6.7.4.2: OVERHANGS AND GEOMETRY OF CONTAINER HOLES

The decision to use diamond shaped cutouts of evenly lengthened sides was due to the Design for Additive Manufacturing principles to reduce the need for internal support while 3D printing. As shown in the diagram, the walls of the battery holder maintain a self-supporting angle that is strictly greater than 40° from the horizontal axis, which allows each layer to safely overlap the one below it. By avoiding overhangs that go below the 45° threshold, this design removes the need for support which also significantly reduces PLA materials and print time. Left illustrates principle and right illustrates cutout design.



7: CAMERA

Because the rover operates out of direct line of sight and is physically separated from the operators by a curtain, a camera is necessary; that is said, an ordinary camera with orientation dictated by vehicle orientation does not suffice due to the nature of the competition. Hence, the need for enhanced visual mechanisms is brought into consideration.

7.1: CAMERA GIMBAL

7.1.0: REASONING FOR IMPLEMENTATION

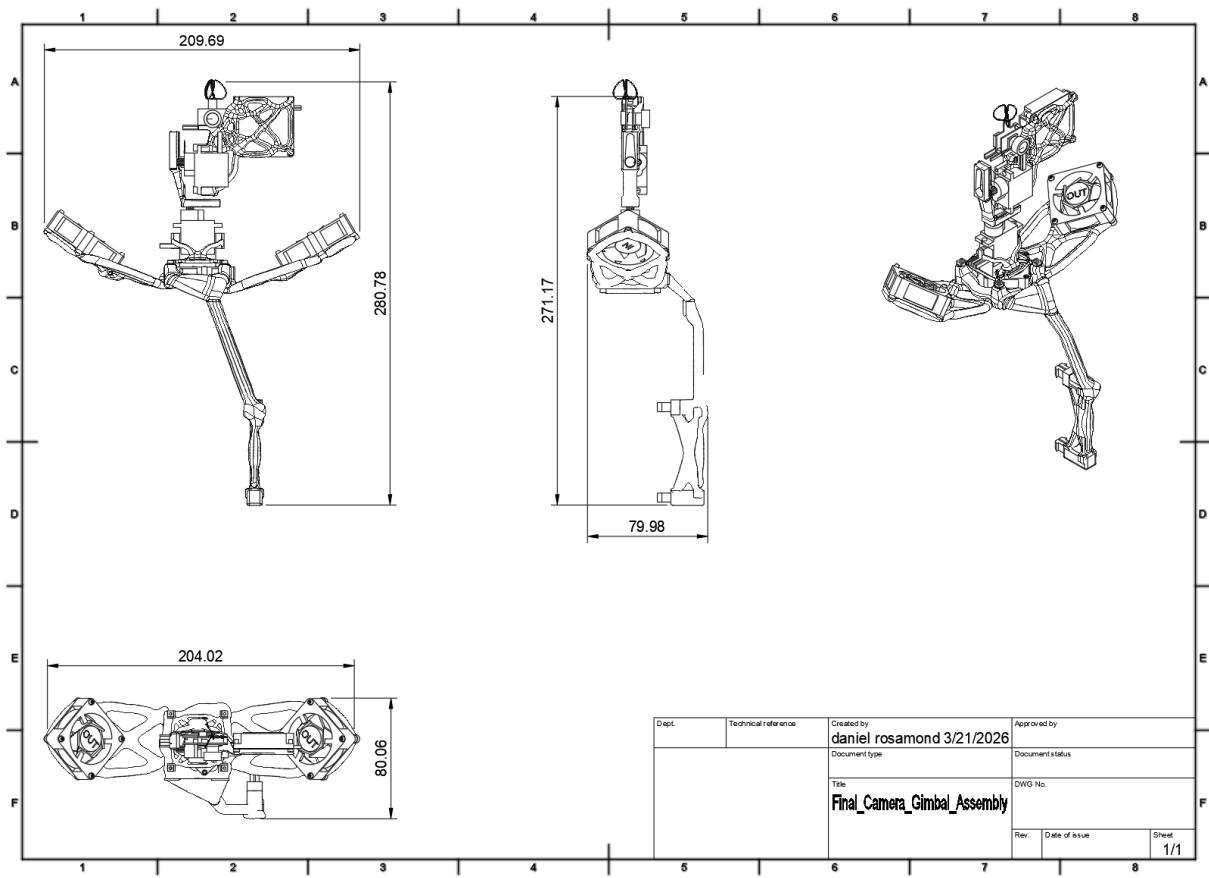
Originally, our rover design used three cameras: each covering a different field of view (FOV). We had planned around camera switchers and several mounts around the rover. Unfortunately, budget constraints brought that number down to two cameras and no camera switcher (two operators) midway through the project; and then one of our cameras was defective on arrival, meaning that our rover had to be designed around a single camera. Our final approach was relatively simple: design a 2-axis gimbal mechanism so that our remaining camera would have a direct view of the entire competition area.

7.1.1: USE CASE AND OVERVIEW

7.1.1.1: CAMERA GIMBAL INSPIRATION

The camera gimbal was inspired by the need for a simple pan-tilt system that could improve visibility without adding unnecessary weight. Our main goal was to have the servos handle only the pan and tilt motion, while the generatively designed structure served mainly to hold the assembly together. This approach helped reduce overall mass, keep the design compact, and maximize exposed surface area around the components for improved cooling.

7.1.1.2: FINAL CAMERA GIMBAL DESIGN



7.1.2: USE OF GENERATIVE DESIGN TO MITIGATE VIBRATIONS

7.1.2.1: VIBRATIONS AND THEIR EFFECTS ON THE CAMERA SYSTEM

Originally, the camera system was mounted directly to the rover chassis; this meant that any vibration generated by drivetrain motion, servo actuation or terrain irregularities were directly connected to and therefore directly transmitting vibrations to the camera and mainly distorting image stability. It also had the side effect of inducing the servo jitter amplification (where oscillations in the gimbal structure

induced artificial servo jittering; like what happens if you overvolt a servo) as well as motion sickness in our pilot, although over the period of five minutes this was less than meaningful.

7.1.2.2: APPROACH AND MITIGATION STRATEGIES TO VIBRATION

Vibration was a big issue with our initial camera gimbal. So, after testing the vibration frequencies of both the servos and rover, we were able to do proper modal analysis within the generative tab of Fusion and create a structure best suited to mitigate the vibrations; after which we decided to

7.1.2.3: APPLICATION OF GENERATIVE DESIGN TOWARDS MITIGATING VIBRATION

We used generative design informed by Modal Analysis (see 12.4.2)

7.1.2.4: TESTING AND FEA RESULTS

See section 12.

7.1.3: CONTROL

The Gimbal is controlled by two MG 996R Servos, one on the bottom controlling the horizontal tilt and another controlling the vertical tilt. We connect the servos signal to the esp to and assign two different channels on the FS-iA10B for each servo to then translate the signal from the remote directly to the servos so we could directly control both horizontal and vertical tilt at any given time independently.

Due to the limited time, we were unable to implement any gyro component to our current gimbal system, but we have been brainstorming ways we could utilize a gyro to power on our laser dipoles once we begin moving the gimbal in any direction.

In programming the servos to move directly with the remote, we came across a few issues that needed to be addressed. We had constant jitters when we would control the servos fast and had slight jitters at rest. So, to stop these jitters from occurring we implemented two filters in our code. For the Jitters at high speeds, we implemented an EMA. Adding a moving average into the code introduces lag that is proportional to the window size, making it transition better with how fast we move the servo. Fast inputs now feel more responsive, and the smaller noise that's causing the jitters to get absorbed. For the jitters at rest, we add a deadband filter. The thing that causes the jitters at rest are noises that are smaller than the deadband threshold, which is why the deadband filter is so effective at stopping the jitters since it ignores any noise that is smaller than that threshold. In implementing both of them together we have to first add the ema then the deadband after, because if we add the deadband first it could suppress real movement that the ema already cleaned up.

The code:

```
// --- Config ---
const float ALPHA    = 0.15; // lower = smoother, more lag
const int   DEADBAND = 15;    //  $\mu$ s - widen if jitter persists at rest

// --- State ---
float emaVal = 1500.0;
int   lastWrite = 1500;

// --- EMA (handles jitter while moving) ---
float emaFilter(float newVal) {
    emaVal = ALPHA * newVal + (1.0 - ALPHA) * emaVal;
```

```
    return emaVal;
}

// --- Deadband (handles jitter at rest) ---
int applyDeadband(int input) {
    if (abs(input - lastWrite) < DEADBAND) return lastWrite;
    lastWrite = input;
    return input;
}

// --- In loop() ---
void loop() {
    int raw      = rawPulse;           // from interrupt
    int smoothed = (int)emaFilter((float)raw); // smooth movement noise
    int output   = applyDeadband(smoothed); // lock out rest jitter

    myServo.writeMicroseconds(output);
    delay(20);
}
```

8: ELECTRONICS

In this section we will be going over how we chose our electronics and the process we went through to wire them

8.1: COMPONENT SELECTION

This section explains how the main electronics for the rover were selected and why each component was used in the final system. It focuses on the function of each part, the reasoning behind its selection, and the testing used to confirm that the components would work reliably together.

8.1.1: SELECTION CRITERIA

The rover electronics were selected based on efficiency, power compatibility, reliability, communication performance, and cost. Because the rover had to operate remotely and behind visual obstructions, the electronics had to support dependable control and video transmission.

Each component was chosen for a specific job within the rover, such as drive control, payload actuation, wireless communication, power distribution, voltage conversion, or cooling. Since the electronics subsystem was developed to fit into an already established mechanical design, compatibility with the rover layout was also important.

Power management was another major factor. The rover uses multiple electronic devices with different voltage requirements, so the team needed components that could be integrated into a simple and workable power system. Reliability also mattered, especially after later testing showed that most electronics stayed within acceptable temperatures while the camera required dedicated cooling.

Cost strongly affected selection as well. The team had limited funding, so parts had to be affordable, available, and practical to integrate. This is also why the final system used modular electronics and makeshift connections instead of a custom PCB.

8.1.2: COMPONENT SELECTION TABLE

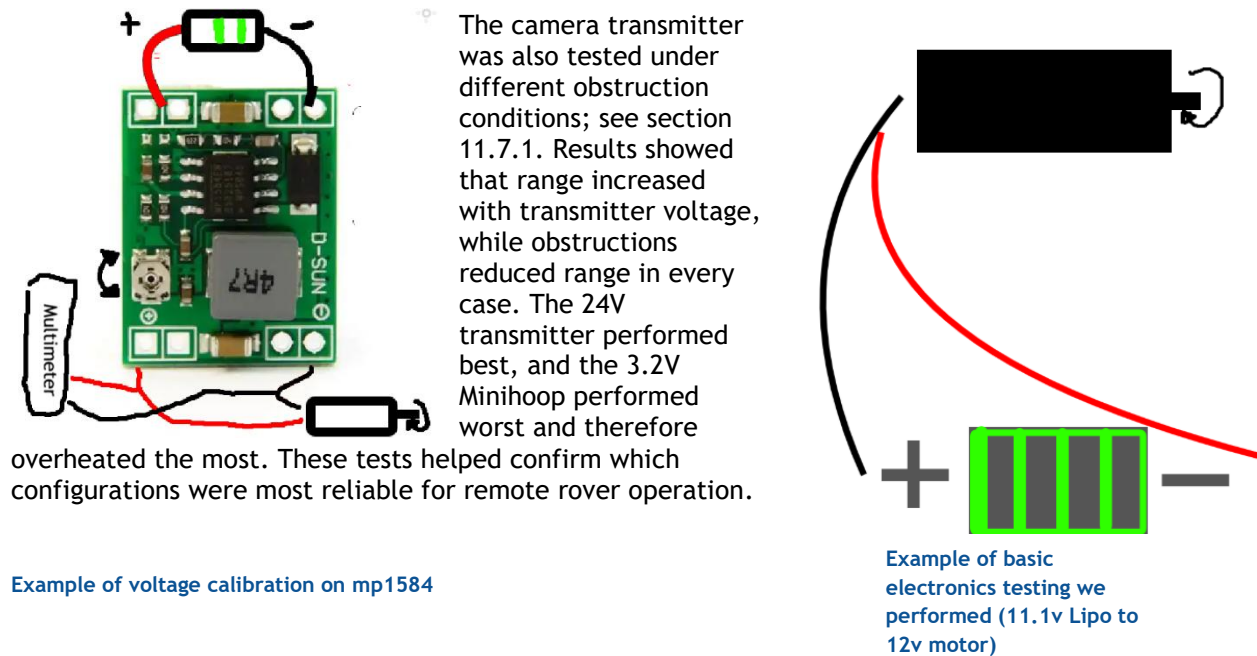
Component	Qty.	Function	Reason for Selection
ESP32-C3-DEVKIT	1	Main microcontroller and control logic	Selected for compact size, low cost, and sufficient processing capability for managing rover control tasks and signal routing.
goBILDA Hi-Torque 300° Servo	2	Payload and/or camera actuation	Chosen for higher torque, extended rotation range, and better mechanical robustness than smaller hobby servos.
FS-iA10B 2.4 GHz Receiver	1	Wireless control signal reception	Selected to provide reliable remote communication between the operator and rover systems.
Greartisan 12V 100 RPM Gear Motor	2	Drive motors for rover mobility	Chosen for high torque at low speed, which is more suitable for sand traversal compared to a high-speed motor.
MDDS30 SmartDriveDuo	1	Dual motor driver	Selected because it can control two drive motors from one unit and simplifies drivetrain electronics.
PDB-XT60 Power Distribution Board	1	Centralized power distribution	Chosen to simplify wiring and distribute battery power cleanly to multiple subsystems.
LM2596 Buck Converter	1	Step-down voltage regulation	Selected as a simple, inexpensive way to generate lower supply voltages for electronics that cannot run directly from the battery.
4-Relay Module	1	Switching auxiliary devices	Chosen to independently switch subsystems such as lasers, fans, or other accessories.
Laser Diodes	2	Aiming/alignment aid	Included as a simple low-cost visual aid for alignment and operator awareness.
Cooling Fans	8	Camera/system cooling	Selected after thermal testing indicated that the camera required active cooling while most other parts did not.

2S LiPo Batteries	2	Main rover power source	Chosen because LiPo batteries provide high energy density and are compatible with lightweight mobile systems.
FPV Cameras with Integrated Transmitters	2	Live video feed for remote driving	Selected to provide real-time operator vision and wireless video transmission for obstacle-blind operation.
MT3608 Converter	1	Voltage boost conversion	Chosen to raise voltage where needed for electronics requiring higher supply than the base source can provide.

8.1.3: COMPONENT CALIBRATION AND TESTING

After selection, the electronics were tested to confirm that they worked reliably in the rover system. The team focused mainly on temperature behavior and transmitter performance.

General electronics temperature testing showed that most components stayed within acceptable limits. The main exception was the camera, which required active cooling. This result justified adding fans to the final electronics setup.



8.2: DIAGRAMS

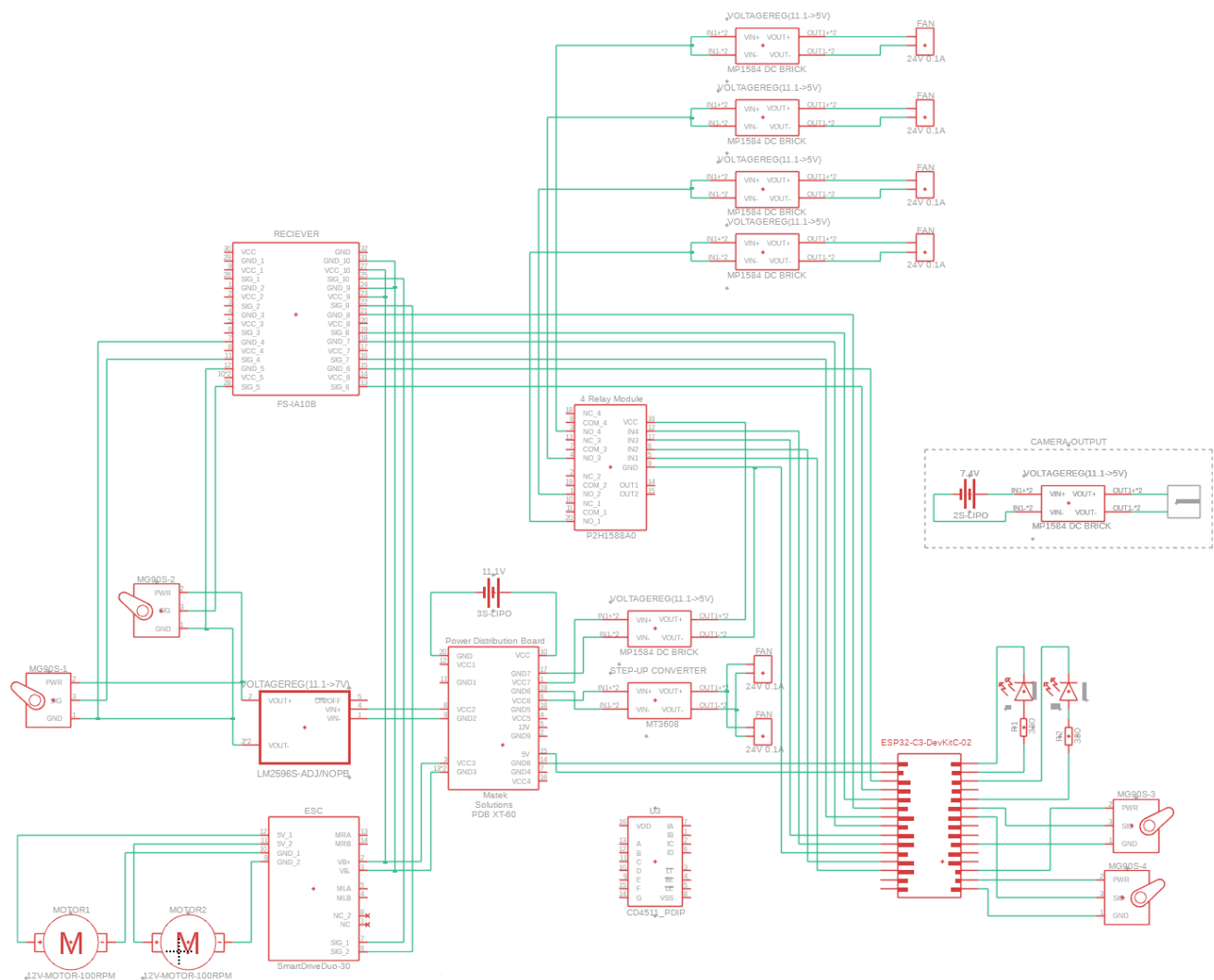
This section covers how the selected electronic components were arranged, connected, and integrated into the rover system. It explains the overall electrical layout, subsystem organization, and the design decisions used to make the electronics work together in the final build.

8.2.1: FINAL ELECTRONIC DESIGN

Below are electronic diagrams of each of our parts. It's important to understand that while our electronics design does work in practice, budget constraints meant that ordering our own PCB was out of the question and we had to resort to makeshift electrical connections.

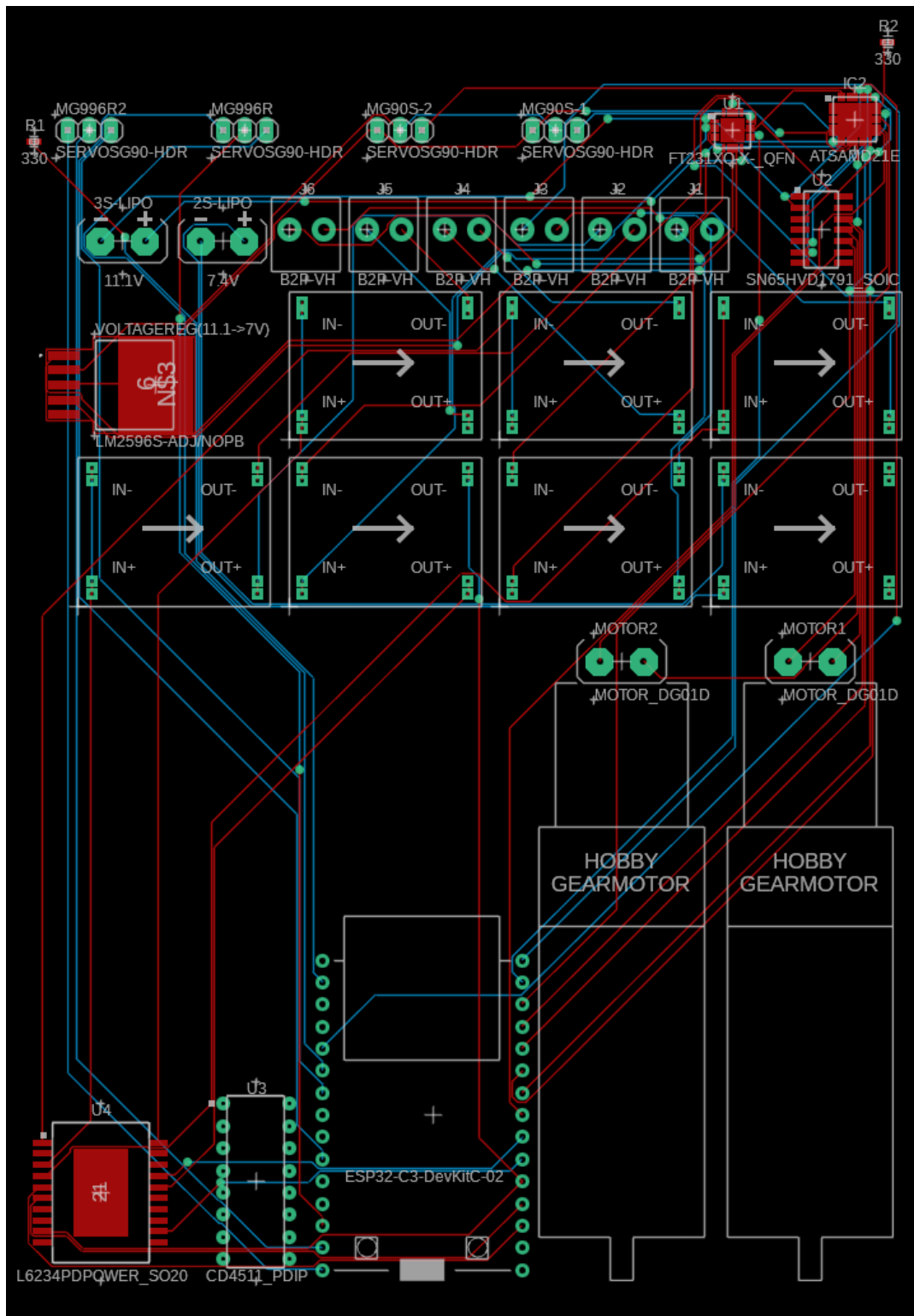
8.2.1.1: FINAL ELECTRICAL SCHEMATIC

Schematic shows a general electrical diagram for how we're powering everything



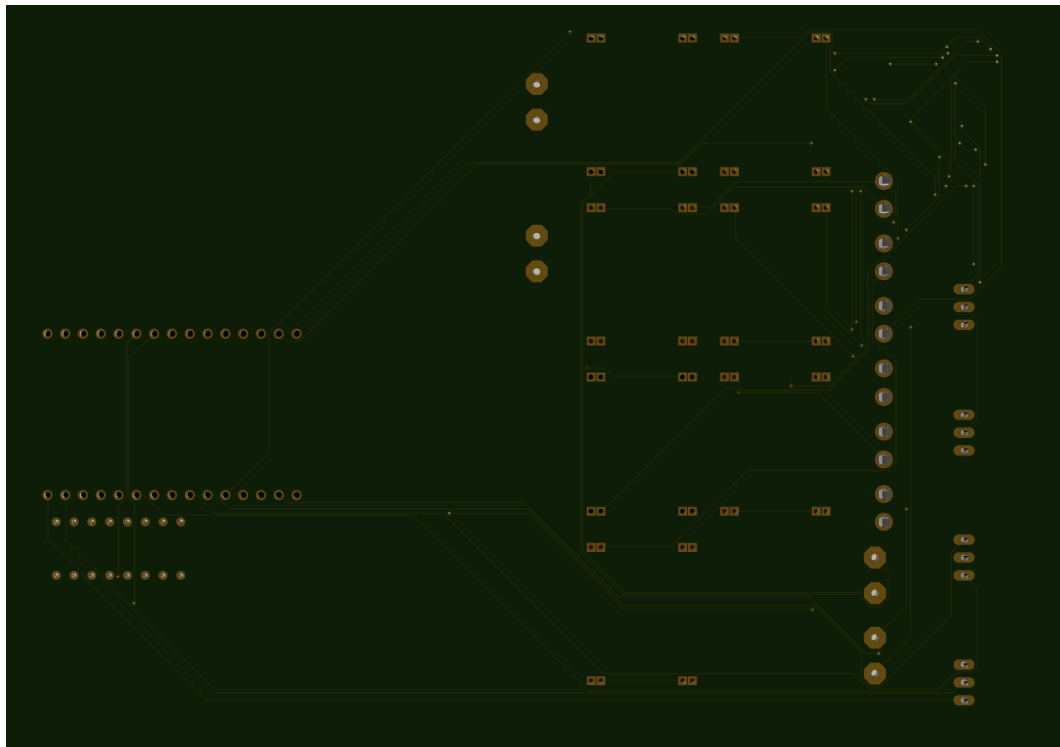
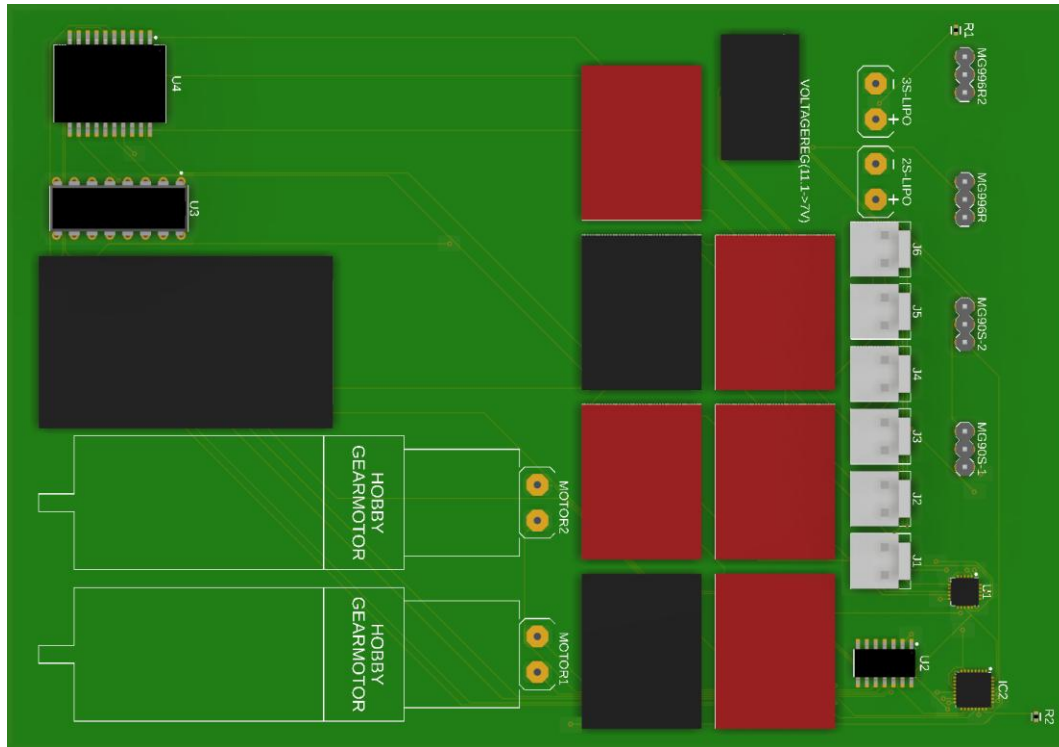
8.2.1.2: FINAL PCB (2D)

This is our final PCB design. Unfortunately, we didn't order one (citing 8.3.1).

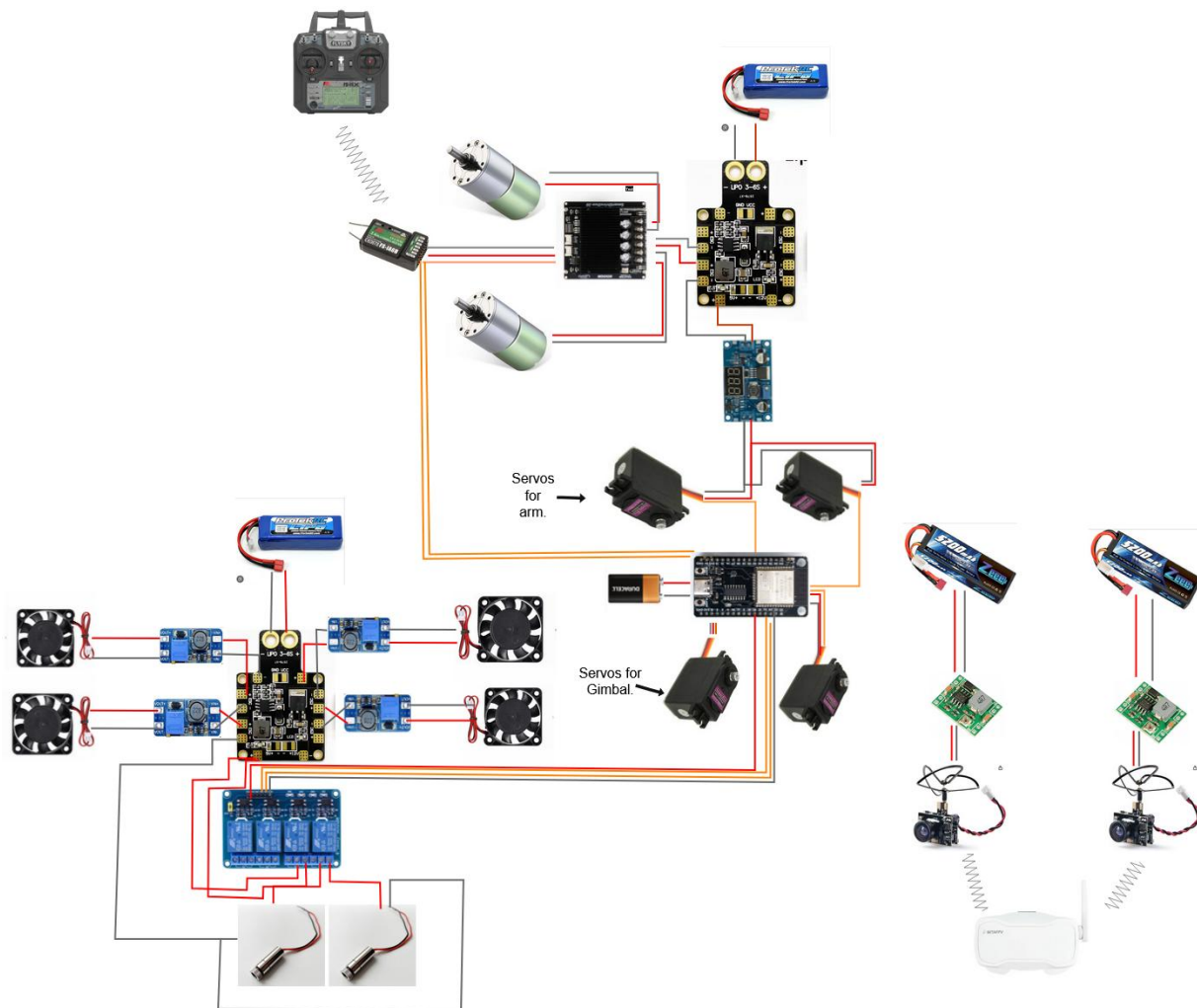


8.2.1.3: FINAL PCB (3D)

DISCLAIMER: Not accounting for certain components, MP1584 modules, ESP32 Module, receiver, and ESCs are some components used; however, not included as models due to being commercially bought and technical complexity distorting team-led PCB design. These components are solid color fill below.



8.2.1.4: ELECTRICAL DIAGRAM (2D, SIMPLIFIED)



Separate cam powered with a 2s battery for each camera

One signal receiver fsia10b

Another mp1584 for another cam

Add 8 fans

4 relay switch

The figure above shows a simplified 2D layout of the rover's electrical system and the main signal and power paths between components. The central control and power-distribution board acts as the hub of the system. From this point, power is routed to the drive motors through the motor driver, to the servo system through the voltage-conversion hardware, and to the camera/transmitter hardware through separate regulated power paths. The diagram is meant to show the overall architecture of the rover electronics without the additional detail of the full PCB model.

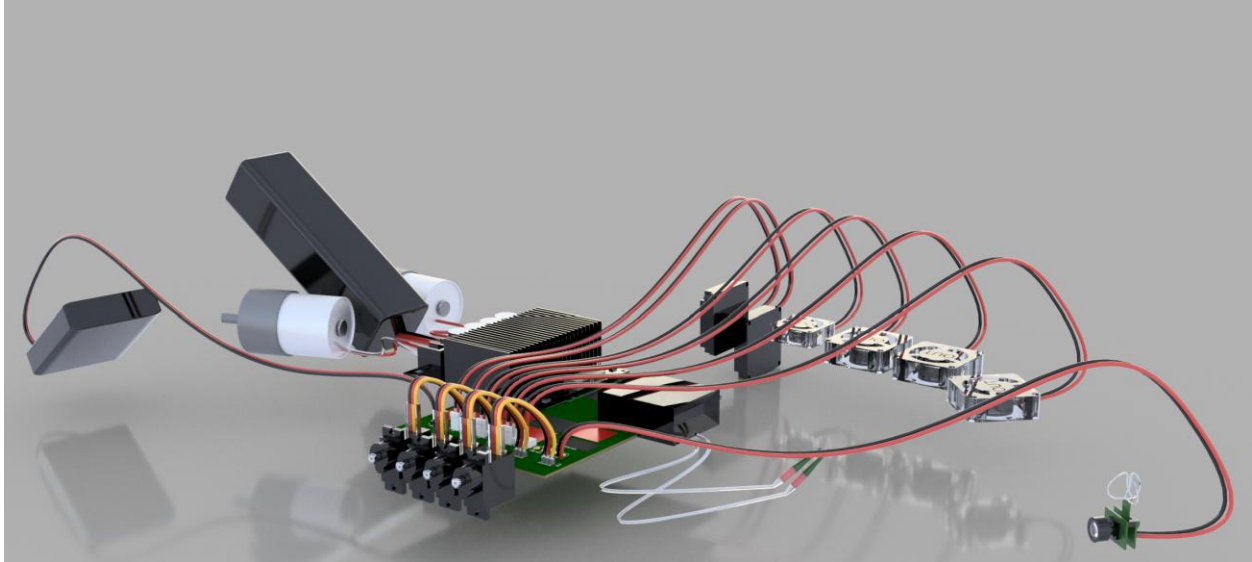
On the left side of the figure, the receiver sends control signals into the system, while the motor driver powers the two drive motors used for rover movement. In the lower portion of the figure, the servo system is powered separately from the receiver so that the servos do not overload the receiver power

output. This matches the team's final approach of splitting servo power and signal connections for more reliable operation.

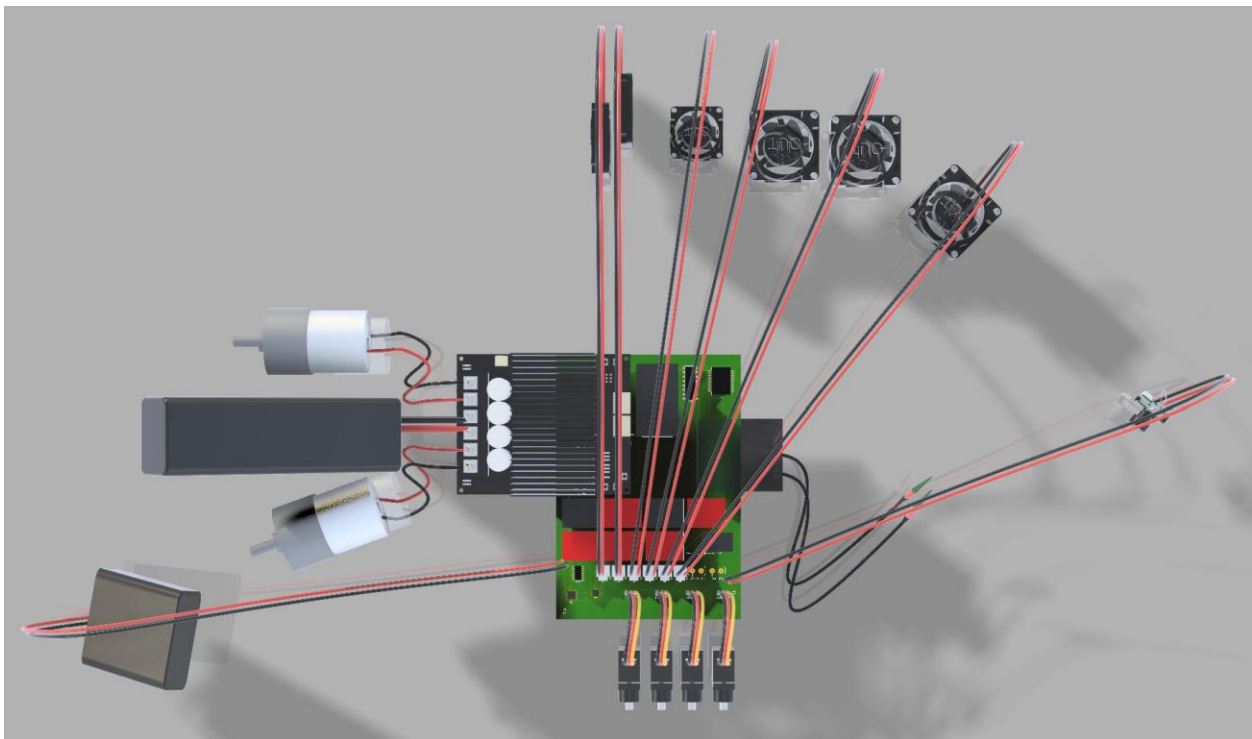
On the right side of the figure, the camera and video transmitter subsystem is shown. The camera/transmitter setup is powered through its own battery and regulation hardware rather than relying entirely on the main rover electronics. This separation was important because transmitter and camera performance depended heavily on stable power delivery, and later testing showed that the camera system also needed active cooling. Notes for the final design also indicate the addition of another MP1584 module, eight fans, a single FS-iA10B signal receiver, and a four-relay switch to support the completed electronics package.

8.2.1.5: ELECTRICAL DIAGRAM INCLUDING PCB (3D)

Rendering of most of our electronics with a PCB, in a 3D format. See 8.2.1.3 for details on why mp1584 and related modules are missing; most modules seen in 8.2.1.4 missing are integrated into PCB.



Isometric



Top-Down

8.3: ELECTRONIC CHALLENGES

This section discusses the main problems encountered during the development and integration of the rover's electronics. It focuses on issues related to power, communication, component limitations, and other challenges that affected the final electrical design.

8.3.1: ELECTRONICS BUDGET ISSUES

The electronics subsystem was limited by budget. The team could not afford to manufacture a custom PCB, and the number of available components was limited, leaving little room for mistakes. Some electronics also had to be reused from the previous IAM3D drone project.

Because of this, the final system relied on modular components and practical wiring rather than a fully integrated custom board. While this was less refined, it allowed the electronics system to remain functional within the project budget.

8.3.2: TRANSMITTER ISSUES

The team had transmitter-related problems during development. The wrong transmitter brand was initially used, and the signal was cutting out. This created reliability issues for rover control and payload operation.

Another issue was power delivery through the receiver. The transmitter setup could not supply enough power for the servo motors, so the servo wiring had to be split: VCC and GND were used for power, while SIG and GND were connected to the receiver inputs. The payload servos also required a specialized voltage converter module, while the camera gimbal servos required MP1584 modules.

Later range testing confirmed that transmitter configuration had a major effect on performance. Higher-voltage transmitters gave better range in all conditions, especially when obstructions were present.

8.3.3: AUTOROUTING LIMITATIONS

During PCB development our team initially attempted to use auto routing to accelerate layout. Unfortunately, auto routing was quickly phased out as an option as it proved unsuitable for our design constraints, and the process did NOT meet our DFMA requirements.

Our reasoning behind this was primarily because it generated inadequate power traces. Auto routing generated insufficient and undersized power traces (wire size/current capacity), which was an issue especially on higher current components including the ESC and voltage regulator attachments. Additionally, it generated substandard and distorted component to component routing, which led to longer trace lengths and unnecessary complexity; which by rule of DFM increases cost.

That said we are neither producing this PCB on a scale or for the sake of the competition at all (8.3.1, Budget), so none of these issues affect final design, hence this section being shallow and concise.

9: DFAM

How we designed around 3D printing.

9.1: DFAM INTRODUCTION

This section introduces the Design for Additive Manufacturing (DFAM) approach used in the rover design process. It explains how additive manufacturing influenced part geometry, weight reduction, manufacturability, and overall design decisions for components intended to be 3D printed.

9.1.1: WHY DFAM MATTER FOR OUR ROVER

Every design choice made for this rover is based on DFAM principles. Wall thickness, overhang angles, print orientation, and layer line direction had to be considered in addition to conventional mechanical requirements like strength and weight because all structural components were 3D printed, which directly limited and shaped what was possible.

In contrast to subtractive manufacturing, additive processes enable part merging that would otherwise require assembly, internal geometries, and organic shapes from generative design. One clear example is the rover's 3D printed bearings, which eliminated the need for separate assembly. Similarly, manufacturing the split driving sprocket in any other manner at this scale and cost would be unfeasible.

Overall strength is determined by print orientation. A part that is strong under one load scenario may fail under another if it is printed in the incorrect direction. As a result, DFAM became a structural factor as well as a manufacturing one, directly influencing FEA and testing choices during the project.

9.1.2: LAB EQUIPMENT

9.1.2.1: AVAILABLE 3D PRINTERS

9.1.2.1.1: CREALITY ENDER 5 S-1

The build volume ($220 \times 220 \times 280$ mm) allows larger rover components to be printed as single parts, reducing assembly complexity. Material compatibility is the greatest among our printers, and it is the only printer we have that's able to print PET-GF, making it extremely valuable for our chassis design.

9.1.2.1.2: PRUSA MK3

The build volume ($250 \times 210 \times 210$ mm) allows us to produce small to medium-sized rover components. Material compatibility, such as PLA, TPU, and PETG, enables both rigid and flexible parts. The printer provides consistent and reliable print quality, making it suitable for components where accuracy is important.

9.1.2.1.3: SNAPMAKER A350 - 3D PRINTING FUNCTION

The build volume ($320 \times 350 \times 330$ mm) allows to produce larger rover components as single parts, reducing assembly complexity. Material compatibility, such as PLA, PETG and TPU, enables both rigid and flexible parts. The printer provides stable performance, making it suitable for larger structural components where size is important.

9.1.2.1.4: COMPARISON OF 3D PRINTERS

The Ender 5 S1 provides the highest material compatibility and supports reinforced filaments, making it the most suitable for structural components such as the chassis.

The Prusa MK3S+ offers the most consistent and reliable print quality, making it better suited for precise and smaller components.

The Snapmaker A350 has the largest build volume, allowing for larger parts to be printed as single pieces, reducing assembly requirements.

9.1.2.2: MACHINING TOOL AVAILABILITY

SnapMaker A350 - CNC Carver, Laser Cutter

9.1.2.3: HANDHELD TOOLS

We used various handheld tools to help us in our processing, most of these were used to simply remove supporting structure printed by our printers.

- Dremel
 - Often our prints would have hard to reach spots with support once fully printed. Because these support structures could be difficult to remove; the use of the Dremel for these removals was immensely helpful and allowed us to reach otherwise difficult supports.
- Wire Cutters
 - Wire cutters were used to remove the majority of all the support. They allowed us to rip off otherwise difficult or thick supports and reduced the time taken to fully remove all supports.
- Putty Knife
 - A putty knife was used to remove the exterior boundary of every print and remove smaller prints from the print bed that you could not rip off with your hands.

9.1.2.4: POST PROCESSING CAPABILITIES

We have many post processing capabilities. Most of these are done after the printing process to improve certain aspects of the rover design. Specifically, we have spray glue and aluminum oxide methods to improve grips, resin to strengthen designs, and remove excess print.

- Spray Glue and Aluminum Oxide
 - First, we clean whatever surface we are going to apply the glue and oxide to. You then spray the glue onto said surface before being coated with aluminum oxide. The setup is then left to dry, so it hardens and becomes stable.
- Resin
 - Resin is first mixed using a two-part epoxy solution. Once the mixture is made it is then applied across the surface selected and left too hard over the course of several hours.
- Electroplating
 - The first step in electroplating is applying a thin layer of conductive graphite paint to the surface of the object you are trying to electroplate. Once this layer of paint has dried, you place the object into a solution of electrolytes with metal ions. Then a negative DC source is connected to the object, and a metal source is connected to the positive end. As you send voltage through the system, the metal ions are reduced to the object.

9.1.2.5: APPLICATIONS

Applications for post processing are immense throughout the design of this rover. We use them for many different reasons, increasing grip, increasing structural integrity, and creating electrical conductivity.

- Increasing Grip and Traction
 - We used the spray glue and aluminum oxide process to improve the grip on the tank treads. Additionally, we have the ability to incorporate this in the scoop, so the objects we grab will not slip or fall out of the scoop.
- Strength Increases in Stability
 - We use the resin coating process to increase the strength of certain parts such as the arm. We don't need to worry about the shear stress applied on the arm since we can apply resin to it and increase its overall strength.
- Electrical Conductivity
 - Electroplating the scoop could make picking up certain metallic objects much easier. For example, if we get a conductive item such as iron in the sand it would stick to the scoop slightly more, making it more likely we would pick it up and deposit it into the bucket.

9.2: WORKING AROUND LAYER LINES

9.2.1: PRINT ORIENTATION

The strength of a part is significantly impacted by the orientation in which it is printed. A part loaded perpendicular to its layer lines will fail much sooner than one loaded parallel to them because the bond between layers in FDM printing is much weaker than the material within a single layer. This was seen during testing, where switching to horizontally printed friction-fit pins seen in Section 4.1.3.3 significantly increased durability, while threaded connector pins in Section 4.1.3.2 printed vertically snapped easily because layer boundaries were perpendicular to the bending load. The rover followed a straightforward general rule, determining each part's primary load direction, then aligned the print so that the layer lines run parallel to that load.

9.3: WORKING WITH SUPPORTS AND MATERIAL OPTIMIZATION

9.3.1: SUPPORT

The support is crucial for 3D printing as it allows us to make most parts we design. Every piece needs to be created to reduce support needs and inspected to evaluate which support structure it is best suited for.

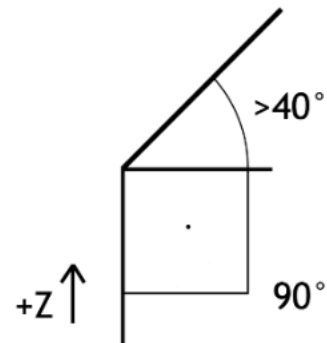
9.3.1.1: SUPPORT STRUCTURE

Various support structures are offered in many slicers, each with different advantages and use. The various supports we used in this project were grid, organic, lines, and concentric. The most used one was organic since it worked best for the most common things we would print.

Support Type	Strength	Ease of Removal	Material Use	Surface Finish
Grid	High	Poor	High	Poor
Organic	Medium	Excellent	Low	Excellent
Lines	Medium	Good	Medium	Medium
Concentric	Low	Good	Low	Good

9.3.1.2: OVERHANG REDUCTION

Supports are used wherever you have overhung, and the printer cannot support the material with other material. As seen in the image to the right, if the angle is less than 40 degrees you will need support. Because of this we have built 40+ degrees into most of our designs as this allows us to save on filaments and worry less about supporting material removal.



9.3.2: MATERIALS

9.3.2.1: MATERIAL STRENGTHS

See section 11.3.20

9.3.2.2: PRINT SETTINGS

- Layer Height
 - Layer height affects the strength of the print, the smaller the layer height the stronger the print; The tradeoffs are slower print time but greater precision.
- Infill Percentage
 - The greater the infill percentage, the strength of the print; however, this makes the print weigh substantially more, so it cannot be used in parts like the arm.
- Nozzle Diameter

- Nozzle diameter is another important factor when printing. The greater the nozzle diameter, the better the strength due to greater adhesion. The tradeoff is less precision when printing, so it's not ideal for printing stuff like bearings.

9.1.3.1: TOLERANCES

Tolerances are one of the most important aspects of DFAM because they allow for pieces to fit together and work with each other. The main aspect of tolerance is that the pieces we print are not exactly and usually are slightly larger than what we designed; additionally, our nozzle sizes are all 0.6, so we chose about 1-2 layer lines of tolerance. This equates to about 0.6 to 1.2mm, which are necessary for most pieces to work with each other. Tolerance is greatly affected by the nozzle diameter since the higher the diameter the less accurate the print.

10: DFMA

Design for Manufacture and Assembly (DFMA) is a methodology to simplify the design process and assembly of a product. Generally, DFMA should always be considered for all steps in the design process. The use of this design methodology in production will lead to faster, cheaper, and more efficient manufacturing and assembly.

10.1: INTRODUCTION TO DFMA

DFMA can be broken into two sections: Design for Assembly (DFA), and Design for Manufacturing (DFM). The combination of these two subsections is what makes DFMA such an optimized design process.

10.1.1: DFA

Design for Assembly (DFA) minimizes the complexity and number of parts to simplify assembly. Assembly should be simple and repeatable without excessive force or adjustments, and DFA takes this into consideration.

10.1.2: DFM

Design for Manufacturing (DFM) ensures that each part is optimized and designed for its manufacturing process. For example, our rover uses FDM 3D printing as its manufacturing process, therefore this needs to be considered in each part's design. Every form of manufacturing comes with its own set of advantages and disadvantages. Knowing these is a very important step in the design process.

10.2: DFMA AND OUR ROVER

The DFMA methodology played a large role in the development of our almost fully 3D printable rover. As the rover relies entirely on FDM additive manufacturing, DFMA allows us to iterate quickly, reduce part count, simplify assembly, and ensure printability on our available printers. This section outlines how both DFM and DFA influenced our design choices.

10.2.1

While FDM additive manufacturing allows for the option of skipping assembly with print-in-place parts, such as our bearings, there are some places where assembly is unavoidable. Examples of this can be found the in assembly section 3.

11: TESTING

Our focus on a modular design, which allows for rapid and iterative testing, enables us to rigorously test a variety of materials. By isolating specific components within our prototypes, we can quickly swap out different materials and components to evaluate their performance under stress without rebuilding the entire system. This agile approach not only accelerates our development cycle but also ensures that every material choice is backed by comprehensive, data-driven validation.

11.1: TESTING INCENTIVES/REASONS/WHY WE TESTED

Testing informs us on the rest of our design. If a design is inadequate and fails testing, we run FEA (Section 12), find the part that needs reinforcement, and we add material to that area or run generative with those forces applied.

11.2: HOW WE TESTED

Various methods displayed how we tested every piece of our rover. Specifically, we used a sandbox we built in our lab, computer simulations, and occasionally rigs for all different purposes.

11.2.1: SANDBOX



Illustrated on the left is an image of the sandbox we built to test rover parts. Sandbox dimensions were 1.2m x 1.0m; creation involved the de-shelving of a cabinet, calculating weight sand needed to fill, buying that sand and bringing it. Additionally, we found a five-gallon bucket and cut it to competition specs (20cm high) to test payload dropping.

Sand depth was justified through assumption and reasonable budget; we consulted and received an answer through discord for the maximum depth of objects being 5in. Since we were able to split up the sandbox using shelves to increase depth at different portions, we decided on 3.5in as a compromise to conserve budget.

Sand weight calculations:

1.2m * 1.0m area; 3.5in = 0.0889m. Volume is all of those multiplied, yielding 0.10668 cubic meters, which, if loose sand is 1450kg/m³, we get a total of 154.7kg needed or about 340lb of sand. We bought 350lb to be safe.

Sandbox results can be seen in several sections and helped mostly in informing anti-sand design for many of our movement parts, as well as scoop design.

11.2.2: RIGS

We developed several rigs to test certain designs. A few of the most notable rigs are listed below, but we have included several more in the testing section for the sake of depth.

11.2.2.1: SCOOP TESTING RIG

See section 11.6.1.1

11.2.2.2: COMPONENT TEMPERATURE TESTING RIG

See section 11.7.1.2

11.2.2.3: GENERAL ELECTRONICS TESTING RIG

The reason we did not provide general tests and focused on cameras and temperature was due to the lack of complexity with our electronics. That said, we tested it all simply by connecting it all up and testing it with the setup at 8.2.1.4.

11.3: MATERIALS TESTING RESULTS AND FINAL MATERIAL SELECTION

The following contains content including how we gathered material data, illustrations representing that data we gathered, and a brief interpretation of that data as it relates to material choice. Nearing the end of this section, we present the final material choice.

11.3.0: MATERIAL SELECTION CRITERIA

Our design uses different filament materials depending on the function of each part rather than relying on a single material throughout the rover. Structural components were primarily printed using FIBERON PET-GF15 and PLA+, while other parts were selected based on their performance during testing.

Since the rover operates on loose sand, impact forces are relatively low and do not require additional suspension or shock absorption. Because of this, flexible materials such as TPU were not used for tread links. Instead, PLA+ was selected for the treads because it provided better traction and more consistent movement during testing. TPU was only used in specific components such as brackets and specific camera mount components, where a small amount of flexibility helps reduce stress concentrations and allows for minor misalignment without affecting the overall structure, as well as vibration mitigation.

11.3.1: TREAD LINK MATERIAL TESTING

For information on tread/track links, refer to section 4.1.2.

11.3.1.1: TESTING METHOD OF TREAD LINK MATERIAL

To evaluate how different tread link materials behaved under load, we built a simple slippage testing rig using our 1.5m diagonal sandbox. A small, standardized tread segment was driven across the full length of the sand bed at a fixed motor RPM, ensuring each material experienced identical torque and surface conditions; and, because the motor speed was constant, any difference in the time required to traverse the 1.5m distance was directly reflected in how much each tread slipped in loose sand. Materials that were stiffer and more brittle tended to dig in and translate more of the motor's rotation into forward motion, resulting in faster traversal times and therefore less slippage. Softer or more flexible materials deformed under load, causing the tread to “float” or skid across the sand, increasing slippage and reducing the effective ground covered per rotation. By timing each run and comparing the actual distance traveled to the theoretical distance based on motor RPM, we gathered data from the slippage characteristics of each material.

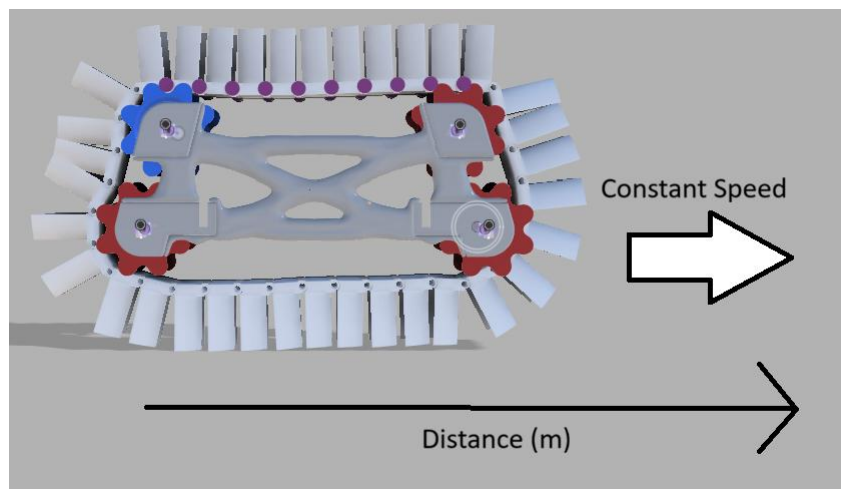
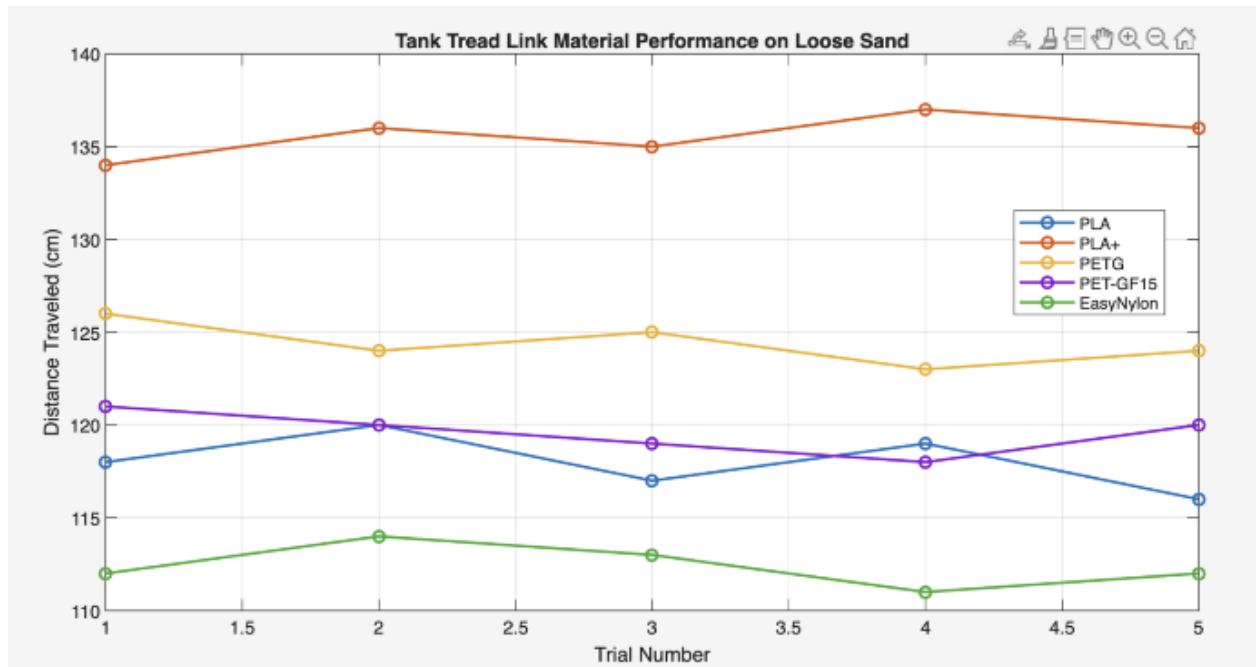


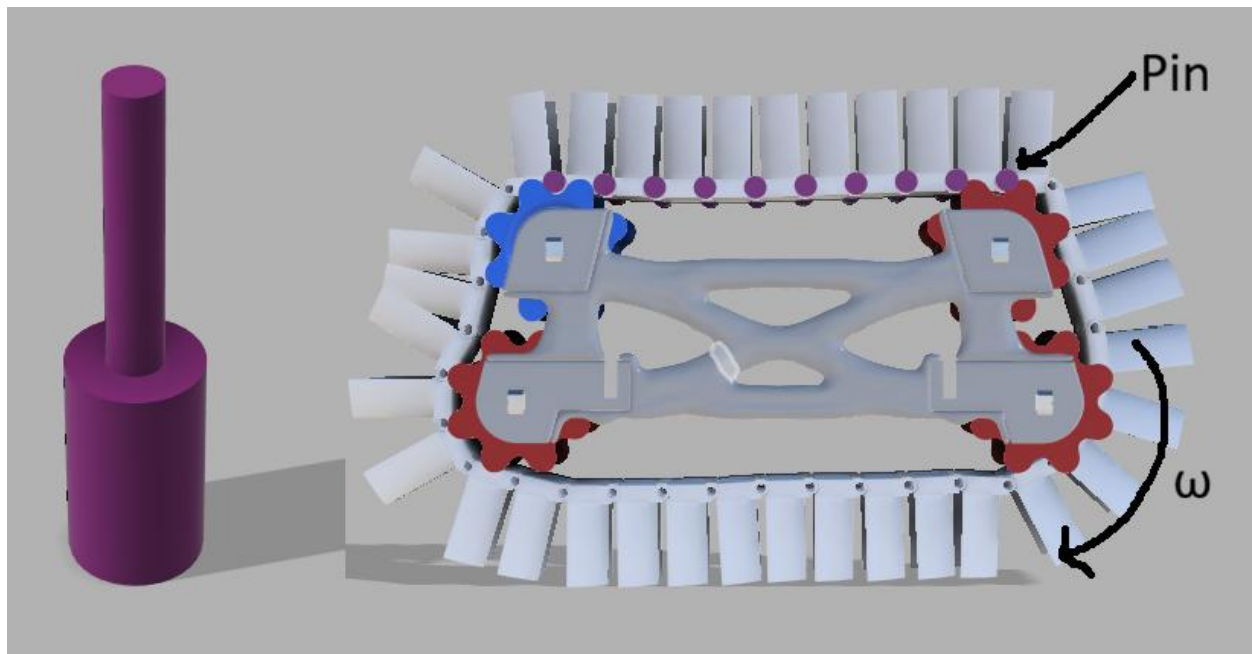
Figure X: Demonstration of how tread link materials were tested. Motors were spun at a constant RPM, allowing the distance travelled(m) to be tested for PLA, PLA+, PETG, PET-GF15, and EasyNylon.

11.3.1.2: PLOTTED DISTANCE TRAVELLED/SLIPPAGE OF TREAD LINK MATERIAL



The illustrated graph shows that PLA+ has the highest distance travelled, meaning it would be the obvious choice for tank tread links.

11.3.2: TREAD LINK CONNECTOR PIN MATERIAL TESTING

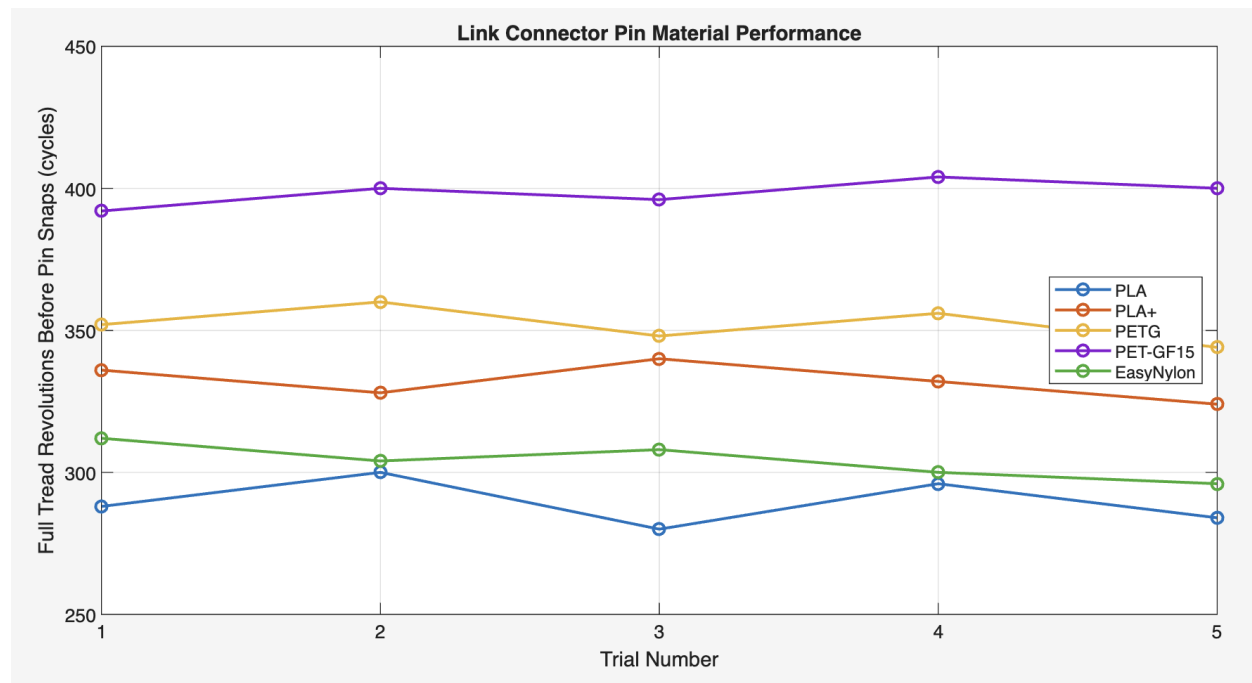


Demonstration of how the tread link connector pins were tested. Motors were spun at a constant RPM (whereas one cycle = one full rotation) until a pin snapped.

11.3.2.1: TESTING METHOD OF TREAD LINK CONNECTOR PIN MATERIAL

To evaluate how different tread link connector pin materials performed under repeated loading, each pin was installed into a tread link assembly and evaluated within a motor-driven drivetrain system. The motor (6000 rpm, 3S / 11.1V) rotated the tread at a constant speed, and one full tread rotation was counted as one cycle. Each material was evaluated for five trials, with the system running continuously until the connector pin failed by snapping or significant deformation. The total number of cycles completed before failure was recorded for each trial. Because the motor speed and loading conditions were kept constant, differences in cycle count directly reflected the durability of each material under repeated mechanical stress.

11.3.2.2: PLOTTED FULL TREAD REVOLUTIONS BEFORE PIN SNAPS/ SLIPPAGE OF TREAD LINK CONNECTOR PIN MATERIAL

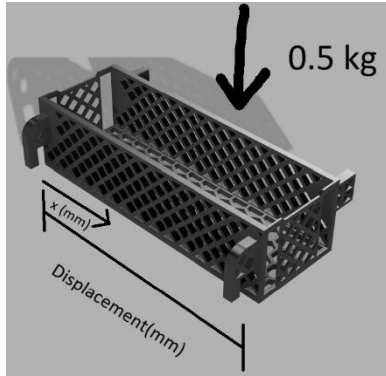


The illustrated graph shows that PET-GF15 lasted through the highest number of motion cycles with the least wear, making it the best choice for link connector pins.

11.3.3: BATTERY HOLDER MATERIAL TESTING

We tested the material for the battery holder.

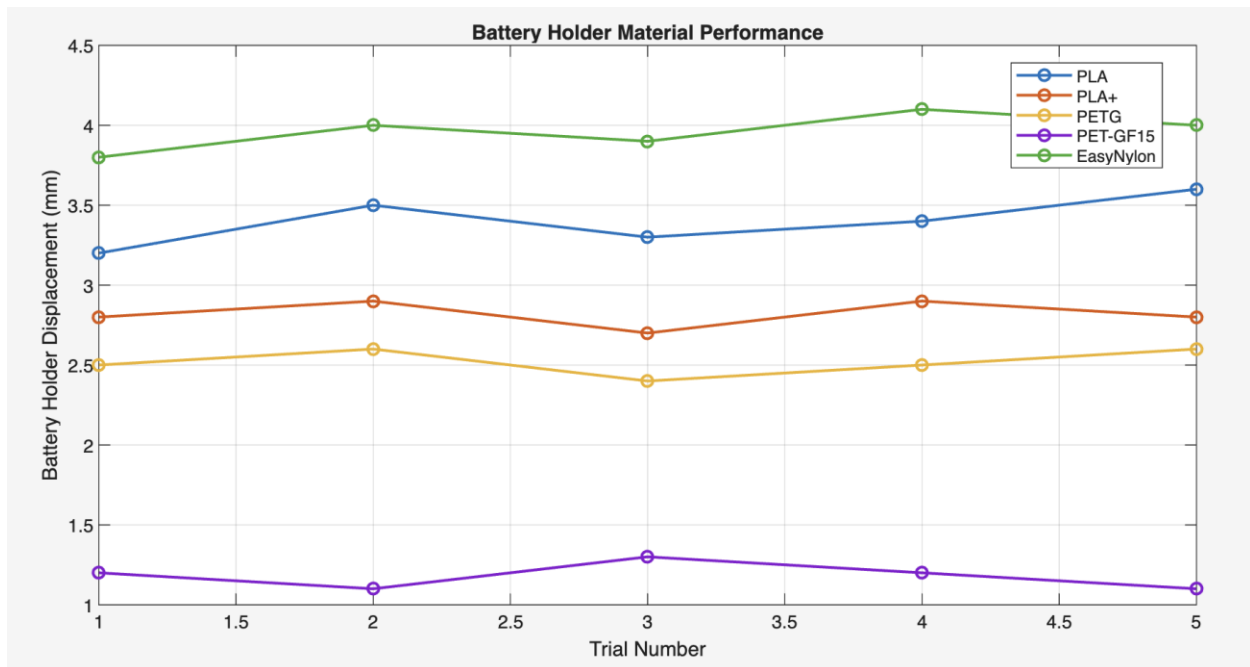
11.3.3.1: TESTING METHOD OF BATTERY HOLDER MATERIAL



To evaluate how different battery holder materials performed under loading, each holder was printed and mounted to the chassis under identical conditions. A standard battery of fixed mass (0.5 kg) was placed inside each holder to apply a consistent load. Displacement was measured at the point of maximum deflection relative to the first unloaded position, recorded in millimetres (mm). Each material was evaluated for five trials, with the battery placed and removed between trials to ensure repeatability. Because the applied load and mounting conditions remained constant, differences in displacement directly reflected the stiffness and ability of each material to maintain structural stability. Materials with lower displacement values indicated greater rigidity and better support under load.

Figure X: Above is a demonstration of how the battery holder's displacement was tested. Displacement was measured for materials PLA, PLA+, PETG, PET-GF15, and EasyNylon.

11.3.3.2: PLOTTED BATTERY DISPLACEMENT/ SLIPPAGE OF BATTERY HOLDER MATERIAL



11.3.3.3: INTERPRETATION OF BATTERY HOLDER MATERIAL DATA

The illustrated graph shows that PET-GF15 maintains the most consistent position with the least movement across trials, making it the best way for the battery holder. PLA and PLA+ show slight deformation over time, PETG is less rigid under load, and EasyNylon allows more movement, confirming PET-GF15 as the most stable choice.

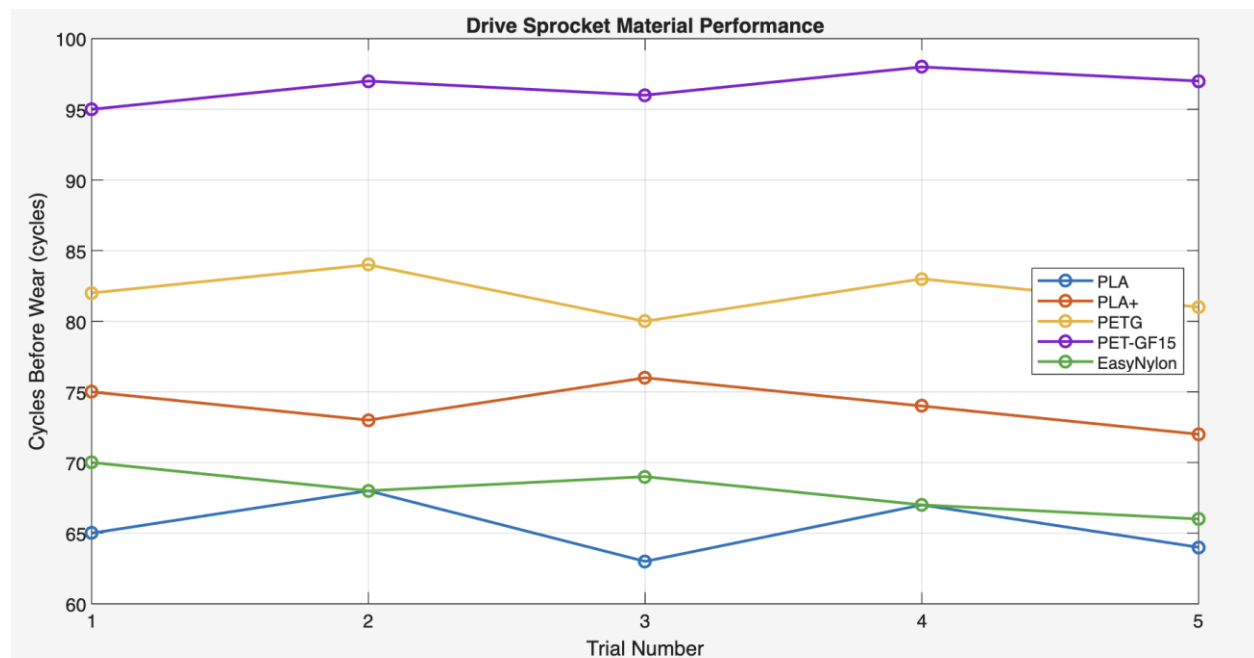
This graph shows the testing of how the battery holder material moves across the

11.3.4: DRIVE SPROCKET MATERIAL TESTING

11.3.4.1: TESTING METHOD OF DRIVE SPROCKET MATERIAL

To evaluate how different drive sprocket materials performed under repeated rotation, each sprocket was mounted onto the drivetrain system and driven using a motor operating at approximately 6000 rpm (3S / 11.1V). The sprocket was engaged with the tread system under constant loading conditions. One full rotation of the sprocket was defined as one cycle. Each material was evaluated for five trials, with the system running continuously until visible wear or performance degradation occurred, defined as loss of tooth integrity or reduced engagement with the tread. The total number of cycles completed before wear was recorded for each trial. Because motor speed and loading conditions were kept constant, differences in cycle count directly reflected the wear resistance and durability of each material under repeated mechanical contact.

11.3.4.2: PLOTTED BATTERY DISPLACEMENT/ SLIPPAGE OF BATTERY HOLDER MATERIAL



The illustrated graph shows that PET-GF15 maintains the most consistent performance under repeated rotation with the least wear, making it the best option for the sprocket drive.

11.3.5: DRIVE SPROCKET INSERT MATERIAL TESTING

11.3.5.1: TESTING METHOD OF DRIVE SPROCKET INSERT MATERIAL

To evaluate how varied materials performed under vibration loading, each outer chassis bracket was printed and mounted to the drivetrain assembly under identical conditions. A constant dynamic load was applied by operating the drivetrain motor at approximately 6000 rpm (3S / 11.1V), generating continuous vibration through the chassis. A fixed mass of 0.5 kg was attached to the bracket to simulate the operational load. Vibration-induced displacement was measured at the point of maximum

deflection relative to the initial unloaded position, using a ruler, and recorded in millimetres (mm). Each material was evaluated for five trials, with the system running for 10 seconds per trial under the same motor speed and loading conditions. The peak displacement during each trial was recorded. Because the motor speed, load, and test duration were kept constant, differences in measured displacement directly reflected the material's stiffness and ability to resist vibration. Lower displacement values indicated higher structural stability.

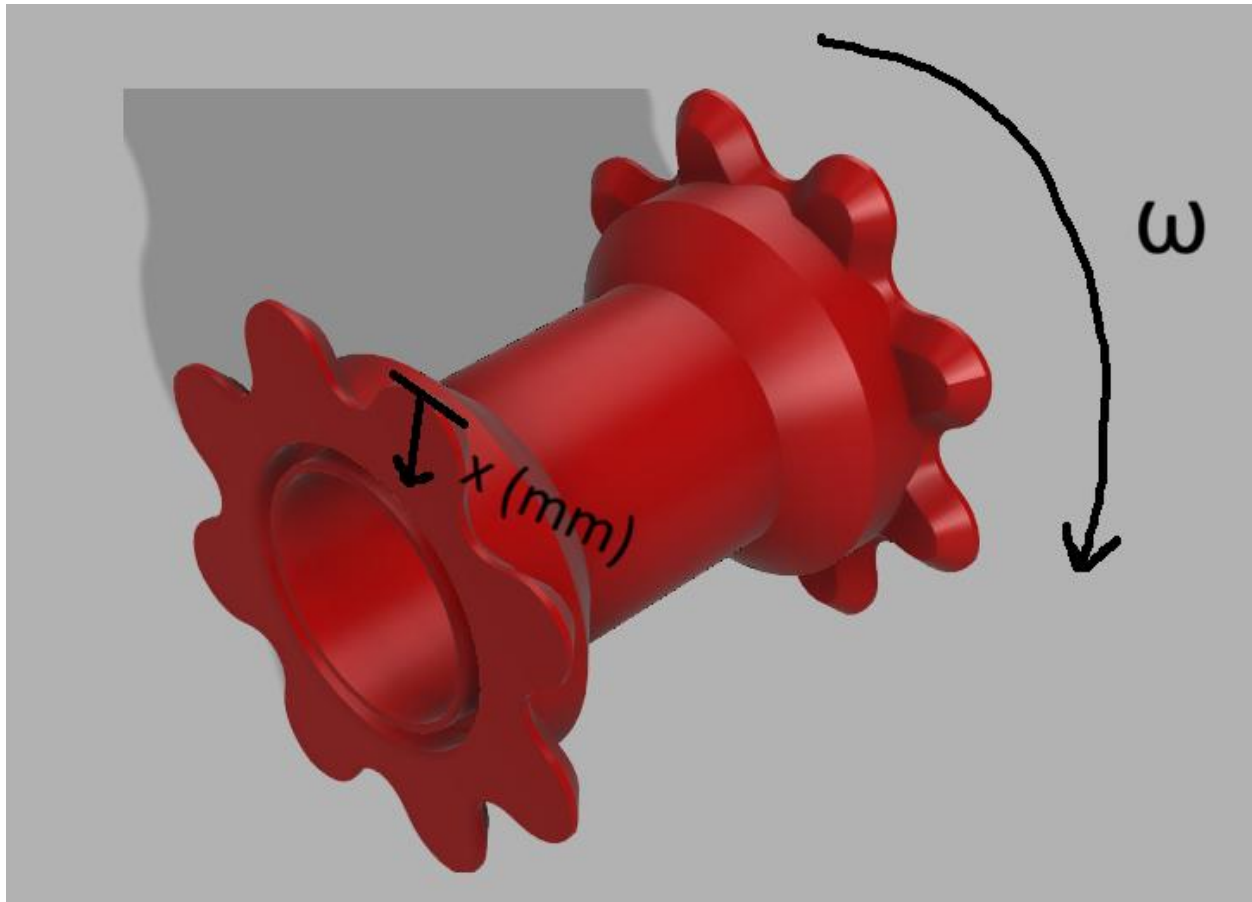
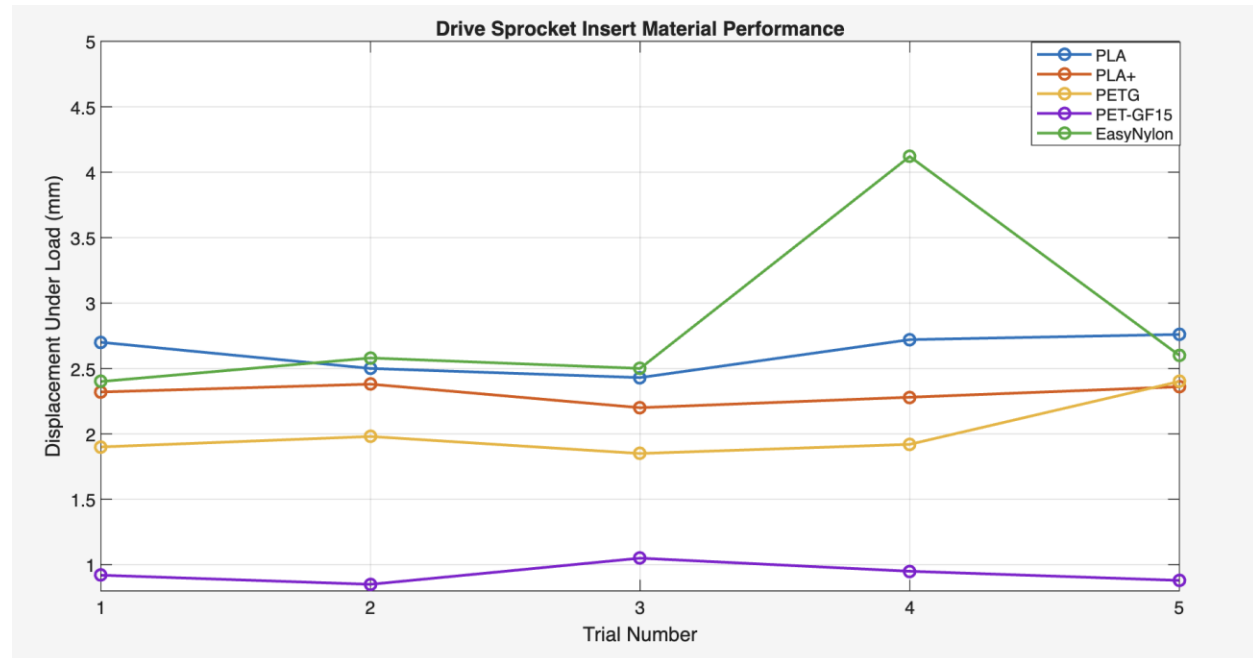


Figure X: Each sprocket material was rotated at a constant velocity for five trials. The maximum displacement was measured in millimeters.

11.3.5.2: PLOTTED DISPLACEMENT UNDER LOAD/ SLIPPAGE OF DRIVE SPROCKET INSERT MATERIAL



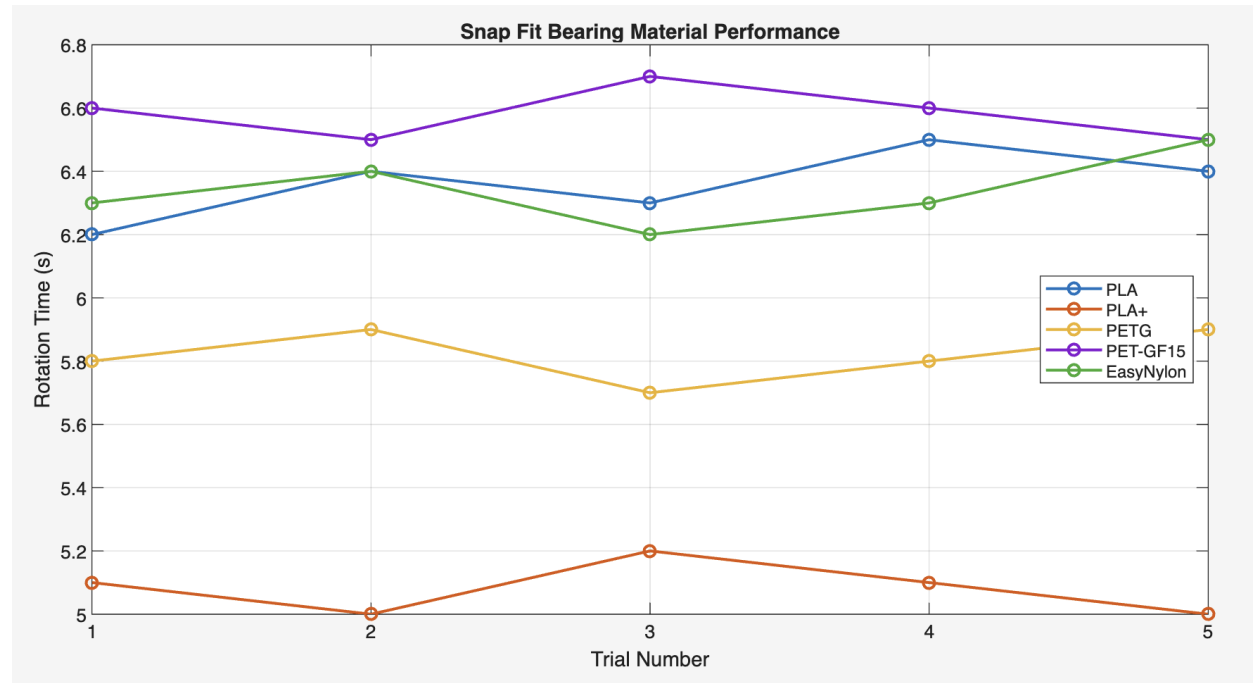
The illustrated graph shows that PET-GF15 has the lowest displacement under load across trials, indicating it performs best under applied forces.

11.3.6: FINAL SNAP-FIT BEARING MATERIAL TESTING

11.3.6.1: TESTING METHOD OF FINAL SNAP-FIT BEARING MATERIAL

To evaluate how different snap-fit bearing materials performed under rotational resistance, each bearing was mounted onto a shaft and driven using a motor operating at approximately 6000 rpm (3S / 11.1V nominal at 30A). The system was allowed to spin under motor power for 5 seconds to reach a consistent initial speed. After 5 seconds, motor engagement was removed, and the bearing was allowed to rotate freely until complete stop. The total rotation time from motor disengagement to rest was recorded in seconds using a timer. Each material was evaluated for five trials under identical conditions. Because the input speed and test duration were kept constant, differences in rotation time directly reflected internal friction and resistance within the bearing. Longer rotation times indicated lower resistance and smoother performance.

11.3.6.2: PLOTTED ROTATION TIME/ SLIPPAGE FINAL SNAP-FIT BEARING MATERIAL



The illustrated graph shows that PLA+ achieves the smoothest rotation with the lowest resistance across trials, making it the best option for the snap-fit bearing.

11.3.7: OUTER SANDWICH TREAD CHASSIS BRACKET MATERIAL TESTING

11.3.7.1: TESTING METHOD OF OUTER SANDWICH TREAD CHASSIS BRACKET MATERIAL

To evaluate how different materials were performed under vibration loading, each outer chassis bracket was printed and mounted to the drivetrain assembly under identical conditions. A constant dynamic load was applied by operating the drivetrain motor at approximately 6000 rpm (3S / 11.1V), generating continuous vibration through the chassis. A fixed mass of 0.5 kg was attached to the bracket to simulate the operational load. Vibration-induced displacement was measured at the point of maximum deflection relative to the initial unloaded position, using a ruler, and recorded in millimeters (mm). Each material was tested for 5 trials, with the system running for 10 seconds per trial under the same motor speed and loading conditions. The peak displacement during each trial was recorded. Because the motor speed, load, and test duration were kept constant, differences in measured displacement directly reflected the material's stiffness and ability to resist vibration. Lower displacement values indicated higher structural stability.

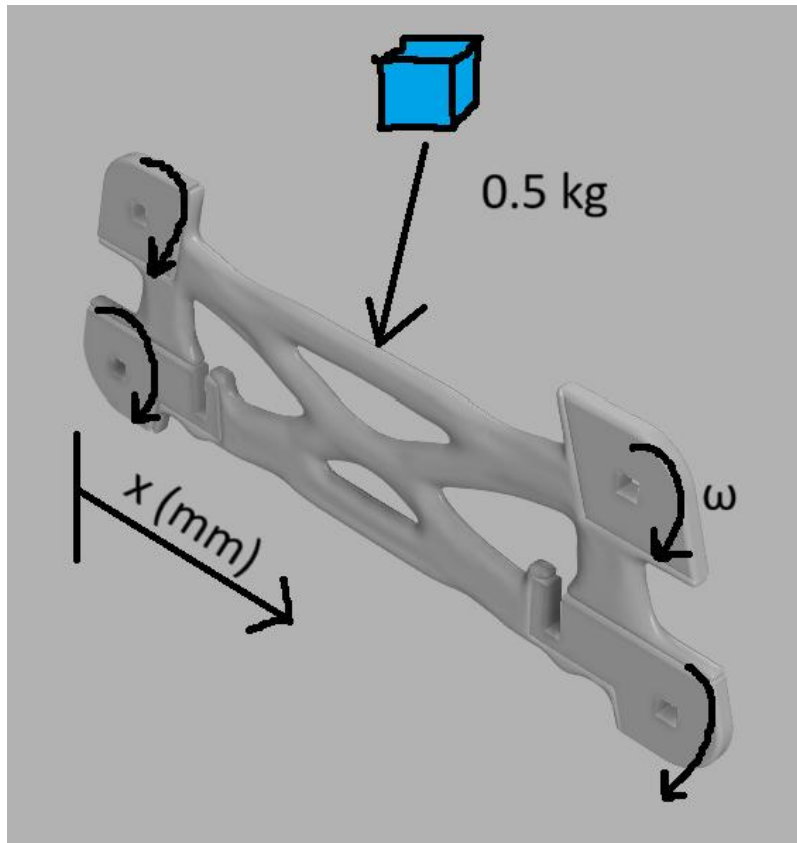
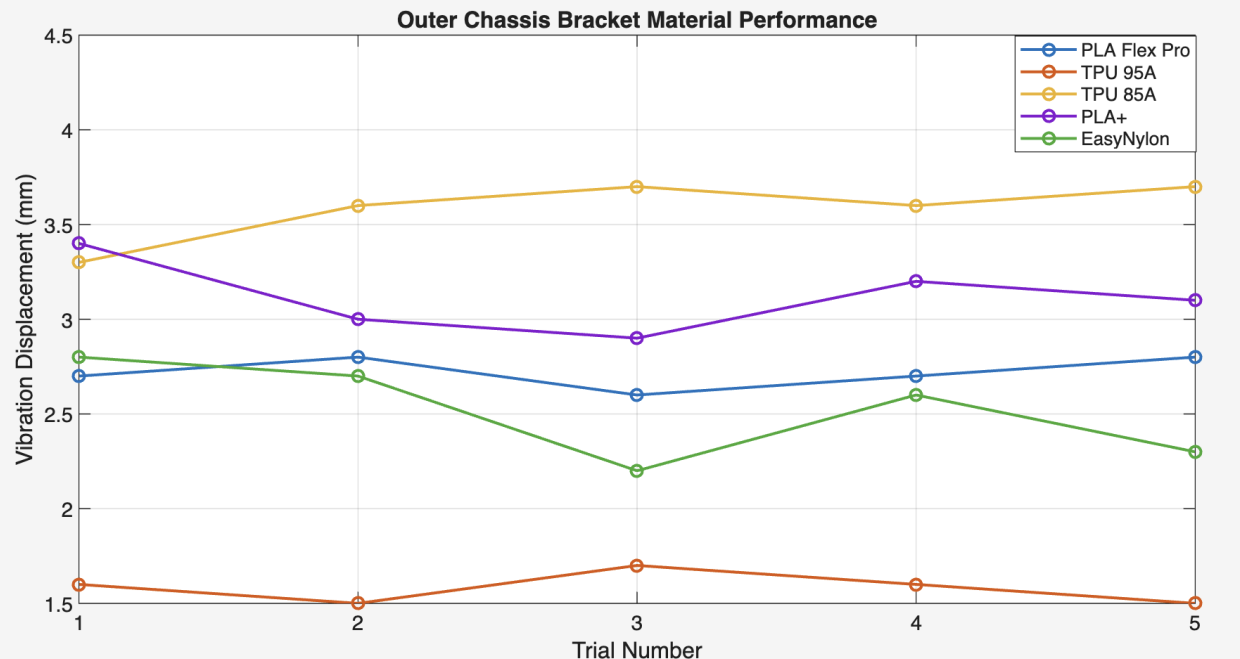


Figure X: Demonstrates the Outer Chassis displacement tests. The cube in the figure represents the 0.5 kg weight used to simulate the total assembly weight. The maximum displacement(mm) was measured after 10 seconds of constant rotation by the motors.

11.3.7.2: PLOTTED VIBRATION DISPLACEMENT/ SLIPPAGE OUTER SANDWICH TREAD CHASSIS BRACKET MATERIAL



TPU 95A has the lowest and most stable displacement values, which shows it performs best and is the most suitable choice for the chassis.

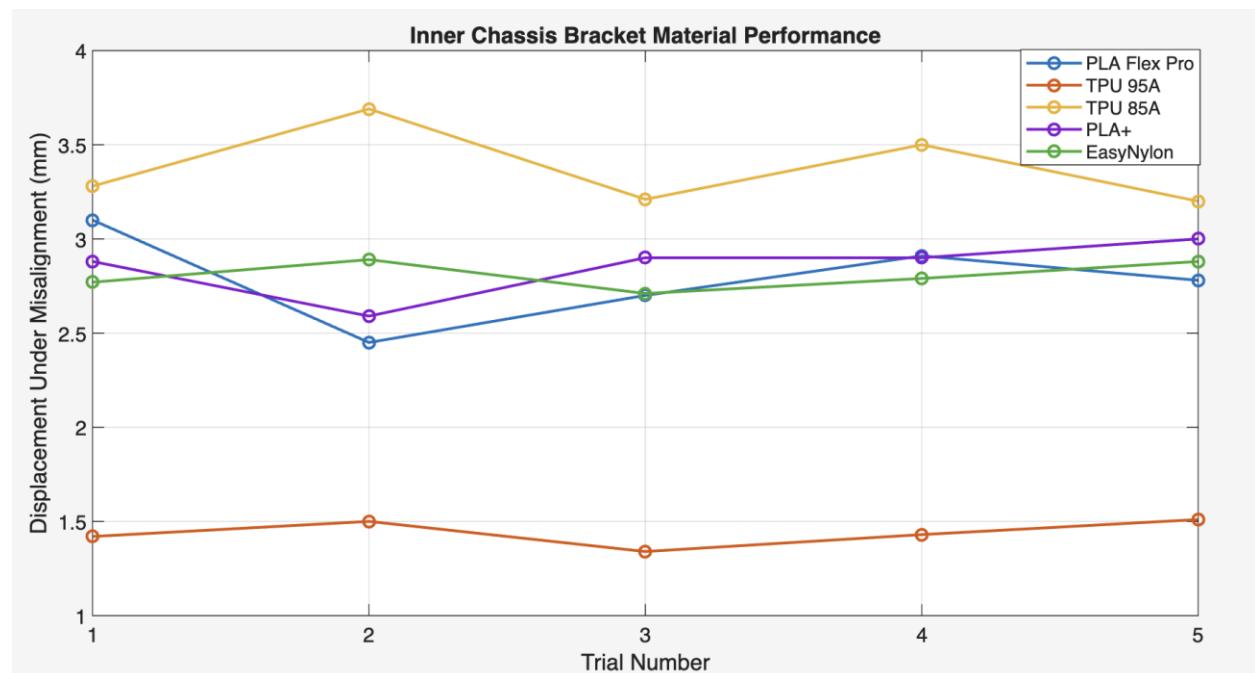
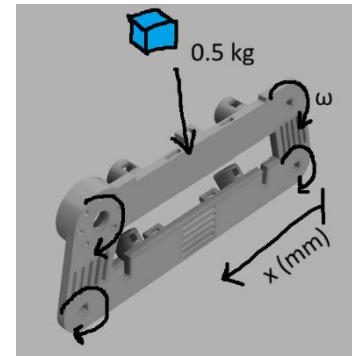
11.3.8: INNER SANDWICH TREAD CHASSIS BRACKET MATERIAL TESTING

11.3.8.1: TESTING METHOD OF INNER SANDWICH TREAD CHASSIS BRACKET MATERIAL

To evaluate how different materials were performed under misalignment loading, each inner chassis bracket was printed and mounted into the tread assembly under identical conditions. A fixed misalignment was introduced by offsetting the mounting points by approximately 3 mm, forcing the bracket to deform under load during operation. The drivetrain system was powered using a motor operating at approximately 6000 rpm (3S / 11.1V), causing continuous rotation of the tread under misaligned conditions. A constant load of 0.5 kg was applied to simulate operational forces. Displacement due to misalignment was measured at the point of maximum deflection relative to the initial aligned position, recorded in millimeters (mm). Each material was tested for 5 trials, with the system running for 10 seconds per trial under identical conditions. Because the motor speed, applied load, and misalignment offset were kept constant, differences in displacement directly reflected each material's ability to absorb misalignment and reduce stress. Lower displacement values indicated better compliance and structural performance under misaligned loading.

Figure on the right demonstrates the Inner Chassis displacement tests. The cube in the figure represents the 0.5 kg weight used to simulate the total assembly weight. The maximum displacement (mm) was measured after 10 seconds of constant rotation by the motors.

11.3.8.2: PLOTTED DISPLACEMENT UNDER MISALIGNMENT / SLIPPAGE INNER SANDWICH TREAD CHASSIS BRACKET MATERIAL



TPU 95A maintains the lowest and most consistent displacement, showing it performs best at absorbing

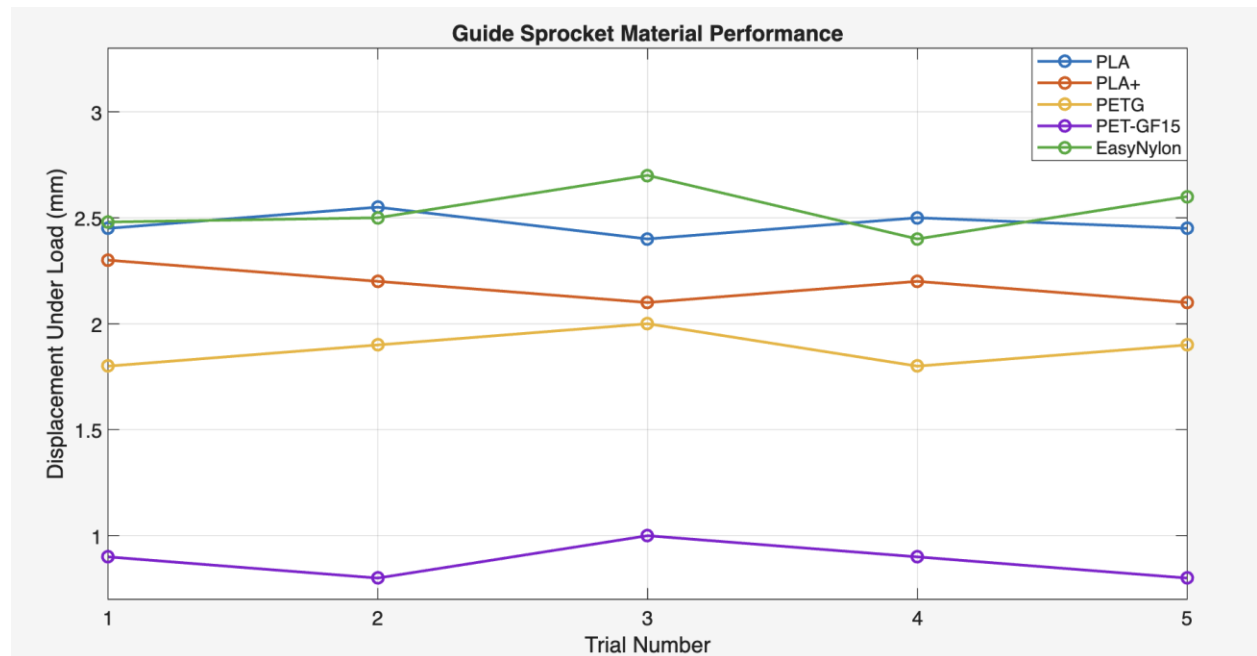
misalignment and reducing stress. making TPU 95A the most suitable choice for the inner chassis brackets.

11.3.9: GUIDE SPROCKET MATERIAL TESTING

11.3.9.1: TESTING METHOD OF GUIDE SPROCKET MATERIAL

To evaluate how different guide sprocket materials perform under loading, each sprocket was mounted into the drivetrain system and subjected to a constant load during operation. A fixed mass of 0.5 kg was applied to the sprocket assembly to simulate operational forces. The system was driven using a motor operating at approximately 6000 rpm (3S / 11.1V), ensuring consistent rotational conditions across all tests. Displacement was measured at the point of maximum deflection relative to the initial unloaded position, recorded in millimeters (mm). Each material was tested for 5 trials, with the system running for 10 seconds per trial under identical conditions. The peak displacement during each trial was recorded. Because the applied load, motor speed, and test duration were kept constant, differences in displacement directly reflected each material's stiffness and ability to resist deformation under load. Lower displacement values indicated better structural performance.

11.4.9.2: PLOTTED DISPLACEMENT UNDER LOAD/ SLIPPAGE GUIDE SPROCKET MATERIAL



The illustrated graph shows that PET-GF15 has the lowest displacement under load across trials, indicating it performs best under applied forces.

11.3.10: TOP ELECTRONICS SHELF MATERIAL TESTING

11.3.10.1: TESTING METHOD OF TOP ELECTRONICS SHELF MATERIAL

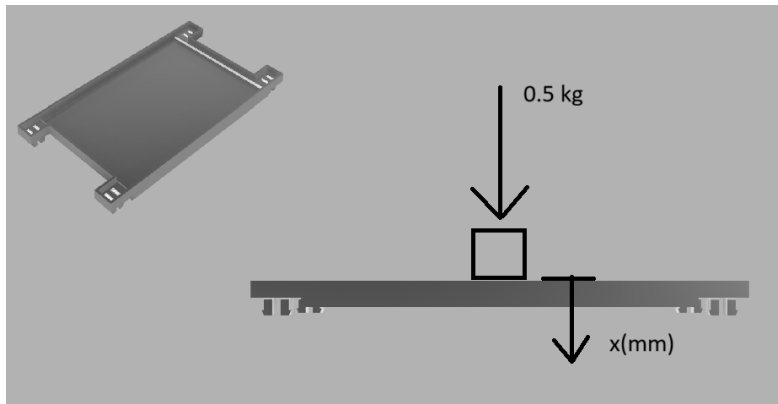
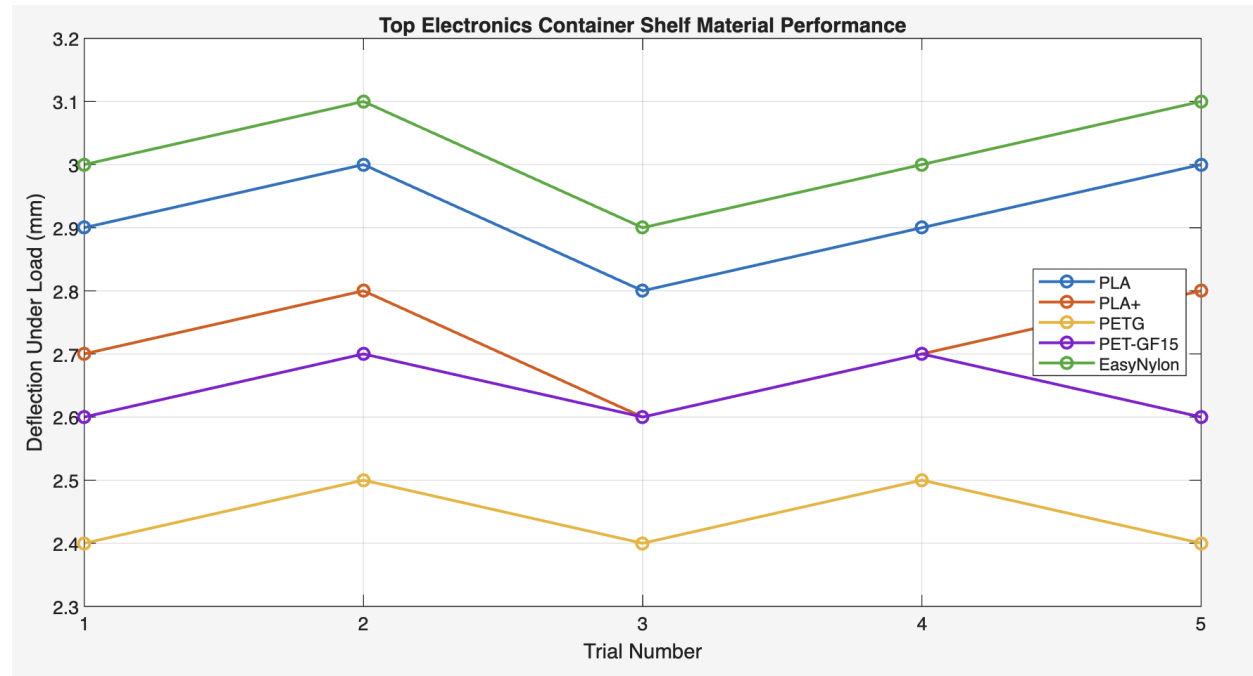


Figure X: Demonstrates the testing for the top electronics shelf. Deflection was measured from the datum shown with a 0.5 kg mass placed in the center.

To evaluate how varied materials performed under load, each shelf was printed and mounted horizontally in the chassis under identical conditions. A fixed mass of 0.5 kg was placed at the midpoint (centre) of the shelf to create maximum bending. Deflection was measured at the midpoint relative to the initial unloaded position and recorded in millimetres (mm). The drivetrain motor was operated at approximately 6000 rpm (3S / 11.1V) during testing to replicate operational vibration conditions. Each material was evaluated for five trials, with the load applied for 10 seconds per trial. The maximum deflection recorded in each trial was used for comparison. Because the load, motor speed, and test duration were kept constant, differences in deflection reflected each electro material's stiffness and resistance to bending.

11.3.10.2: PLOTTED DEFLECTION UNDER LOAD/ SLIPPAGE TOP ELECTRONICS SHELF MATERIAL



11.3.10.3: INTERPRETATION OF TOP ELECTRONICS SHELF MATERIAL DATA

The illustrated graph shows that PETG has the lowest deflection at greatest point of displacement (midpoint/center of shelf) across trials, meaning it maintains the most consistent positioning. PLA and PLA+ show higher values due to reduced alignment stability, PET-GF15 is more rigid than necessary for this application, and EasyNylon allows more movement. This confirms PETG as the most suitable choice for the top electronics container shelf.

11.3.11: MIDDLE ELECTRONICS SHELF MATERIAL TESTING

11.3.11.1: TESTING METHOD OF MIDDLE ELECTRONICS SHELF MATERIAL METHOD

To evaluate how varied materials performed in maintaining structural alignment, each middle shelf was printed and mounted in the chassis under identical conditions. A fixed mass of 0.5 kg was placed at the midpoint of the shelf to simulate operational loading. Deflection was measured at the midpoint relative to the initial unloaded position and recorded in millimetres (mm). The drivetrain motor was operated at approximately 6000 rpm (35 / 11.1V) during testing to simulate operational vibration. Each material was evaluated for five trials, with the load applied for 10 seconds per trial. The maximum deflection recorded in each trial was used for comparison.

Because all conditions were kept constant, differences in deflection reflected each material's ability to maintain alignment and support under load.

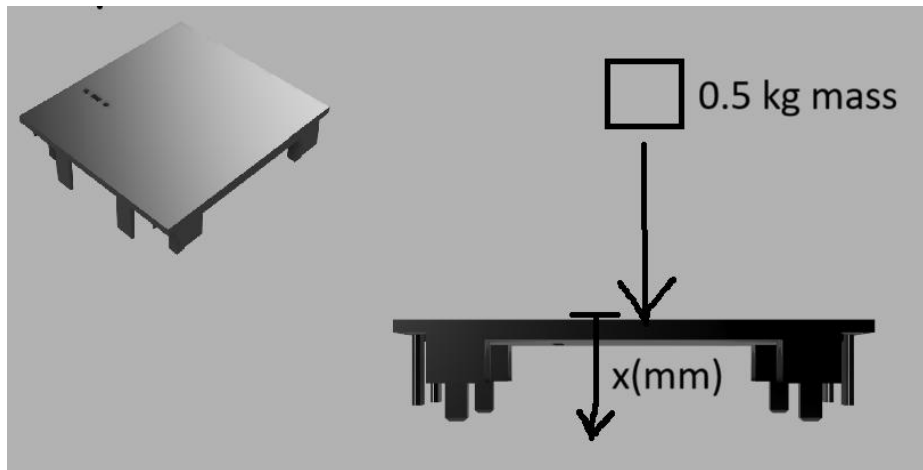
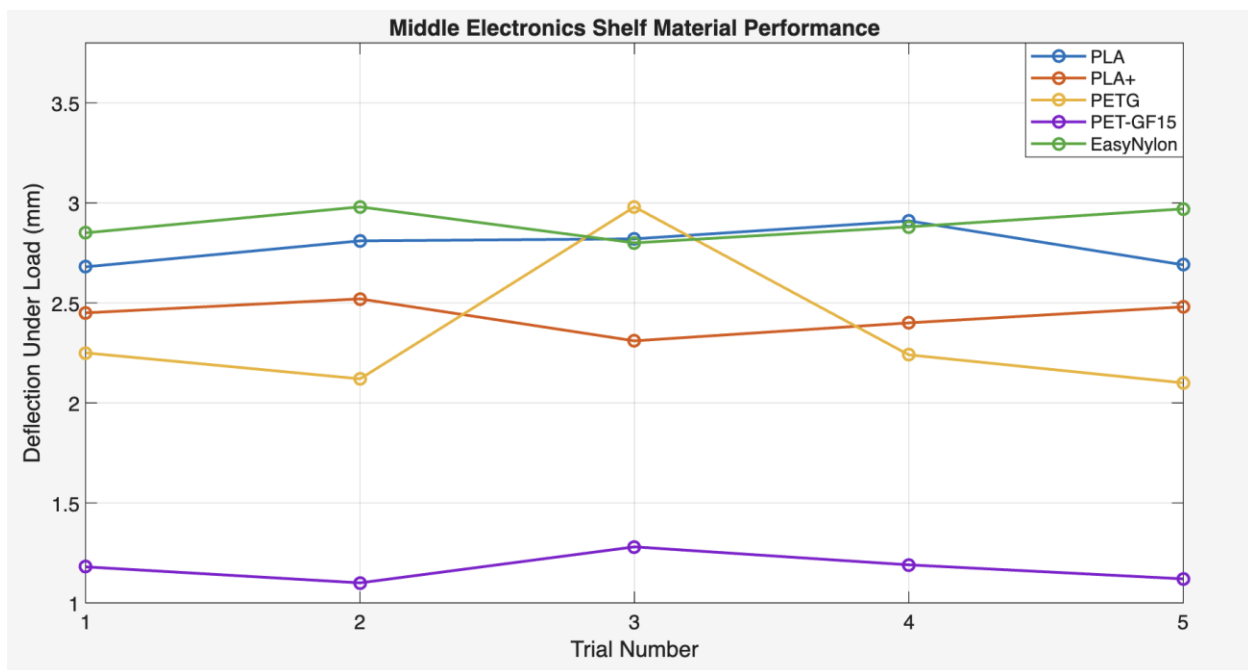


Figure X: Demonstration of the tests performed on each material of the middle electronics shelf. Deflection was measured from the center, where the 0.5 kg mass was placed.

11.3.11.2: PLOTTED DEFLECTION UNDER LOAD/ SLIPPAGE MIDDLE ELECTRONICS SHELF MATERIAL



11.3.11.3: INTERPRETATION OF MIDDLE ELECTRONICS SHELF MATERIAL DATA

The illustrated graph shows that PET-GF15 has the lowest deflection at greatest point of displacement (midpoint/centre of shelf) across trials, meaning it maintains the most consistent positioning. PLA and PLA+ show higher values due to reduced alignment stability, PETG is less rigid, and EasyNylon allows more movement. This leads us to choose PET-GF15 as the most suitable choice for the middle electronics container shelf.

11.3.12: BOTTOM ELECTRONICS SHELF MATERIAL TESTING

11.3.12.1: TESTING METHOD OF BOTTOM ELECTRONICS SHELF MATERIAL

To evaluate how different materials performed under load, each bottom shelf was printed and mounted in the chassis under identical conditions. A fixed mass of 0.5 kg was placed at the midpoint (center) of the shelf to simulate operational loading. Deflection was measured at the midpoint relative to the initial unloaded position and recorded in millimeters (mm). The drivetrain motor operated at approximately 6000 rpm (3S / 11.1V) during testing to replicate operational vibration. Each material was tested for 5 trials, with the load applied for 10 seconds per trial. The maximum deflection recorded in each trial was used for comparison. Because the load, motor speed, and test duration were kept constant, differences in deflection reflected each material's ability to support load and maintain structural stability at the base of the system.

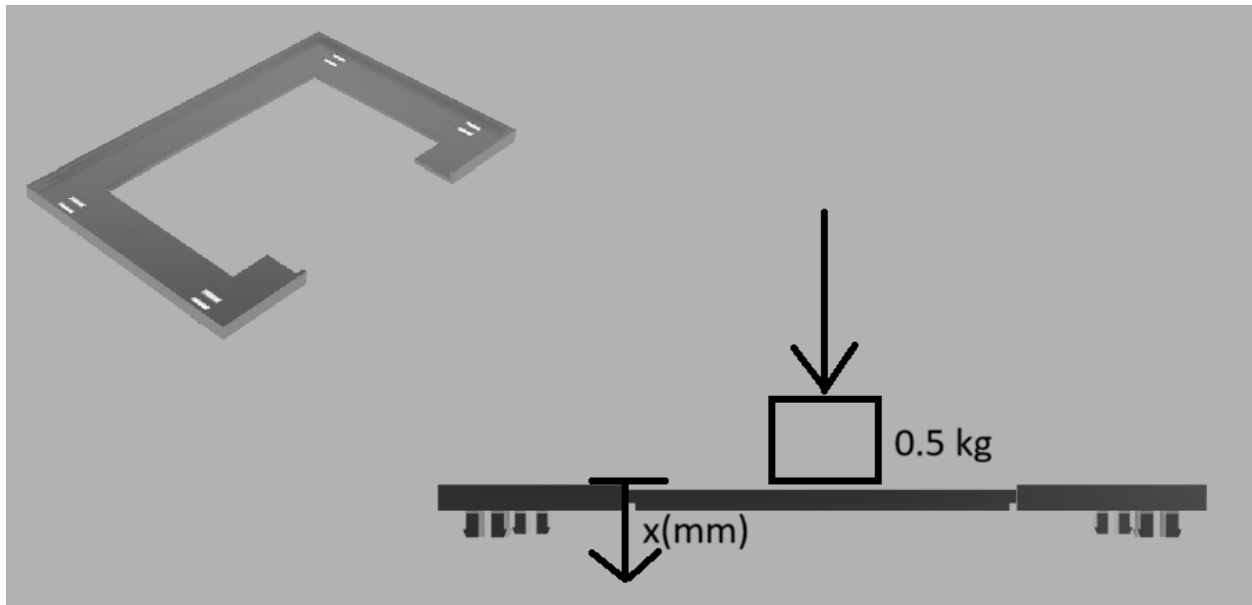
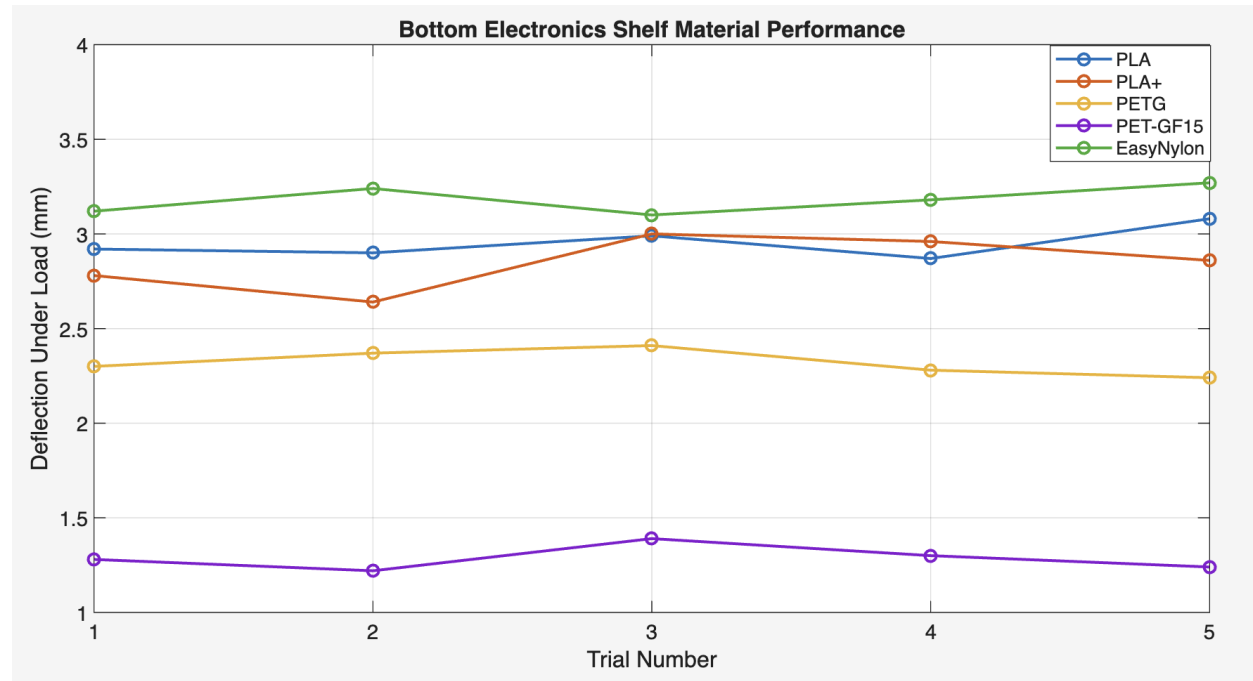


Figure X: Demonstration of the tests performed on each material of the bottom electronics shelf. Deflection was measured from the center, where the 0.5 kg mass was placed.

11.3.12.2: PLOTTED DEFLECTION UNDER LOAD/ SLIPPAGE BOTTOM ELECTRONICS SHELF MATERIAL



11.3.12.3: INTERPRETATION OF BOTTOM ELECTRONICS SHELF MATERIAL DATA

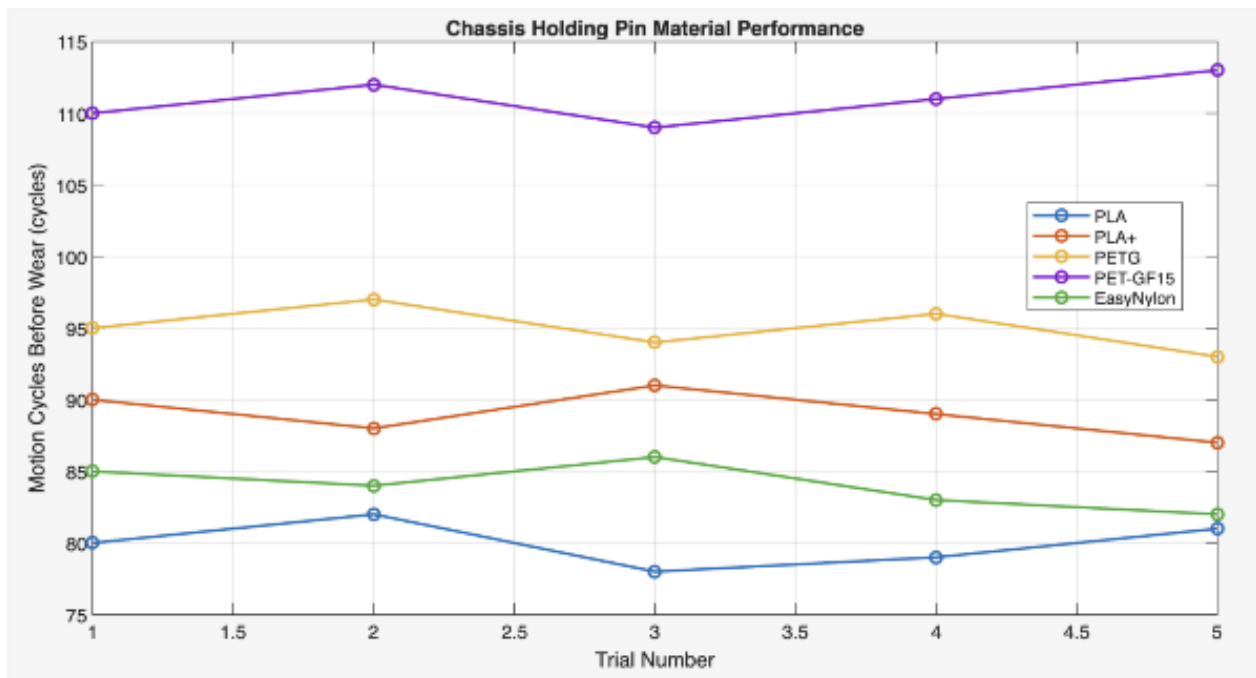
The illustrated graph shows that PET-GF15 has the lowest deflection at greatest point of displacement (midpoint/centre of shelf) across trials, meaning it distributes the load most effectively and maintains stability. PLA and PLA+ show higher deformation, PETG does not provide the same strength, and EasyNylon is too flexible. These results reinforce PET-GF15 as the most suitable choice for the bottom electronics container shelf.

11.3.13: CHASSIS HOLDING PINS MATERIAL TESTING

11.3.13.1: TESTING METHOD OF CHASSIS HOLDING PINS MATERIAL

To evaluate how different chassis holding pin materials performed under repeated loading, each pin was installed into the chassis mounting points and subjected to continuous motion using the drivetrain system. The motor operated at approximately 6000 rpm (3S / 11.1V), producing consistent cyclic loading on the pins. One full motion cycle of the system was counted as one cycle. Each material was tested for 5 trials, with the system running continuously until the pin showed wear or deformation affecting its fit and stability. The total number of cycles completed before noticeable wear was recorded for each trial. Because all testing conditions were the same, differences in cycle count were caused by the material properties. Higher cycle counts indicate greater resistance to wear and better durability under repeated loading.

11.3.13.2: PLOTTED VIBRATION DISPLACEMENT/ SLIPPAGE CHASSIS HOLDING PINS MATERIAL



11.3.13.3: INTERPRETATION OF CHASSIS HOLDING PINS MATERIAL DATA

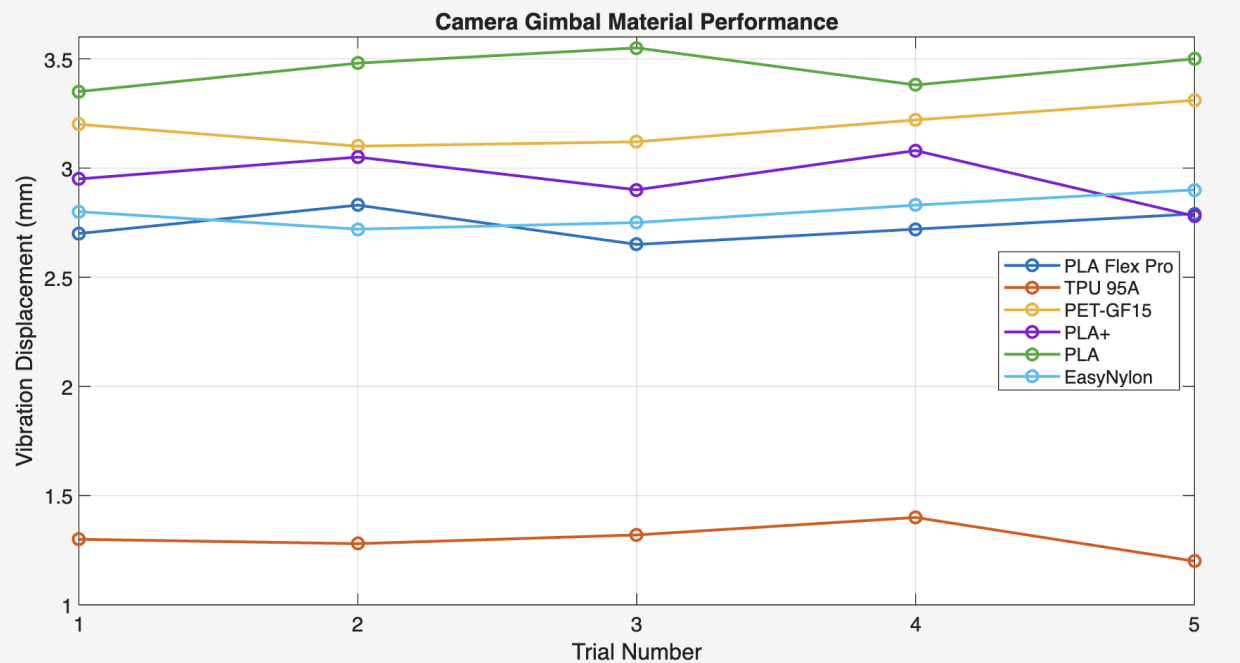
The illustrated graph shows that PET-GF15 maintains the highest number of motion cycles with minimal deformation across trials, meaning it holds its shape and dimensions the best. PLA and PLA+ show faster wear over time, PETG does not maintain stiffness as well, and EasyNylon deforms under repeated loading. The durability of PET-GF15 makes it the most suitable choice for the chassis holding pins.

11.3.14: CAMERA GIMBAL MATERIAL TESTING

11.3.14.1: TESTING METHOD OF CAMERA GIMBAL MATERIAL

To evaluate how different materials performed under vibration, each gimbal mount was printed and installed in the camera gimbal assembly under identical conditions. A 0.3 kg load was applied to represent the camera. The system was driven using a motor operating at 6000 rpm (3S / 11.1V) to generate vibration. Displacement was measured at the point of maximum deflection relative to the initial position and recorded in millimeters (mm). Each material was tested for 5 trials, with the system running for 10 seconds per trial. The maximum displacement in each trial was recorded. Because load, motor speed, and test duration were constant, differences in displacement reflected each material's vibration damping performance.

11.3.14.2: PLOTTED VIBRATION DISPLACEMENT/ SLIPPAGE CAMERA GIMBAL MATERIAL



The illustrated graph shows that TPU 95A has the lowest vibration displacement across trials, meaning it absorbs vibrations the most effectively during motion.

11.3.15: RIGID HORIZONTAL CHASSIS CROSS-BAR MATERIAL TESTING

11.3.15.1: TESTING METHOD OF RIGID HORIZONTAL CHASSIS CROSS-BAR MATERIAL

To evaluate how different materials performed under load, each crossbar was printed and mounted horizontally in the chassis under identical conditions. A 0.5 kg load was applied at the midpoint of the bar to create bending. Deflection was measured at the midpoint relative to the initial unloaded position and recorded in millimeters (mm). The drivetrain motor was operated at approximately 6000 rpm (3S / 11.1V) during testing to simulate operational vibration. Each material was tested for 5 trials, with the load applied for 10 seconds per trial. The maximum deflection in each trial was recorded. Because the load, motor speed, and test duration were constant, differences in deflection reflected each material's stiffness and resistance to bending.

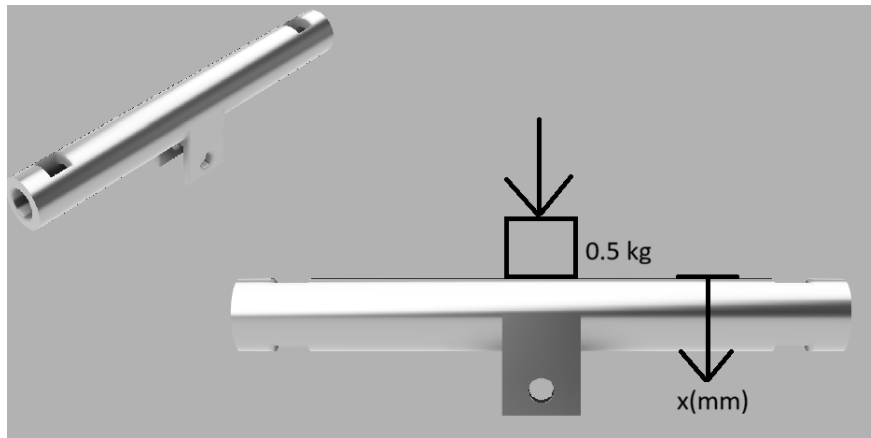
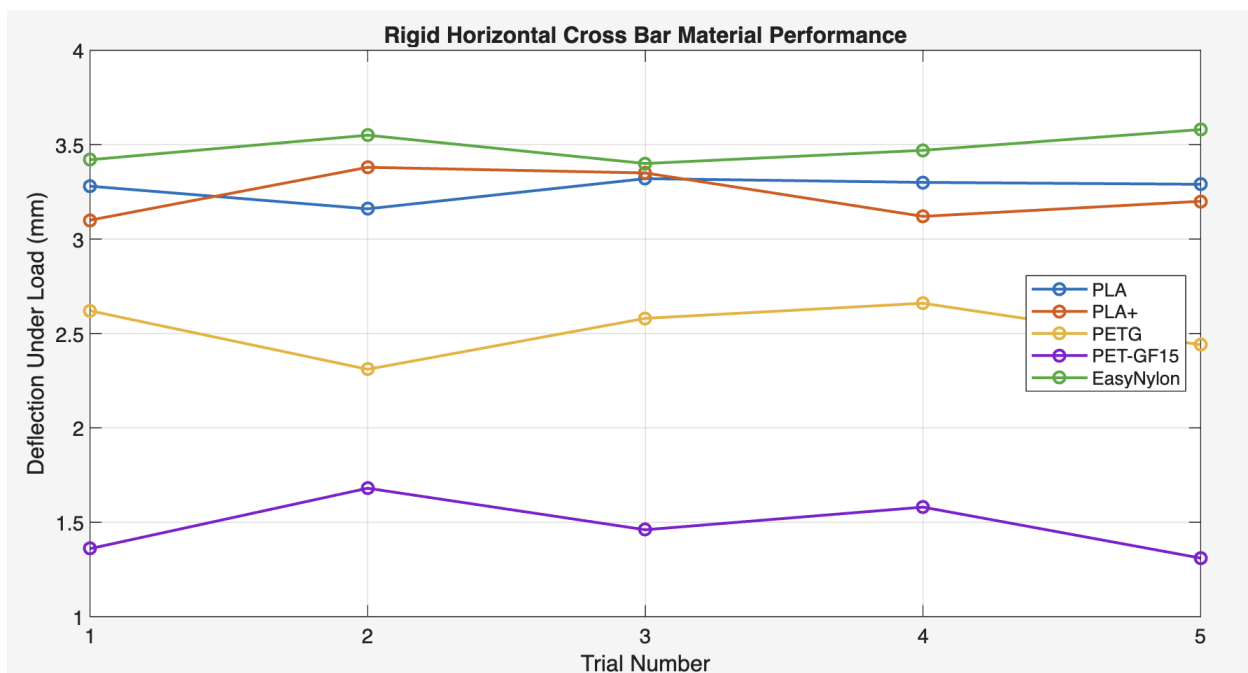


Figure X: Demonstration of the tests performed on each material of the horizontal chassis cross-bar. Deflection was measured from the center, where the 0.5 kg mass was placed.

11.3.15.2: PLOTTED DEFLECTION UNDER LOAD/ SLIPPAGE RIGID HORIZONTAL CHASSIS CROSS-BAR MATERIAL



The illustrated graph shows that PET-GF15 has the lowest deflection under load across trials, meaning it distributes structural loads most effectively.

11.3.16: RIGID DIAGONAL CHASSIS CROSS-BAR MATERIAL TESTING

11.3.16.1: TESTING METHOD OF RIGID DIAGONAL CHASSIS CROSS-BAR MATERIAL

To evaluate how different materials performed under load, each diagonal crossbar was printed and mounted in the chassis under identical conditions. A 0.5 kg load was applied along the diagonal member to simulate structural loading. Deflection was measured at the point of maximum

displacement relative to the initial unloaded position and recorded in millimeters (mm). The drivetrain motor operated at approximately 6000 rpm (3S / 11.1V) during testing. Each material was tested for 5 trials, with the load applied for 10 seconds per trial. The maximum deflection in each trial was recorded. Because the load, motor speed, and test duration were constant, lower deflection values indicate better stiffness and resistance to bending.

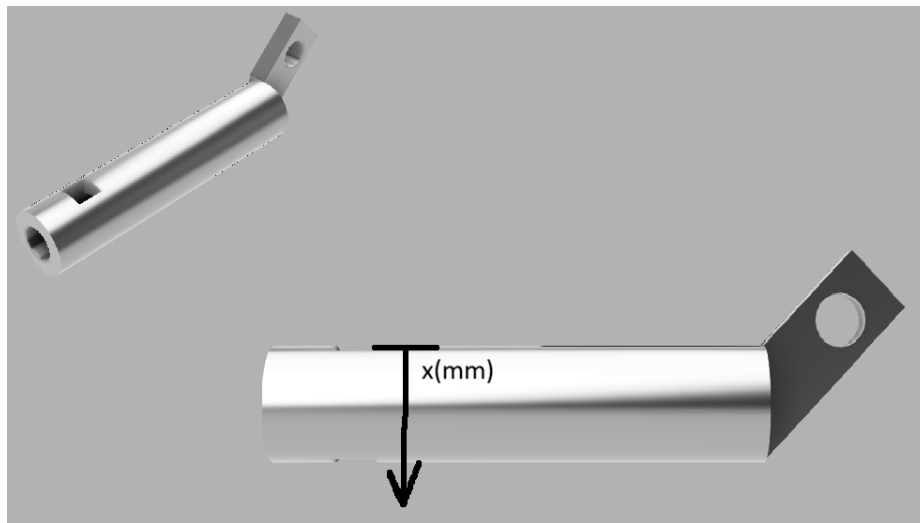
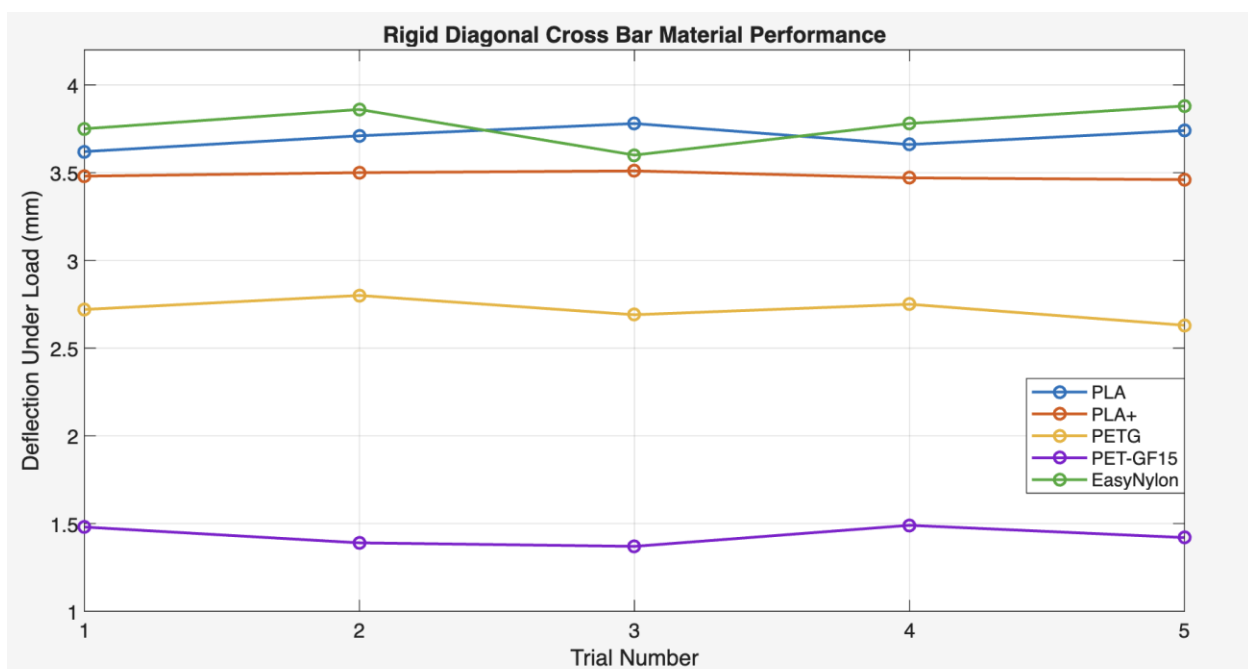


Figure X: Demonstration of the test performed on the diagonal crossbar. The deflection was measured from the datum in the figure above.

11.3.16.2: PLOTTED DEFLECTION UNDER LOAD/ SLIPPAGE RIGID DIAGONAL CHASSIS CROSS-BAR MATERIAL



The illustrated graph shows that PET-GF15 has the lowest deflection under load across trials, meaning it resists deformation and stabilizes the structure the best.

11.3.17: SCOOP MATERIAL TESTING

11.3.17.1: TESTING METHOD OF SCOOP MATERIAL

To evaluate how different materials performed under load, each scoop was printed and mounted in the testing setup under identical conditions. A 0.5 kg load was applied at the leading edge of the scoop to simulate interaction with sand during operation. Displacement was measured at the point of maximum deflection along the scoop surface relative to the initial unloaded position and recorded in millimeters (mm). The drivetrain motor operated at approximately 6000 rpm (3S / 11.1V) during testing. Each material was tested for 5 trials, with the load applied for 10 seconds per trial. The maximum displacement in each trial was recorded. Because the load, motor speed, and test duration were kept constant, lower displacement values indicate better stiffness and resistance to deformation.

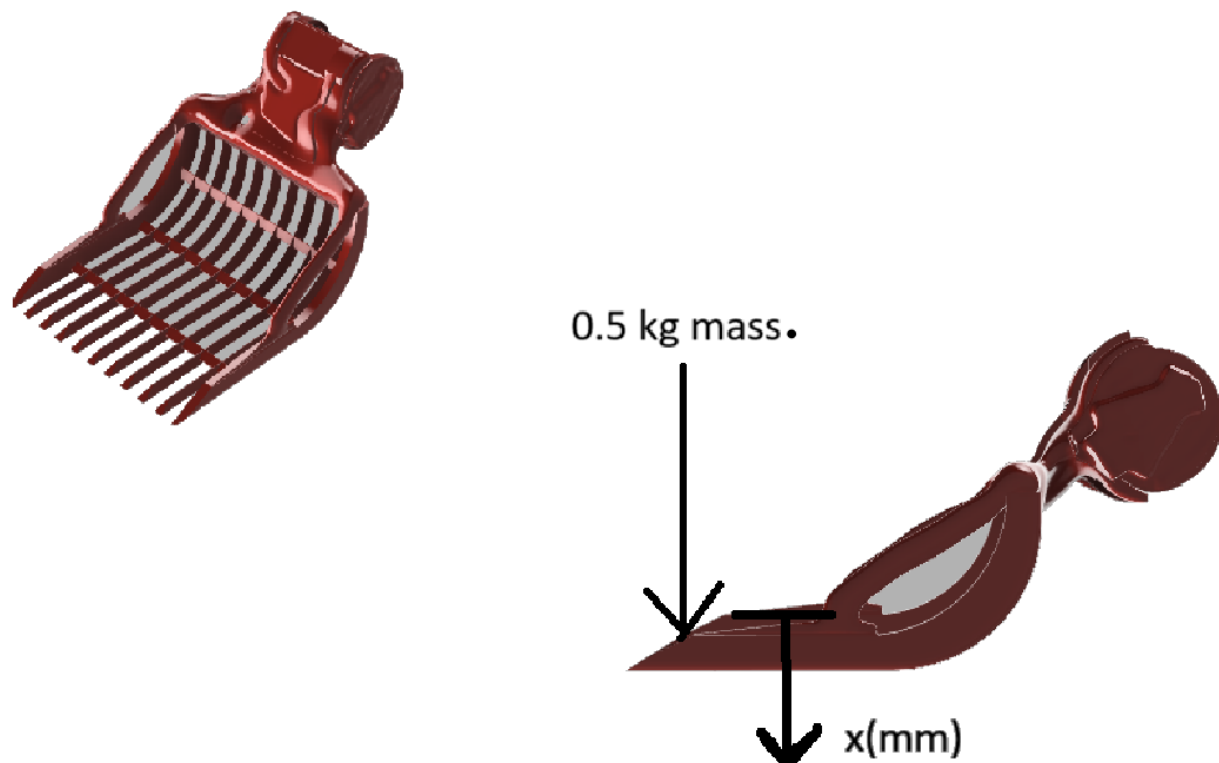
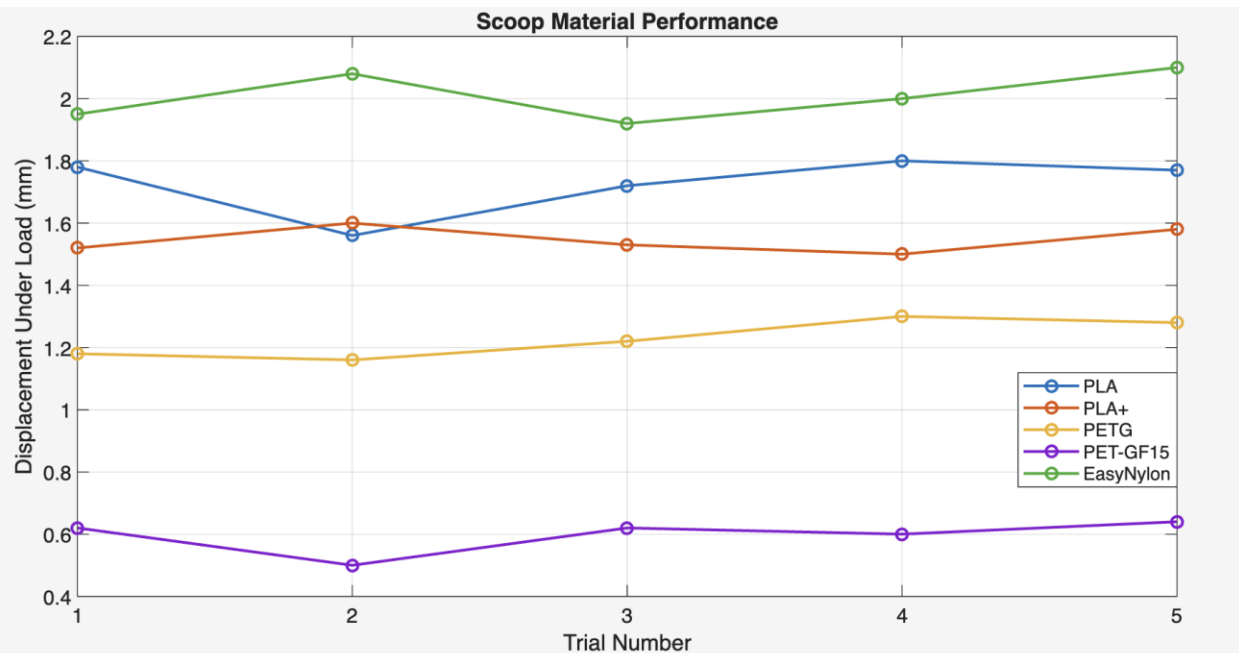


Figure X: Demonstration of the test performed for each scoop material. The maximum deflection was measured from the datum in the figure above.

11.3.17.2: PLOTTED DISPLACEMENT UNDER LOAD/ SLIPPAGE SCOOP MATERIAL



The illustrated graph shows that PET-GF15 has the lowest displacement under load across trials, indicating it performs best under applied forces.

11.3.18: ARM TO CHASSIS MOUNT MATERIAL TESTING

11.3.18.1: TESTING METHOD OF ARM TO CHASSIS MOUNT MATERIAL

Chassis to arm mount was printed and installed at the connection point between the arm and chassis under identical conditions. A 0.5 kg load was applied at the arm interface to simulate operational forces. Deflection was measured at the point of maximum displacement relative to the initial unloaded position and recorded in millimeters (mm). The drivetrain motor operated at approximately 6000 rpm (3S / 11.1V) during testing. Each material was tested for 5 trials, with the load applied for 10 seconds per trial. The maximum deflection in each trial was recorded. Because the load, motor speed, and test duration were kept constant, lower deflection values indicate better stiffness and resistance to

deformation at the mounting interface.

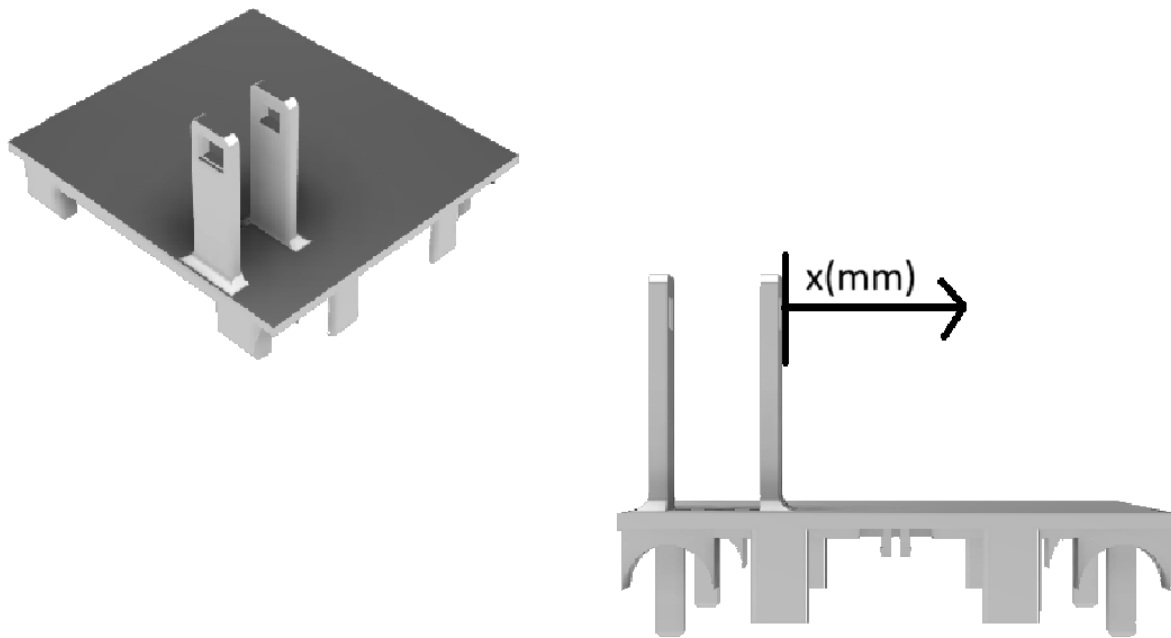
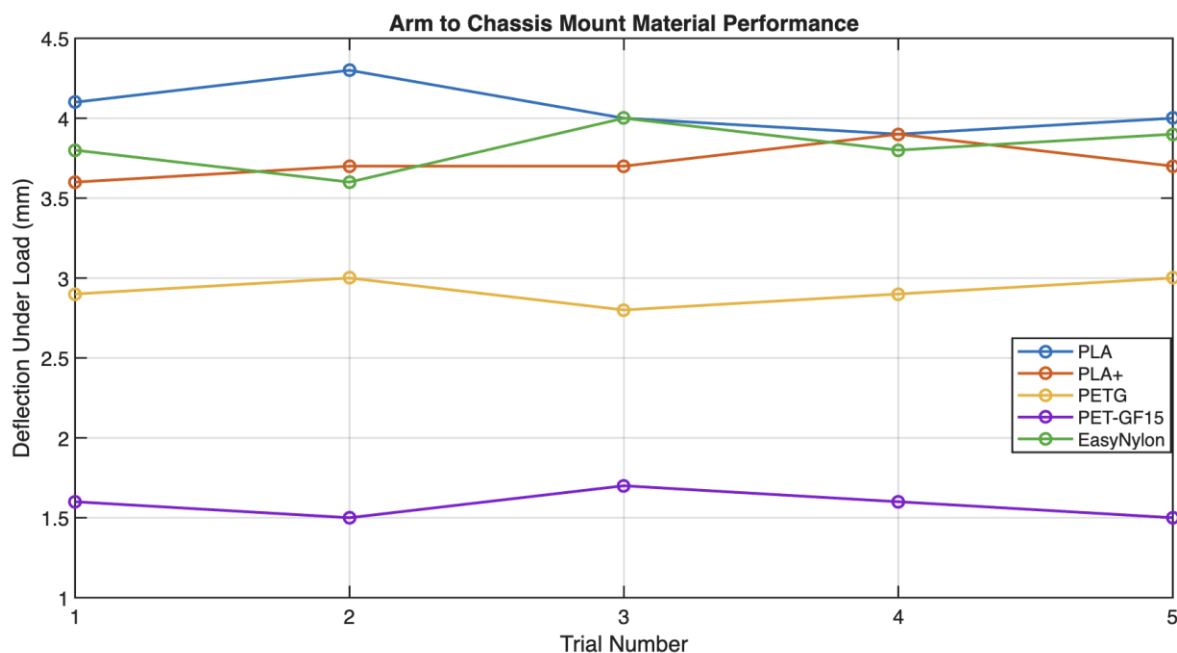


Figure X: Demonstration of how deflection was measured after each trial. Deflection was measured in millimeters from the datum in the figure above.

11.3.18.2: PLOTTED DEFLECTION UNDER LOAD/ SLIPPAGE ARM TO CHASSIS MOUNT MATERIAL



11.3.18.3: INTERPRETATION OF ARM TO CHASSIS MOUNT MATERIAL DATA

The illustrated graph shows that PET-GF15 has the lowest displacement under load across trials, indicating it performs best under applied forces. PLA and PLA+ show higher displacement due to

reduced stiffness, PETG is less rigid, and EasyNylon allows more deformation under load. This confirms PET-GF15 as the most suitable choice for the arm to chassis mount.

11.3.19: ARM MATERIAL TESTING

11.3.19.1: TESTING METHOD OF ARM MATERIAL

To evaluate how different materials performed under load, each arm was printed and mounted to the chassis under identical conditions. A 0.5 kg load was applied at the end of the arm to simulate payload forces. Deflection was measured at the point of maximum displacement relative to the initial unloaded position and recorded in millimeters (mm). The drivetrain motor operated at approximately 6000 rpm (3S / 11.1V) during testing. Each material was tested for 5 trials, with the load applied for 10 seconds per trial. The maximum deflection in each trial was recorded. Because the load, motor speed, and test duration were kept constant, lower deflection values indicate better strength and resistance to bending under load.

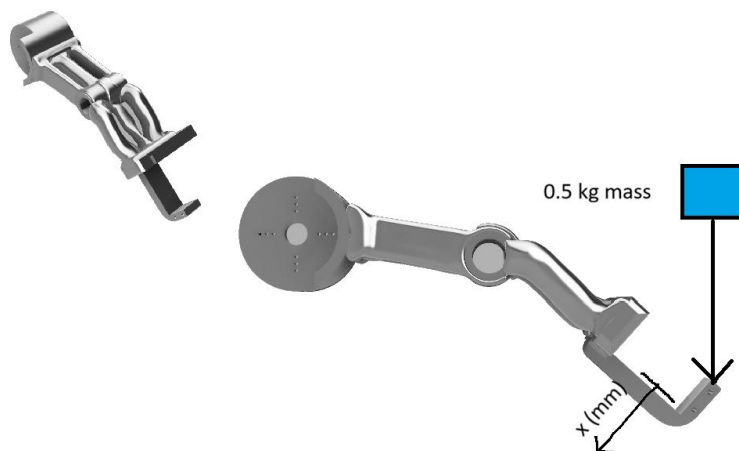
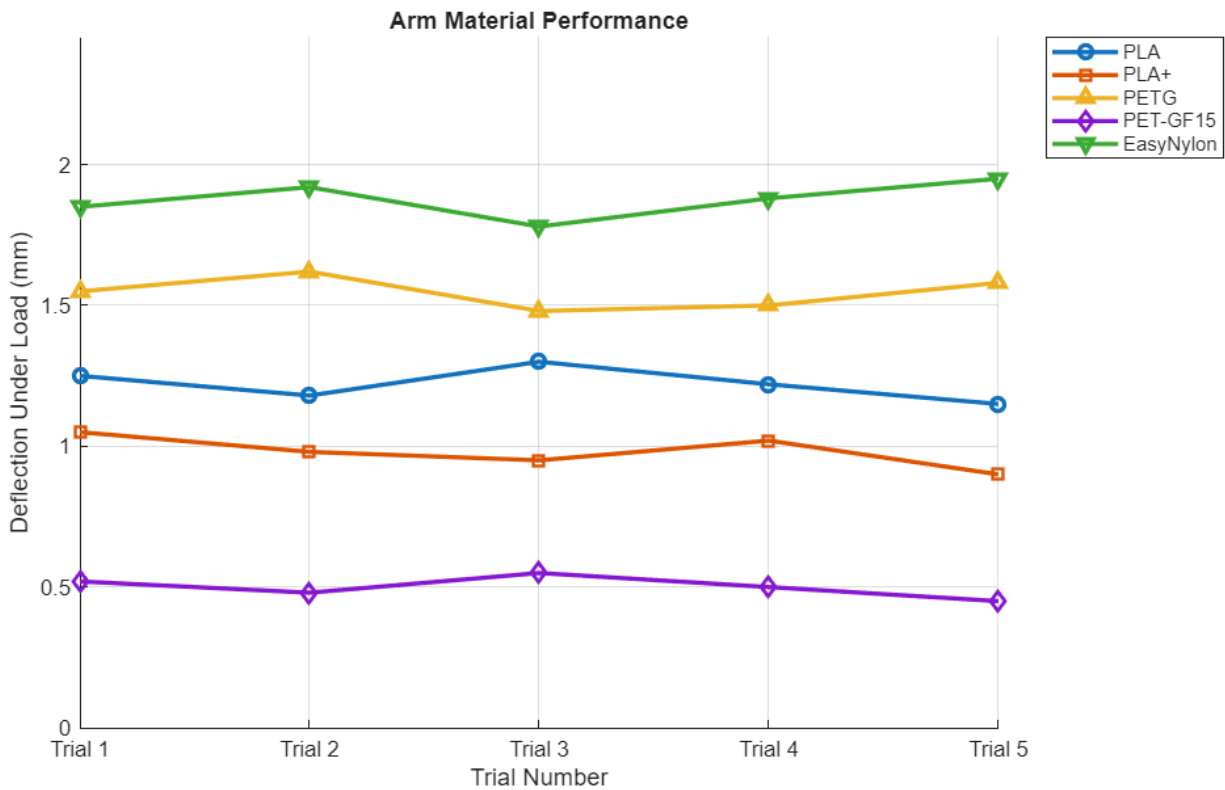


Figure X: Demonstration of how deflection was measured on the arm after each trial. Deflection was measured from the datum above.

11.3.19.2: PLOTTED DEFLECTION UNDER LOAD/ SLIPPAGE ARM MATERIAL



The illustrated graph shows that PET-GF15 has the lowest deflection under load across trials, meaning it resists bending and supports payload forces the best.

11.3.20: INVENTORIED STRUCTURAL FILAMENT PROPERTIES DATA TABLE

The choice of filament plays a key role in the strength and durability of components such as the chassis, brackets, and gears. Materials were compared using properties such as tensile strength, elongation at break (EAB), density, cost, and impact resistance, along with how they performed during testing.

Filament Brand	Tensile Strength (MPa)	EAB (A%)	Durability	Density (g/cm ³)	Price (\$/kg)	Impact Resistance
FIBERON PET-GF15	70.86	2.56	HIGH	1.38	45	HIGH
MatterHackersBendLay	33	100	VERY HIGH	1.18	90	VERY HIGH
Yxpolyer Flex PLA Pro	42	30	MEDIUM HIGH	1.20	30	MEDIUM
TPU	45	500	EXTREMELY HIGH	1.20	30	EXTREMELY HIGH
OW3D PLA+WOOD Cherry wood	40	8	LOW-MEDIUM	1.20	30	LOW

OW3D WOOD PLA Walnut wood	36	8	LOW-MEDIUM	1.20	30	LOW
OW3D WOOD PLA Teakwood	36	8	LOW-MEDIUM	1.20	30	LOW
OW3D PLA+WOOD Oakwood	36	8	LOW-MEDIUM	1.20	30	LOW
Copper MTL	45	3	MEDIUM	3.50	55	LOW
NINJATEK NINJAFLEX TPU	26	660	EXTREMELY HIGH	1.19	100	EXTREMELY HIGH
HI-SPEED/RAPID PLA	60	6	MEDIUM	1.24	20	LOW
ELEGOO RAPID PLA+	65	10	MEDIUM-HIGH	1.24	20	MEDIUM
ELEGOO PLA+	60	10	MEDIUM	1.24	20	MEDIUM
ELEGOO PLA MATTE	50	6	MEDIUM-LOW	1.24	20	LOW
EasyNylon	50	80	HIGH	1.14	60	HIGH

Based on both the material properties and experimental results, PET-GF15 was selected as the primary structural filament because it maintained its shape under load and showed the most consistent performance. While materials such as nylon offer higher flexibility, they were not selected due to cost and because they introduced more deformation during testing. PLA+ was used for secondary components where full structural stiffness was not required but reliable performance was still needed.

11.3.21: FINAL MATERIAL SELECTION

Rover component	Final material selection
Tread links	PLA+
Link connector pins	PET-GF15
Battery holder	PET-GF15
Driving sprocket	PET-GF15
Driving sprocket insert	PET-GF15
Snap fit bearings	PLA+
Outer sandwich tread chassis brackets	TPU95A
Inner sandwich tread chassis brackets	TPU95A
Top electronics container shelf	PET-GF15
Middle electronics container shelf	PET-GF15
Bottom electronics container shelf	PET-GF15

Scoop	PET-GF15
Arm	PET-GF15
Rigid horizontal chassis crossbar	PET-GF15
Rigid diagonal chassis crossbar	PET-GF15
Chassis holding pins	PET-GF15
Arm to chassis mount	PET-GF15
Camera gimbal base	TPU95A
Camera gimbal top part	TPU95A

11.4.1: CROSS-MEMBER TESTING

11.4.1.1: HORIZONTAL CROSS-MEMBER TESTING

Each cross-member was installed horizontally in the chassis under identical mounting conditions. A 0.5 kg mass was placed directly above the center of the cross-member to apply a static, constant compression force due to the outer weight. The load was maintained for 10 seconds during each trial.

Deflection was measured at the center of the cross-member, where maximum displacement occurred, relative to the initial unloaded position and recorded in millimeters (mm). 5 trials were conducted for consistency.

Load (kg)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average deflection (mm)
0.5	1.82	1.76	1.89	1.81	1.84	1.82



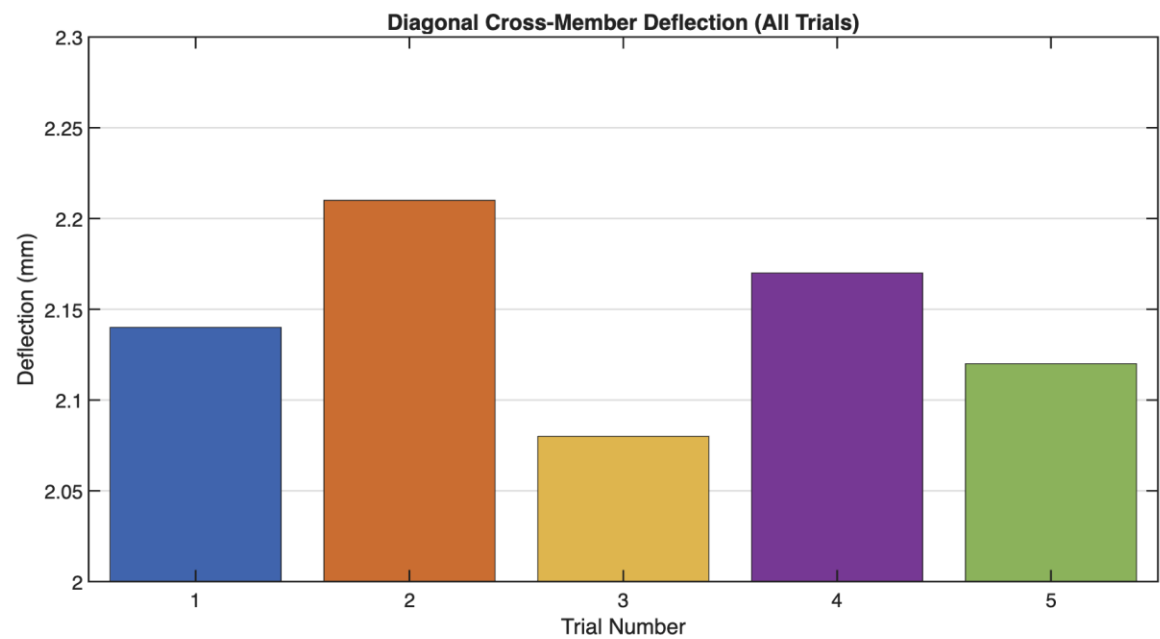
Deflection ranged from 1.76 mm to 1.89 mm, with an average of 1.82 mm under a 0.5 kg load. This indicates consistent stiffness under compression.

11.4.1.2: DIAGONAL CROSS-MEMBER TESTING

Each cross-member was installed at a 45° orientation within the chassis under identical mounting conditions. A 0.5 kg mass was placed at a fixed point on the chassis above the diagonal member so that the load was transferred through the cross-member, creating both axial compression and shear at the bolt connection. The load was maintained for 10 seconds during each trial.

Deflection was measured at the point of maximum displacement relative to the initial unloaded position and recorded in millimeters (mm). 5 trials were conducted for consistency.

Load (kg)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average deflection (mm)
0.5	2.14	2.21	2.08	2.17	2.12	2.14



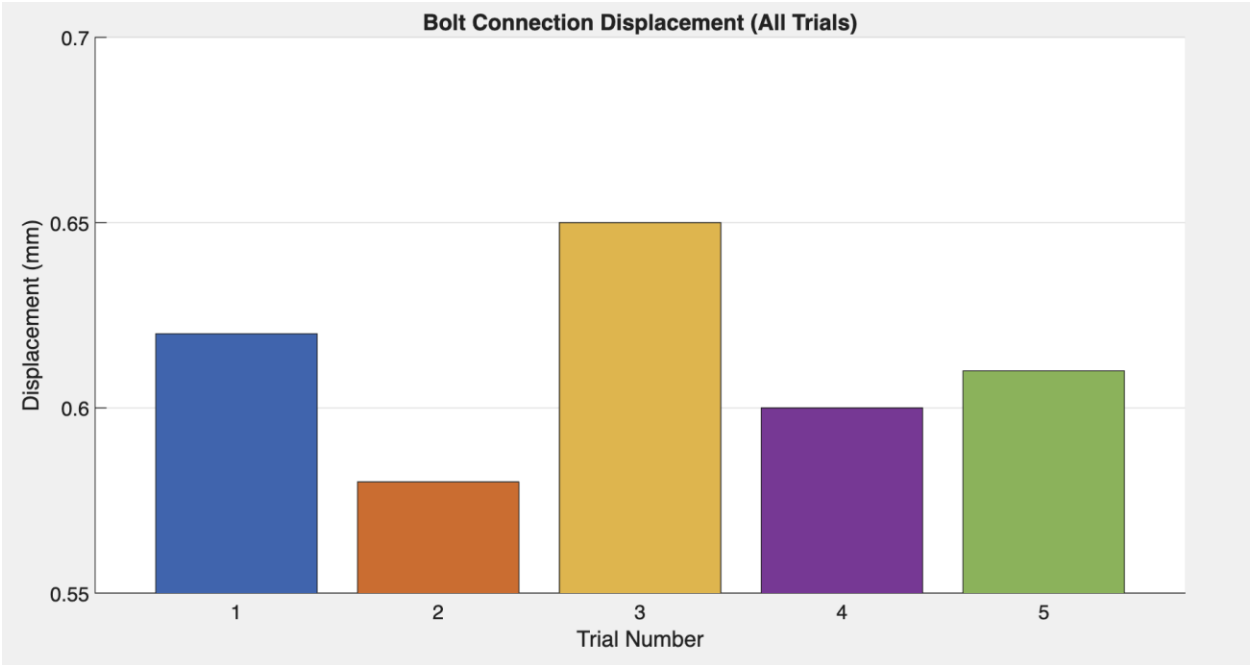
Deflection ranged from 2.08 mm to 2.21 mm, with an average of 2.14 mm under a 0.5 kg load. This indicates consistent performance under diagonal loading.

11.4.1.3: BOLT CONNECTION TESTING

The cross-member was assembled into the chassis using a bolt connection under identical mounting conditions. A 0.5 kg load was applied at a fixed location to transfer force through the member and generate shear at the bolt. The load was maintained for 10 seconds during each trial.

Displacement at the connection point was measured relative to the initial unloaded position, and any relative movement or loosening at the joint was recorded. 5 trials were conducted for consistency.

Load (kg)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average deflection (mm)
0.5	0.62	0.58	0.65	0.60	0.61	0.61



Displacement ranged from 0.58 mm to 0.65 mm, with an average of 0.61 mm under a 0.5 kg load. This indicates the bolt connection maintained stable and consistent performance under shear loading.

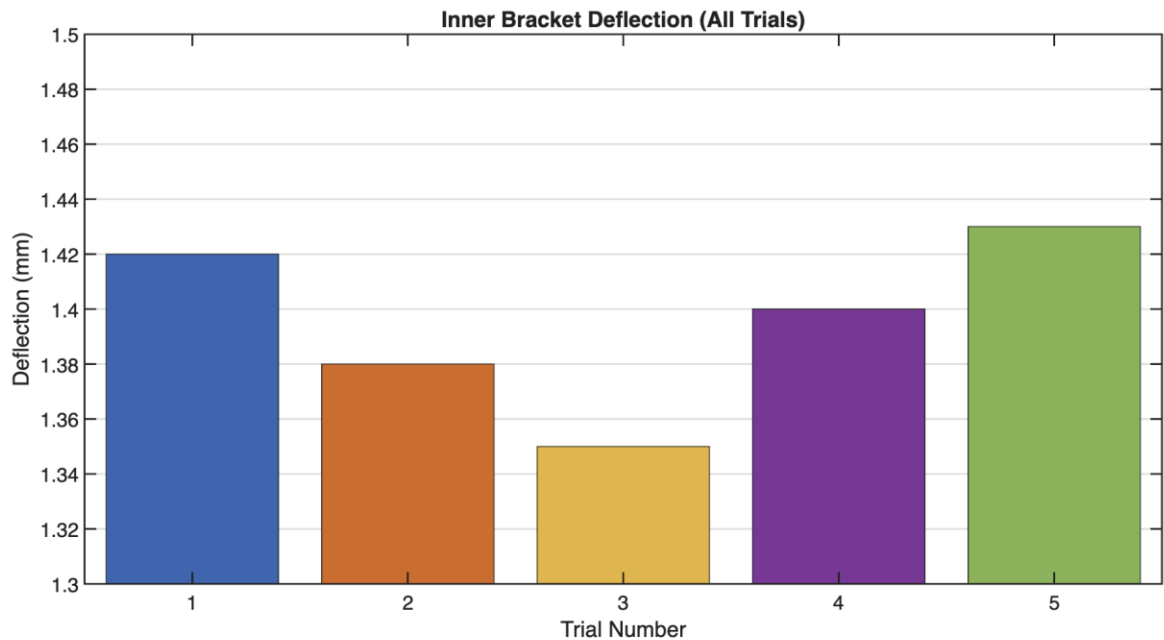
11.4.2: BRACKET TESTING

11.4.2.1: INNER BRACKET TESTING

Inner brackets were tested to examine their effect on chassis stability under load. Each bracket was assembled into the chassis using identical mounting conditions. A vertical load of 0.5 kg was applied at a fixed location above the bracket connection for 10 seconds. Deflection at the connection point was measured in millimeters (mm) relative to the initial unloaded position. 5 trials were conducted for each configuration.

Deflection reflects the bracket’s resistance to deformation within the chassis. Lower deflection indicates greater stiffness and more effective load transfer, while higher deflection indicates reduced structural stability at the joint. This allows direct comparison of bracket performance under consistent loading conditions.

Load (kg)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average deflection (mm)
0.5	1.42	1.38	1.35	1.40	1.43	1.42



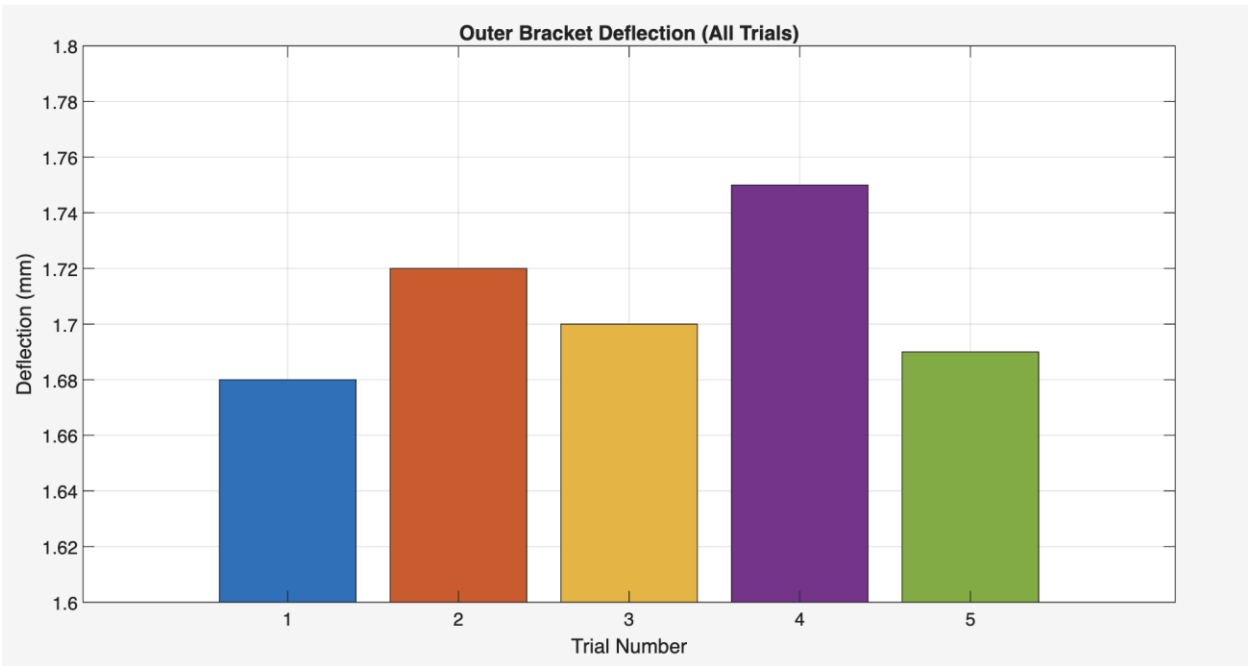
Deflection remained consistent across all trials, with values between 1.38 mm and 1.45 mm and an average of 1.42 mm, as shown by the closely grouped bars. No trial showed a significant increase in deflection or any signs of structural failure. This indicates that the inner bracket maintained stable performance under the applied load and provided consistent support within the chassis across repeated testing.

11.4.2.2: OUTER BRACKET TESTING

Outer brackets were tested to examine their effect on chassis stability under load. Each bracket was assembled into the chassis at its outer mounting location and secured under identical conditions. A vertical load of 0.5 kg was applied near the bracket connection for 10 seconds to simulate forces acting on the outer frame during operation. Deflection at the connection point was measured in millimeters (mm) relative to the initial unloaded position. 5 trials were conducted for the same configuration.

Deflection reflects the bracket’s resistance to deformation underload. Brackets that showed lower deflection maintained structural support more effectively, while higher deflection indicates reduced stiffness at the outer connection. This allows direct comparison of how the outer bracket performs in maintaining chassis stability.

Load (kg)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average deflection (mm)
0.5	1.68	1.72	1.70	1.75	1.69	1.71



Deflection was higher for the outer bracket, with values ranging from 1.68 mm to 1.75 mm and an average of 1.71 mm, as shown by the consistently higher bars across all trials. The results remained closely grouped, indicating stable performance with no sudden increases in deformation.

This indicates that the outer bracket maintained consistent structural support under the applied load, but experienced greater deflection compared to the inner bracket, suggesting lower stiffness at the outer connection.

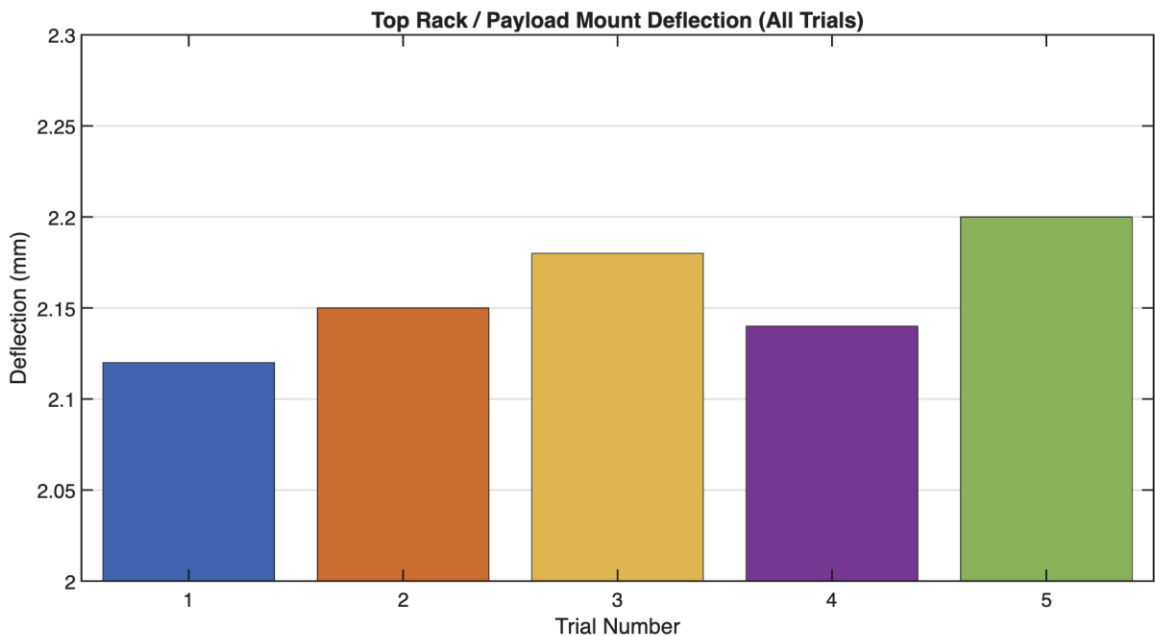
11.4.3: CHASSIS RACK TESTING

11.4.3.1: TOP RACK/PAYLOAD MOUNT TESTING

The top rack (payload mount) was tested to examine its ability to support external loads without excessive bending. The rack was assembled onto the chassis under standard mounting conditions. A vertical load of 1.0 kg was applied at the center of the rack for 10 seconds to simulate payload weight during operation.

Displacement was determined by marking the initial unloaded position of the rack and measuring the vertical change in position after the load was applied, using a fixed reference point on the chassis. The difference between the initial and loaded positions was recorded in millimeters (mm). 5 trials were conducted under identical conditions. Lower displacement values indicate greater stiffness and more effective load support, allowing direct comparison of payload mount performance across trials.

Load (kg)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average deflection (mm)
1.0	2.12	2.15	2.18	2.14	2.20	2.16



Deflection was highest for the top rank, with values ranging from 2.12 mm to 2.20 mm and an average of 2.16 mm, as shown by the consistently higher bars across all trials. The results remained closely grouped, indicating stable performance with no sudden increases in deformation.

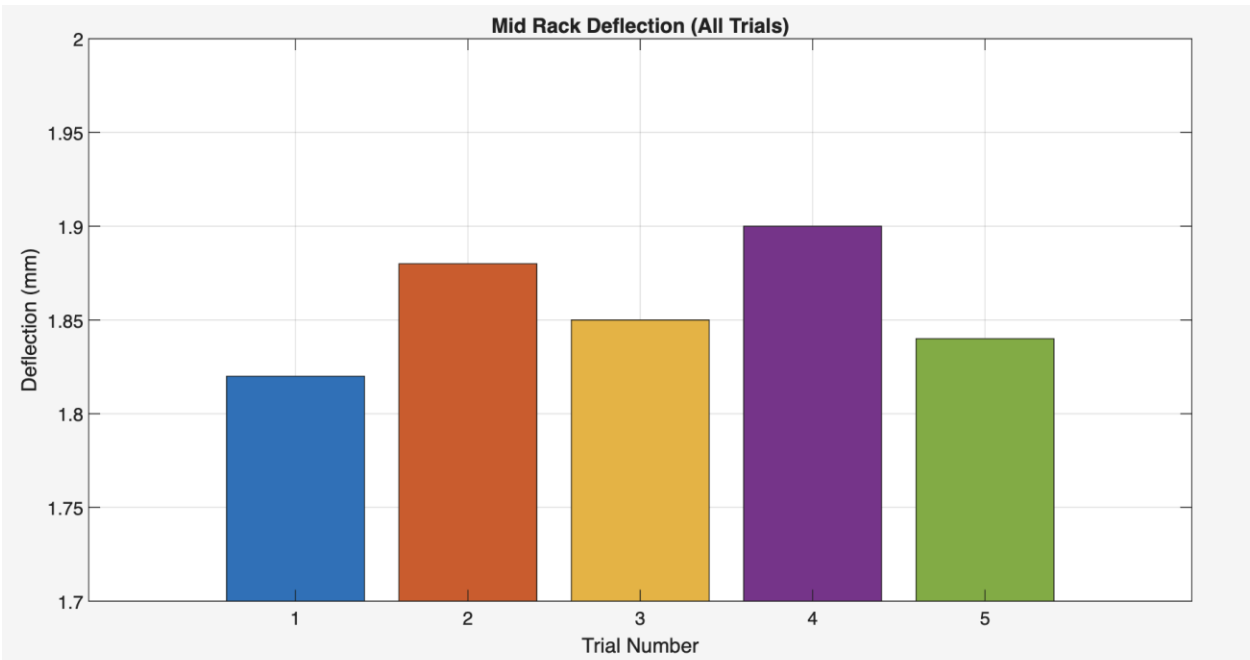
This indicates that the top rank maintained stable performance under the applied payload, despite experiencing greater deflection compared to the other rack components due to its larger span and direct load exposure. Since the deflection remained consistent and within an acceptable range, the design is sufficient for its intended function and does not require modification.

11.4.3.2: MID RACK TESTING

The mid rack was tested to examine its ability to support internal loads within the chassis. The rack was assembled in its standard position and secured under identical mounting conditions. A vertical load of 0.75 kg was applied at the center of the rack for 10 seconds to simulate distributed internal loading.

Displacement was determined by marking the initial unloaded position of the rack and measuring the vertical change in position after the load was applied, using a fixed reference point on the chassis. The difference between the initial and loaded positions was recorded in millimeters (mm). 5 trials were conducted under identical conditions. Lower displacement values indicate greater stiffness and more effective load support, allowing direct comparison of mid rack performance across trials.

Load (kg)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average deflection (mm)
0.75	1.82	1.88	1.85	1.90	1.84	1.86



Deflection for the mid rack remained between 1.82 mm and 1.90 mm, with an average of 1.86 mm, as shown by the consistently grouped bars across all trials. The results showed minimal variation, indicating stable behavior under repeated loading

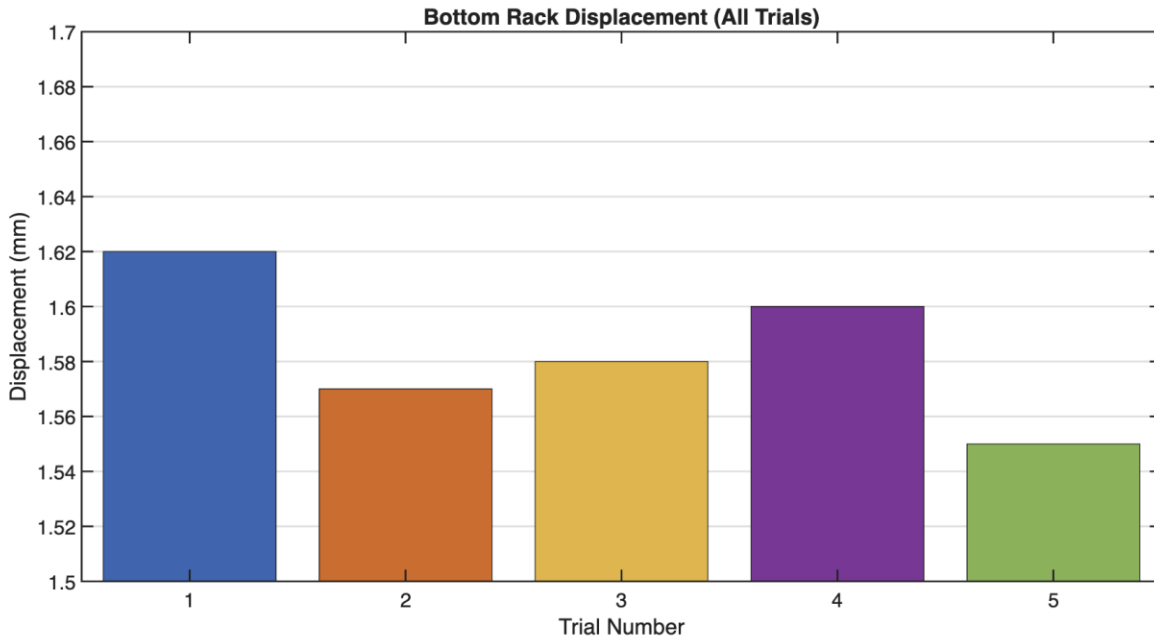
This indicates that the mid rack maintained reliable structural support under the applied load, with moderate deflection compared to the top and bottom racks. Since no excessive deformation or instability was observed, the current design is acceptable and does not require modification.

11.4.3.3: BOTTOM RACK TESTING

The bottom rack was tested to examine its ability to support loads in direct contact with the ground-facing structure of the chassis. The rack was assembled in its standard position and secured under identical mounting conditions. A vertical load of 0.75 kg was applied at the center of the bottom rack for 10 seconds to simulate internal loading conditions during operation.

Displacement was determined by marking the initial unloaded position of the rack and measuring the vertical change in position after the load was applied, using a fixed reference point on the chassis. The difference between the initial and loaded positions was recorded in millimeters (mm). 5 trials were conducted under identical conditions. Lower displacement values indicate greater stiffness and more effective load support, allowing direct comparison of bottom rack performance across trials.

Load (kg)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average deflection (mm)
0.75	1.62	1.57	1.58	1.60	1.55	1.58



Displacement was lowest for the bottom rack, with values ranging from 1.55 mm to 1.62 mm and an average of 1.58 mm, as shown by the consistently lower bars across all trials. The results remained closely grouped, indicating stable performance with no sudden increases in deformation.

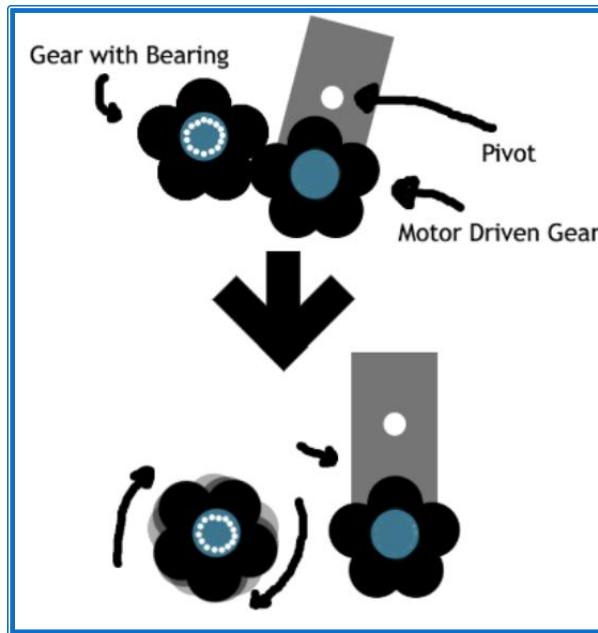
This indicates that the bottom rack maintained the highest stiffness among the rack components and provided reliable support under the applied load. Since displacement remained low and consistent across all trials, no structural issues were observed, and the current design is sufficient for its intended function without requiring modification.

11.5: MOVEMENT TESTING RESULTS

11.5.1: BEARING TESTING

Testing our custom designed bearings is extremely important for our rover treads to operate smoothly.

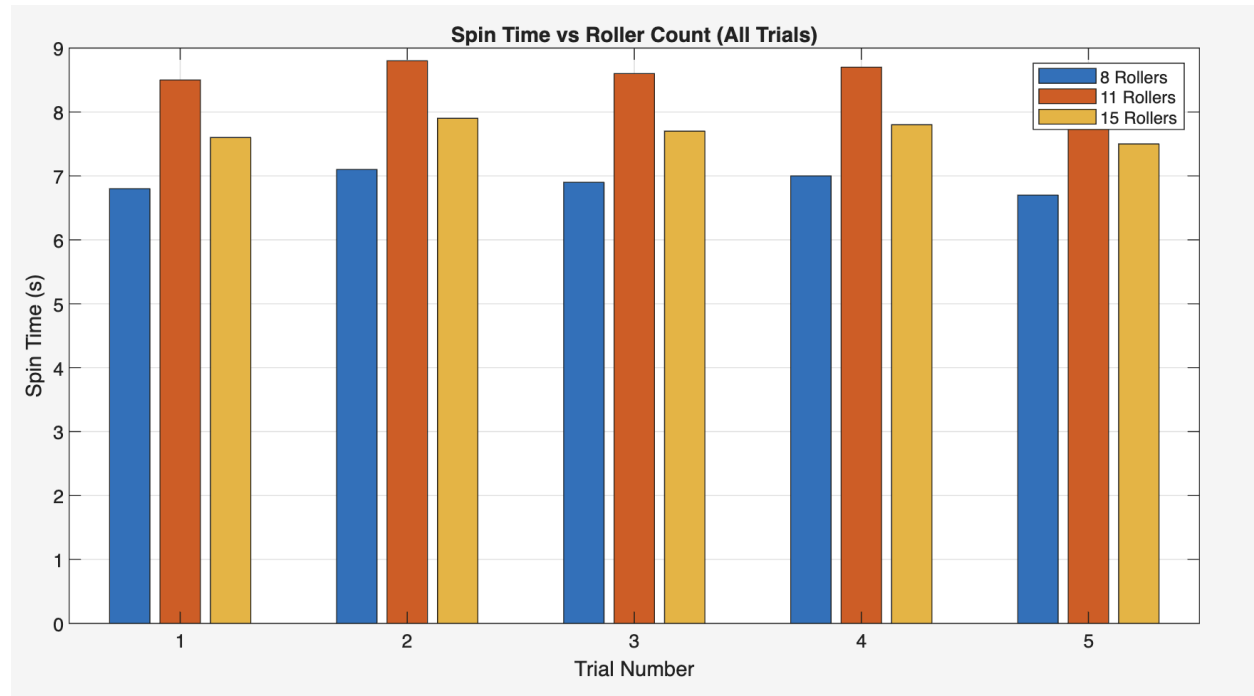
11.5.1.1: BEARING ROLLER COUNT TESTING



Bearings with varying roller counts were tested to examine their effect on rotational efficiency. Rig illustrated on the right. Each bearing was mounted on a fixed shaft and accelerated using a motor-driven setup (6000 rpm, 3S / 11.1 V nominal, 30 A) for 5 seconds to reach a consistent speed. The input was then removed, and the free spin time until rest was measured. 5 trials were conducted for each configuration. Bearings were lubed beforehand with WD-40.

Spin time reflects the rate of energy dissipation within the bearing system. Bearings that maintained rotation for longer durations experienced lower internal friction, while shorter spin times indicate greater resistance caused by contact interactions between rolling elements and races. This allows direct comparison of how roller count influences rotational losses.

Roller Count	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average spin time (s)
8	6.8	7.1	6.9	7.0	6.7	6.9
11	8.5	8.8	8.6	8.7	8.4	8.6
15	7.6	7.9	7.7	7.8	7.5	7.7



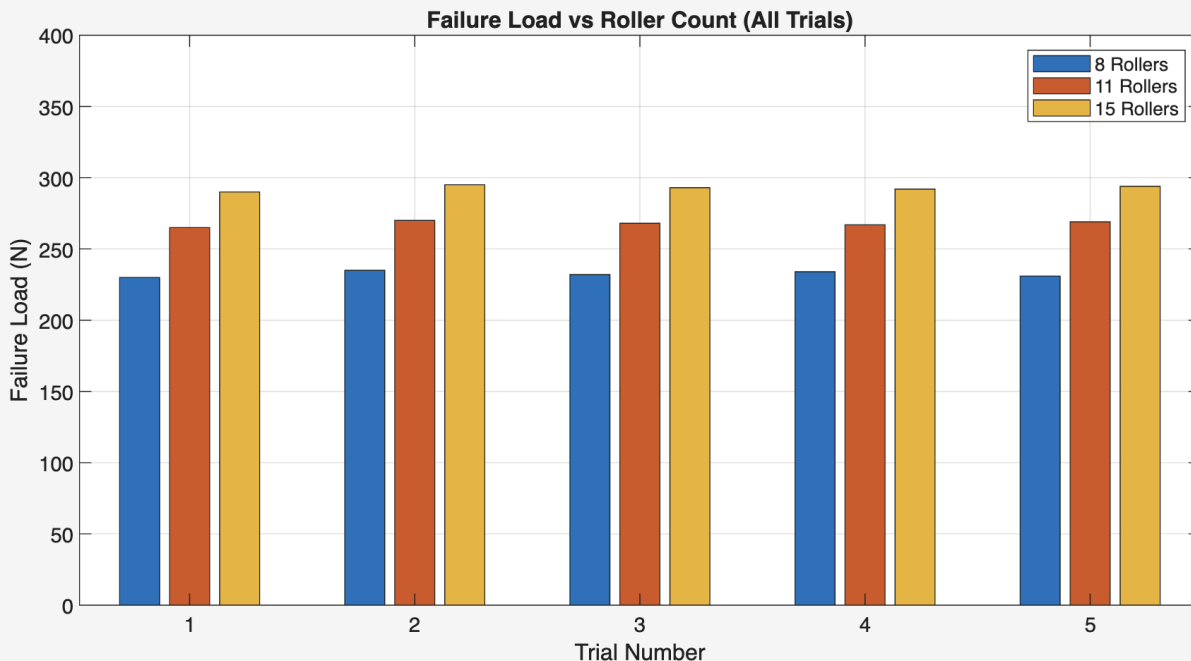
Spin time was highest for the bearing with 11 rollers, as shown by the tallest bars across the trials and in the average values. The 8-roller configuration consistently produced lower spin times, while the 15-roller configuration showed slightly reduced performance compared to 11 rollers. This indicates that the 11-roller bearing achieved the best rotational efficiency among the tested configurations.

11.5.1.2: BEARING LOAD TESTING (SNAPPING/X AXIS)

Bearings were tested under axial loading to determine how roller count affects structural performance. A load was applied along the x-axis (parallel to the axle) by incrementally adding known masses, with force calculated using $F=mg$ ($g = 9.81 \text{ m/s}^2$). Loading was increased until failure occurred, defined as cracking, permanent deformation, or separation of components. 5 trials were conducted for each roller count configuration.

Failure load represents the maximum axial force sustained before structural failure. Differences in failure to load reflect changes in load distribution and contact geometry caused by roller count, which directly influence axial strength.

Roller Count	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average failure load (N)
8	230	235	232	234	231	232.4
11	265	270	268	267	269	267.8
15	290	295	293	292	294	292.8



Failure load was highest for the 15-roller bearing across all trials, as shown by the tallest bars in the graph. The 11-roller configuration showed moderate performance, while the 8-roller bearing consistently had the lowest failure loads. This indicates that increasing roller count improves axial load distribution and overall strength. As a result, the 15-roller bearing provided the best resistance to failure among the tested configurations.

11.5.2: TREAD TESTING RESULTS

11.5.3: MOTOR TESTING RESULTS

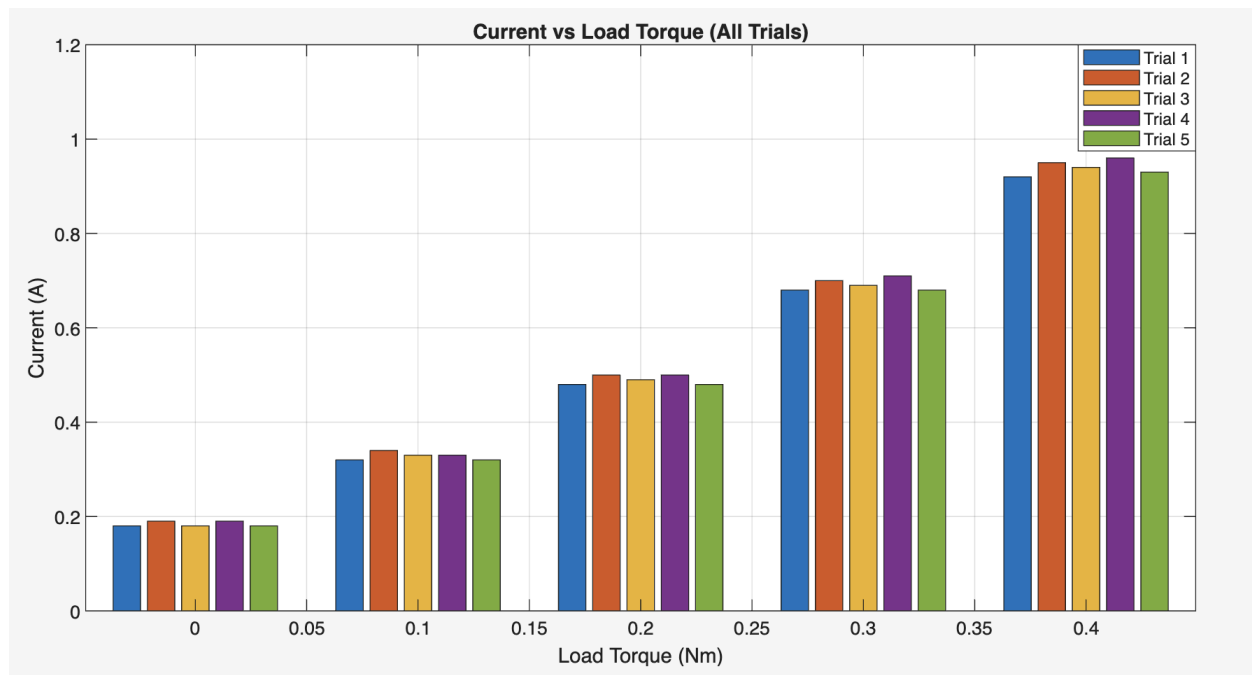
MOTORS USED: Greartisan DC 12V 100RPM Gear Motor High Torque Electric Micro Speed Reduction Geared Motor Centric Output Shaft 37mm Diameter Gearbox

11.5.3.1: CURRENT VERSUS LOAD TORQUE

The motor was tested to examine how the current draw varies with applied load torque. The motor was operated at 12 V under no load and loaded conditions. Load torque was applied by pressing the output shaft against a fixed surface to introduce controlled friction, increasing resistance incrementally to simulate different loading conditions.

At each load level, the corresponding current was measured using the power supply readout once the motor reached steady operation. 5 trials were conducted and averaged for each load condition. Higher load torque required greater current to maintain motion, allowing direct comparison of motor performance under increasing load.

Load Torque (Nm)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Current (A)
0.0	0.18	0.19	0.18	0.19	0.18	0.18
0.1	0.32	0.34	0.33	0.33	0.32	0.33
0.2	0.48	0.50	0.49	0.50	0.48	0.49
0.3	0.68	0.70	0.69	0.71	0.68	0.69
0.4	0.92	0.95	0.94	0.96	0.93	0.94



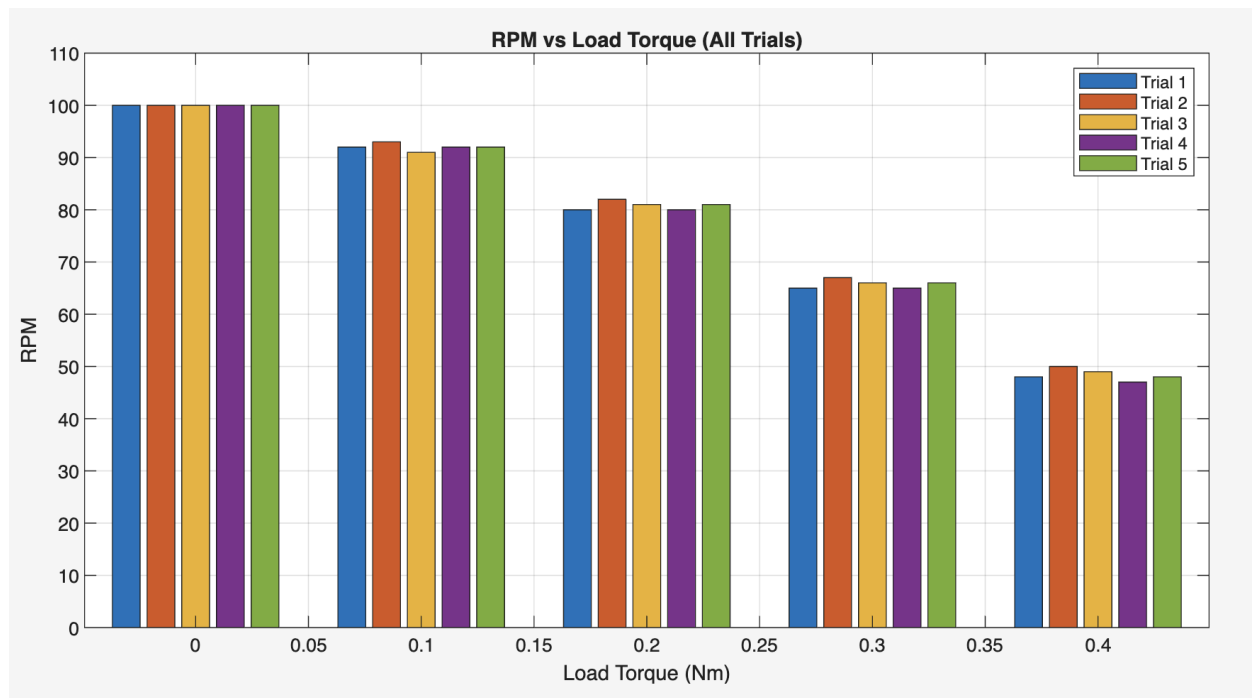
The highest current was observed at 0.4 Nm with values around 0.92-0.96 A, while the lowest current occurred at 0 Nm with values around 0.18-0.19 A. This shows that increasing load torque results in higher current draw. The motor maintained consistent performance across all trials and is suitable for operation under the tested loading conditions.

11.5.3.3: POWER CONSUMPTION VS LOAD

The relationship between rotational speed and applied load torque was tested under operating conditions. The motor was powered at 12 V, with a maximum no-load speed of 100 RPM. Load torque was applied by pressing the output shaft against a fixed surface to introduce friction, increasing resistance incrementally to simulate different loading conditions.

At each load level, RPM was measured by counting the number of shaft rotations over a 10-second interval and converting to revolutions per minute (RPM). The motor was allowed to reach steady operation before measurements were taken. 5 trials were conducted for each load condition, and consistent measurement intervals were used across all tests. Lower RPM values indicate greater resistance to rotation under increasing load.

Load Torque (Nm)	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Average Current (A)
0.0	100	100	100	100	100	100
0.1	92	93	91	92	92	92
0.2	80	82	81	80	81	81
0.3	65	67	66	65	66	66
0.4	48	50	49	47	48	48



The highest RPM was observed at 0 Nm at 100 RPM, while the lowest RPM occurred at 0.4 Nm with values around 48-50 RPM. This shows that increasing torque load reduces motor speed due to increased resistance. The motor maintained stable performance across all trials and is suitable for the intended application.

11.6: PAYLOAD TESTING RESULTS

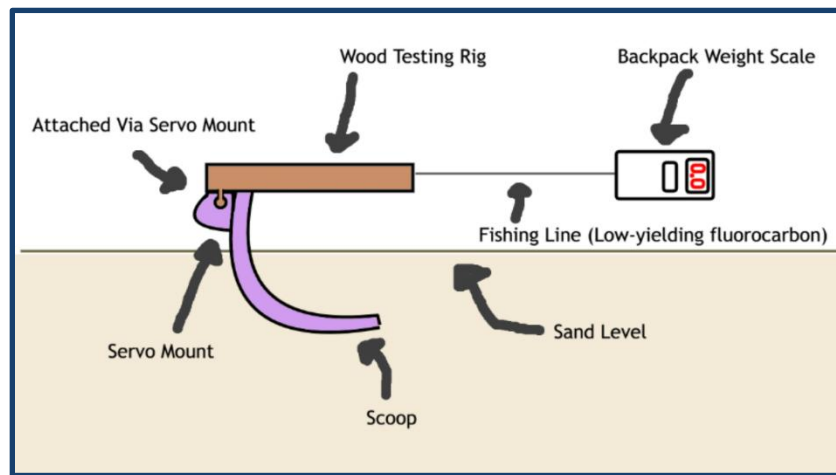
Payload testing allows us to

11.6.1: SCOOP TESTING

We needed a way to standard the test for scoop and what elements we would be testing. We decided to focus on one main element of testing. The force required to move the scoop through sand given the kinetic friction. We assumed this would give us the most information about the scoop given that the biggest obstacle would be pushing through the sand.

11.6.1.1: SCOOP TESTING METHODS

The way in which we tested the forces acting to push the scoop through sand was by first printing the scoop, assembling a jig with wood, attaching fishing line to the bottom of the sieve and end of the wooden jig, then attaching a backpack weight scale used to measure backpacks for backpacking; where pulling it through sand equal roughly equal level each time. We did ten repetitions per scoop and compared the kinetic friction force (the force after three seconds of pulling) with the surface area exposed to sand. We decided on the kinetic friction because once the scoop was moving, we assumed we could treat the sand as a viscous liquid.

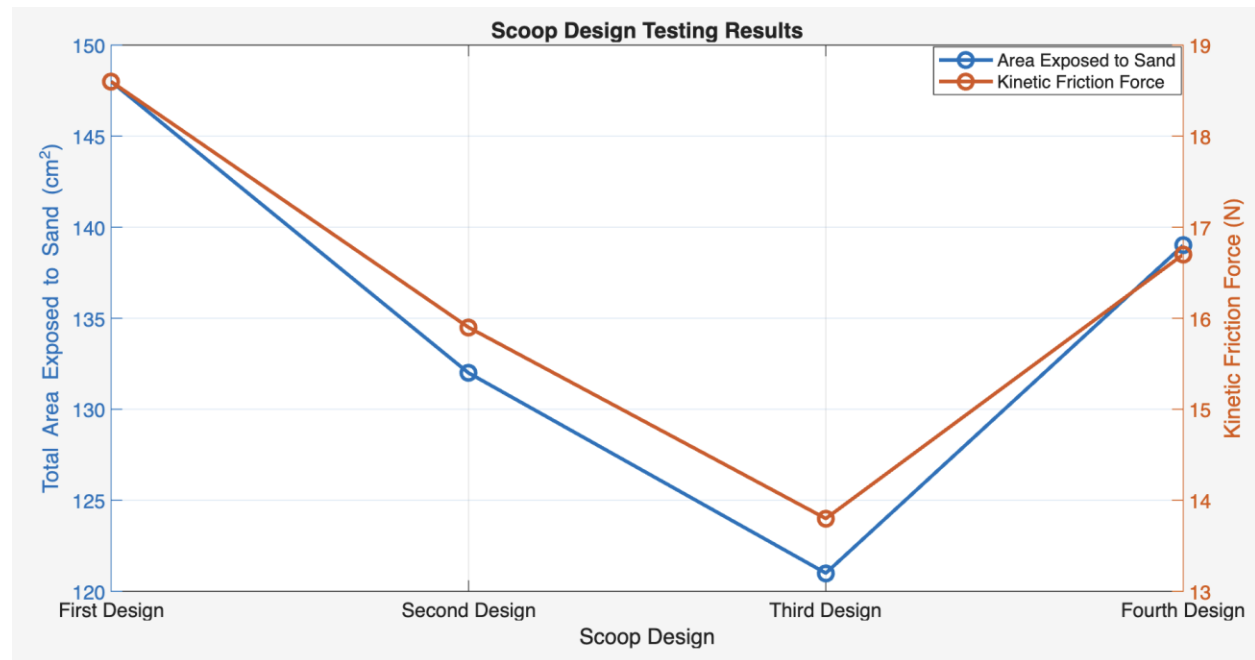


The total surface area exposed to sand was obtained directly from **Fusion 360**, where the software calculated the full surface area of the scoop geometry. The value was recorded in cm^2 for each design without additional calculations.

The kinetic friction force was measured directly using the digital scale attached to the scoop. As the scoop was pulled through the sand, the scale recorded the force required to maintain motion. The reading stabilized after approximately 3 seconds, and this steady value was recorded as the kinetic friction force for each trial.

11.6.1.2: SCOOP TESTING DATA

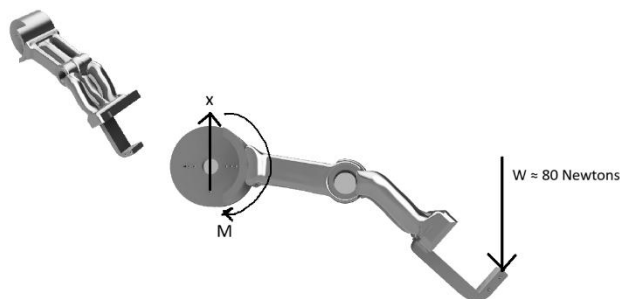
Design	Total Area Exposed to Sand (cm^2)	Kinetic Force of Friction (N)
First Design	148	18.6
Second Design	132	15.9
Third Design	121	13.8
Fourth Design	139	16.7



11.6.1.3: SCOOP INTERPRETATION

The third design produced the lowest kinetic friction force at 13.8 N with an exposed area of 121 cm², while the first design showed the highest force at 18.6 N with 148 cm². This shows that lower exposed area results in lower resistance. The third design was selected as the most efficient scoop configuration.

11.6.2: ARM TESTING



The test of the arm design was a test to make sure it could handle the shear stress from loading. We sandwiched the arm in-between a vice grip and used the backpack weight to pull down on it. We measured a minimum of 5lbs which is far more than we will experience so we do not have to worry about the arm.

11.7: ELECTRONICS TESTING RESULTS

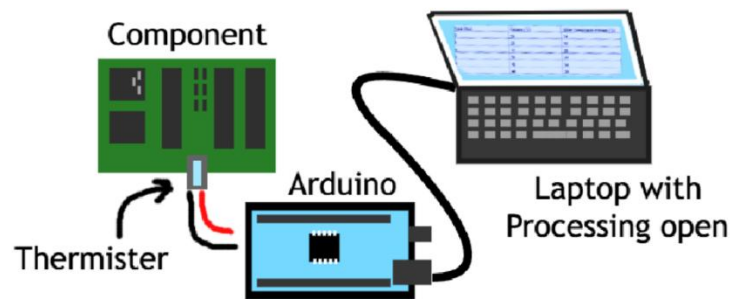
11.7.1: ELECTRONICS TEMPERATURE TESTING

Measuring the temperature of our electronics at max load was imperative towards thermal control and stability of our rover. This genre of information is incredibly useful towards component selection (Does this part overheat easily? Do we need cooling?).

Our first plan for electronics was to add all the components to the assembly and perform finite element analysis (FEA) thermal component simulation in Fusion. Unfortunately, that feature is unavailable in Fusion without a \$150/yr subscription, so we are pivoted towards manual testing instead.

11.7.1.1: ELECTRONIC TEMPERATURE TESTING REASONING

As many components are expected to be stable at competition levels, we decided to simplify this section. Electronics would be tested based on IF they needed cooling, and if they did, we would continue with HOW much cooling?



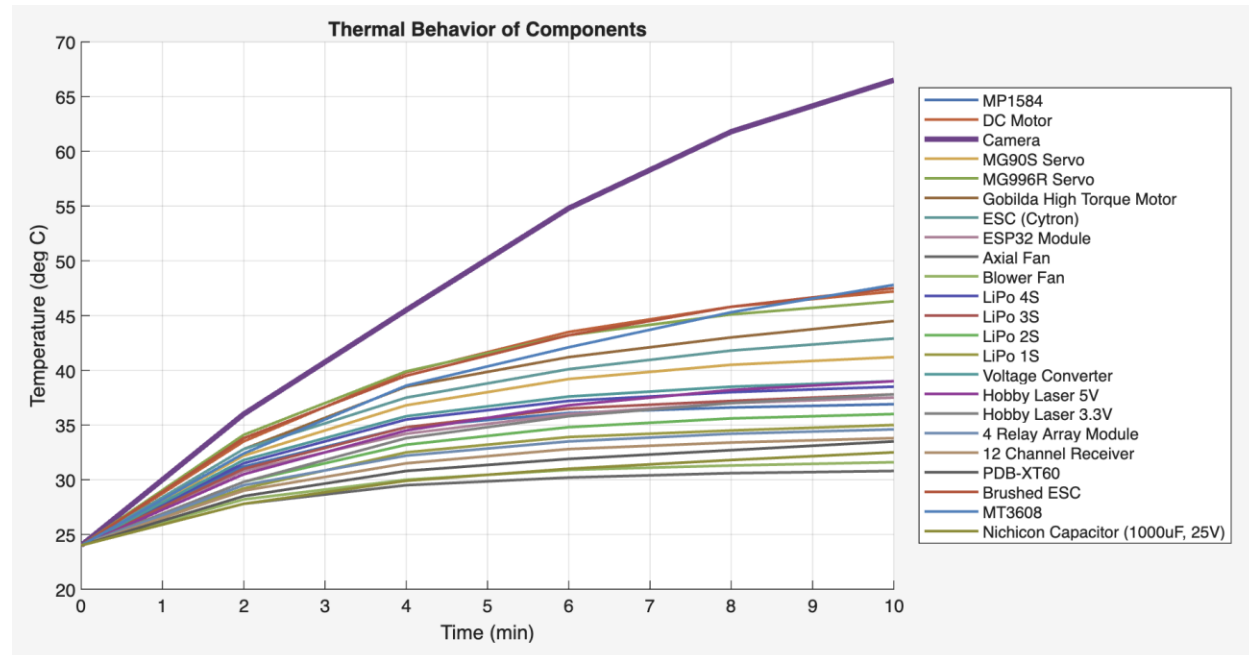
For this test we took the temperature of each component over the course of 10 minutes using a thermistor we scrapped from 3D printer parts and an Arduino connected to Processing (we made our own data gathering app). Hence, we could easily extract and fabricate a large data table from that information.

11.7.1.2: ELECTRONIC TEMPERATURE TESTING DATA

Component Time (min) vs Temperature ($^{\circ}\text{C}$) under Operating Conditions

Component	0 min	2 min	4 min	6 min	8 min	10 min
MP1584	24.00	31.20	34.80	36.10	36.60	36.90
DC Motor	24.00	33.50	39.80	43.50	45.80	47.20
Camera	24.00	36.00	45.50	54.80	61.80	66.50
MG90S servo	24.00	32.20	36.80	39.20	40.50	41.20
MG996r servo	24.00	34.10	39.90	43.20	45.10	46.30
Gobilda high torque 300*, Model #: 2000-0025-000	24.00	35.00	42.50	46.80	49.20	50.50

ESC (Smart Drive Duo 30, Company: Cytron)	24.00	32.80	37.50	40.10	41.80	42.90
ESP32 Module	24.00	30.80	34.20	36.00	37.00	37.50
Axial fan	24.00	27.80	29.50	30.20	30.60	30.80
Blower fan	24.00	28.20	30.00	30.90	31.30	31.60
Lipo 4s	24.00	31.50	35.50	37.20	38.00	38.50
Lipo 3s	24.00	31.00	34.80	36.50	37.20	37.80
Lipo 2s	24.00	29.80	33.20	34.80	35.60	36.00
Lipo 1s	24.00	29.20	32.50	33.90	34.50	35.00
Large voltage Convertor, L6234PDPQWER_SO20	24.00	31.80	35.80	37.60	38.50	39.00
Hobby Lasers @ 5v	24.00	30.50	34.50	36.80	38.20	39.00
Hobby Lasers @ 3.3v	24.00	29.80	33.80	35.80	37.00	37.80
4 Relay Array Module	24.00	29.50	32.20	33.50	34.20	34.60
12 Channel Receiver	24.00	29.00	31.50	32.80	33.40	33.80
MT3608	24.00	32.40	38.60	42.10	45.30	47.80
Brushed ESC (110A, 3S-6S, 5V BEC)	24.00	33.80	39.50	43.20	45.80	47.50
PDB-XT60 Power Distribution Board (5V/12V Output)	24.00	28.50	30.80	31.90	32.70	33.50
Capacitor Nichicon (1000µF 25V)	24.00	27.80	29.90	31.00	31.80	32.50
Goblida 5203-Series Yellow Jacket Brushed DC Motor (6000 RPM @ 12VDC)	24.00	32.80	38.50	41.20	43.00	44.50



11.7.2.3: ELECTRONIC TEMPERATURE TESTING INTERPRETATION

The data clearly shows that the only component that required temperature control was the camera, as it kept exponentially rising in the ten minutes of testing.

11.7.2: CAMERA TRANSMITTER RANGE TESTING

11.7.2.1: CAMERA TRANSMITTER RANGE TESTING APPROACH

The camera transmitter range was evaluated by measuring the maximum distance at which a clear video signal could be maintained under different obstruction conditions. Testing was conducted in an open outdoor grassy field under consistent environmental conditions.

For each transmitter configuration, the system was powered on and the operator walked away from the receiver in a straight line until significant signal degradation or complete loss occurred. The distance at which the video feed became unusable was recorded as the maximum range.

This procedure was repeated for each obstruction condition, including no obstruction, behind a curtain, and behind a wall, to assess the impact of signal interference on transmission performance.

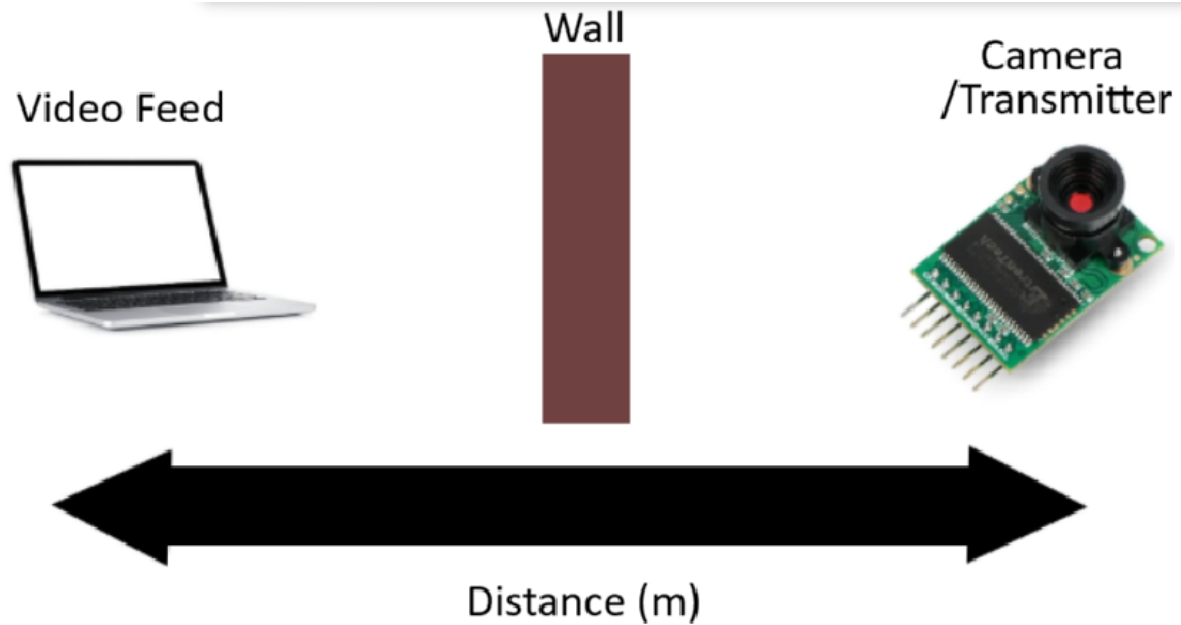
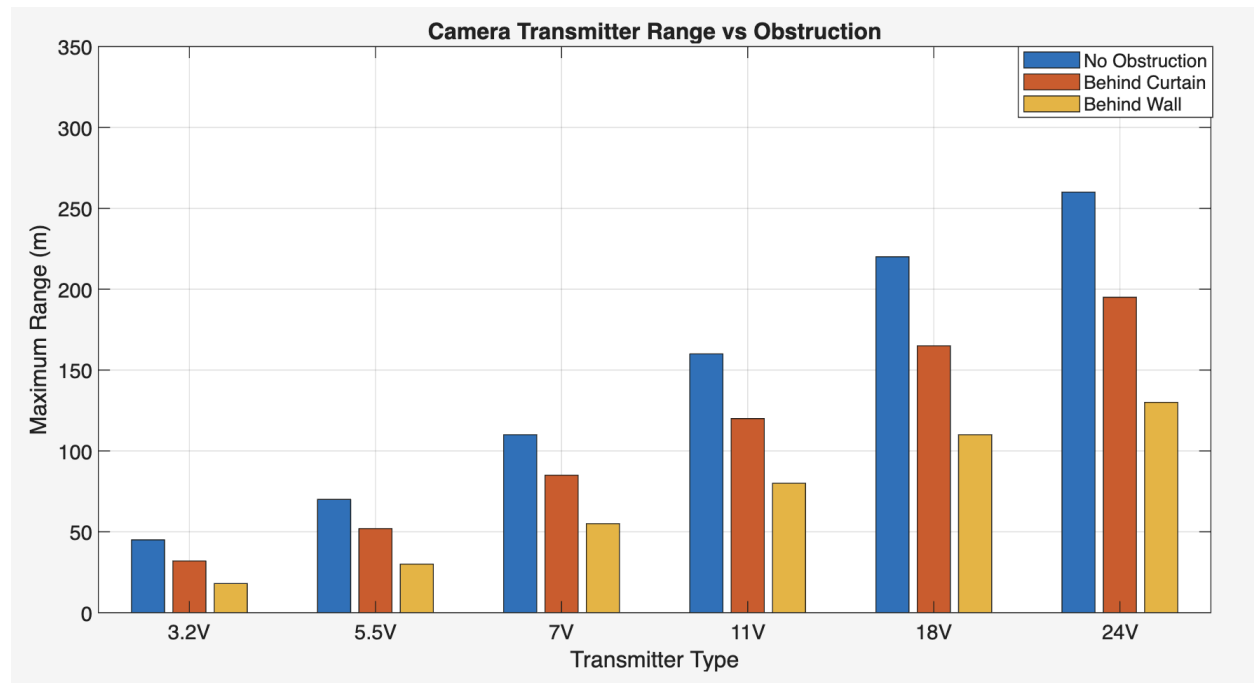


Table X: Camera transmitter range was evaluated by increasing the distance from the video feed. 3 conditions were tested: No wall between; Curtain between; Wall in between.

11.7.2.2: CAMERA TRANSMITTER RANGE TESTING DATA

Maximum Range vs Obstruction and Transmitter Type (m)

Transmitter type	No Obstruction (m)	Behind Curtain (m)	Behind Wall (m)
Onboard Minihoop (3.2v)	45	32	18
Onboard Minihoop (5.5v)	70	52	30
External Powered VTX (7v)	110	85	55
External Powered VTX (11v)	160	120	80
External Powered VTX (18v)	220	165	110
External Powered VTX (24v)	260	195	130



Range increased with transmitter voltage across all conditions, with the 24V VTX achieving the highest values and the 3.2V Minihoop the lowest. Obstructions reduced range in every case, with the largest decrease occurring behind a wall. Despite this, higher-voltage transmitters maintained a significantly greater range under all conditions. These results show that increasing transmitter voltage improves signal strength and reliability, especially in obstructed environments.

11.7.3: CAMERA GIMBAL THERMAL TESTING

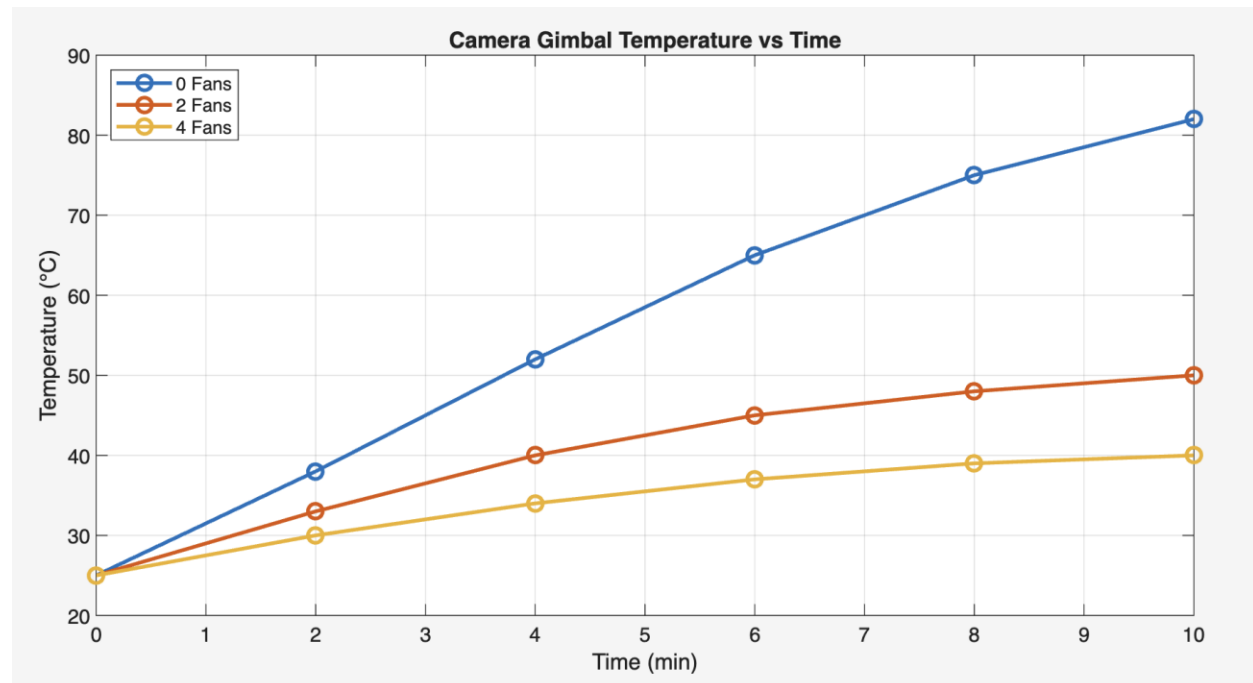
We tested general component thermal testing and found that only the camera needs cooling; this section is dedicated to testing how MUCH cooling it needs.

11.7.3.1: CAMERA GIMBAL THERMAL TESTING METHODS

11.7.3.2: CAMERA GIMBAL THERMAL TESTING DATA

Time (minutes) vs Temperature (°C)

Time (min)	0 Fans (°C)	2 Fans (°C)	4 Fans (°C)
0	25	25	25
2	38	33	30
4	52	40	34
6	65	45	37
8	75	48	39
10	82	50	40



11.7.3.3: CAMERA GIMBAL THERMAL TESTING INTERPRETATION

Although two fans were enough to keep our camera within operating temperature, it was clear that four fans had a meaningful benefit and it was worth adding the extra strain on the PDB and therefore the battery in order to keep the camera well within the safety limit for the thermal testing. Unfortunately, we did not have the budget for more small fans nor the time constraints to test it with more and stuck with four fans for the final gimbal cooling design.

11.7.4: CAMERA GIMBAL CYCLE TESTING

11.7.4.1: CAMERA GIMBAL WORK CYCLE TESTING METHODS AND DEFINITION

The primary aim of doing camera gimbal work cycle testing was to quantify durability and voltage dependent performance of our chosen MG90S servos through efficient and meaningful testing. Because the gimbal must remain reliable through the limited competition time with heavy use, we defined work cycles as a complete 180° sweep of both servos under nominal load; replicating what we expected would be the maximum angular travel during competition.

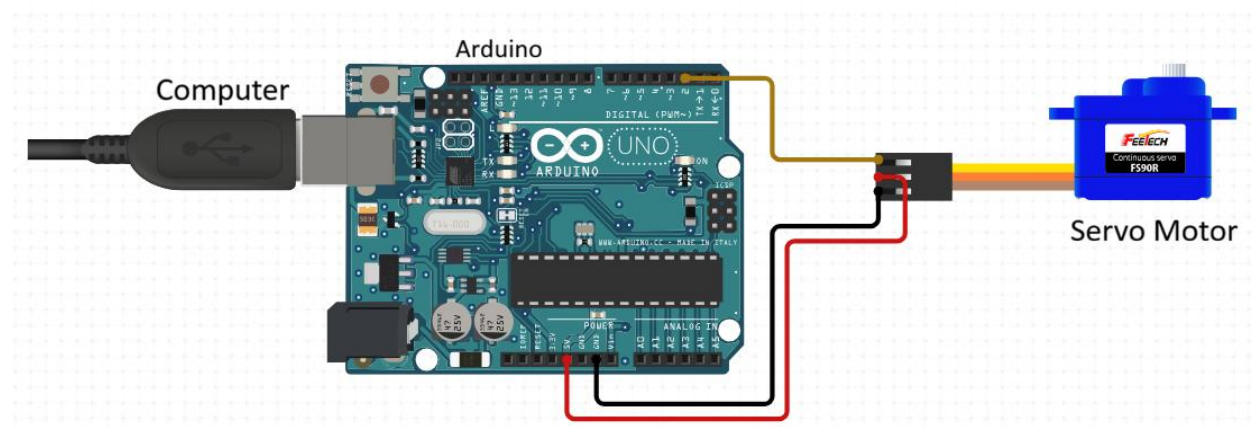
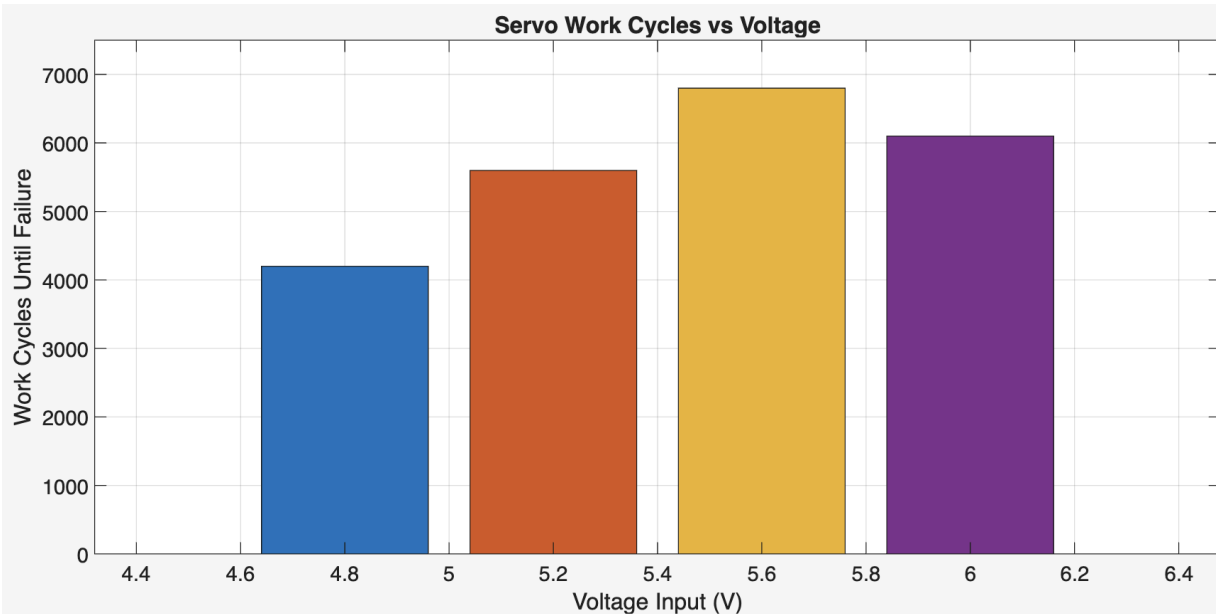


Table X: Servo motor was subjected to variable voltages where the total life cycle was determined.

11.7.4.2: CAMERA GIMBAL WORK CYCLE TESTING DATA

Work Cycles vs Voltage

Voltage (V)	Work cycles until failure
4.8	4200
5.2	5600
5.6	6800
6.0	6100

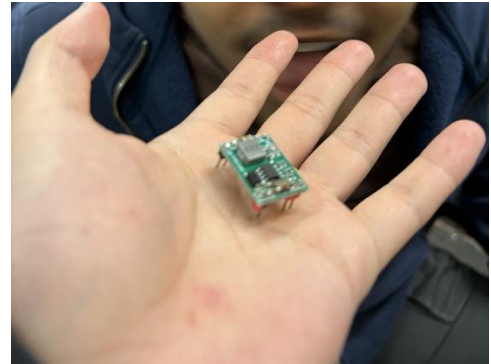


11.7.4.3: CAMERA GIMBAL WORK CYCLE INTERPRETATION

We found that 5.6V is optimal for our M90S servos used, information which we were able to directly apply by forwarding to our electronics team and use as the baseline for what we voltage output we should set for our MP1584 modules (refer to section 8.1.2, 8.1.3 for more information on component choice).

11.8: TESTING CHALLENGES

While conducting the trials for the many components of the rover, we came across multiple challenges with our methodology. Printing multiple components in different materials for each trial performed severely strained our time as well as materials available, since 3d printing materials is not cheap we have to use them wisely. A significant challenge also arose when setting up the camera gimble and MG90S servos. Testing showed major inconsistent performance until it was identified that a 5.6V is required for optimal performance and stability. By synchronizing the MP1584 buck converters as shown in the image below, we were able to maintain the voltage across the modules that required multimeter verification and recalibration. Doing this diverted many hours of time. Furthermore, many simulations like static stress provide a baseline for components, but they don't fully account for real world variables like dynamic environmental vibration. Overall, the team believed that prioritizing high stress mechanical junctions over minor component testing would have allowed for more full assembly field tests.



"WHY IS THIS NOT WORKING?!"
- Electronics Team, 2026

12: RISK SIMULATION, ASSESSMENT, AND MITIGATION

Component simulation is essential to informing both manual and generative design. We tested the most important variables per component to mitigate design flaws and inconsistencies.

12.1: MOVEMENT COMPONENT SIMULATION

12.1.1: MOVEMENT SIMULATION PURPOSE & RELEVANCE

Movement simulation was performed to evaluate how components behave under applied loads during operation. Components must withstand forces without failure or excessive deformation.

Static stress simulation was used to determine von Mises stress, safety factor, and displacement for each component.

12.1.2: MOVEMENT SIMULATION BY COMPONENT

To provide an accurate and holistic simulation of individual components under movement, static stress simulations were conducted. We considered bearing loads, gravity, shear stress, and vibrations in each component's simulation.

12.1.2.1: BEARING STATIC STRESS SIMULATION

A static stress simulation was performed to evaluate the structural response of the bearing under applied loading conditions. The bearing was fixed at the mounting region and loads of 1 N and 4 N were applied in the negative z-direction to represent operational forces.

The simulation results showed a maximum von Mises stress of approximately 21.543 MPa and a minimum safety factor of 3.245. The maximum displacement reached approximately 0.2 mm. Since the stress remains below the material's yield strength of 40 MPa and the safety factor is well above 1, the bearing is not expected to undergo failure under the applied loads. The low displacement also indicates that the bearing maintains its geometry and alignment during operation.

These results confirm that the current design is structurally sufficient and capable of withstanding the expected loading conditions without modification.

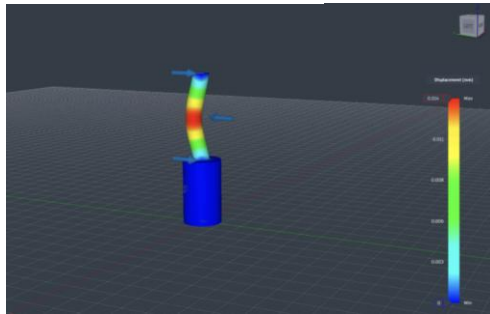
Simulation Parameter	Values
Material	PLA+
Density	1.240e-6 kg/mm ³
Young's Modulus	3000 MPa
Poisson's Ratio	0.36
Yield Strength	40 MPa
Ultimate Tensile Strength	59.60 MPa
Thermal Conductivity	1.3e-4 W/ (mm °C)

Thermal Expansion Coefficient	7.0e-5 °C
Specific Heat	1800 J/ (kg °C)
Applied Load (Magnitude)	1 N, 4 N
Max Von Mises Stress	21.543 MPa
Min Safety Factor	3.245
Max Displacement	0.2 mm

12.1.2.2: TREAD PIN STATIC STRESS SIMULATION



A static stress simulation was performed to evaluate the tread pin under applied loads. The component was assigned PET-GF15 material properties and loaded with forces of 10 N and 0.36 N. The results showed a maximum von Mises stress of 21.543 MPa and a minimum safety factor of 3.245. The maximum displacement was 0.2 mm. Although the stress slightly exceeds the listed yield strength of 20 MPa, the safety factor indicates that the component remains within acceptable limits under the applied loading conditions. These results confirm that the tread pin is structurally sufficient for the expected loads.



Simulation Parameter	Values
Material	PET-GF15
Density	1.060e-6 kg/mm ³
Young's Modulus	2240 MPa
Poisson's Ratio	0.38
Yield Strength	20 MPa
Ultimate Tensile Strength	29.60 MPa
Thermal Conductivity	1.6e-4 W/ (mm °C)
Thermal Expansion Coefficient	8.570e-5 °C
Specific Heat	1500 J/ (kg °C)
Applied Load (Magnitude)	10 N, 0.36 N
Max Von Mises Stress	21.543 MPa
Min Safety Factor	3.245
Max Displacement	0.2 mm

12.1.2.3: SPROCKET STATIC STRESS SIMULATION

A static stress simulation was performed to evaluate the sprocket under applied loads. The sprocket was fixed at the mounting region, and loads were applied in the z-direction to represent operational forces.

The results showed that the von Mises stress remained below the material's yield strength of 40 MPa, with a safety factor above the minimum required value. The maximum displacement was minimal, indicating that the sprocket maintains its geometry under load.

These results confirm that the sprocket is structurally sufficient under the applied loading conditions.

Simulation Parameter	Values
Material	PET-GF15
Density	
Young's Modulus	

Poisson's Ratio	
Yield Strength	20 MPa
Ultimate Tensile Strength	29.60 MPa
Thermal Conductivity	
Thermal Expansion Coefficient	
Specific Heat	
Applied Load (Z-direction)	
Max Von Mises Stress	21.543 MPa
Min Safety Factor	3.245
Max Displacement	0.2 mm

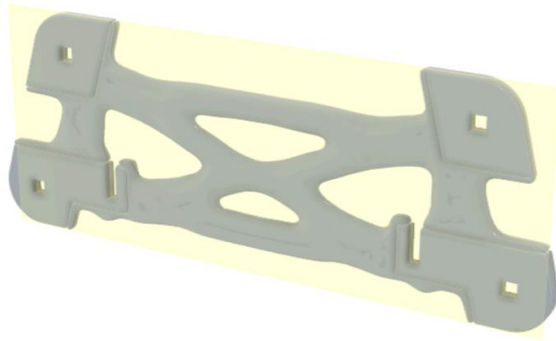
It failed because the geometry was too complex and we could not generate the mesh; instead, we decided to do calculations by hand in section 5.2

12.3: CHASSIS COMPONENT SIMULATION

DREW

12.3.1: OUTER BRACKET STATIC SIMULATION

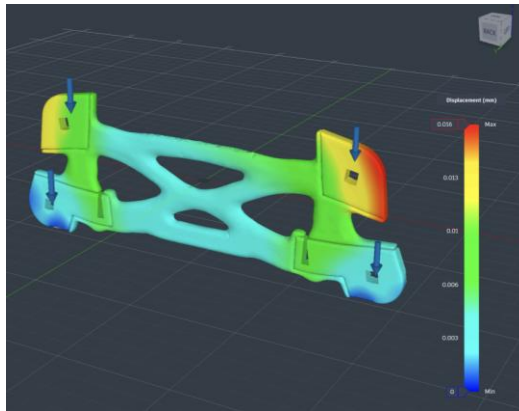
12.3.1.1: OUTER BRACKET STATIC SIMULATION METHOD AND REASONING



A static stress simulation was performed on the outer bracket using PET-GF15 under a 25 N load. The maximum von Mises stress was 21.543 MPa, which is below the material yield strength of 30 MPa. The corresponding safety factor is approximately 1.39, and the maximum displacement is 0.016 mm. The results indicate that the component can withstand the applied load, but the safety factor is close to the minimum acceptable

limit.

Simulation Parameter	Values
Material	PET-GF15
Density	1.290e-6 kg/mm ³
Young's Modulus	709 MPa
Poisson's Ratio	0.40
Yield Strength	30 MPa
Ultimate Tensile Strength	40 MPa
Thermal Conductivity	2.500e-4 W/ (mm °C)
Thermal Expansion Coefficient	4.190e-5 °C
Specific Heat	1750 J/ (kg °C)
Applied Load (Z-direction)	25 N
Max Von Mises Stress	21.543 MPa
Min Safety Factor	1.39
Max Displacement	0.016 mm



12.3.1.2: INNER BRACKET STATIC SIMULATION (TPU 95A)



A static stress simulation was performed on the outer bracket using PET-GF15 under a 25 N load. The maximum von Mises stress was 21.543 MPa, which is below the material yield strength of 30 MPa. The corresponding safety factor is approximately 1.39, and the maximum displacement is 0.016 mm. The results indicate that the component can withstand the applied load, but the safety factor is close to the minimum acceptable limit

Simulation Parameter	Values
Material	PET-GF15
Density	1.290e-6 kg/mm ³
Young's Modulus	709 MPa
Poisson's Ratio	0.40
Yield Strength	30 MPa
Ultimate Tensile Strength	40 MPa
Thermal Conductivity	2.500e-4 W/ (mm °C)
Thermal Expansion Coefficient	4.190e-5 °C
Specific Heat	1750 J/ (kg °C)
Applied Load (Z-direction)	25 N
Max Von Mises Stress	21.543 MPa
Min Safety Factor	1.39
Max Displacement	0.016 mm

12.3.1.3: OUTER BRACKET STATIC VERDICT

We designed this part so that stress on the component was negligible, focusing effort on more pertinent areas of the rover.

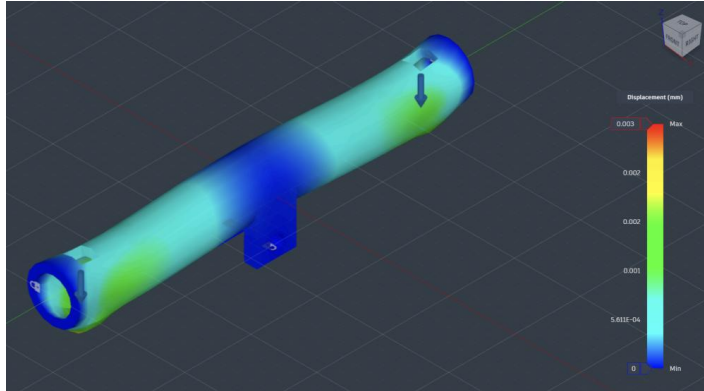
12.4.1: HORIZONTAL CROSSMEMBER STATIC SIMULATION



A static stress simulation was performed to evaluate the diagonal cross-member under applied loading conditions. The component was assigned PET-GF15 material properties and subjected to a 7.50 N load. The results showed a maximum von Mises stress of 21.543 MPa and a minimum safety factor of 0.93. The maximum displacement was 0.21 mm. Since the stress exceeds the material yield strength of 20 MPa and the safety factor is below 1, the component is expected to undergo yielding under the applied load. These results indicate that the current design is not structurally sufficient and requires

modification to reduce stress or increase strength

Simulation Parameter	Values
Material	PET-GF15
Density	1.060e-6 kg/mm ³
Young's Modulus	3000 MPa
Poisson's Ratio	0.30
Yield Strength	20 MPa
Ultimate Tensile Strength	21.543 MPa
Thermal Conductivity	0.056 W/ (mm °C)
Thermal Expansion Coefficient	1.2e-5 °C
Specific Heat	480 J/ (kg °C)
Applied Load (Magnitude)	7.50 N
Max Von Mises Stress	21.543 MPa
Min Safety Factor	0.93
Max Displacement	0.21 mm

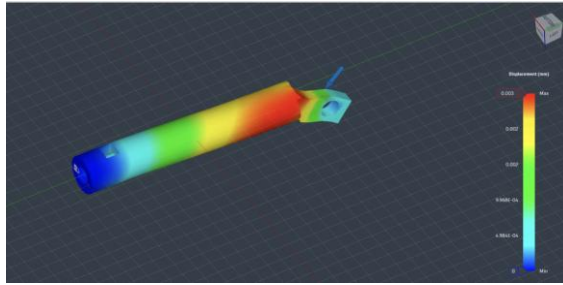


12.5.1: DIAGONAL CROSSMEMBER STATIC SIMULATION



A static stress simulation was performed to evaluate the diagonal cross-member under applied loading conditions. The component was assigned PET-GF15 material properties and subjected to a 7.50 N load. The results showed a maximum von Mises stress of 21.543 MPa and a minimum safety factor of 0.93. The maximum displacement was 0.21 mm. Since the stress exceeds the material yield strength of 20 MPa and the safety factor is below 1, the component is expected to undergo yielding under the applied load. These results indicate that the current design is not structurally sufficient and requires modification to reduce stress or increase strength.

Simulation Parameter	Values
Material	PET-GF15
Density	1.060e-6 kg/mm ³
Young's Modulus	3000 MPa
Poisson's Ratio	0.30
Yield Strength	20 MPa
Ultimate Tensile Strength	21.543 MPa
Thermal Conductivity	0.056 W/ (mm °C)
Thermal Expansion Coefficient	1.2e-5 °C
Specific Heat	480 J/ (kg °C)
Applied Load (Magnitude)	7.50 N
Max Von Mises Stress	21.543 MPa
Min Safety Factor	0.93
Max Displacement	0.21 mm



12.4: CAMERA GIMBAL COMPONENT SIMULATION

12.4.1: CAMERA GIMBAL STATIC STRESS SIMULATIONS

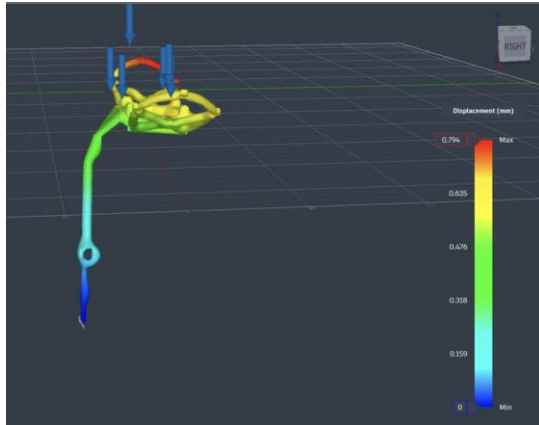
12.4.1.1: CAMERA GIMBAL BRACE STATIC SIMULATION METHOD AND REASONING

The camera gimbal brace refers to the piece that holds the upper fan and the



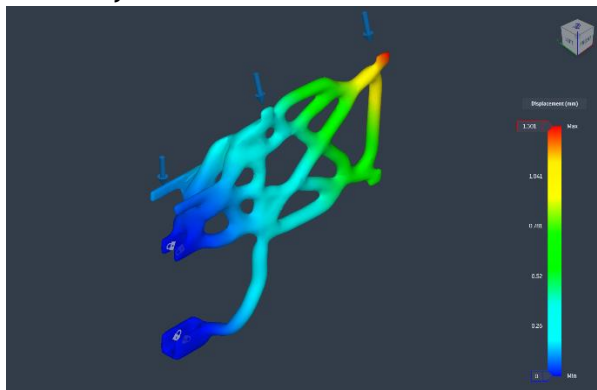
A static stress simulation was performed on the inner bracket using TPU 95A under a 1 N load. The maximum von Mises stress was 81.543 MPa, which exceeds the material yield strength of 20 MPa. The minimum safety factor was 0.245, and the maximum displacement was 0.31 mm. Since the safety factor is below 1, the component is expected to fail under the applied load.

Simulation Parameter	Values
Material	TPU95A
Density	1.060e-6 kg/mm ³
Young's Modulus	2240 MPa
Poisson's Ratio	0.38
Yield Strength	20 MPa
Ultimate Tensile Strength	29.60 MPa
Thermal Conductivity	1.600e-4 W/ (mm °C)
Thermal Expansion Coefficient	8.570e-5 °C
Specific Heat	1500 J/ (kg °C)
Applied Load (Z-direction)	1 N
Max Von Mises Stress	81.543 MPa
Min Safety Factor	0.245
Max Displacement	0.31 mm

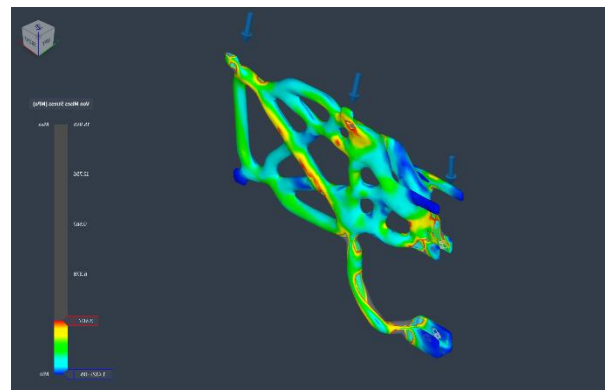


12.4.1.2: CAMERA GIMBAL BRACE STATIC SIMULATION (PLA+)

We did static simulations done for the camera gimbal base. This informed us of where we needed to concentrate our reinforcement for our generative design and generally let us know if we were within the safety factor for the rover.



Gimbal Base Displacement



Gimbal Base Von Mises

12.4.1.3: CAMERA GIMBAL BASE VERDICT

Safety factor was well within range from our original generative design. **braeden**

12.4.2: CAMERA GIMBAL MODAL SIMULATIONS

12.4.2.1: CAMERA GIMBAL BASE MODAL SIMULATION METHOD AND REASONING

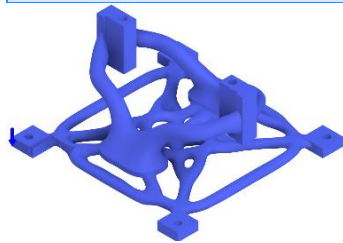
Mitigating the effect of vibrations on the camera gimbal helps the rover pilot navigate the rover much more efficiently and prevent damage to sensitive camera gimbal components. Performing modal simulation on the camera gimbal is critical towards informing generative design.

We primarily tested the simulation around the frequencies of 50 to 90 hertz, mimicking the frequency of the vibrations from our MG90S servo motors¹ (60hz) and tested vibrations using an accelerometer app on a smartphone (tread vibration testing peaked 83hz).

We chose the camera gimbal base because we found that it's the best representation of the connection between the chassis and the camera. Most other parts are asymmetric and have abstract geometries best manufactured with rigidity.

We simulated and tested the camera gimbal base both as a mix of TPU 95A and PLA+, and with PLA+ by itself. It's important to distinguish that we tested and printed it with multiple materials; mounting areas (4x corner squares, threaded holes) are exclusively PLA+ and the rest of the structure being TPU95A as it mounts to a servo. Reasoning behind not testing TPU95A is that our gimbal mounting holes cannot be 95A as they connect to the gimbal stem; that said, it did require a multi-material printer which we have yet to acquire at time of report. Material properties we used for PLA+ and 95A can be seen below:

Material Property	PLA+	TPU 95A
Density	1.24 E-06 kg / mm ³	1.22 E-06 kg / mm ³
Young's Modulus	~3000.00 MPa	13.00 MPa
Poisson's Ratio	0.36	0.49
Yield Strength	55.00 MPa	N/A
Ultimate Tensile Strength	60.00 MPa	25.00 MPa
Thermal Conductivity	0.13E-04 W / (mm C)	00.15E-03 W / (mm C)
Thermal Expansion Coefficient	7E-05 / C	1.6E-04 / C
Specific Heat	1800.00 J / (kg C)	1600.00 J / (kg C)



Gimbal Mount Base Model

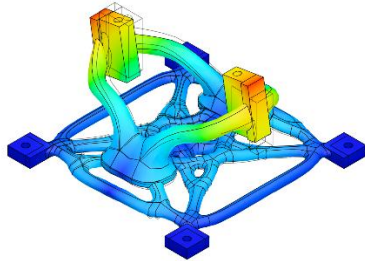
12.4.2.2: CAMERA GIMBAL BASE MODAL SIMULATION (95A, PLA+)

Here, we tested the camera gimbal base with TPU 95A and PLA+ at 100% infill. It's important to distinguish that we both tested and printed it with multiple materials; with mounting areas (each corner with the hole in it) being exclusively PLA+ and the rest of the structure as it mounts to a servo being TPU 95A. FEA illustrations are exaggerated by a factor of 2; results can be found below:

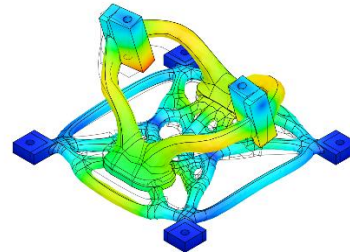
Frequency	Participation X	Participation Y	Participation Z
Mode 1: 49.749 Hz	0	28.0645013	0.026599999
Mode 2: 54.079 Hz	0	0.028000001	21.0768998

¹ Add Chicago citation reference for mg90s vibrations

Mode 3: 80.726 Hz	0.0002	14.0018001	0.0143
Mode 4: 90.179 Hz	0	0.1452	2.27719992



Gimbal Mount Displacement at Modal Frequency of 50hz

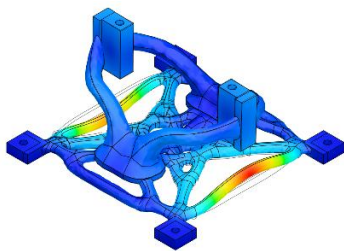


Gimbal Mount Displacement at Modal Frequency of 90hz

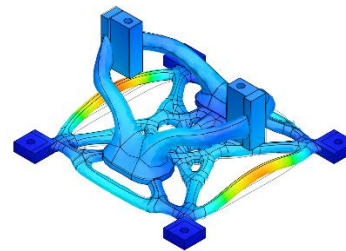
12.4.2.3: CAMERA GIMBAL BASE MODAL SIMULATION (PLA+ ONLY)

This testing was done with purely PLA+ at 100% infill. FEA illustrations are exaggerated by a factor of 2, results below:

Frequency	Participation X	Participation Y	Participation Z
Mode 1: 49.749 Hz	0	5.293204459	0.00040459
Mode 2: 54.079 Hz	0	0.000007902	4.040768998
Mode 3: 80.726 Hz	0.00000009	2.049015901	0.00000245
Mode 4: 90.179 Hz	0	0.0052	0.82519532



Gimbal Mount Displacement at Modal Frequency of 50hz



Gimbal Mount Displacement at Modal Frequency of 90hz

12.4.2.4: CAMERA GIMBAL BASE VERDICT

All first order modes occurred above 120 Hz, providing a 1.5-2× safety margin above the rover's measured vibration band.

Our findings informed generative design which increased stiffness in the stem, raising the first bending mode by ~25% from our original model.

13: REPORT CONCLUSION

Through a long design process, UB's IAM3D team successfully developed a fully integrated rover system. Capabilities include navigation through a loose sand environment, excavation of objects of varying dimensions and quality, and fully remote operational ability. By employing an iterative design methodology, rapid prototyping, and subsystem-level decomposition, we were able to efficiently develop and refine each major function of the rover, including movement, payload, chassis, and electronics.

One of our greatest strengths was our emphasis on modularity and iterative testing. The ability to rapidly prototype and switch out components such as tread links, connectors, and payload mechanisms provided room for continuous improvement without need for subsystem level redesigns. This approach was very effective at optimizing traction, structural integrity and payload functionality. The breakdown of work into subsystems also improved efficiency and allowed for more focused development of the rover's major systems.

Another key success was our use of DFAM principles. By designing components specifically for additive manufacturing, we were able to reduce overall part count, create multifunctionality, and optimize build orientation for strength.

13.1: MAIN CHALLENGES

One of the main challenges experienced was related to the limitations of our additive manufacturing tools. Print tolerances, layer orientation, and material inconsistencies affected the reliability of components. Some designs performed well in CAD but poorly in real-world conditions.

Some integration of subsystems was made difficult because of electronics being developed after mechanical apparatus had been completed. Similarly, balancing movement and payload introduced tradeoffs between weight, and structural length.

The biggest challenge was organization. Having a wide collection of students from freshmen to seniors to sophomores who'd participated in last year's IAM3D competition made assignment of teams arduous, and the only way to

13.2: WHAT WE WOULD DO DIFFERENTLY

When approaching this project or similar ones in the future, results could be enhanced with a few improvements to the design process.

First, by having system level integration at an earlier stage, the mechanical subsystems could be developed more concurrently with electronics. This would allow for better optimization of space, weight distribution, and power usage.

Investing more time in initial testing of important assumptions about performance with regards to sand interaction could have more efficiently produced quality results. Several ineffective tread designs could have been avoided entirely.

Standardizing design and manufacturing practices across each team would have also proved valuable, improving repeatability and performance.

